



FormWorks 2021 Q4

Administration - Editor

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Applicability

This document applies to Formworks 2021 Q4

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Contents

Getting Started.....	12
Key Terms	12
File Descriptions.....	16
Documented Components	17
<i>Process Configuration Modules</i>	<i>17</i>
<i>Resource Configuration Modules (PCM)</i>	<i>18</i>
Overview of FWEditor Window	19
<i>The FWD Tree</i>	<i>20</i>
<i>FWD Tree Component Terms.....</i>	<i>21</i>
<i>The Configuration Window.....</i>	<i>22</i>
<i>The Message Window.....</i>	<i>23</i>
Plugin Configuration Module(s).....	26
<i>PCM Resource Types.....</i>	<i>26</i>
<i>PCM Process Types</i>	<i>26</i>
<i>Events</i>	<i>27</i>
<i>Information Types</i>	<i>27</i>
Information about Building a New System	28
License Information.....	29
License Key Utility	29
Licence Key File (Structure).....	29
<i>CustomerID Attribute.....</i>	<i>29</i>
<i>[FWEdit] Attribute</i>	<i>29</i>
<i>Key Attribute.....</i>	<i>29</i>

<i>License Key Generation</i>	29
Register the License Key in FWEditor	30
Open A FormWorks Database	30
<i>Create a FormWorks Editor Icon</i>	31
<i>Set a Default Database</i>	32
Build a .CFD	32
To Build A CFD	34
<i>Edit the System Name</i>	37
<i>.SFD Lock File</i>	38
<i>SFD Backup File(s)</i>	38
Create 4df files	39
<i>To Generate 4df Files</i>	40
Resources	44
Resource Folders Creation	44
<i>To Create A Resource Type Folder</i>	44
<i>To Manage Folders</i>	45
Internationalization	46
<i>Editor Configurations for Internationalization</i>	47
<i>DateFormat Resource</i>	54
<i>Entry Grid Resource</i>	56
Fileref Resource	63
<i>To Create a Fileref</i>	63
Function Resource	64
Create and Configure an Inventory Resource	64

Inventory Resource	66
<i>Connection</i>	66
<i>Inventory Server</i>	66
<i>Inventory Database</i>	67
<i>Server Extension DLLs</i>	67
<i>Client</i>	67
<i>Attributes</i>	70
<i>To Create an Attribute</i>	71
DCN Groups	77
<i>To Create A DCN Group</i>	78
Option Groups	79
<i>Option Groups - Create A Decision Tree</i>	79
<i>Add a New Group</i>	80
<i>To Add an Option to a Group</i>	82
<i>To Enable Options</i>	83
<i>To Associate a Batch with an Option</i>	84
Batch Level Inventory Options	84
<i>General Tab</i>	85
<i>Enabled Attributes Tab</i>	85
<i>Default Overrides Tab</i>	86
<i>DCN Tab</i>	87
<i>The Print Option</i>	87
<i>Print Script</i>	88
<i>Version</i>	89

OCRepair Color	89
<i>Remove Color</i>	91
<i>Add Custom Color</i>	91
OCRepairFieldClass	92
<i>To Create a New Field Class</i>	93
Output Driver Resource	94
<i>Add/Remove Output Drivers</i>	95
<i>Rule DLL Resource</i>	112
<i>Store DLL Resource</i>	114
<i>Table Resources</i>	115
<i>Indexed File Tables</i>	120
<i>TRK Resources</i>	122
Pages	127
Create Pages	128
<i>Copy a Page</i>	128
<i>Create New Page</i>	128
Create/Add a Graphical Variant to a Page	130
Select/Replace Variant Image Template	132
Add Fields to a Page	133
<i>Field Lists - Overview</i>	133
<i>Shortcuts in the Field Lists</i>	134
<i>Text Fields</i>	134
<i>Optical Mark Recognition Fields (OMR)</i>	135
Advanced Options	138

<i>Dynamic Field Creation</i>	138
<i>Deleting a Field</i>	139
<i>To Delete an OMR Subfield from a Page</i>	140
<i>To Remove a Field from a Variant</i>	142
<i>To Remove a Subfield from a Variant</i>	143
<i>Reordering Fields and Subfields</i>	143
<i>Rename a Field</i>	144
Define Bounding Rectangles.....	145
<i>To Define Bounding Rectangles</i>	145
<i>Enable/Disable Snap Feature</i>	148
<i>Tips for Defining Bounding Rectangles</i>	149
Add Page Variants	149
<i>To Copy an Existing Variant</i>	149
<i>Rename Pages and Variants</i>	150
<i>Delete Pages and Variants</i>	151
Working with Documents	153
Create a New Document.....	153
<i>To Create a New Document</i>	153
<i>Add Document Level Fields</i>	154
<i>Add and Remove Document Pages</i>	155
<i>Delete a Document</i>	155
<i>Rename A Document</i>	156
Batches.....	157
Create a New Batch	157

<i>To Create a New Batch</i>	158
Adding and Removing Documents.....	159
Processes.....	160
Create A New Processing Step.....	160
<i>To Create A New Processing Step</i>	161
Configure A Process	161
<i>AC Process</i>	162
<i>Collator Process</i>	165
<i>FIP Process</i>	171
<i>Inventory Process (Optional)</i>	173
<i>OCR Process</i>	176
<i>Store Process</i>	184
TwainScan Process.....	191
<i>TWAINScan Process Configuration</i>	192
Configuration Page and Page Variant Processing.....	209
Open a Configuration Window (Page Level).....	209
AC: Page Processing	209
<i>To View Rules for a Page</i>	210
<i>To View Rules that Reference a Specific Field</i>	210
Adding New Rules.....	212
Adding New Sub-lists	214
<i>To Add A New Sub-List</i>	214
<i>Deletion of Rules (or Sub-Lists)</i>	216
<i>Disable a Rule</i>	216

<i>Move a Rule</i>	217
<i>Replace Fields, Tables, or Sources (Filerefs) in Rules</i>	217
<i>Unused Functions/Options</i>	218
FIP: Page Field Processing.....	218
<i>To Open The FIP: Page Field Window</i>	218
<i>Filter</i>	223
Configure Regions for the Optical Character Recognition (OCR) Worker	225
<i>Create an OCR Process</i>	225
<i>Manage Regions - OCR</i>	225
<i>Selection of Regions</i>	227
OCR: Page Field Processing	228
<i>To Open the OCR Page Field Window</i>	228
<i>Number of Lines</i>	234
<i>Regions Settings</i>	234
<i>OCRepair - Page Processing</i>	235
<i>Store: Page Processing</i>	237
<i>Page Variant Processing</i>	237
<i>FIP: Variant Page Processing</i>	238
FIP: Variant Image Layout	242
<i>How to Define Non-Dropout Forms</i>	244
<i>Dropout Regions</i>	245
<i>Other Form Definition Options</i>	247
<i>FIP: Variant Field Processing</i>	251
<i>OCR: Variant Field Processing</i>	253

Configuration of Document and Batch Processing	256
Collator: Document Processing.....	256
AC: Document Processing	258
<i>To Open the AC Configuration Window.....</i>	<i>258</i>
Store: Document Processing	259
Collator: Batch Processing.....	261
FIP: Batch Processing.....	267
FIP Batch Script	267
Description of Typical FIP Steps.....	270
FIP Batch Processing (PageID)	273
Scan: Batch Processing	274
Scan Batch Configuration Category/Item Definitions.....	276
Appendix A: New System Checklist, Tips And Troubleshooting	283
New System Checklist.....	283
Preliminary System Setup	284
System-Specific Configuration	286
Create Pages and Variants	287
Create Documents	288
Create Batches	289
Tips for Configuring a System.....	289
Defining Bounding Rectangles.....	289
FIP.....	290
OCR	291
Troubleshooting.....	291

Looping..... 291

Getting Started

The FormWorks Editor is a GUI (Graphical User Interface) tool that is used to define objects (pages, documents, and batches) in conjunction with the processing steps and resources associated with those objects in a FormWorks 5.0 system. The Editor provides a framework for maintaining common configuration information for each object; and the information is stored in the FormWorks database (.FWD). Process-specific and resource-specific configurations are maintained by multiple Plugin Configuration Modules (PCMs) which are automatically loaded by the Editor when it is opened.

Note: The encryption protocols, keys and strategies have been upgraded for the FormWorks system in this release for increased data security. Refer to System Administration user guide for more information

Note: Key Terms, Definitions, Configurations, Process and Functions described within this document are in reference to the base FWEdit functionality.

PCM Modules may alternately be referenced by their individual process name within this document

Key Terms

Term	Definition
AC	Auto Capture, is a specialized version of the current Data Validation (DV) worker. It consists of a rule processing engine; plug in rules, tables, and a worker shell. The same engine plugs into KE and should be able to be plugged into virtually any worker. The AC Worker may be used for both validation and for auto capture processing. The difference between auto capture and data validation are the types of rules implemented and the position in the system route. Generally, FormWorks 5 systems are structured so that references to ACI are to Auto Capture before OCRepCrit and ACII refer to Auto Capture after OCRepCrit.
AEG (OpenText)	Software character recognition engine (company originally Advanced Engineering from Germany, now AEG is a product of OpenText).
Agent	The System Administration service that is part of FormWorks 5. Agent provides functionality not provided in prior FormWorks versions but also replaces several contrasting utilities (JumpStart, FMD, Table Updater, and Config Server) with a single application. Functionally, Agent is a central management point of a FormWorks 5 system. Agent controls file and configuration updates to the individual machines and also runs as a service allowing Navigator to control remote workers and provide access to remote log files. <i>Note: On an individual's machine where a non-web process may run, LaunchPad will connect to the Agent to request for processes to start and stop (SQA Viewer, Inventory Client, and Scan).</i>

Term	Definition
Batch	A FormWorks Batch is a group of documents that can have one or more types of documents. An operator enters a batch into the system using an Inventory client to specify various options, and to enter values for batch attributes. These options and attributes are configured through the Inventory Resource PCM. Typically, a batch is also the grouping of paper placed into a scanner as a single unit of work.
BWS	Background Worker Shell is an executable capable of loading and running a custom worker DLL. The DLL contains all of the information necessary to process its work; the BWS only oversees the management of resources. The BWS is by design to be utilized only by approved FormWorks support personnel.
CFD	Compressed FormWorks Database; distributed to each worker in a Formworks system. The CFD is the file loaded and run by those workers with a PCM in a FormWorks system. The CFD contains the business rules and other worker settings specific to a particular system.
Collator	This worker assembles the batch into document(s) by grouping pages scanned with the same DCN together. The information is written to the work record of the batch.
Component	A unit of software that is produced by the FormWorks Development Team for the FormWorks application.
Data Validation	The role of Data Validation (DV) is to evaluate field contents in context and determine whether or not the data is logical.
DCN	Document Control Number - A unique number assigned to each document in a system, usually imprinted on the form as its scanned. The DCN is reserved by Inventory and assigned by Scan (for scanned documents).
Deskew	When scanning in documents one thing that could happen is the possibility of pages not feeding properly, so some of the images will be appear slanted. Deskew is a feature that resolves this problem. After scanning in a document, if the document's text or images are at a slight angle, the FormWorks software will auto-rotate the document so that the alignment is corrected.
Document Fields	Document level fields are fields that are attached to the document as a whole, rather than to a specific page of a document. Document level fields are all non-imaged, since they could occur anywhere within the document.
FIP	FIP (Forms Image Processing) is the FormWorks component that identifies forms and assigns PageIDs. FIP also reads any checkboxes (also known as OMR or Optical Mark Recognition), removal (form instructions, lines), and deskews scanned images. FIP manipulates images to have the best quality for OCR (reads text on the page) and performs the mark recognition. It is the responsibility of the FIP worker to register and clean-up images. The FIP worker generates the Removed Image, OCR, and History file for each document.

Term	Definition
FWD	FormWorks Database; contains configuration information for a FormWorks system. The FWD is the uncompressed FormWorks database used by FWEEdit. FWEEdit is responsible for creating the .CFD database format.
FMD	The File Migration Daemon is a service imbedded in the Agent. When the LWM receives work records, the FMD is responsible for moving all required batch files (images, OCR, and history) to the local host. <i>Note: The FMD was a standalone process in FormWorks 4 and prior versions.</i>
FWEEdit, FWE; Editor	FormWorks Editor drives much of the system and uses many flexible functions to assist in creating rules. It is through FWEEdit that users may configure the system.
FW5	FormWorks 5
JDAY	Julian Date, consisting of a year and a day number starting at 1 for each new year and incrementing for each subsequent day throughout the year. For example, February 1, 2012 is the 32 nd day of the year, so the JDay would be 12032.
KE	This worker presents any fields that were rejected by a business rule to an operator for correction via a graphical user interface. An operator may then correct the data, override the reject, or reject the document. Any table lookup that could not auto plug is to be completed by the operator choosing an entry from a selection list. Most often, KE is the second keying step (after KFI) within a typical system configuration.
KFI	Key From Image, a type of KE. The interface is identical to KE except there is no information in the data window from OCR. KFI is used to key images that failed to register properly at FIP. At KFI, the entire form is keyed. <i>Note: While not common, in some systems there is not an actual KFI step, instead, formats that fail FIP will be Keyed From Image in a normal KE step.</i>
LWM	The Local Work Manager component runs as a service on all FormWorks hosts (each host will have its own LWM). The LWM receives work records from the MWM and distributes the records to the appropriate worker on the host on which the LWM runs. The LWM service must be running for any worker to run.
MWM	Master Work Manager is a component run on the QManager machine. It is responsible for assigning batches to workers, tracking the status, location of all batches, and monitoring the status of all hosts and workers. The MWM must be running for work to move through the system. For a given system, there may be only one MWM.
OCR	Optical Character Recognition, this worker attempts to read the data in a monochrome TIFF image. The system stores read data and confidences of the data to the OCR file. OCR occurs after FIP.
OCRepair	An interactive worker to supplement the OCR engine in the Data Capture process. Any character "guess" from OCR that does not meet the Confidence

Term	Definition
	Threshold or is less than the Confidence Span from the next "best guess" is presented to an operator for correction via a GUI.
OCRepCrit	OCRepair Critical, an optional stage (located between ACI and ACII) where an implementation of OCRepair is used to repair the "critical" fields, (a "critical" field usually means that the field is an essential part of a larger lookup). For example, the Member ID, DOB and Last Name are repaired. The system matches this information within the database negating the need to repair the address, first name, etc.
OCRepSig	OCRepair Signature is for entering signatures only. It is most often used to capture Box 31 on a HCFA for database matching purpose. OCRepSig is not part of a standard FW5 configuration.
OMR	Optical Mark Recognition; OMR fields are fields that consist of checkboxes instead of text. OMR analyzes the image of a form and determines if a checkbox has been filled in. OMR is performed by FIP.
PCM	Plugin Configuration Module or sometimes called the Process Configuration Module; a PCM module is loaded by the Editor to manage the configuration of a process or a resource in a FormWorks system. Essentially, a PCM is a DLL with specific entry points the Editor uses to load and invoke the module. A single PCM may only support one Processing type, but it may support any number of processes or resources of that type.
Processing Step	Each processing step is a distinct stage, in which some operation is performed on the data.
Processing Type	The Processing Type describes the nature of the work being performed at a given step. The Processing Type determines which FormWorks worker is used at a particular step, and which PCM is used to maintain the configuration for that step.
RCM	Resource Configuration Module; a PCM for a Resource (see PCM).
Resource	A configuration object that is shared by one or more processes. Each resource has a basic type that defines and describes what the resource is (Fileref, Function, Table, etc.).
Resource Type	Every resource in a FWD has a resource type that describes what it is (Fileref, Table, etc.). The resource type determines which RCM is responsible for maintaining its configuration.
SFD/STC Plus	The SFD is the compressed version of the FWD (also referred to as STC Plus) intended to increase the speed of loading, saving, exiting the FormWorks Editor) and when called is used to build or fulfill specific functions. The FWEditor may always access the SFD. The STC library is responsible for reading and writing all three forms of the FW database (SFD/CFD/FWD) and also for providing a generic interface to each.
Slot	A slot is a location for a piece of data to output. Slots are defined/named in a template set resource.

Term	Definition
Store	A background worker responsible for the output of a system. Store may be configured to output flat files, 837 files, database output, XML, .TIFF (and other formats). A batch's state is typically changed to "Retired" after a batch has moved through the Store worker.
Variant	A graphically different version of a page. Each page object in the FWD may have a single variant or multiple variants.
Worker (Background Worker)	Worker/Background Worker Workers refer to interactive workers involving some degree of human interaction. Examples of Workers are KE, KFI (and others). Background Worker (modules) are unattended workers (no human interaction), typically running at all times (may have user configured sleep times). Background workers include FIP, OCR, Collator, AC-I, AC-II, DV, Store, Historian, SQAC, StatStore, Inventory Server (among others).
Word Record	A unit of work that passes through the FormWorks system. A Batch can be composed of one or more Work Records and may be split by client needs based upon: Number of Documents. Particular routing needs or requirements. Exception processes.

File Descriptions

Term	Definition
Work Record File	Serves as a "Master File" for each Work Record; every Work Record has one Work Record File(WRF). The Work Record Files contain information at the record level followed by a series of Items and information about those items. Record – General batch information such as the Batch Type, Inventory information, and Routing information. Item – Typically each page of the work record DCN (Document Control Number) and PageID, Deskew information and other page or document attributes. Item 0 represents the Work Record itself. Subsequent Items contain information about the contents of the Work Record (documents, pages, etc.), A typical collated work record of single-page documents encompasses the following item structure.

Term	Definition
	Item 0 – The Work Record. Item 1 – Document 1. Item 2 – Page 1 of Document 1Page 1 of Document 2. Item 5 – Document 3. Etc.
OCR File	Created at FIP. The OCR file contains the claim data and is modified by any worker that reads or changes data (including workers that perform table lookups). Additionally, OCR files (as well as images and history) move only when the batch is assigned to a new worker. Files move to the local host "d:\rri\systemname\fmdfs" directory.
Scanned Image (Original)	The image produced by the scanner.
Removed Image	Deskewed image that results from FIP. The removed image is used by OCR and Keying applications.
History File	Each page has one history file. Each worker appends the changes it has made to form an audit trail at the end of the process. The Historian worker consolidates histories for archiving and later retrieval by History Viewer. A typical configuration is to store history one history file per JDAY.

Documented Components

This document discusses using the Editor and PCMs to create and maintain common configuration information for objects in a FormWorks system. The following table lists the PCM versions covered in this document.

Process Configuration Modules

Note: PCMs and their version numbers defined in the following table are the minimal required versions to properly configure and run FWEdit FormWorks 5.16.1.0.

Component/PCM	Version	Description/Version Detail
FormWorks Editor	Version 1.5	Build 329
AC	1.1	AC Config (Version 12.1.1 Export 365)
AEG	5.0	AEG + CGK/océ Configuration (Export 53)
Collator	1.5	Collator PCM Version 1.2.0 Build 101
FIP	1.0	FIP configuration (Export 62)

Component/PCM	Version	Description/Version Detail
GenPickup2	1.5	GenPickup2 PCM Version 1.0.0 Build 40
Inventory	1.5	Inventory PCM Version: 1.5.0 Build 122
KE (Key Entry)	1.1	For Configuring KE5
OCRepair3	0.0	Edits OCRepair3 Configurations (Export 36)
Render	2.5	Render Config (Version:24)
Scan	1	SCAN Configuration (beta 13)
TWAINSCAN	1.1	For Configuring TWAINSCAN
Store	1.1	Store Config Plugin (Version: 12.1.1 Export: 365)

Resource Configuration Modules (PCM)

Component/PCM	Version	Description - Version Detail
DateFormat	0.0	Date Format (Version: 12.1.1)
EntryGrid	0.0	Entry Grid Config for KE5
FileRef	1.5	FileRef Resource configuration (Export 35)
Function	1.1	User Functions (Version 12.1.1 Export 365)
InputDriver	1.5	InputDriver Resource Configuration (Export 17)
Inventory	1.5	InventoryRCM Version 1.5.0 Build 122
Language	1.0	Language Resource Configuration (Export 10)
OCRepairColor	0.0	OCRepair3 Color Resource Configuration (Export 36)
OCRepairFieldClass	0.0	OCRepair3 Field Class Resource Configuration (Export 36)

Component/PCM	Version	Description - Version Detail
OutputDriver	2.0	Store2 Output Driver Configuration (Export 92)
RuleDLL	1.1	Rule DLLs (Version 12.1.1, Export 365)
StoreDLL	2.0	Store2 Output Driver DLL (Export 92)
Table	1.1	Tables Config (Version 12.1.1, Export 365)
TRK	0.0	TRK Resource Configuration (Export 20)

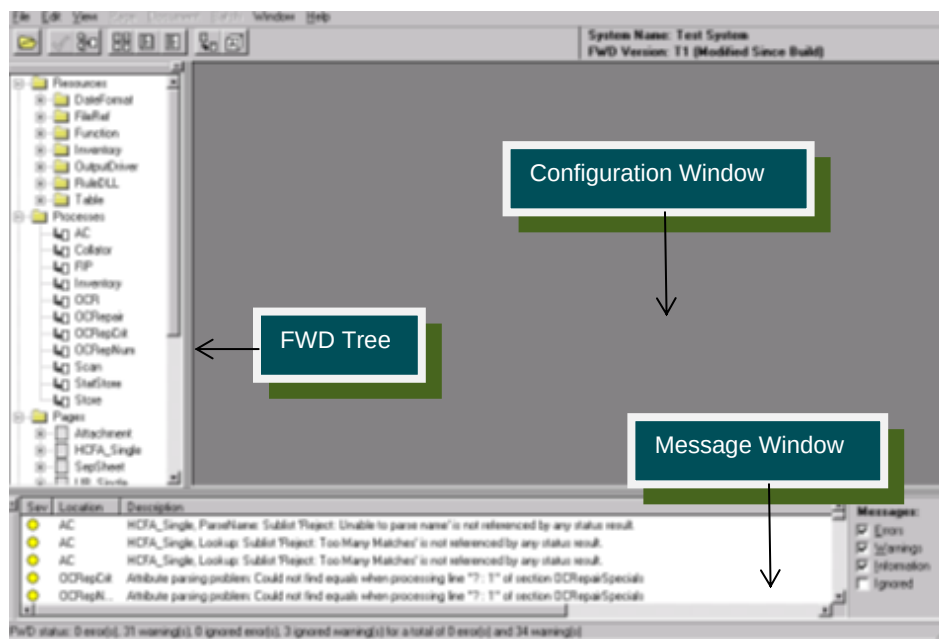
Note: A single PCM may control only a single process type. However, a PCM may control multiple processes.

For a new system setup and configuration refer to [Appendix A - New System Checklist, Tips And Troubleshooting](#).

Overview of FWEditor Window

When opening the FormWorks Editor, a three-part window is displayed.

1. FWD Tree
2. Configuration Window
3. Message Window

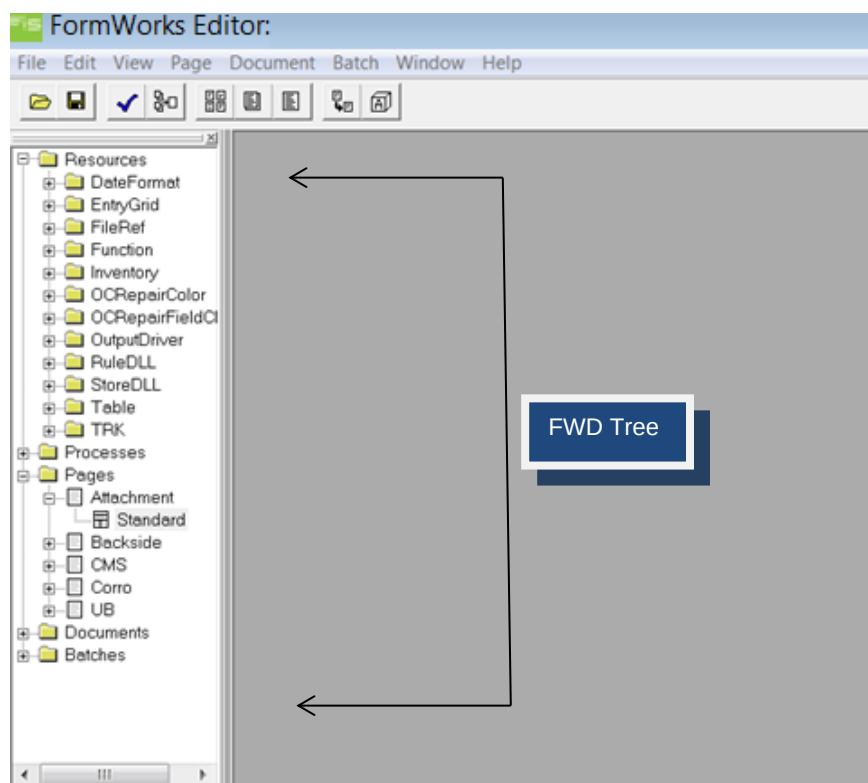


Selection of the Close Window icon  from within numerous configuration windows and fields (drop down boxes, etc.) will in many instances automatically save the contents of the window.

A best practice is to note the individual window and field selections in the event the original settings should require restoration to their prior settings.

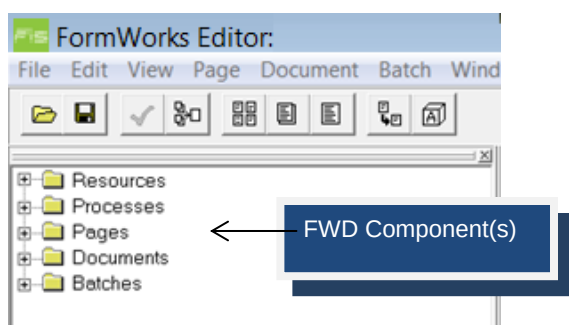
The FWD Tree

Every FormWorks database or FWD has common components that are managed through the FormWorks Editor. Each managed component is listed in the FWD Tree in the Editor window.



*Note: To Show or Hide the FWD Tree, press **CTRL+T***

To access a component, double-click the component category in the tree to access the Configuration Window. Or, select the  sign next to a folder to expand the tree view for more detail.

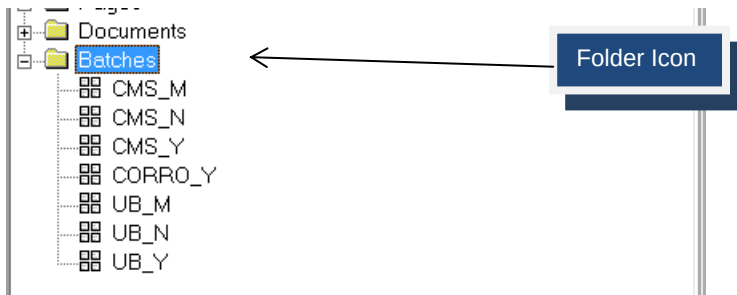


FWD Tree Component Terms

Term	Definition
Variant	A graphically different version of a page. Each page object in the FWD may have a single variant or multiple variants.
Resource(s)	A resource is a configuration that may be shared by one or more processes. Each resource has a basic type that defines and describes what the resource is (Fileref, Function, Table, etc.).
Processes	Each processing step is a distinct stage, in which some operation is performed on the document. The FWD may contain Inventory, Scan, FIP, OCR, Collator, AC, OCRRepair, StatStore, Store processes and so on. There may be more than one process of a given type. Each process has a system level configuration, as well as one or more of the following sub-configurations: - Resource - Page - Page Variant - Document - Batch <i>Note: While referred to by the system as "Batch", in actuality this is a Work Record.</i> As an example, Collator is configured at the document and batch levels; FIP operates on page, page variant, and batch level data.
Page(s)	A page contains fields, the corresponding location of fields on a form, and processing steps for the page. A page may have one or more variants, where each variant is graphically different, but contains the same fields and page processing information.
Document(s)	A document consists of one or more pages, processing steps for the document, and may contain additional fields at the document level. Different

Term	Definition
	documents may share some pages; for example, a HCFA document and UB-92 document may both contain an Attachment page.
Batch(es)	A batch consists of documents and processing rules for the batch.

Double-click any **Category** (represented by a folder icon) to open a separate window listing all objects of the selected type. For example, double-click **Batches** to view a list of batch types and their descriptions.



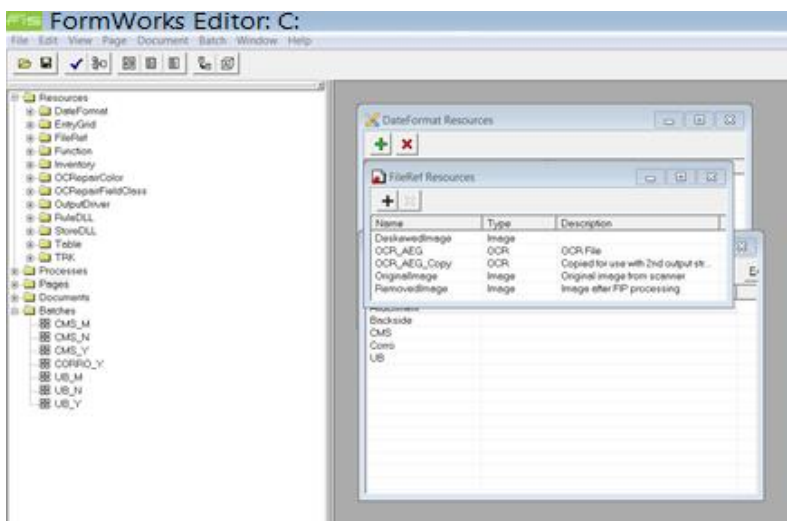
*Note that a few of the Resources do not open a window in the manner described above; for these occurrences, expand the folder (Click the + sign), then **Double-click** a specific item in the resource list.*

The Configuration Window

When selecting a Component, Object, Resource or Processing Step in the FWD tree, a configuration window for that particular item opens.

Note: Users may have multiple configuration windows open at once.

To switch between multiple Configuration windows, simply select the **open window** or press **CTRL + Tab** for the next window to become active. The window selected will be promoted to the top of all open windows. In the example below, the "FileRef Resources" has been selected and promoted to the top of the two other windows currently open on the desktop. The "Window" menu at the top can also be used to quickly switch to a specific configuration window instead of cycling through each one.



The Message Window

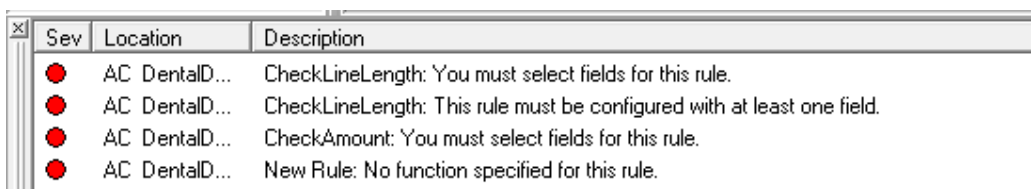
Errors, warnings, and other information about the FWD are listed in the message window at the bottom of the Editor window.

Each time a database is opened, the Check Configuration button  is selected, or a CFD is built or rebuilt, the messages are updated. See [The FWEditor Message Window](#) for a graphic detailing the location of the Message Window.

Message Types

- **Error** – An error, marked with a red bullet in the "Sev" field, indicates a problem with the existing configuration that prevents proper operation.

*Errors are problems that **MUST** be resolved. The FormWorks Editor prevents the creation of a release version .CFD if ANY errors remain unresolved.*

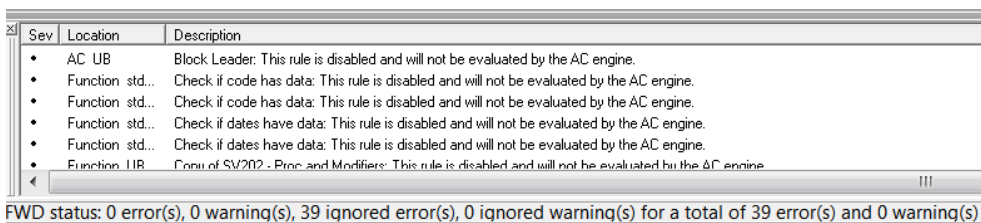


Sev	Location	Description
●	AC DentalD...	CheckLineLength: You must select fields for this rule.
●	AC DentalD...	CheckLineLength: This rule must be configured with at least one field.
●	AC DentalD...	CheckAmount: You must select fields for this rule.
●	AC DentalD...	New Rule: No function specified for this rule.

A warning, marked with a red bullet in the "Sev" field, alerts the user to a possible problem with the configuration.

Note: The CFD may 'compile' and work (as configured), even if either the Editor or a PCM think the configuration does not make logical sense.

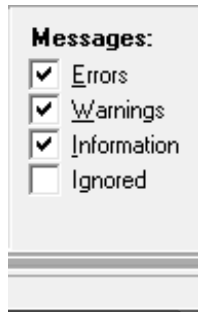
- **Information** – These messages are informational only (do not indicate problems in the configuration), and are marked with a black bullet in the "Sev" field. This category of messages includes general status information, and new (optional) feature information.



Sev	Location	Description
•	AC UB	Block Leader: This rule is disabled and will not be evaluated by the AC engine.
•	Function std...	Check if code has data: This rule is disabled and will not be evaluated by the AC engine.
•	Function std...	Check if code has data: This rule is disabled and will not be evaluated by the AC engine.
•	Function std...	Check if dates have data: This rule is disabled and will not be evaluated by the AC engine.
•	Function std...	Check if dates have data: This rule is disabled and will not be evaluated by the AC engine.
•	Function LIB	Conn of SV202 - Proc and Modifiers: This rule is disabled and will not be evaluated by the AC engine.

FWD status: 0 error(s), 0 warning(s), 39 ignored error(s), 0 ignored warning(s) for a total of 39 error(s) and 0 warning(s)

- Use the checkboxes on the right side of the window (Messages) to control which messages appear in the window.



- The Editor by default displays Errors, Warnings, and Information when Ignore is selected. See [Ignoring Messages](#) below for more information.

Note: The option Ignored is not selected by default.

Each line in the message window is composed of three parts:

1. Severity Error – Warning, or Information, (denoted by color).
2. Location – The resource or processing step involved.
3. Description – Error message, warning message, or information, including the page, document, batch, rule, or function name as applicable.

In many cases, users may select an error or warning message to open the appropriate configuration window needed to fix the problem.

Important: Selecting a message may open a configuration window at the variant level (field level), rather than the page level. For example, for a field that has no character set defined, FWEditor opens the OCR: Variant Processing window. In most cases, it is better to set options at the page level. See [Configuring Page and Page Variant Processing](#).

How to Ignore/unignore Messages

Users have the option to filter selected messages from the window by ignoring them.

To Ignore Messages

- Select Message(s):
 - Click to select a **single message**.

- SHIFT + click to select **consecutive messages**.
- CTRL + click to select **non-consecutive messages**.
- Right-click the **selected message(s)**, and select Ignore from the menu.
- To view the ignored messages, select the **Ignored** checkbox at the right side of the message window.

To Unignore a Message

- View the ignored messages.
- Select messages using the methods defined in [To Ignore Messages](#).
- Right-click the **selected message(s)** and select Don't Ignore from the menu.
- Select the **appropriate message type** (Error, Warning, or Information) to view the message.

Sort Messages

Click the **column heading** to sort alphabetically (Severity is arranged with Errors first, then Warnings, then Information). To reverse the sort order, Press **CTRL + click**.

Copy Messages

On occasion, users may need to copy a message, or messages, to an email or other application window.

- Select **Message(s)** – Click to select a **single message**; SHIFT+ click to select **consecutive messages**; CTRL + click to select **non-consecutive messages**.
- Right-click the **selected message(s)**. Choose **Copy from the Menu** and **Paste** in the desired application. Or, just press **CTRL + C** to copy the selected messages.

Message Shortcuts

- Press **CTRL + I** to ignore selected messages.
- Press **CTRL + A** to select all messages.
- Press **CTRL + U** to un-ignore selected messages.

Print Messages

To print messages, select messages using the methods listed above, Right-click, then select **Print**

Note: Users cannot choose a printer; messages will be printed to your default printer.

Plugin Configuration Module(s)

Important: The processing steps and workflow models presented in this document are representative of a typical FWEEditor Express environment. Users of this document are to note that it is anticipated that FormWorks production processing environments will differ from the representative FWEEditor Express environment illustrated in this document.

While FormWorks Editor allows the user to directly manipulate process-independent information in the FWD, Plugin Configuration Modules (PCMs) that are dynamically loaded into FormWorks Editor are used to edit all process-specific and resource-specific information. This design enables new versions of workers and their configuration objects to be introduced without having to update the Editor.

When the FormWorks Editor is started, all of the available PCMs are loaded. Each of the PCM objects are queried to determine the processing type supported, as well as what type of FWD information the PCM operates on. For example, the FIP PCM may indicate that it operates on batch, page, and page variant data, while the Store PCM registers as operating on page and document data.

PCM Resource Types

PCM Name	Resource Type
Outdriverconfig	StoreDLL
Oudriverconfig	OutputDriver
Acconfig	Table

PCM Process Types

PCM Name	Process Type
Aegconfig	AEG
ocrepair3	OCRepair3

When setting up an FWD using the Editor, users designate the processing steps the data will go through. Each of these steps has a base processing type that is specified by a loaded object. If the database contains a processing step that uses a base type not defined by a PCM, the Editor will not allow users to edit the configuration of that particular processing step.

By using a two-tier approach of processing steps and base processing type, users may have multiple workers of one type. For example, many systems have OCRepNum (to repair numeric fields) and OCRepair (to repair alphanumeric fields). Both of these steps have the base process type of OCRepair and their configurations are maintained by the same PCM.

Note: For programming level information pertaining to PCM, refer to the FW Plugin Configuration Module (PCM) Developers Guide.

Events

For each type of processing (Batch, Document, Page, and Page Variant, etc.) the PCM that supports the processing type receives edit events which are generated when editing the process configuration for any FWD object.

Events are also generated when the information in the public section (Pages/Batches) of the FWD is changed (such as deleting an object, or changing a field's base type). Each PCM for all of the processing steps is notified of the change, giving each configured process an opportunity to update its configuration accordingly.

Finally, before the deletion of any public data (Pages/Batches), each PCM is given a chance to indicate the consequences of the deletion. If a PCM indicates that there may be a problem, users have a chance to cancel the deletion. If the user elects to continue, the PCM receives a deletion event.

For example, if a field is deleted from a page, the AC object may display a warning that the field is used in two rules.

Information Types

Process configuration objects can register for the following types of information editing:

Information	Description	Example
Variant Field	This is the lowest level of information tracked in FormWorks Editor	An OCR worker configuration for segmenting the field.
Variant	This level allows configuration of processes for a specific graphic variant and its field list.	OMR configuration that maps checkboxes to a given page field.
Page Field	The configuration for a field common to the page but not specific to a graphic variant.	An AC rule that verifies the page field's value is within a stated maximum size.
Page	This is a higher level of configuration that allows configuration of a process for a page and its field list, independent of the graphic variations.	An AC rule that breaks out an imaged field into multiple non-imaged fields. One example, parsing a full name into a lastname and firstname.

Information	Description	Example
Document Field	A field which belongs to a document, but not a specific page in the document.	An AC rule that verifies the document field's value is within a stated maximum size. Validating the total dollar amount of an entire document across all line items on all pages.
Document	This level of configuration encompasses the entire document and its information.	AC rules that fill in document fields based on information from multiple pages, such as a "CheckID" field to verify that an ID number is the same on all the pages of a document.
Batch	Information specific to the processing of the given batch type.	Collation rules for the batch.
System	System-specific information for a process.	This contains worker configuration information such as default log levels, port numbers, and so on.

Information about Building a New System

Typically pieces of the FWD are constructed from smallest to largest (i.e. create fields on pages and their variants, then designate what pages are in a document, then what documents are in a batch).

Users should plan a new system by:

- Determining what batches are first needed.
- Decide what documents may be included in each batch.
 - Type of pages in each document.
 - Type of fields (field variant) for each page.

A system checklist, covering the steps (best practice) to building a new system may be viewed in [Appendix A - New System Checklist, Tips And Troubleshooting](#).

System Engineers: When building a new system, utilize the "FWExpress" build as a base system.

Note: Specific functions are not configured within FWEditor and must be setup separately.

- *Table Update (configured in tableupdater.cfg)*
- *Site files*
- *Agent files*

- [Routes](#)
- [Common.cfg](#)

Contact a FIS systems engineer for details regarding these (and any other) functions for which assistance may be required..

License Information

FWEdit FormWorks 5 includes (built within) the licensing module. This license must be authorized in order to edit the .FWD within the FWEditor.

Important: Request assistance from a FIS technical representative before beginning the installation, registration or configuration of the FWEdit license. The following information is an outline regarding the steps required to install FWEdit licensing.

License Key Utility

The License Key utility requires the license file to be structured in the following format.

```
[LicenseInputs]
CustomerID = "a unique string to ID the customer or user"
```

```
[FWEdit]
Key = ""
```

Licence Key File (Structure)

CustomerID Attribute

The "CustomerID" attribute ensures that each key is unique and traceable to the proper owner.

- For production systems – The Customer ID should uniquely identify the system.
- For developers and testers – The CustomerID should be structured as "FIS FormWorks Dev" followed by the developer or tester full name.

Note: "FIS" is not an acceptable value for the CustomerID attribute

[FWEdit] Attribute

Represents the name of the component to be secured.

Key Attribute

The "Key" attribute is to remain empty (blank) as shown in the License Key utility format.

License Key Generation

To generate the license key, execute the LicenseKeyGen utility by first specifying the name and path of the [License Key File](#) previously created.

```
<drive>:\apps>LicenseKeyGen.exe D:\rri\<systemname>\bin\License.cfg
```

After execution, the LicenseKeyGen utility will generate the FWEditor key and update the License.cfg file. At this time, the license file will be structured similar to the following sample.

License.cfg Sample

```
[LicenseInputs]
CustomerID = "FIS"

[FWEdit]
Key = "c41d69eff200b9fc00d45ad3f06a6af5"
```

Important: Request assistance from a FIS technical representative before beginning the installation, registration or configuration of the FWEdit license. The following information is an outline regarding the steps required to install FWEdit licensing.

Register the License Key in FWEditor

The FWD license is a separate key generated by FWEdit and stored in the .CFD file. This key is used by Background Worker Shell (BWS) to verify that the .CFD file was produced from a licensed FWEdit.

To verify that FWEdit is properly licensed and secure, verify that the license.cfg file is present in a known location for the system. On load of the FWEditor, an open dialog box is displayed, this screen prompts for the license file with the default extension of ".cfg" selected.

Note: It is not necessary that the extension of the license file be ".cfg" and there is an option to select a file with any extension.

Once the license key has been validated and the FWEditor loads, the user may proceed with working with the FWEditor application. When all the changes are done and the user chooses to build the CFD file, the editor then takes the CustomerID, .FWD build version and generates a unique code to build a separate license key that is stored in the .CFD file. When the BWS executable loads the .FWD for several of the workers, the license key is validated against the information in the .CFD file. If the license key is found to be valid, then the RRIBWS executable continues to load the worker. If the license key is found to be invalid, then an error message stating that the FWEdit license was invalid is logged into the log file for that worker and an error alert is also issued.

Open A FormWorks Database

Databases may be interfaced via the FWEditor menu options. For convenience, it is recommended to create a FormWorks Editor icon (below) by creating a Windows shortcut on the desktop.



Create a FormWorks Editor Icon

A sample shortcut target is as follows:

```
D:\rri\systemname\bin\fwedit.exe ../fwd.sfd
```

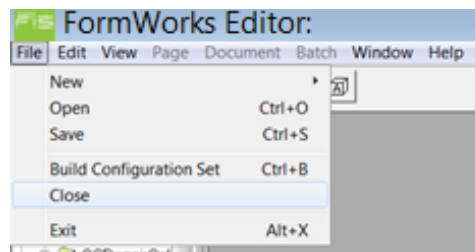
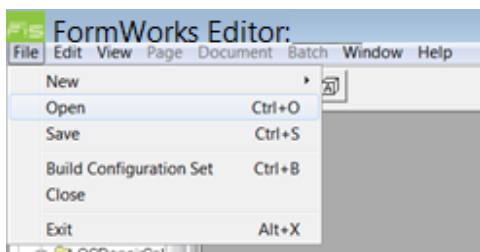
If preferred, the FormWorks Editor may also be started from the command line:

```
\\systemname> start bin\fwedit fwd.sfd
```

Within Windows Explorer, the "fwedit.exe" file appears as follows.

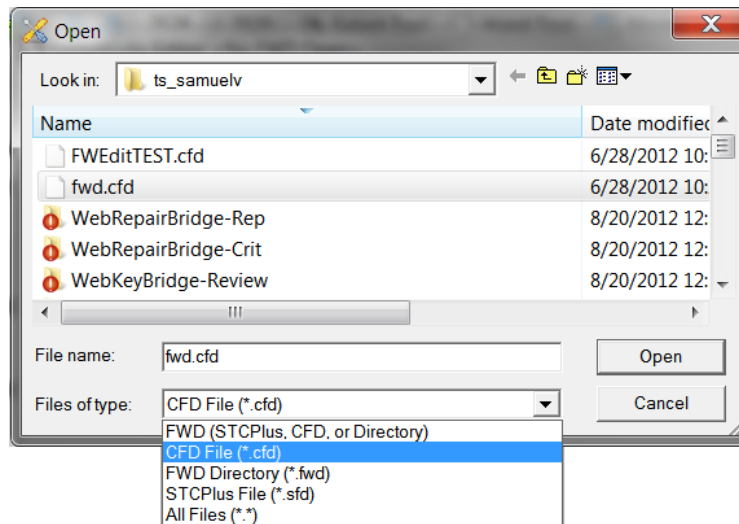
 fwedit.exe	2/15/2017 3:10 PM	Application	1,541 KB
--	-------------------	-------------	----------

Users may also open and close a database in the FWEditor window by utilizing the **File + Open** and **File + Close** function(s).



.CFD, .FWD and .SFD files may also be opened directly from the Editor.

Note: When selecting a .CFD to open, FormWorks Editor prompts the user to specify a name for the FWD directory, which it creates. The Editor then decompresses the .CFD to a .SFD and opens.



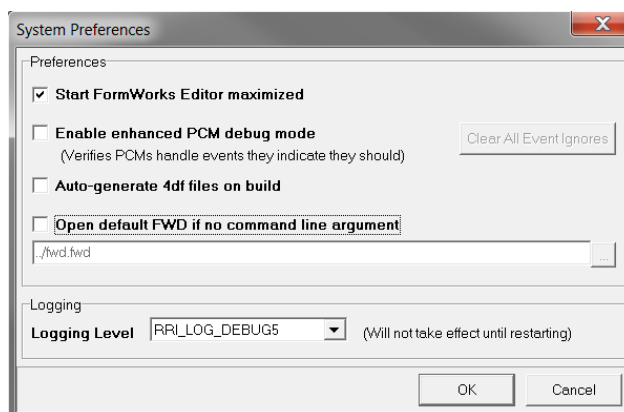
Set a Default Database

Select **Edit + Preferences** to Open the Preferences window. Use the browse button to specify the default FWD.

By default, the following two check boxes may be selected.

- Start **FormWorks Editor** maximized – This is a user preference, starts the FWEditor in full screen mode if checked.
- Auto-generated 4df files on build – Uncheck (deselect) **Auto-generate 4df files** on build. As a best practice, for new builds or configurations this feature should not be enabled.

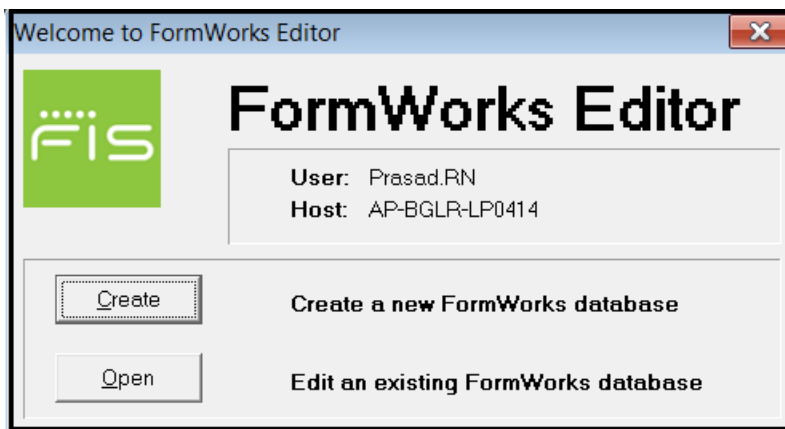
Note: 4df files are not required by new system builds that use KE 5.x or greater, or RemoteKeying.



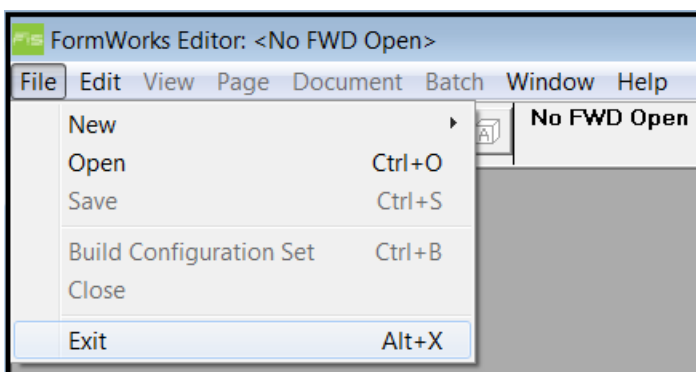
Build a .CFD

Before building a .CFD (Compiled FormWorks Database), first verify that no errors in the .SFD are present. To do this:

- From the FWEEditor (upon startup), the "Welcome to FormWorks Editor" screen is displayed. select Open.

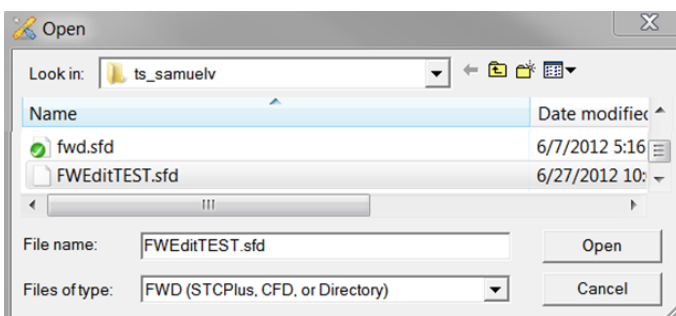


Or, open the .SFD from the FWEEditor menu.



Note: The FWEEditor is designed to only work with a single database file (.FWD/.CFD/.SFD) at a time.

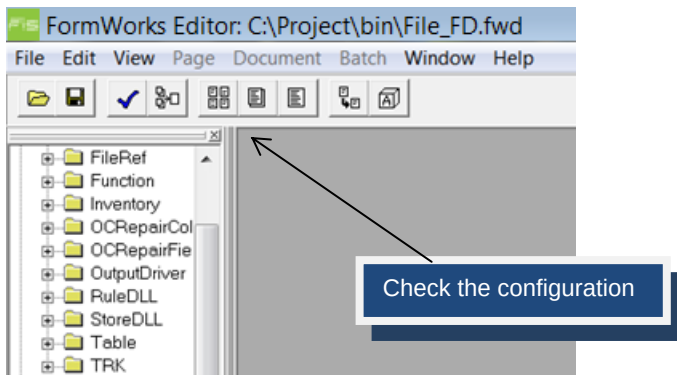
- Select the .SFD file



Once the database is open the Message window will display that the "Check configuration has not been run" message.

Sev	Location	Description
●	Editor	Check configuration has not been run.

- To check the configuration, select the Check displayed on the toolbar.



After check configuration has run, errors, warning and other messages are reported in the [Message Window](#). To trace the source of an error, double-click the error.

The status bar at the bottom of the FWEditor window displays to the user a summary of error(s), warning(s), ignored error(s), ignored warning(s) along with total error(s) and warning(s).

FWD status: 0 error(s), 0 warning(s), 39 ignored error(s), 0 ignored warning(s) for a total of 39 error(s) and 0 warning(s)

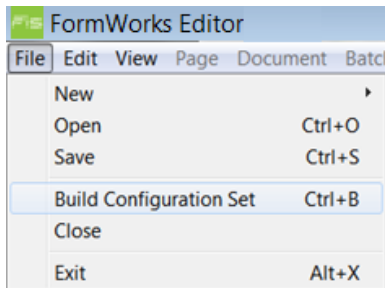
The FWEditor does allow users to save a test build that contains errors, but does not allow a Release build (.CFD) to be saved until the FWD is free of errors. Additionally, if the status bar at the bottom of the FWEditor window states "FWD status: The FWD needs to be checked" this indicates that there have been changes since the last execution of check configuration.

To Build A CFD

After the Check Configuration has been run and all errors, warnings, and messages are resolved the .CFD file may be built. This is done via the "Build Configuration Set" function.

*Note: **Ctrl+B** key command executes the Build Configuration function.*

- First, select File + Build Configuration Set (Ctrl + B) or select the Build Configuration set  button located on the toolbar..

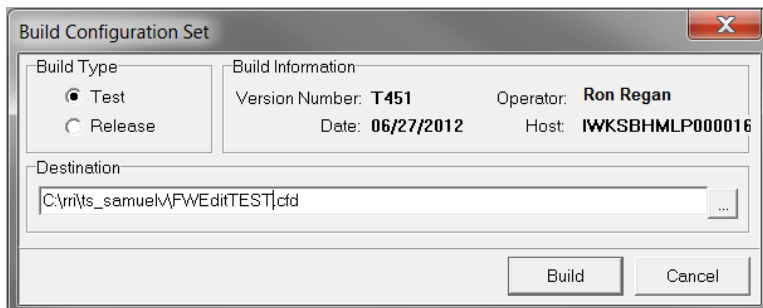


- Editor checks the configuration for any changes and if any errors or warnings are found in the FWD a warning message is displayed. If no errors are found, the Build Configuration Set window opens.
- Select a Build Type (Test or Release).



For the first instance of build (either Test or Release), specify the Destination directory and filename. Click Build, the system prompts the user to enter a system name.

Note: The last CFD build location and filename are stored; Editor continues to use the stored location and filename until a new name and location is specified. Additionally, when a new path is selected the filename previously populated in the window remains and may require modification before executing the "Build".



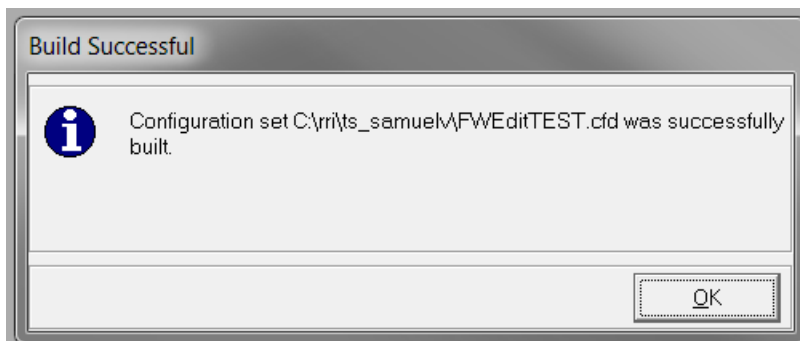
A version number is assigned for each build and is incremented for subsequent builds.

- Test build types are preceded by the letter T.
- Release build types are preceded by the letter R.

Eventually builds T1, T2, R3, T4, R2, R3, etc. (or as in the example above T451) may be created and numbered. Build numbers are sequential and may not be reset.

Note: In a running system, the build number is noted in worker log files and History for later reference.

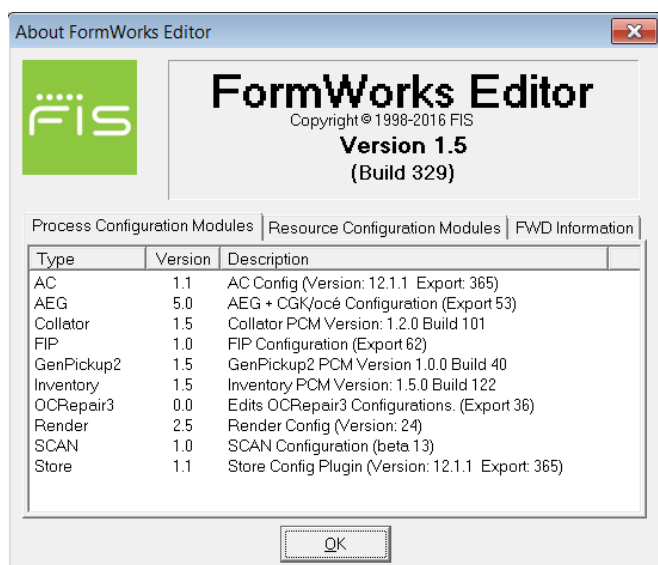
When the build is complete, the "Build Successful" message is displayed. Select **OK** to close the window.



The system name and version are displayed in the top right portion of the Editor window.



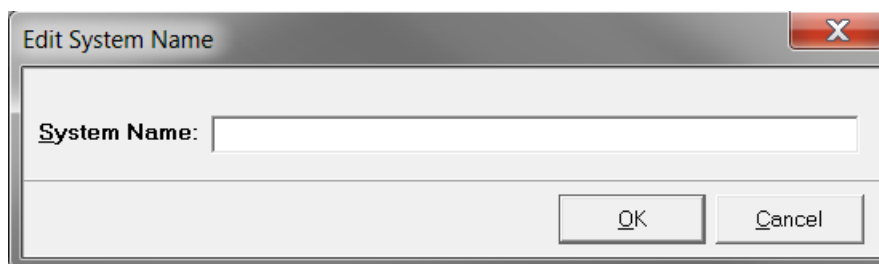
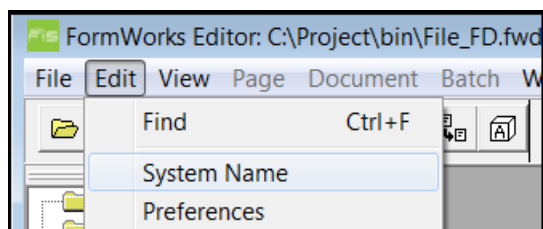
The build date, host name and operator name displayed in the Build CFD Window are all stored by the Editor. These items may be viewed by selecting **Help > About > FWD Information**. Switch to the FWD Information tab for this screenshot?



Edit the System Name

The system name specified when a build is created is embedded in the resulting CFD and may be edited by selecting **Edit > System Name**.

Important: If changing the system name, remember to update the system name with the Build Configuration Set destination when generating the next Test or Release build if future builds are to have the new naming convention.

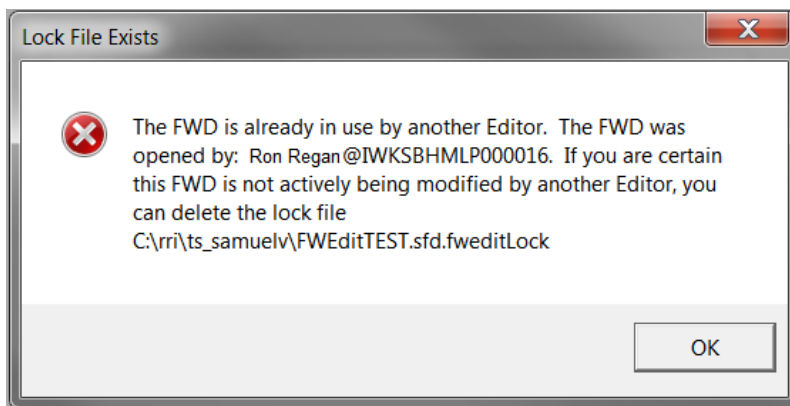


.SFD Lock File

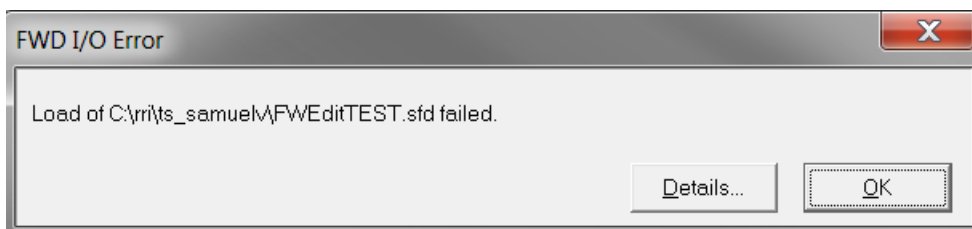
When a .SFD file is opened from FWEditor the file is placed in a locked state. The locked file (Type FWEDITLOCK FILE) is denoted by the "fweditLock" extension to the file name (see example below).

Name	Date modified	Type	Size
 FWEditTEST.sfd.fweditLock	6/27/2012 11:38 A...	FWEDITLOCK File	1 KB

If an attempt to open the file is made while the file is locked, the following message is displayed.



After selecting **OK**, the FWD I/O Error window displays



Select **OK**

To unlock the file, exit the **FWEdit application**, delete the <FWDfilename>.fweditLock file, located in the same directory as the .SFD file, after which return to the FWEdit application. Upon reentry to FWEdit the applicable .SFD file should be accessible.

SFD Backup File(s)

Each time that a .SFD file is opened from FWEdit the system automatically creates a backup file (.BAK extension).

The backup function is controlled by FWD_Open making a call to STC_ContainerOpen, attribute "STCPLUS_MaxBackupcount" The default value is specified at twelve. It is strongly recommended that this setting have a value of not less than six.

Backup files are numbered and automatically renumbered sequentially with the lowest numbered extension representing the most recent backup.

In the following sample, FWEditTEST.sfd.001.bak was created at 11:36 a.m. while FWEditTEST.sfd.012.bak was created at 10:40 a.m.

Date modified	Name	Type
 6/27/2012 11:18 AM	FWEditTEST.cfd.bak	BAK File
 6/27/2012 11:36 AM	FWEditTEST.sfd.001.bak	BAK File
 6/27/2012 11:34 AM	FWEditTEST.sfd.002.bak	BAK File
 6/27/2012 11:18 AM	FWEditTEST.sfd.003.bak	BAK File
 6/27/2012 10:50 AM	FWEditTEST.sfd.005.bak	BAK File
 6/27/2012 10:49 AM	FWEditTEST.sfd.006.bak	BAK File
 6/27/2012 10:49 AM	FWEditTEST.sfd.007.bak	BAK File
 6/27/2012 10:48 AM	FWEditTEST.sfd.008.bak	BAK File
 6/27/2012 10:48 AM	FWEditTEST.sfd.009.bak	BAK File
 6/27/2012 10:44 AM	FWEditTEST.sfd.010.bak	BAK File
 6/27/2012 10:44 AM	FWEditTEST.sfd.011.bak	BAK File
 6/27/2012 10:40 AM	FWEditTEST.sfd.012.bak	BAK File

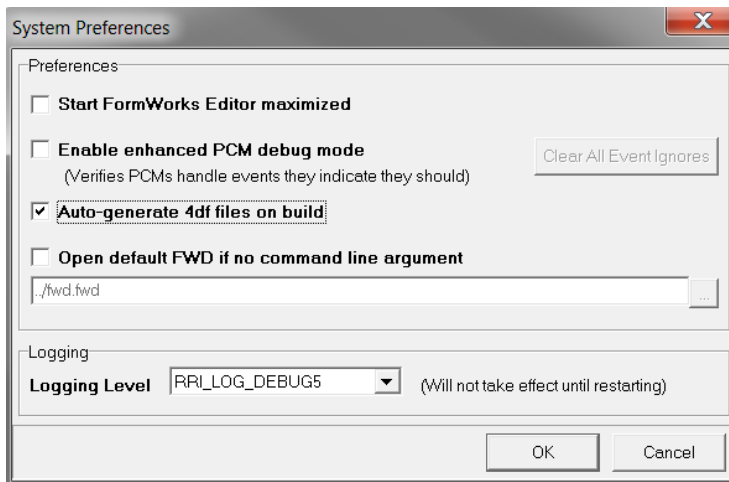
Create 4df files

4df files are used by older KEWorkers that record the location and type of every field on a page. The following procedure writes the page and variant fields to the 4df file.

From Definition files (4df) are used by KE4 and are not generally applicable to the FormWorks 5 environment. The following information regarding the generation of 4df files is not applicable to FW5 builds or installations or FW4 builds using KE5.

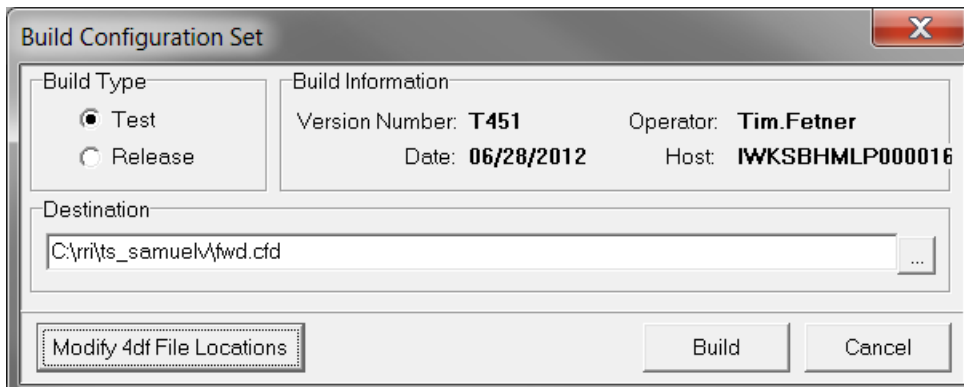
To Generate 4df Files

In the System Preferences window (**Edit > Preferences**), select **Auto Generate 4df files on build**.

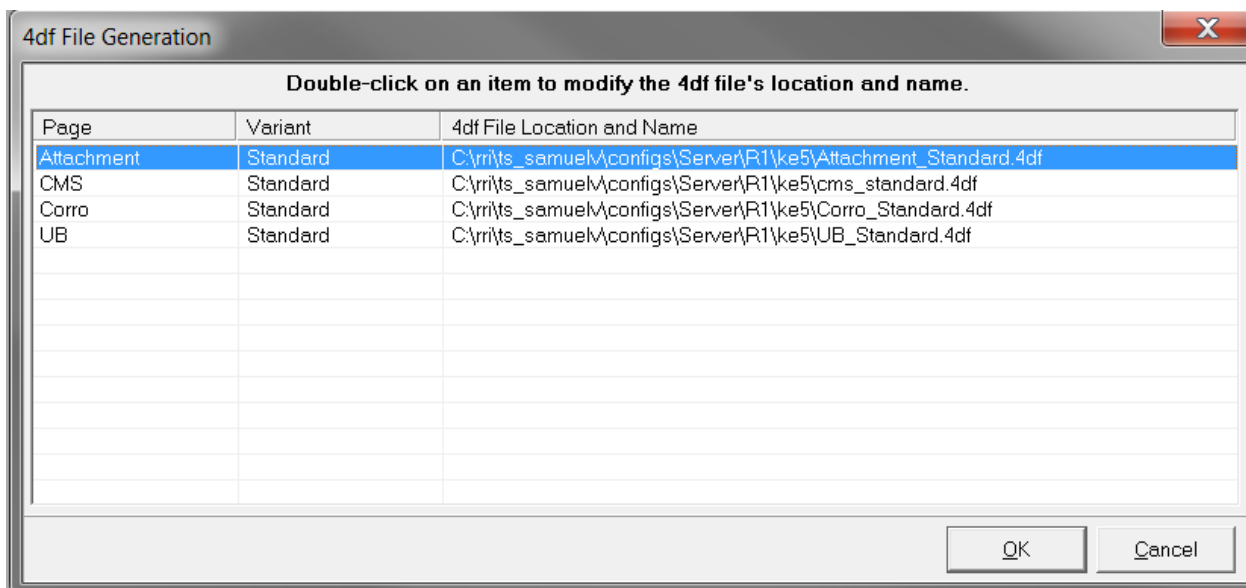


Next, select **Build Configuration Set**.

From the Build Configuration Set window, modify the 4df file locations as required. This may be done by selecting the Modify 4df File Locations button.

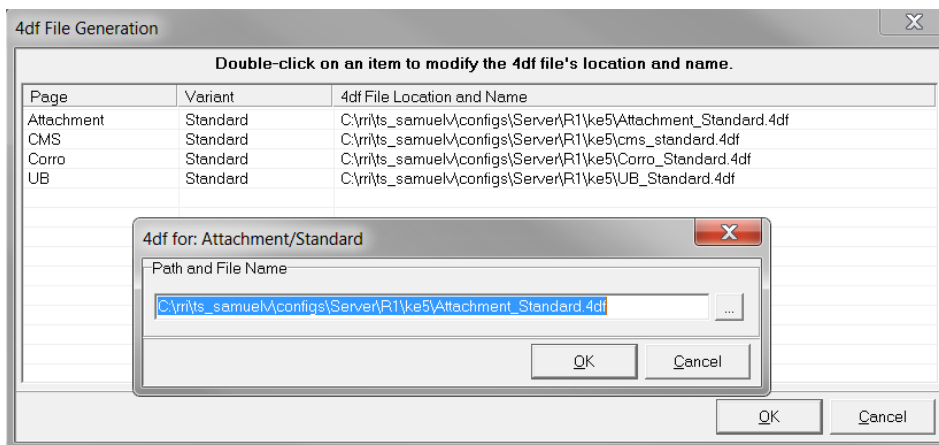


Upon selection of the Modify 4df File Locations, the 4df File Generation is displayed.



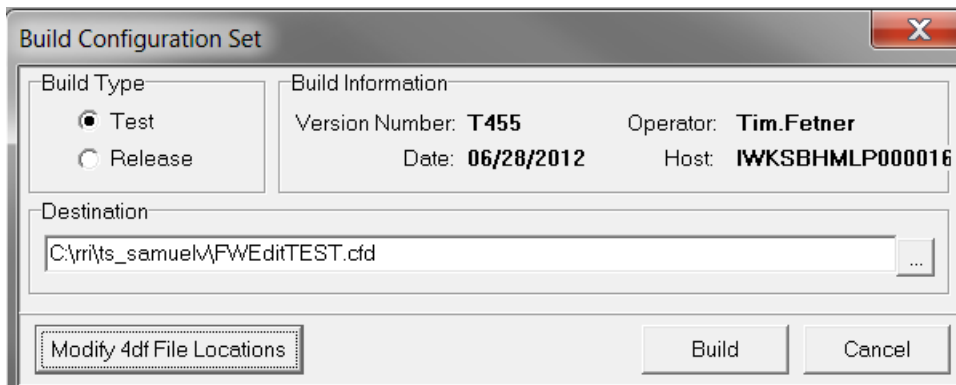
Select the **specific page** to display the 4df for: Attachment/Standard window and edit the Path and file Name with the desired path.

Note: Verify with the appropriate IT personnel what drive mapping and naming conventions are appropriate.

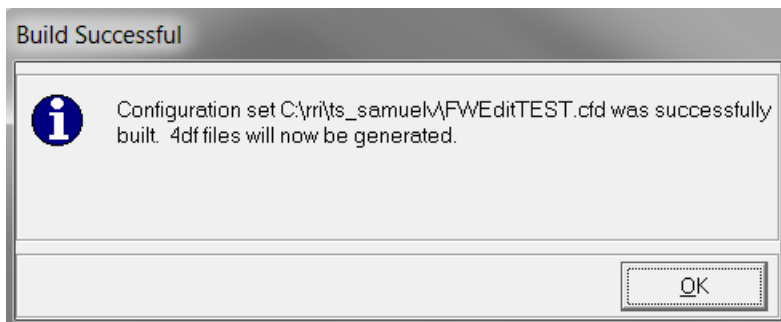


Select **OK** to save and exit the window.

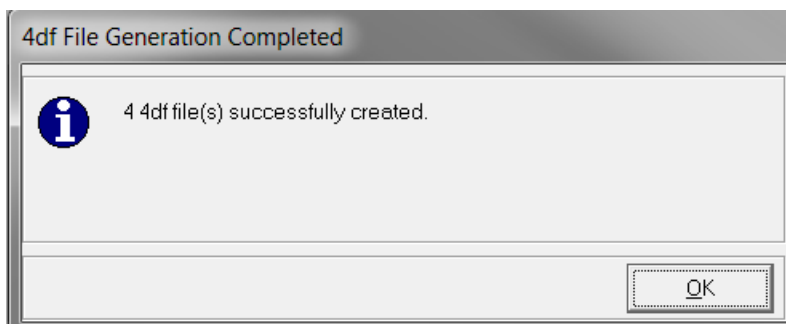
Next, choose the appropriate **Build Type** and select **Build**



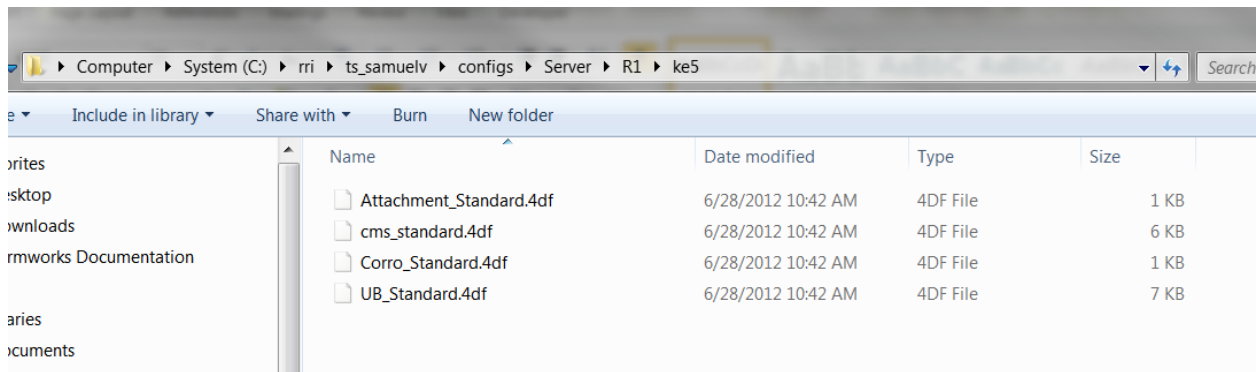
A Build Successful message is displayed. Select **OK** to build the 4df files.



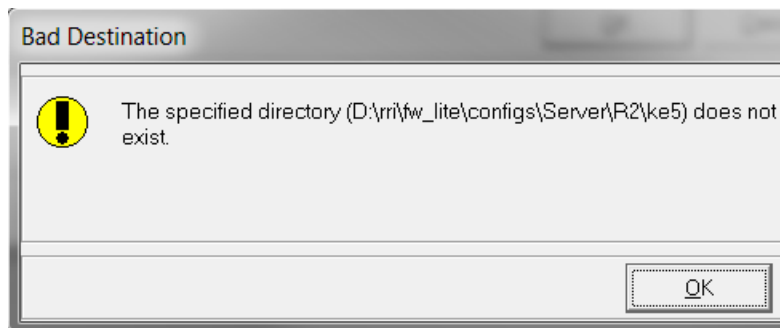
A confirmation message is displayed indicating that the 4df files were successfully created.



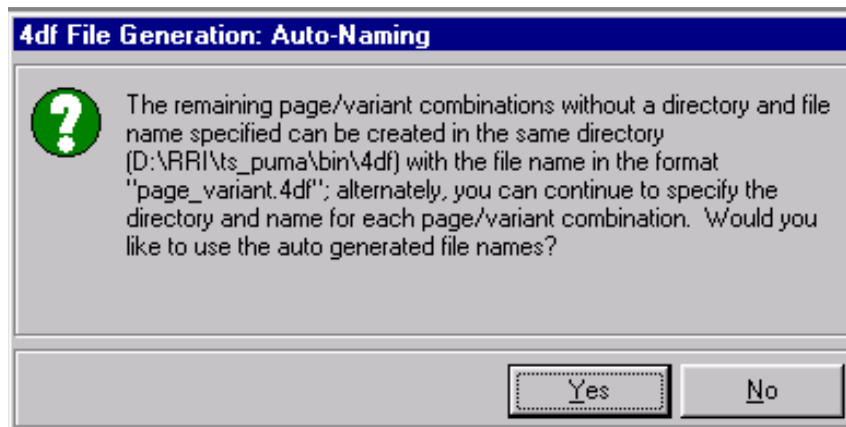
The newly created 4df files now reside in the assigned directory.



If the directory/path entered into the 4df File Location and Name field is invalid, the directory and path must be manually created. If an invalid path entered into the Path and File Name field should be entered, the system displayed the following message.



Note: If modifying 4df File Locations, the FWEditor prompts for the locations when the CFD file is built. Once the location for the first file has been set, Editor presents a window message allowing Locations and Filenames to be set individually, or allows FWEditor to store all information in the same location, and in turn assign filenames based on the Page and Variant name.

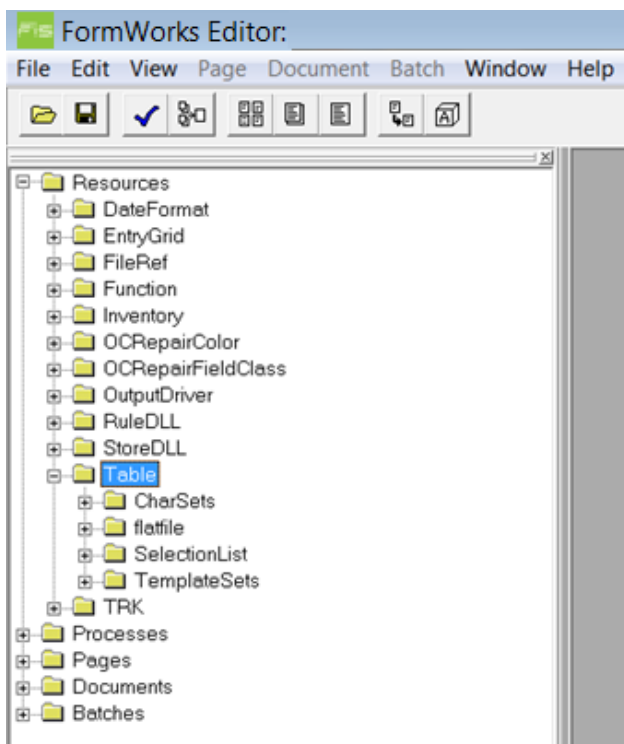


Resources

Important: PCMs are updated frequently. For a current listing of applicable PCMs (FormWorks 5 Express) at the time of document publication, see [Documented Components](#).

Resource Folders Creation

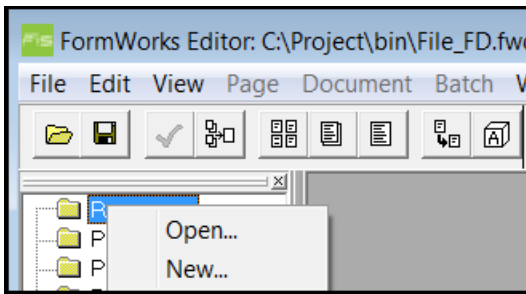
To view a list of all resources of any given type in a separate window, select the resource type name in the FWD tree or select the  sign.



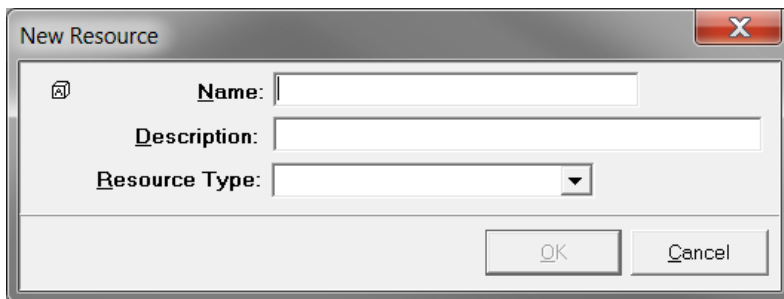
Some resource types may have many items in the list. For example, the Table resource type includes CharSets, Template Sets, SelectionList and other tables. As a best practice, use folders for any resource type to keep the lists manageable.

To Create A Resource Type Folder

- Right-click a Resource in the Resource list select New from the menu.



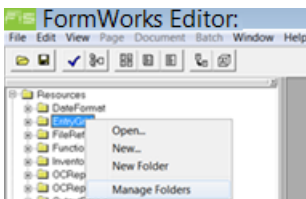
- The New Resource window displays.



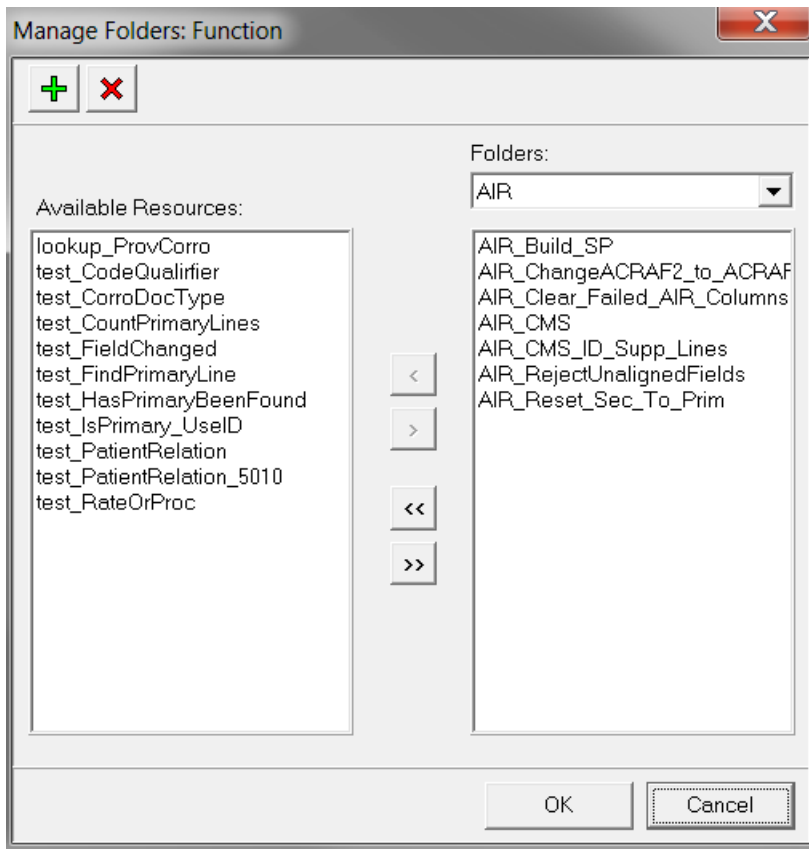
- Enter the **Name** and **Description** of the New Resource.
- Select the new **Resource Type** from the drop down list.
 - DateFormat
 - EntryGrid
 - FileRef
 - Function
 - InputDriver
 - Inventory
 - OCRepairColor
 - OCRepairFieldClass
 - OutputDriver
 - RuleDLL
 - StoreDLL
 - Table
 - TRK
 - Language
 -
- Select **OK**.

To Manage Folders

- Right-click the appropriate Resource type in the resource list, select Manage Folders from the menu.



- In the Manage Folders Function dialog box, select the Folder Name from the Folders list.



- From the Available Resources list, select the Items to be put into the selected folder. Use SHIFT to select consecutive items, CTRL for non-consecutive items.
- Click the right pointer button  to move the selected items into the folder.
- To remove items from the folder, select from the right side, and use the left pointer button  to move the selected items out of the folder.

Note: New folders may be created in the Manage Folders dialog box by selecting the green plus button  at the top of the window. Delete the current folder by selecting the red X  button. When a folder is deleted, all of its contents are moved back to the Available Resources list.

Internationalization

Users can configure Western European Languages that are supported by the Windows-1252 character set in addition to English for reject messages. Some of the languages that can be configured for reject messages are English, Spanish, French, German, Italian and Portuguese. Refer to [Editor Configurations For Internationalization](#) for configuration information.

Editor Configurations for Internationalization

This section provides information to users on the steps involved to configure the Language and other screens in the FormWorks Editor to support keyers facing internationalization.

Configure Language Resources

The languages can be configured using the following steps:

1. Right Click on the Resources in the Treeview and select 'New' from the menu items.
2. A new resource creation dialog box is displayed. When the language resource screen is successfully loaded, a new Resource Type called Language would be displayed as below.



3. Provide a Name to the Language resource and click OK, and new dialog box would be displayed as below.



4. Six languages are currently defined and the user is allowed to create one language resource for each language. Contact your FIS Account Manager if other languages are desired.
5. Close the dialog box after selecting the desired language for the language resource to be created.

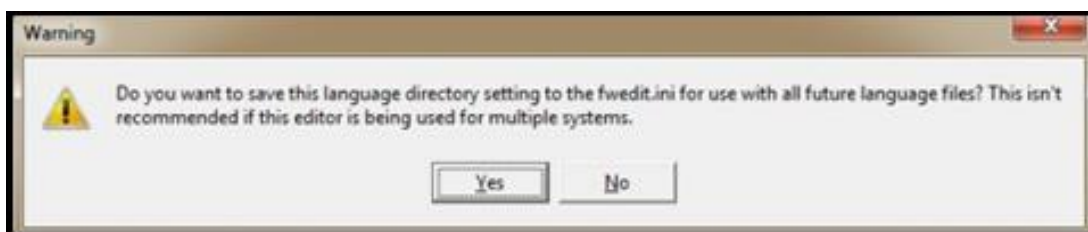
Select a Directory to Save the Language Resources (.RESX Files)

Use the following steps to save the language resource file in a directory that is selected by the user.

1. The Editor prompts the user to select the directory upon creating a language resource. This directory location will be later used by the Editor to save all reject messages to an external resource file (.resx).



- Once the directory is selected, the Editor will prompt you to specify whether to save the directory location in an ini file or to maintain the directory location for the current Editor session.

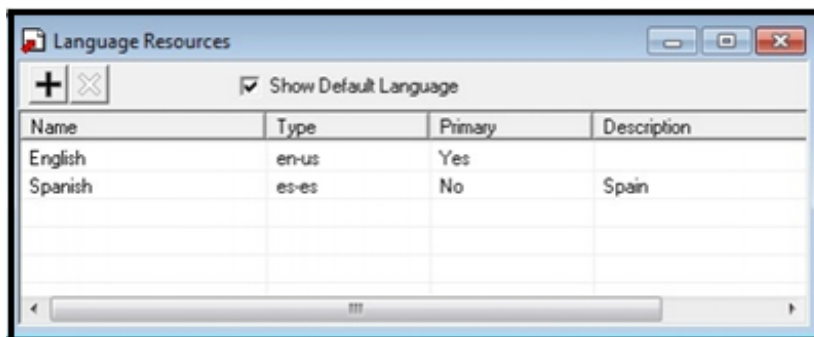


- Click 'Yes' to save the directory location into the ini file.

Note: Clicking 'No' will prompt the user to select the language directory path every time Editor is opened.

To View the List of Language Resources

- Upon successfully creation of the language resources in the Editor, the user can view the list of languages, by double clicking on the Language folder in the Treeview.



- The user can also create and remove a language resource from this dialog box if desired.

To View the Default Language

1. The show default language option allows the user to choose whether to display the 'Default' item to be available in the drop-down list that is displayed in the IRejectFields.

*Note: If there are no languages defined for the FWD, then the 'Default' item is always displayed in the drop-down list irrespective of the selection for the **Show Default Language** checkbox.*



Primary Language and Copy from Template

On opening a language resource for the purpose of editing, **Primary** and **Copy From Template** options that were unavailable during the creation of the resource are now displayed.



Primary: This option enforces a strict rule on the system to have non-blank reject reasons. This ensures that for a language that has been defined as primary, all the reject messages are entered.

Note: Only one language can be defined as primary, and the primary can be changed from one language to another. For example: to change the primary language from English to Spanish, select the Primary checkbox in the relevant Language Resource window by checking the Primary option there and the primary option in English will be automatically cleared.

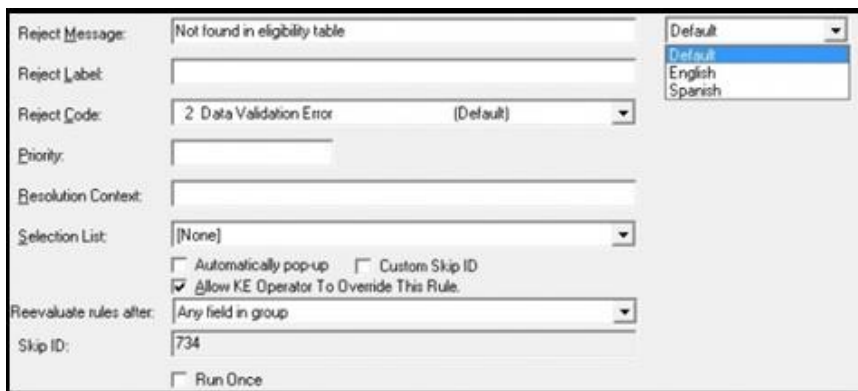
Copy From Template: This option allows the user to populate the external resource file containing error messages with the contents of the existing messages. All reject messages are copied into the language resource for which the user has selected this option.

Modify the IRejectFields

Once the language resources are created as described above, the language resources would be available for the **IRejectFields** rule to use them.

The language resources for **IRejectFields** can be configured as below:

1. Open an IRejectFields rule for which the language is to be configured.
2. A new drop-down list beside the Reject Message box is available, which lists all the languages that were configured earlier.



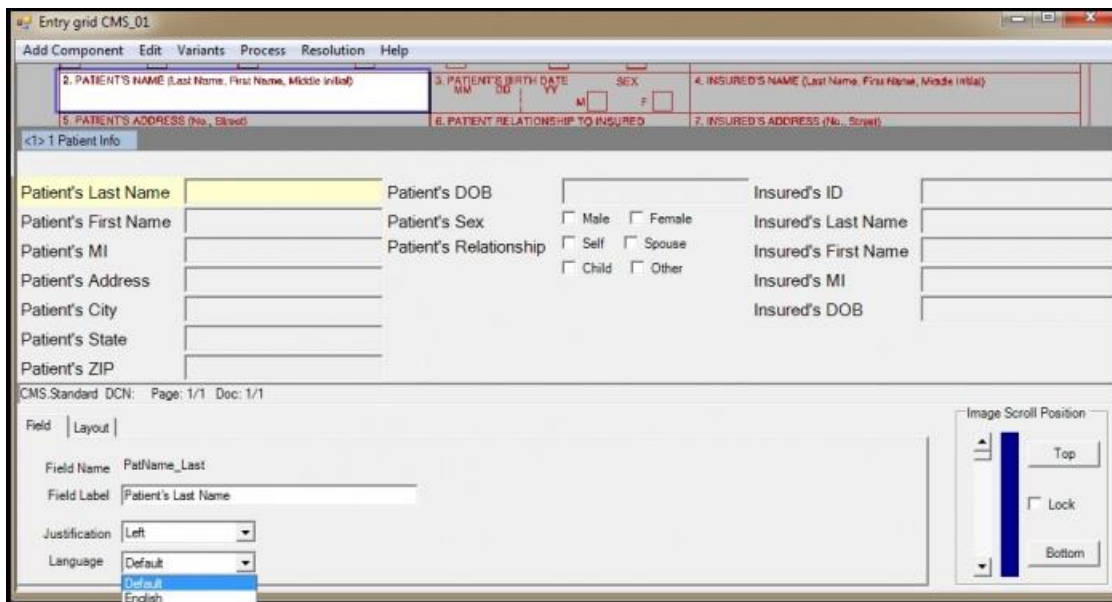
3. The 'Default' option in the drop-down list allows the user to save the Reject Message into the FWD, whereas selecting the languages from the drop-down list will save the Reject Messages to an external resource file (.resx).
4. If 'Show Default Language' checkbox is unchecked in the language resources, then the IRejectField rule will appear as below:



Modify the Entrygrids

Once the language resources are created, they are available for all Entrygrid resources. The language resources for Entrygrids can be configured as follows:

1. Open an Entrygrid and select the field label for which to configure the language.
2. A new drop-down list below the 'Justification' dropdown listing all the languages configured earlier is now displayed.



Entry grid CMS_01

Add Component Edit Variants Process Resolution Help

2. PATIENT'S NAME (Last Name, First Name, Middle Initial) 3. PATIENT'S BIRTH DATE MM DD YY SEX M F 4. INSURED'S NAME (Last Name, First Name, Middle Initial)

5. PATIENT'S ADDRESS (No., Street) 6. PATIENT RELATIONSHIP TO INSURED 7. INSURED'S ADDRESS (No., Street)

<1> 1 Patient Info

Patient's Last Name Patient's DOB Patient's Sex Male Female Insured's ID

Patient's First Name Patient's Relationship Self Spouse Insured's Last Name

Patient's MI Child Other Insured's First Name

Patient's Address Insured's MI

Patient's City Insured's DOB

Patient's State

Patient's ZIP

CMS.Standard DCN: Page: 1/1 Doc: 1/1

Field Layout

Field Name PatName_Last

Field Label Patient's Last Name

Justification Left

Language Default

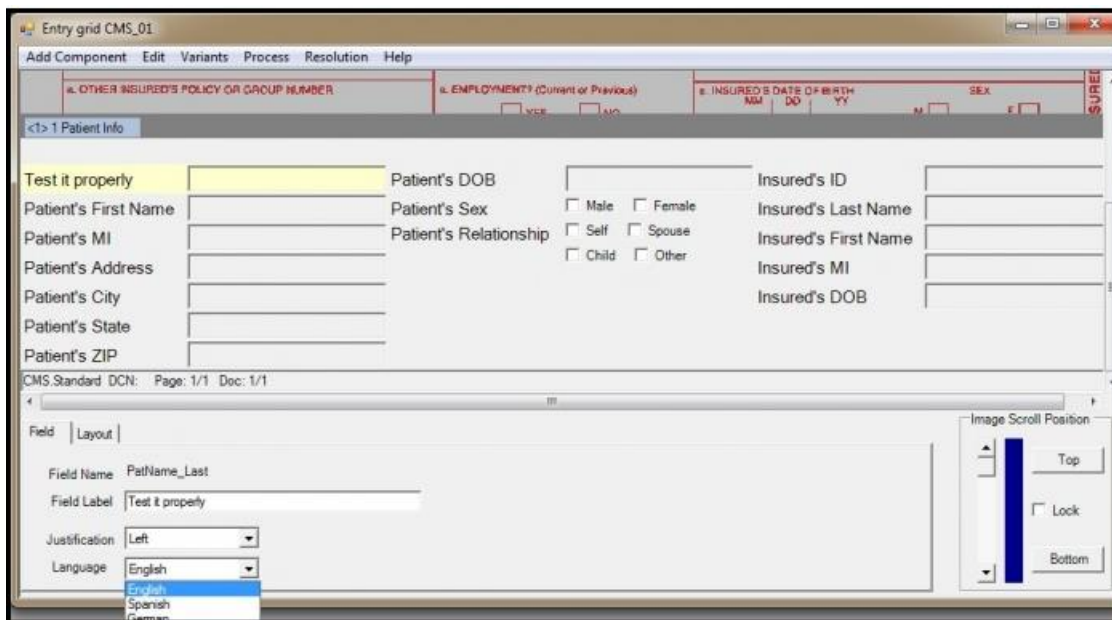
Image Scroll Position

Top

Lock

Bottom

- The "Default" option in the drop-down list allows you to save the field label into the FWD, whereas selecting the languages from the drop-down list will save the field label to an external resource file(.resx).
- If the 'Show Default Language' is unchecked in the language resources, then the Entrygrids will appear as below:



Entry grid CMS_01

Add Component Edit Variants Process Resolution Help

8. OTHER INSURED'S POLICY OR GROUP NUMBER 9. EMPLOYMENT? (Current or Previous) YES NO 10. INSURED'S DATE OF BIRTH MM DD YY SEX M F

<1> 1 Patient Info

Test it properly Patient's DOB Patient's Sex Male Female Insured's ID

Patient's First Name Patient's Relationship Self Spouse Insured's Last Name

Patient's MI Child Other Insured's First Name

Patient's Address Insured's MI

Patient's City Insured's DOB

Patient's State

Patient's ZIP

CMS.Standard DCN: Page: 1/1 Doc: 1/1

Field Layout

Field Name PatName_Last

Field Label Test it properly

Justification Left

Language English

Image Scroll Position

Top

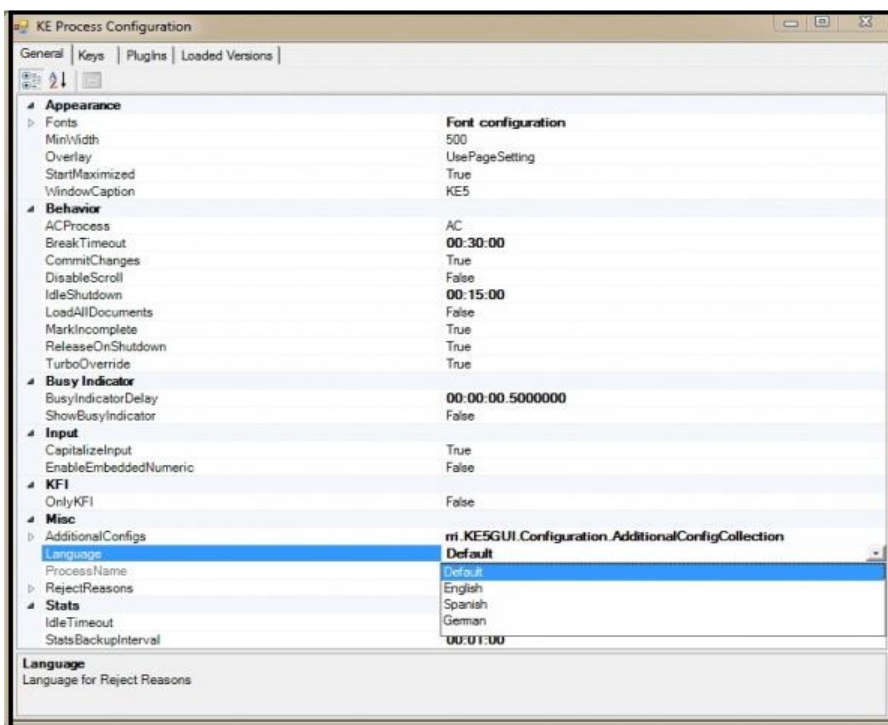
Lock

Bottom

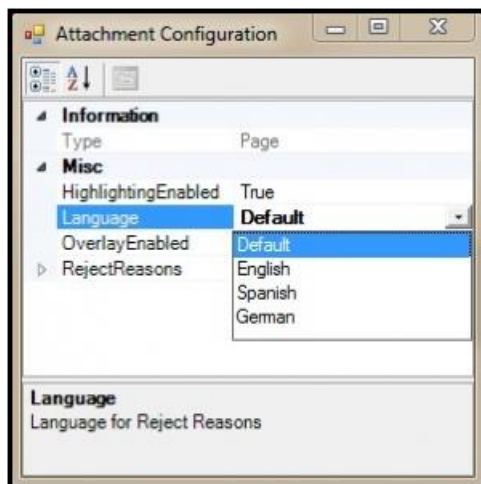
Modify the KE Process Reject Reasons

Once the language resources are created, they are available for all the KE processes for use. The language resources for KE processes can be configured as below:

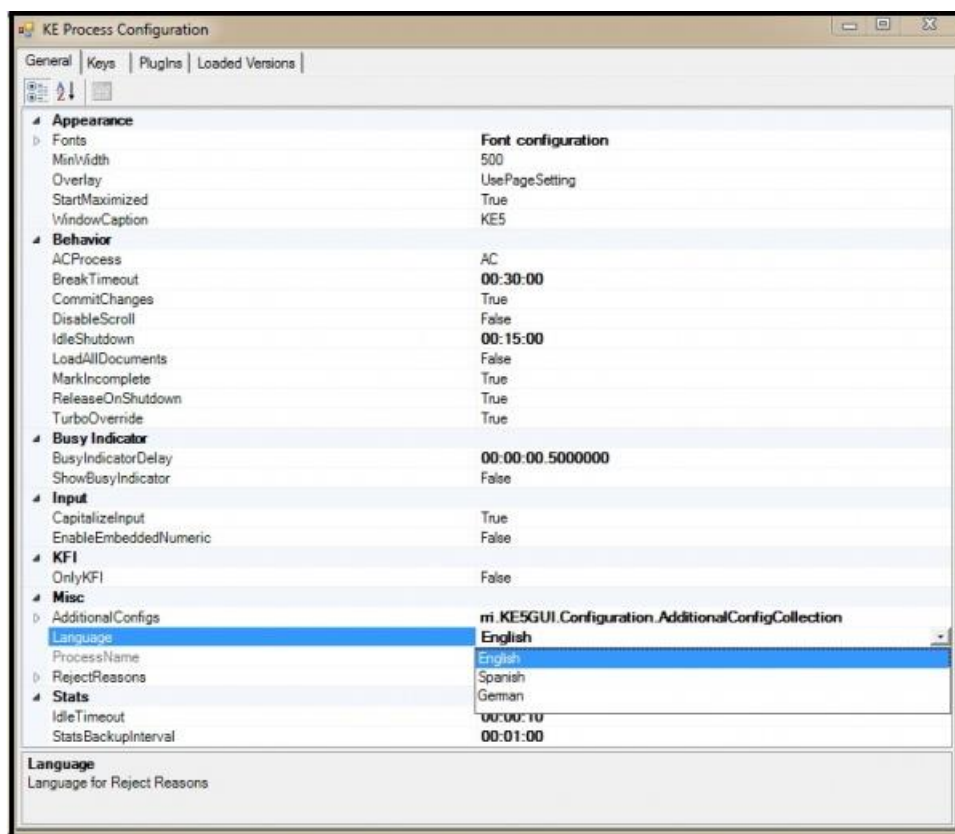
1. Open a KE process for which to configure the language.
2. A new property called Language which lists all the languages that had been configured earlier is available.



3. Right click on Page/Variant/Document/Batch, select Processes and then select KE.



4. The 'Default' option in the drop-down list allows the user to save the reject reasons into the FWD, whereas selecting the languages from the drop-down list will save the reject reasons to an external resource file (.resx).
5. If 'Show Default Language' is unchecked in the language resources then the KE process configuration will appear as below upon selecting Processes >> KE.



6. Right click on Page/Document/Batch, select Processes and then select KE.

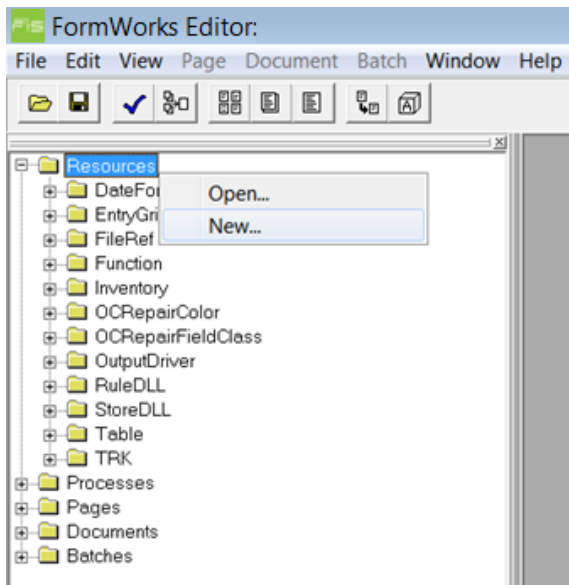


DateFormat Resource

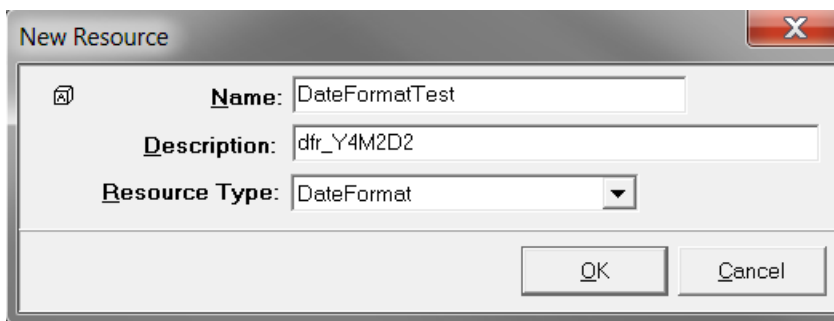
The DateFormat resources created are the date format selections available in all functions involving dates.

To Create A Date Format Resource

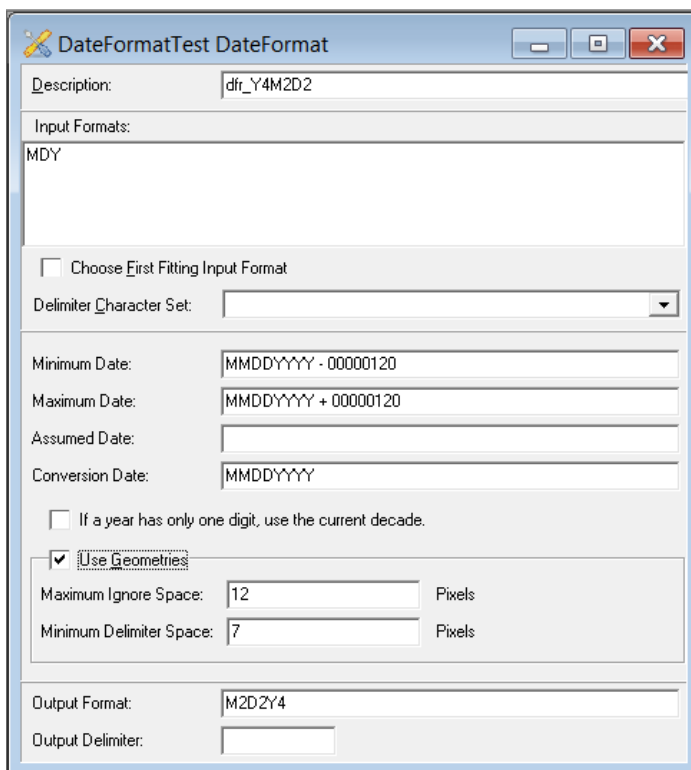
- Right-click **Resources** in the FWD tree (or DateFormat in the Resources list), Select **New**.



- In the New Resource dialog box enter a **Name and Description** for the resource, and select **DateFormat** from the list of resource types. The name entered should be descriptive of the format itself; use a name such as dfr_Y4M2D2 for an 8-digit date, Select **OK**.



- The DateFormat dialog box opens.



- Each group of available options within the DateFormat window is described below.

Input Formats

- **Format:** Enter the format(s) that will be read, using a date format expression to indicate date parts. For example, M2D2Y2 indicates a 2-digit month, 2-digit day, and 2-digit year; MDY24 is a month of 1-2 digits, day of 1-2 digits, and year of 2 or 4 digits.
- **Choose First Fitting Input Format:** When selected, a date that could be interpreted in multiple ways will use the first format that fits.

- **Delimiter Character Set:** Select the appropriate delimiter character set for the input date (if it contains delimiters). If the defined characters are found, the characters are treated as delimiters.

Note: If no character set is specified, the date is interpreted as having no delimiters.

- **Date Constraints**
 - **Min/Max Dates** – These dates refer to the minimum and maximum reason-able allowable dates, and are used to limit the range of dates. As a best practice, these are normally set to +/-120 or less depending on the specific type of date being modeled. In the example above, a valid date would be one that is today's date +/- 120 years.
 - **Assumed Date** – (optional) Use the Assumed Date to fill in values for dates with missing parts. The assumed date MM-01YYYY will fill in the last day of the month if the input field is missing the day.
 - **Conversion Date** – Determines the century for an input date with two digits. It plugs the largest century value that can be used without making the date in the future.
- **Current Decade** – Check box to enable verification that if a year in one of the Date Constraint fields (not the input date) should have only one digit, the system, by default will use current decade.

Use Geometries

If selected and geometries are available, geometries are used to determine groupings of numbers.

- **Maximum Ignore Space** – The maximum number of pixels between two characters that can be ignored (not considered a delimiter).
- **Minimum Delimiter Space** – The minimum number of pixels between two characters that can be considered a delimiter.

Output

- **Format** – Enter a date format expression to indicate the format of the output date.
- **Delimiter** – If the output date should contain delimiters, enter the delimiter character.

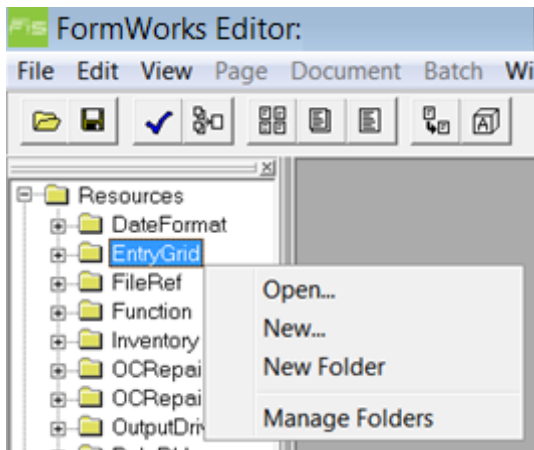
Entry Grid Resource

The Entry Grid editing dialog allows the configuration of an Entry Grid for KE5. The main part of the entry grid dialog is dedicated to displaying the actual entry grid, while the bottom is reserved for editing individual components.

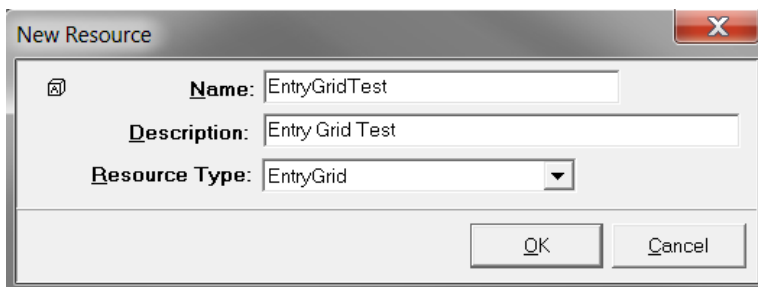
Note: Entry Grid is identified within FW5 also as EntryGrid.

Create A New Entry Grid.

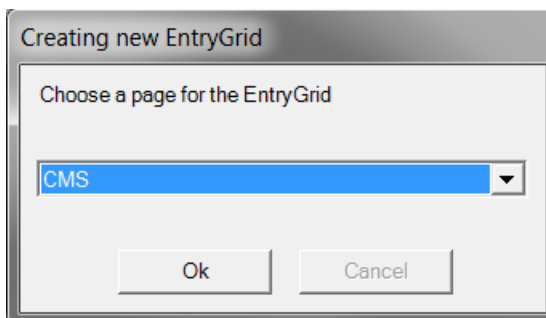
- Right-click Resources in the FWD tree (or EntryGrid in the resources list), select New.



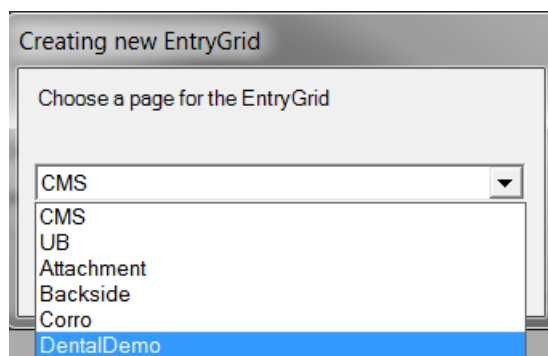
- In the New Resource dialog box enter a Name and Description for the Resource, and select EntryGrid from the list of Resource Types, select OK.



The Creating New EntryGrid window is displayed.

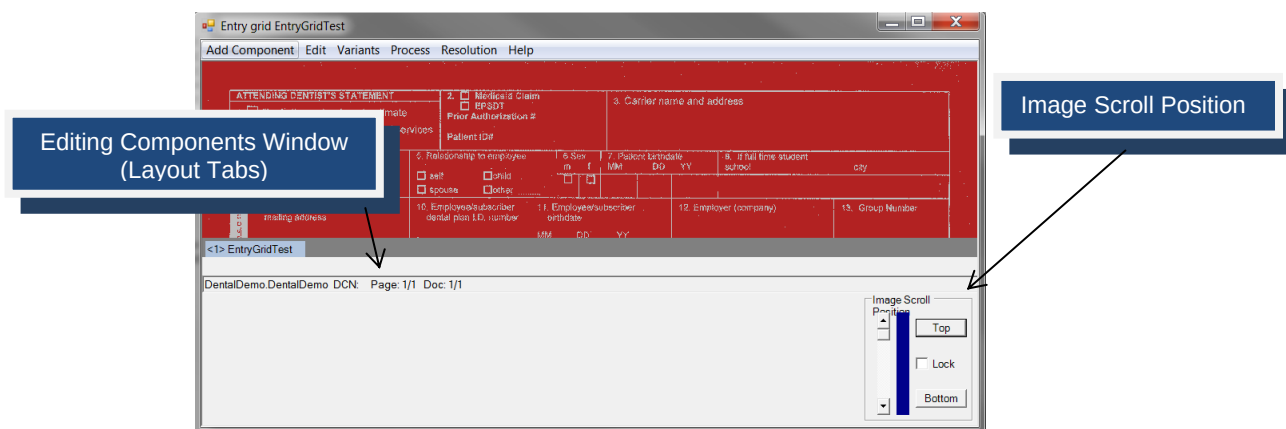


- From the drop-down box within the Creating New EntryGrid window, select the Page for which the entry grid is to be applied. Select OK from the Creating New EntryGrid window.



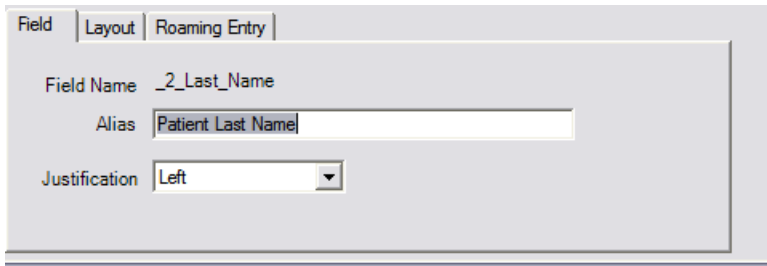
Warning – The Cancel button within the Creating New EntryGrid window is disabled. If the user wishes to cancel the creating of an entry grid, select OK, then delete the grid from the FWEdit Tree. A resolution to this issue is in place and applicable to future releases.

The Entry Grid window displays.



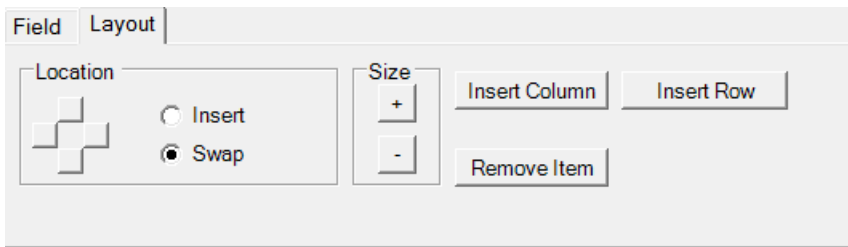
Field Tab

The Field tab displays when a field is selected and is a part of a Multi-Line field group. The Field tab allows the user to configure properties relating to the group for the particular field, as well as overall properties for the group.



- **Alias** – Optional, enter an alias name for the field. As a best practice, aliases should be concise yet descriptive. This is the field label that displays to the keyer.
- **Justification** – From the drop down box, select either Left or Right. This controls how the data will be justified within this field. For example, detail line charges in a column are typically right-justified.

Layout Tab



The Layout Tab moves and resizes components once they have been added to the entry grid. When the current component is a Multi-Line field (MLF) control, the layout operations operate on the group as a whole.

- **Location** – Moves components around on the entry grid. The four buttons  correspond to the four directions in which a component can be moved.
- **Insert** –  **Insert** When moving a component left or right, the component currently occupying the space is shifted down to make room for the new component.

Note: This shift can push other components down as well. When moving up or down, the components are swapped.

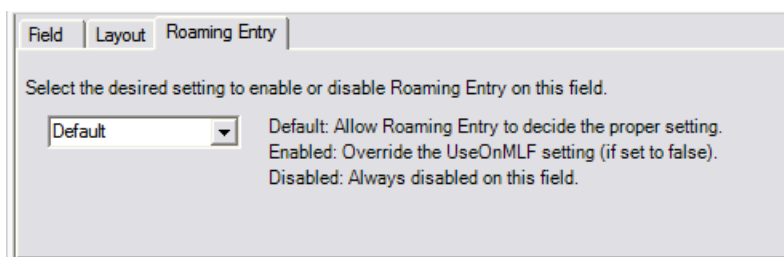
- **Swap** –  **Swap** The adjacent component swaps positions with the selected component.

Note: When a collision involves a component with a width greater than 1, the move is not allowed. As a best practice, shrink the components in order to move them successfully.

Roaming Entry

This tab allows editing of the roaming entry options for the field.

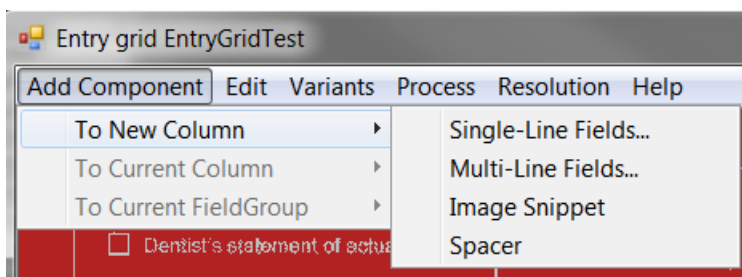
Note: The Roaming Entry tab will only display if the Roaming Entry Plugin in the KE Process Configuration is set up. Refer to KE Process Configuration for details.



Entry Grid - Add Component Menu

The Add Component menu allows the addition and assignment of components to the entry grid.

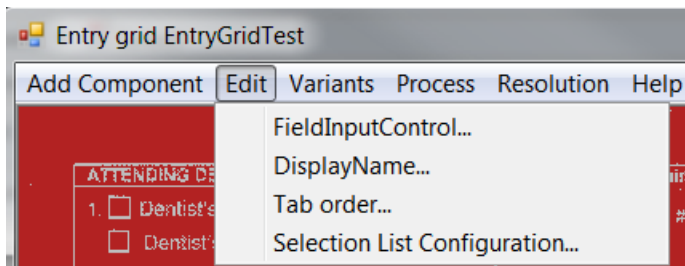
Note: See [Create A New Entry Grid](#) regarding information relating to creating an entry grid.



- To New Column – Starts a new column to the right of all the existing columns, beginning with the placement of components in the first row, after which a new row is added for each component. Column widths are adjusted to fit all columns.
- To Current Column – Becomes enabled once a component is selected. When initiated, places components at the bottom of the column, adding a new row for each component.
- To Current FieldGroup – Becomes enabled once a field is selected in a [MLF](#) Group. Adds a new field to that group. The only component available for selection in the To Current FieldGroup is a Multi-Line field.
 - Types of Components that may be added.
 - Single-Line Fields – Add Single-Line or [OMR](#) fields to the entry grid.
 - Multi-Line Fields – Adds Multi-Line fields in a FieldGroup.
 - Image Snippet – Shows a portion of the image within the Entry Grid. (Only visible in the desktop KE client),
 - Spacer – Used for aligning rows or adding space in columns.

Entry Grid – Edit Menu

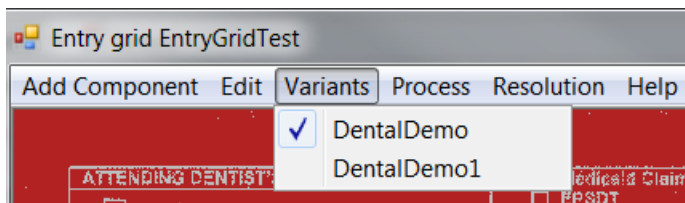
Allows the user to edit existing fields and data.



- FieldInputControl – Allows the modification of settings for the FieldInputControl.
- DisplayName – Changes the name of the entry grid as it appears in the EntryGridListDisplay.

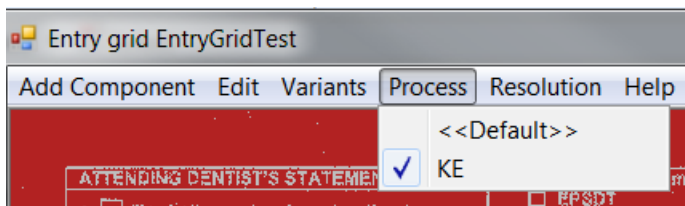
Entry Grid – Variants Menu

Displays a list of available variants, the selection affects which overlay (page variant image) is displayed and the geometry for the field highlighting. This option is cosmetic for the PCM only.



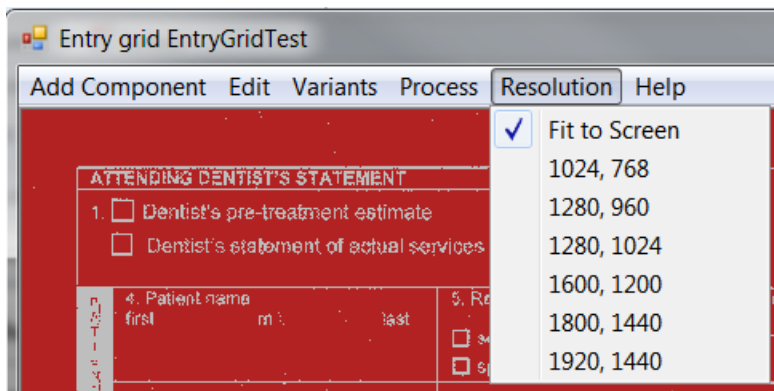
Entry Grid – Process Menu

A list of available processes. Select the KE5 process to be used to display the entry grid. The selection affects the [StatusBar](#), which reads its configuration from the process.



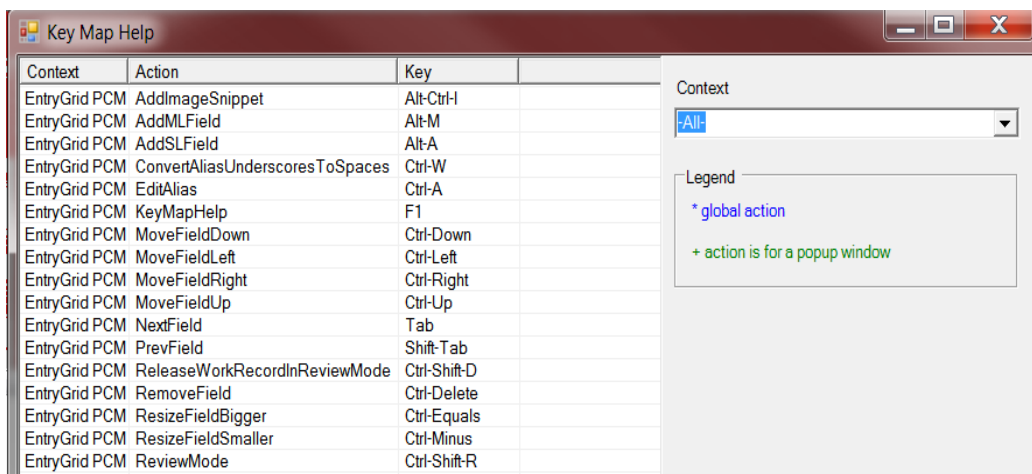
Entry Grid – Resolution Menu

Select the resolution at which to preview the entry grid. As a best practice, it is recommended to select the "Fit To Screen" option which automatically resizes the Entry Grid to fit inside the editing window.



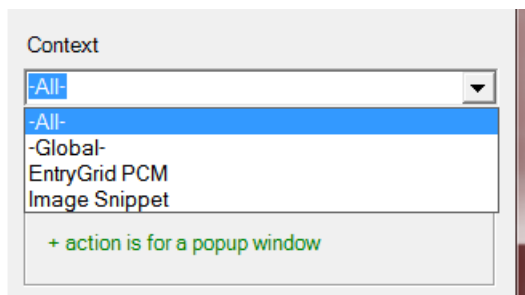
Entry Grid – Help Menu

Selection of Help displays the Key Map Help window.



From the Context drop down, select the list of help keys applicable to your need.

Note: Selection of All displays the complete list of Key Map actions and short-cut keys.



Entry Grid Hotkeys

See [Entry Grid - Help Menu](#) for a current list.

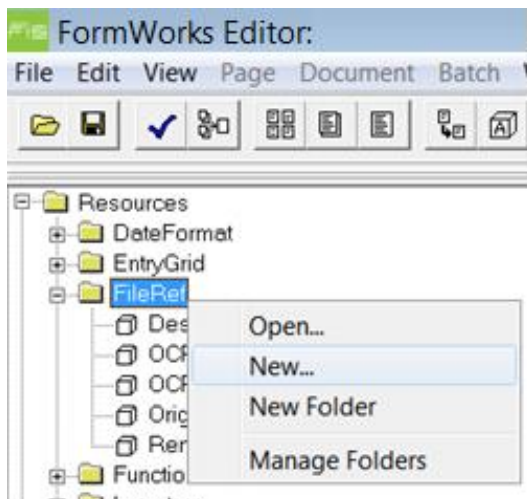
Fileref Resource

Filerefs are the names of image and OCR files, and are used throughout the system. Filerefs standard to the FWEEdit Express configuration are:

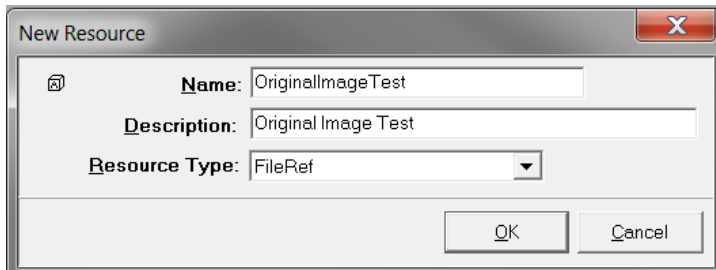
- OriginalImage – The image produced by a scanner.
- RemovedImage – The image after FIP processing.
- OCR_AEG – The OCR data file.
- DeskewedImage – FIP output. Deskews function is to straighten the image.
- OCR_AEG_Copy – Used for dual output of 4010 and 5010 X12/837 output.
 - This is a copy of the OCR_AEG fileref, created to support output of both 4010 and 5010 837 output. Without it, changes occurring during the first type of output creation are retained and affect the 2nd type of output. If only one version of 837 is generated, this fileref is not needed.

To Create a Fileref

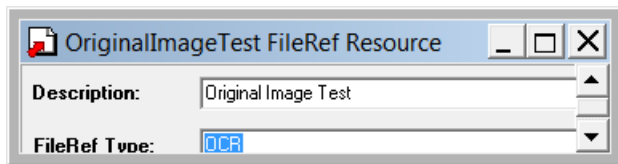
- Right-click Resources in the FWD tree (or Fileref in the resources list), select New.



- In the New Resource dialog box enter a Name and Description for the Resource, and select Fileref from the list of Resource Types, select OK.



- Click in the FileRef Type dialog box; select the appropriate FileRef Type by using the arrow keys to scroll through the selections.



- OCR
- Image

Function Resource

Function Resource constitutes the “User Defined” function category for AC and Store rules, and is typically used for complex rules, shared/reused algorithms, or rules that need to be run multiple times in a loop. Within the Function resource, named field list parameters and status results are defined.

Note: For information related to troubleshooting looping see [Troubleshooting Looping](#) within Appendix A.

In general, a User Defined Function (UDF) is a rule list, in which the rules refer to named field list parameters rather than any specific fields. These field lists are expanded against the list of fields passed to the user defined function when it is called by an AC or Store rule.

The interface for creating a function resource is identical to the AC and Store rules interface. Any function may be called, including other user defined functions, from a function resource. See [AC Page Processing](#) for more information.

Create and Configure an Inventory Resource

The Inventory resource contains configuration options for the Inventory Server (which receives information from Scan and Store Workers configured in the Processes list) and Client.

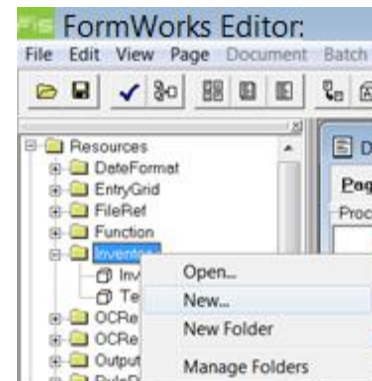
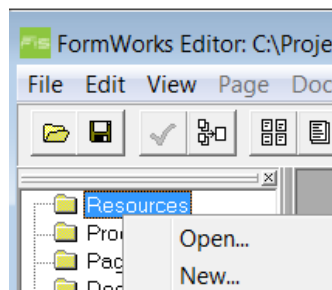
The Inventory Resource configuration is constructed primarily through three modules.

- Inventory Server:
 - Controls the creation of batches
 - Enforces DCN integrity

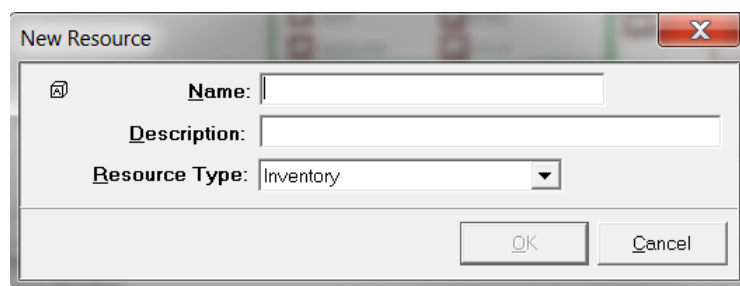
- Retires batches from a FormWorks system
- Inventory Client(s)
 - Provide a way to gather information about batches from an operator
 - Retrieve information from the server
 - Retire batches manually
- Inventory PCM
 - Provides a way to configure batch level settings
 - Such as selecting a DCN group
 - Enabling user attributes.

Note: The Inventory Server (worker) has configuration options in both the Inventory Resource and Processes list (under the Store process).

- Right-click Resources in the FWD tree (or Inventory in the resources list), select New.



- In the New Resource dialog box, enter a Name and Description for the Resource, and select Inventory from the list of Resource Types.



Inventory Resource

As a best practice, it is recommended that all batch types should be created before the Inventory Resource is configured. All batch types are needed to fully configure Option Groups within Inventory.

- Double-click the **Inventory Resource** name to open its configuration dialog box. Each set of options is discussed in the following sections.

Connection

The first tab in the Inventory Resource dialog box deals with connection information.

The screenshot shows the 'InventoryResourceSample Resource' dialog box with the 'Connection' tab selected. The dialog has several tabs: 'Connection', 'Client', 'Attributes', 'DCN Groups', 'Option Groups', and 'Printing'. The 'Inventory Server' section contains fields for 'Host' (LOCALHOST), 'Port' (1693), and 'Titlebar' (Inventory Server). There is a checkbox for 'Use an authentication database' which is currently unchecked. The 'Inventory Database' section contains fields for 'DSN', 'DBMS' (Microsoft SQL Server), 'UserID' (Administrator), 'Password' (masked with asterisks), and 'Schema (Optional)'. There is also a checkbox for 'Attempt to synchronize database' which is unchecked. The 'Server Extension DLLs' section contains fields for 'PreOp' and 'PostOp'.

Inventory Server

- **Host** – The host on which the inventory server is located (default value is “localhost”).
- **Port** – Port number for the inventory server (default value is 1693).
- **Authentication Database** – Select this when using validation for inventory server connections then enter the pathname to the authentication database. If no database is specified, no validation is performed.

Inventory Database

- DSN – The ODBC data source name of the database the server uses, on the host listed above.
- DBMS – Type of database. Currently supported options are Access, SQLServer, Oracle, and DB2.
- UserID and Password – The userID and password the inventory server uses to log in to the ODBC datasource. Both default to empty strings.

Important: Typically, Host, Port, DSN, User ID, and Passwords are typically set in the site.cfg files. Settings in the site.cfg files override any field configured here.

Server Extension DLLs

- *PreOp – multipurpose_inv.dll
- *PostOp – multipurpose_inv.dll

*Server extension DLLs may be required for the following reasons:

- Customer batch indexing (starting DCN) requirements call for a "customer batch number" in the DCN. This is the "BBB" in a DCN of: YYJJJ.BBB.000.
- To format attributes to overcome shortcomings in downstream components. The most common example of this is when an integer attribute with a value of "3" cannot be printed on the batch header sheet as "003" without reformatting it to a string via the server-side plugin dll.
- To extract information from other customer, FormWorks tables associated with the batch, but not directly in the USERATTRS table. If required, this can be done on the 'Post by calling the BatchAttrsGet callback. This makes it appear as if the procured information was in the USERATTRS table.

Client

Inventory Client configuration options are shown.

The screenshot shows a Windows-style dialog box titled "InventoryResourceSample Resource". It has a tabbed interface with the "Client" tab selected. The tabs are "Connection", "Client", "Attributes", "DCN Groups", "Option Groups", and "Printing". The "Client" tab contains several configuration fields: "Caption" (text box with "Inventory Client Utility"), "Retire Status" (text box with "1"), a checked checkbox "Allow client to retire batches that are in the entered state (", "Left Panel Width" (text box with "170"), a "Receipt Date" section with "Receipt Format" (dropdown with "MM/DD/YY"), "Receipt Date" (dropdown with "Any Valid Date"), "Attribute" (dropdown), and a checkbox "Hide controls for receipt date from user" (unchecked). Below this is a "Logging" section with "Logging Level" (dropdown with "Debug"), "Logging Style" (dropdown with "daily"), "Log Location" (text box with "./logs"), "Logfile Suffix" (text box with "inv"), and "Logfile Max Age" (text box with "30").

- Caption – The title that appears in the inventory client main window.
- Retire Status – The code (number) to use in the database for retired batches.

Note: As a best practice, Do Not Change without first consulting a Sr. FormWorks engineer.

- Allow Client To Retire Batches – Check box, defines whether or not the Inventory Client application can be used to manually retire batches.
- Left Panel Width – The width of the left panel in pixels of the Client application. This should be adjusted to allow names and other options to fit correctly without truncation or excessive space.
- Receipt Format – The format for the receipt date. A variety of formats are available

- Receipt Date – Date restriction for receipt date entered at the Inventory Worker. Valid selections are:
 - Any Valid Date
 - Today and Any Date in Past
 - Today and Any Date in Future
 - Use date from user attribute (enter attribute into the "Attribute" box)

Note that Receipt Date also supports entering the date in the JDay format.

- Logging Options – These are the typical worker logging options usually found in configuration files.

Note that these are the options for Inventory Client workers, not the Inventory server.

- Logging Level – Defines level of logging to be used in order to control logging output.
 - Fatal – Serious errors or messages that should always be output.
 - Warning – Warning or important messages that may be ignored.
 - Notice – Messages that give processing information.
 - Debug – Debugging messages.
 - Pedantic – All available debugging messages.

Note: As a best practice, Pedantic is not recommended. Selection of Pedantic logs all details. While purposeful for troubleshooting, Pedantic should generally not be selected for daily operation(s).

- Logging Style –
 - Default
 - Daily

- Null
- Log Location – Location of log files, user definable
- Logfile Max Age – As a best practice, recommend a value of 30 (days).

Attributes

Attributes refers to user attributes which may be entered at an Inventory Client Worker by an operator. The operator may elect to set the attributes to a default value, or an override value based on batch type. Attributes may be any available information that is required by the customer. For example, region, line of business, comment, or various dates. The attributes created on the Attributes tab appear in the batch configuration window, which is accessed through Option Groups (once batch types have been assigned to options). See [Option Groups](#) for additional information.

InventoryResourceSample Resource

Connection | Client | **Attributes** | DCN Groups | Option Groups | Printing | Print Script | Version

Production
DocsProcessed
DocsProcessedKFI
DocsRejected

Attribute Details

Name: Production

Load From File: ?

Caption: Production

Type: Captioned String

☒ Hide from view in client

Default: Y

☒ Use radio buttons rather than a combobox

Value	Caption
Y	Yes
N	No

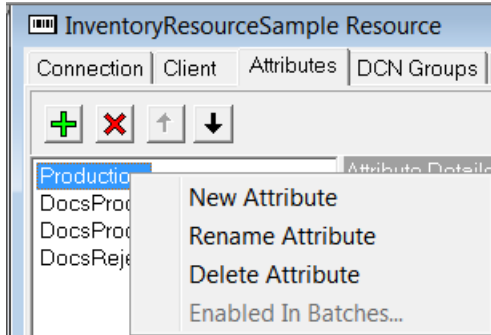
Enable In Batches

Each attribute can be enabled for specific batches. Attributes may be configured as editable, hidden or view only. Additionally, an override value may be set for an attribute at the batch level.

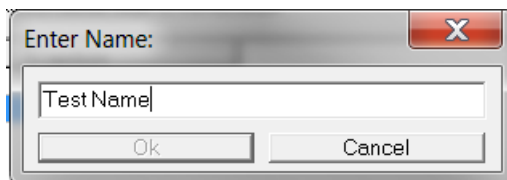
The basic process to create any of the five types of attribute is the same, but each attribute has unique properties which must be configured to process work as intended.

To Create an Attribute

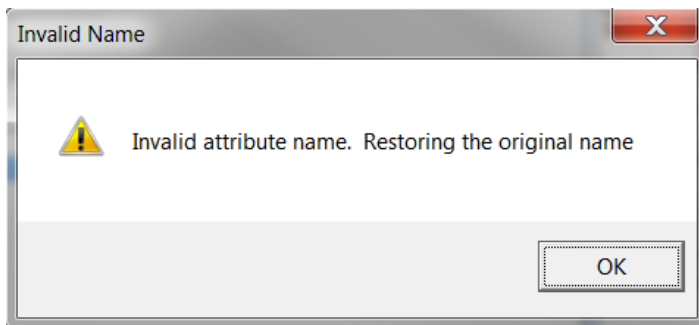
- Click the Add button  at the top of the window or right-click, select New Attribute (from within the Attributes list).



- Enter an Attribute name in the dialog box, (Attribute names may not include spaces. In the event an invalid Attribute name is entered the system will automatically deselect the OK button, not allowing the invalid naming convention to be saved).



- In the event the user attempts to change a current Attribute name with an invalid entry, the system displays the following. Select OK to return the Attribute name to the original entry.



- The name automatically populates the Name field.
 - Load From File – The configuration for the attribute is loaded from this file. To configure the Attribute using the PCM, Clear this option.
 - Caption – In the Attribute Details, Specify a Caption. The Caption displays differently than the name.

Note: The Caption is the label that is presented to the user.

Attribute Details

Name: Test

Load From File: [] ?

Caption: Displayed To User

- Type – The following illustrates the properties of each attribute.
 - Signed Integer
 - String
 - Formatted Date String
 - Captioned Signed Integer
 - Captioned String

Features Applicable to All Attribute Types

- Hide from view in client – (applicable to multiple types) – When this option is selected for an Attribute, the system will simply record the default value (or the override value, if present) for the Attribute. However, a user may still enable a hidden Attribute for selected batches in the Option > Groups settings.

Inventory Resource

Connection | Client | Attributes | DCN Groups | Option Groups | Printing | Print Script | Version

+ - < >

JDay

Production

DocsProcessed

DocsProcessedKFI

DocsRejected

RouteSelect

Priority

DocType

ClientName

InvDuplexOverride

Attribute Details

Name: JDay

Load From File: [] ?

Caption: JDay

Type: Formatted Date String

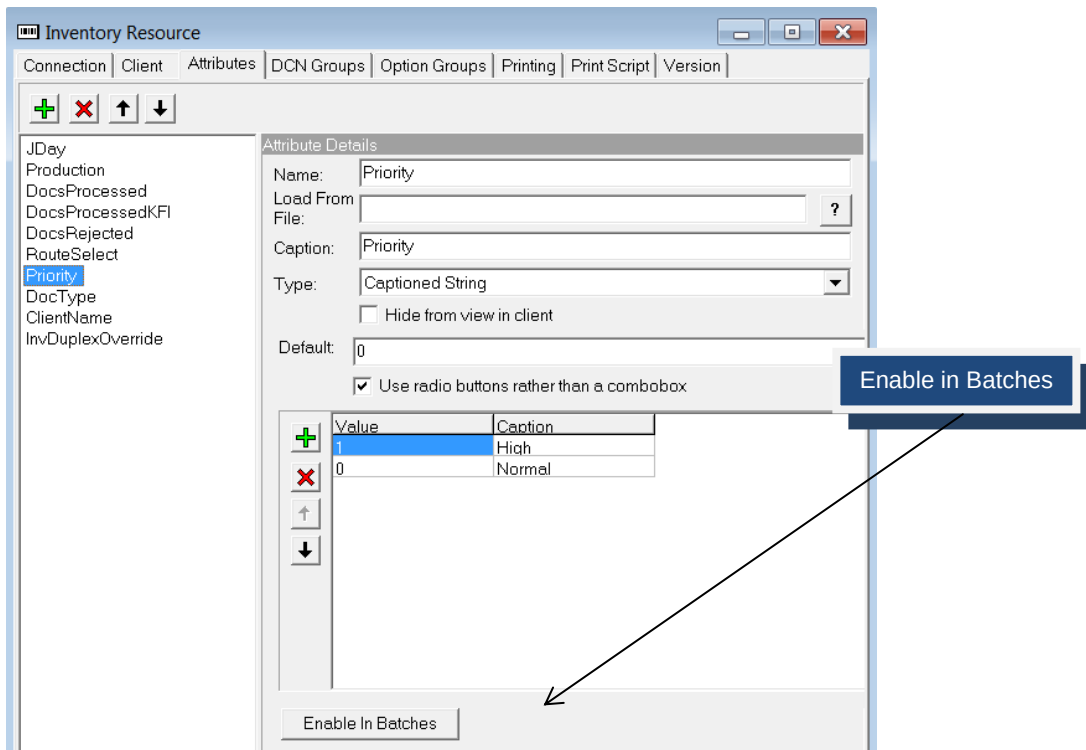
☐ Hide from view in client

Date Format: YYDDD

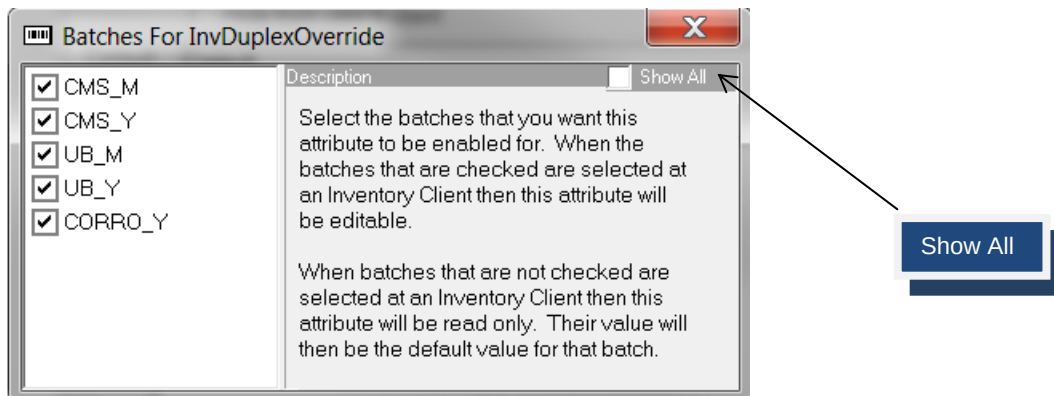
Hide from view in client

Note: Selecting the Hide from view in client option (checkbox) disables the Enable In Batches button at the bottom of the Attribute Details screen.

- Enable in Batches (applicable to multiple types) – If the attribute is to be visible in the Inventory Client, select the batch(es) for which an operator should enter information for this attribute.



- Click Show All to view a complete list of batch types.



*Note: For unselected batches, the attribute value appears as read only in the Inventory Client. For the attribute to not appear in the client window, select **Hide From View in Client** on the **Attributes Tab**.*

Formatted Date String Attribute

Any dates needed by the customer, such as a due date, can be created as Inventory attributes. The following figure illustrates a completed Formatted Date String attribute.

The screenshot shows the 'Inventory Resource' window with the 'Attributes' tab selected. On the left, a list of attributes is shown, including 'JDay', 'Production', 'DocsProcessed', 'DocsProcessedKFI', 'DocsRejected', 'RWT_KeyLocation', 'RWT_UniqueID', 'RWT_OtherBatchNum', 'DocInputType', 'DailyBatchNum', 'Scan_Hostname', 'DailyBatchNum04', 'RealStartingDCN', 'RealStartingPCN', and 'ScanOrientation'. The 'JDay' attribute is selected, and its details are shown on the right. The details include: Name: JDay, Load From File: (empty), Caption: JDay, Type: Formatted Date String, Date Format: YYDDD, and various date-related settings like Strip delimiters, Default Date, and date limits.

Note: Do not create a Receipt Date attribute; as a best practice this should be configured on the Client Settings tab.

- **Date Format** – Select a date format from the list. Month/Day/Year (M_D_Y) and Julian date formats are available (~20 different date formats are available).

The screenshot shows a list of date formats. The list includes: MM_DD_YY, MM_DD_YY_hh_mm, MM_DD_YYYY, MM_DD_YYYY_hh_mm, DD_MM_YY, DD_MM_YY_hh_mm, DD_MM_YYYY, DD_MM_YYYY_hh_mm, DD_MM_YYYY_hh_mm, YYYYDDD, YYDDD, YYDDD, DDDYYYY, DDDYY, DDDYY, DDDYY, DDDYY, YY_MM_DD, and YYYY_MM_DD. The 'MM_DD_YY' format is highlighted.

- **Strip delimiters on when writing to database** – Characters used to specify the boundary between regions or data streams are removed.
- **Default Date** – Set a default relative to today's date, i.e. the number of days in the future or past.
- **Lower Date Limit** – Set the earliest date allowed as the number of days in the future or past.

Note: If this setting is disabled, there is no lower date limitation.

- Upper Date Limit – Set the latest date allowed as the number of days in the future or past.

Note: If this setting is disabled, there is no upper date limitation.

- Warn User Settings – If needed, create a second date range for which the dates are allowable, but the operator must be confident about the entry. This is set as a number of days in the future or past (in the Warn user settings). For example, the upper and lower date limits might be 30 days in the past or future, but you want to issue a warning to the operator if the date entered is more than 15 days in the past or future. Disable these settings to omit the warning message.

Note: Two checkboxes, one check box states "Warn user if they enter a date too far in the past". The second check box states, Warn user if they enter a date too far in the future". These functions may be selected either individually or in combination.

Signed Integer Attribute

The Signed Integer attribute is similar to the Captioned Signed integer except that the Captioned Signed Integer attribute associates an alphanumeric string with a caption. Use the Signed Integer attribute to set a numeric value for a user attribute.

The screenshot shows the 'Inventory Resource' window with the 'Attributes' tab selected. On the left is a list of attributes: JDay, Production, DocsProcessed, DocsProcessedKFI, DocsRejected, RouteSelect, Priority, DocType, ClientName, and InvDuplexOverride. The 'DocsRejected' attribute is selected. The 'Attribute Details' panel on the right shows the following configuration:

- Name: DocsRejected
- Load From: (empty field with a question mark icon)
- File: (empty field)
- Caption: DocsRejected
- Type: Signed Integer (dropdown menu)
- ☒ Hide from view in client
- Maximum: 1
- Minimum: 0
- Default: 0

At the bottom of the panel is a button labeled 'Enable In Batches'.

- Maximum – Enter maximum allowable value.
- Minimum – Enter the minimum allowable value.
- Default – Enter a default value. By default the value is automatically set to the value entered into the Minimum field when the Attribute is created.

Captioned String - Captioned Signed Integer

As both attribute types are similar in functionality (screen Attribute layout) only the Captioned String attribute is defined below. Use these types to present an operator with a selection list of text values. The system stores the numeric value associated with each text option.

Captioned String	Captioned Signed Integer

- Default – Enter a default value (string or integer). In the next step, the caption associated with the default value will be entered.
- Value/Caption – Add options to the Value/Caption list. Click the Add Option button next to add rows to the Value/Caption list (right side of the window).
- A new option (line) appears in the list, with placeholder values (Value = 0, Caption = “New Caption”).

Value	Caption
0	New Caption 0

- Click the applicable field and enter the new Value and Caption.
- Existing options may be edited by simply selecting the option, then entering the new values.

Value	Caption
0	New Caption 0
0	New Caption 1
0	New Caption 2

String

Use a string type attribute for any text value to be entered by an operator or set by the system.

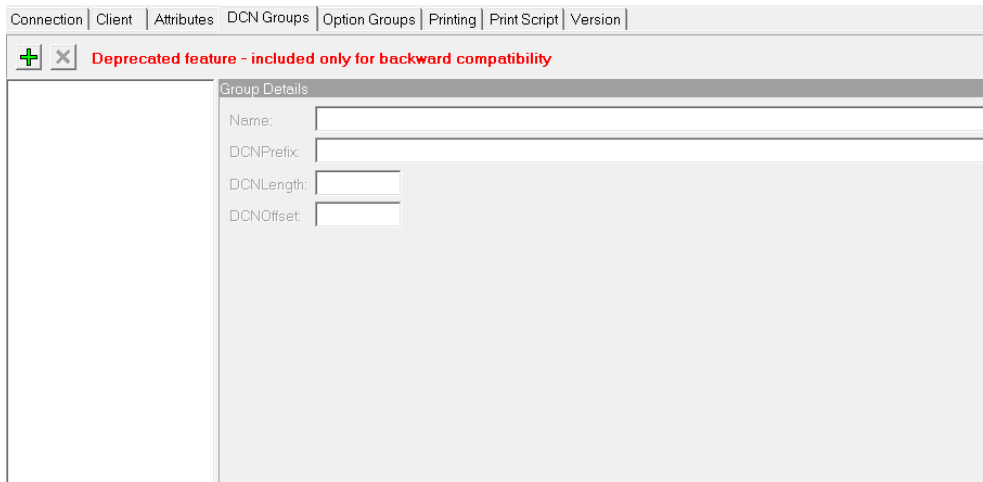
The screenshot shows the 'Attribute Details' tab in the FormWorks 2021 Q4 Editor Administrator Guide. The 'Name' field is 'Test'. The 'Load From' field is empty. The 'Caption' field is 'Displayed To User'. The 'Type' field is 'String'. The 'Hide from view in client' checkbox is unchecked. The 'Default' field is empty. The 'Maximum Length' field is '10'. The 'Minimum Length' field is '1'. The 'Force Uppercase' dropdown is set to 'False'. There is an 'Enable In Batches' button at the bottom.

- Default – Enter a **default value**. Make sure the default does not exceed the Maximum Length value (next field).
- Maximum Length – Enter the **Maximum Length** (number of characters).
- Minimum Length – Enter the **Minimum Length** if the Maximum Length is defined greater than zero.
- If the string value is to be forced to uppercase, select **True**, else select False.

DCN Groups

The Inventory Server guarantees that each document has a unique DCN. Each system typically has different requirements for DCNs, but in general, DCNs are composed of two parts.

Important: DCN Groups (tab) is NOT a deprecated feature. The deprecated message displayed within the DCN Groups tab is erroneous. This message will not be displayed in future versions of FWEdit



- Prefix – Information entered at the client by an operator or default values for attributes based on the batch type (such as Julian Date, BatchNum, Region, and other attributes).
- Sequence number – The customer may require a particular starting number.

To Create A DCN Group

- Click the Add button .
- Enter a Group Name in the dialog box.
- Enter a Prefix. This is a string that is passed to a print attribute to format information in the desired way. For example, you may enter a string to have just the Julian Date or concatenate attributes like Julian Date, Region, and Batch Type.
- Enter a DCN length. As a general guideline scanners limit the endorsed DCN to 12 digits. For verification, contact your Sr. FormWorks engineer for verification.

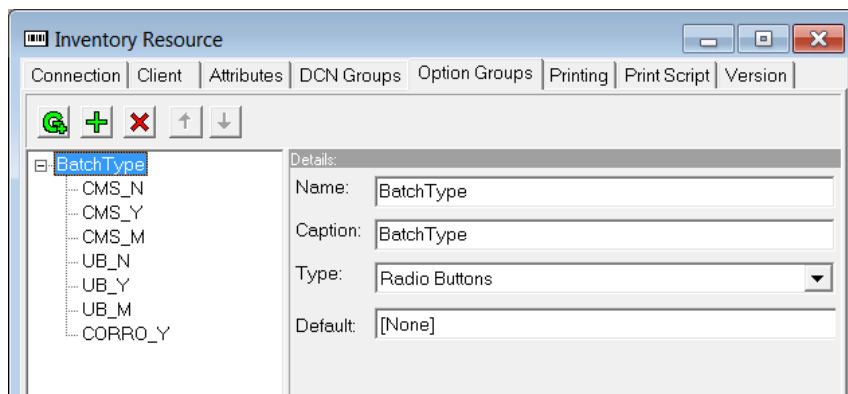
Important: Make sure that the number of digits in the DCN Sequence Number plus the number of digits in the DCN Prefix do not exceed the stated length. No check is performed on the DCN length in FormWorks Editor. In the event there are too many digits defined, the Inventory Client worker generates an error.

- Enter the DCN Offset. This is the starting sequence number for each day; usually set to 0 or 1, but the customer may have other requirements.

Option Groups

Option Groups are the selections presented to an operator at an Inventory Client worker. These groups are designed as a decision tree, starting with the most general option, and narrowing down to the most specific, limiting the options presented to the operator. When the client is opened, the user is presented with the first option group only (such as Form Type).

Note: Once a selection is made, only the appropriate options from the next group are presented.

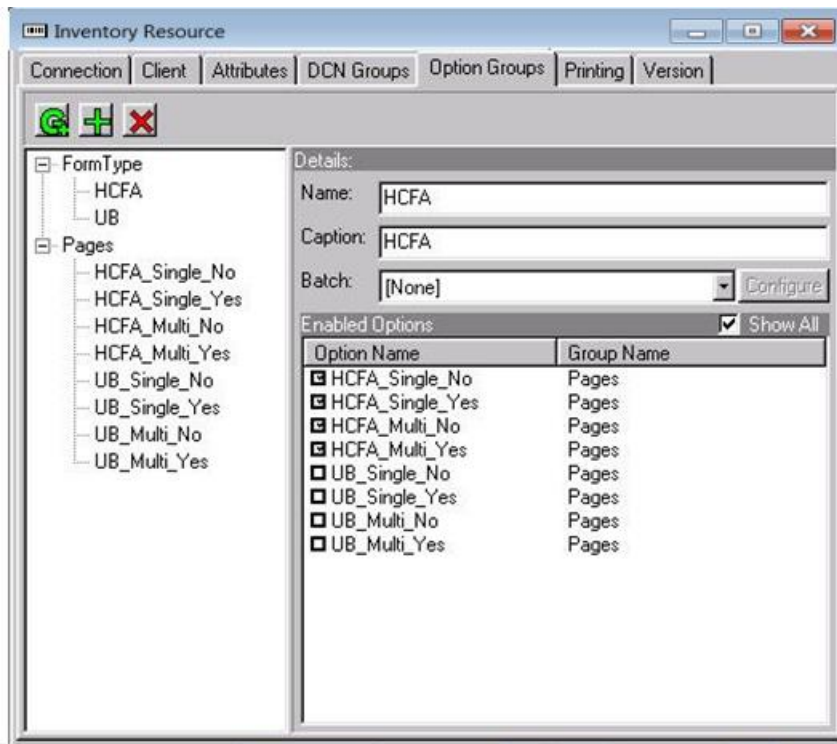


The general selection of form type is presented first. The form type selected (HCFA or UB) controls which page options are presented; the operator sees only the relevant pages for any particular form type.

Once the most specific level of options have been reached, Batches (in the following example), assign a specific batch to the option. After the selection(s) have been made, select the **Configure** button to list the batch level options (see [Batch Level Inventory Options](#) for details).

Option Groups - Create A Decision Tree

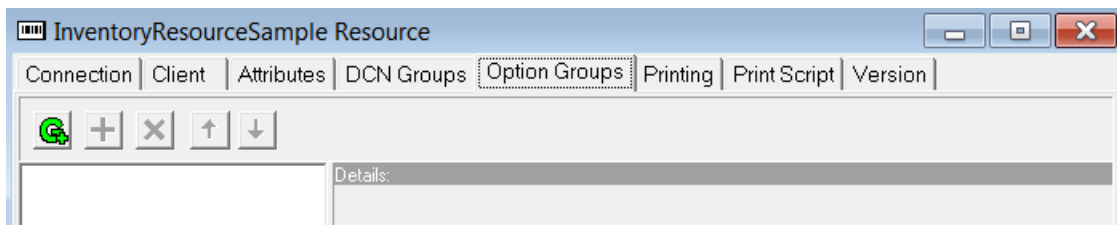
The following sections describe how to create the example shown in the following example.



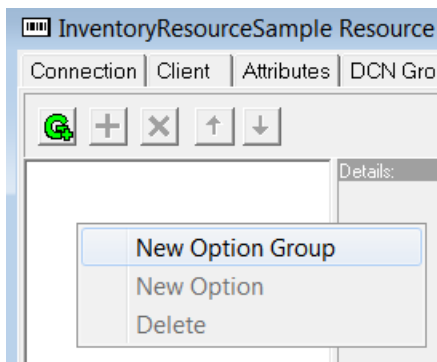
Note: Individual systems may vary in the availability of options and groups.

Add a New Group

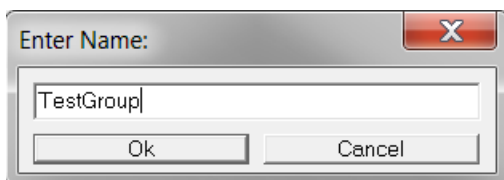
From the Option Groups tab



- Click the **New Group** button , or Right-click in the **Options** list, and select **New Option Group**.

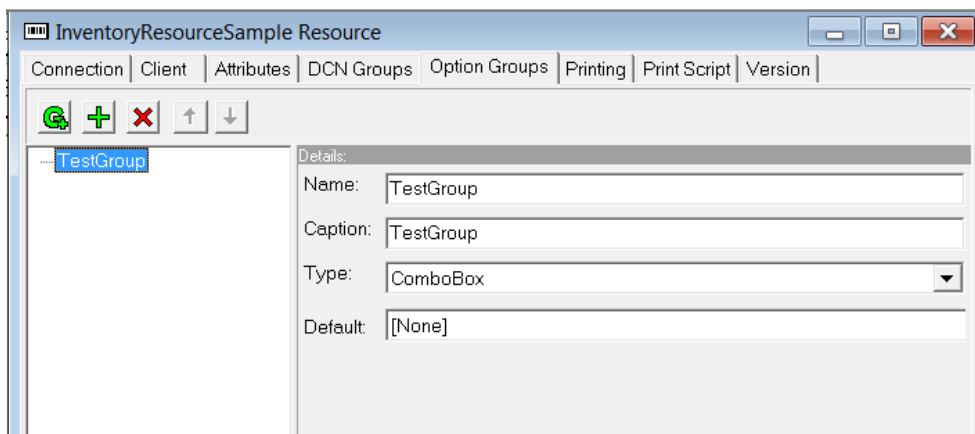


- Enter a **Name** for the group in the dialog box (FormType), Select **OK**.

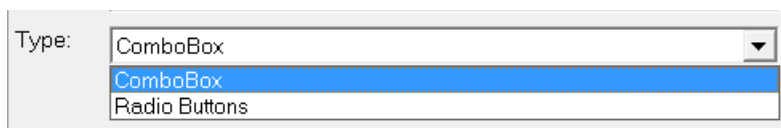


- Enter a **Caption** for the group, if different from the group name (Form Type). The caption is the text that appears in the Inventory Client window when creating a batch, and may include spaces.

Note: Users never see the group name.



- Select the entry type, either ComboBox or Radio Buttons.



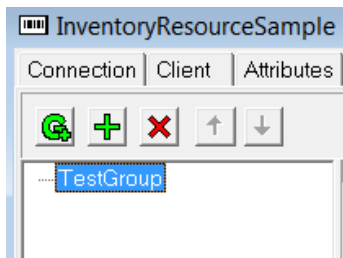
Note: A best practice is to use ComboBox for the option type as Radio Buttons utilize more screen real estate and are often found to be confusing to the majority of users.

- By design, the value for the default field is None. If a default value is required, enter a value into the default field.

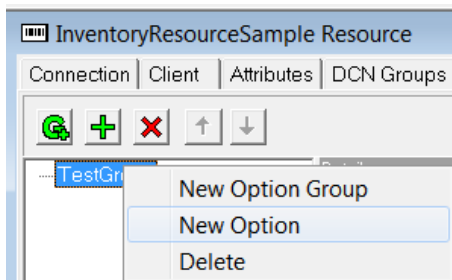
Default:

To Add an Option to a Group

- Select the Option Group name in the options list (left side of the window).

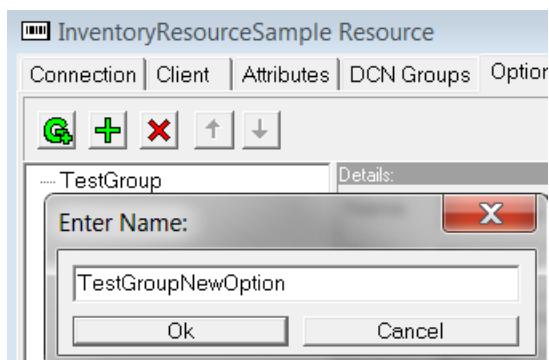


- Right-click Option button , or right-click the Group Name and select New Option.



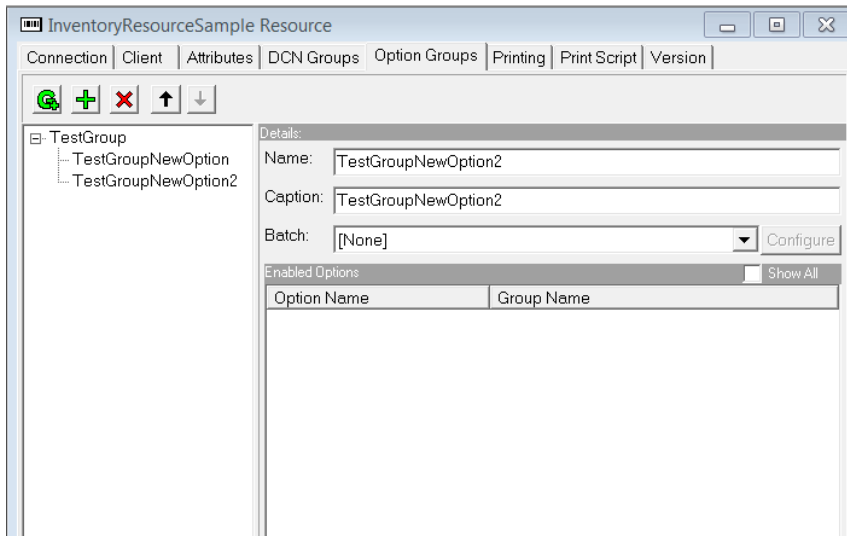
- Enter a Name for the option in the dialog box.

Note: Names may not include spaces, punctuation or extraneous characters



- Enter a Caption for the option, if different from the name. The caption is the label that appears in the Inventory Client window when creating a batch.
- Add the second option to the group. Click the Add Option button , or right-click the Group Name and select New Option.
- Enter a Name for the option in the dialog box.

The Option Group now appears as follows.



Note: When creating the first group, there are not any available options to select in the Enabled Options section. Set up all the necessary option names in the first group, then create the second group, and finally return to the first group and select the appropriate options to be enabled.

In the majority of production environments, many systems only have one option group for BatchType, and no enabled options are defined.

To Enable Options

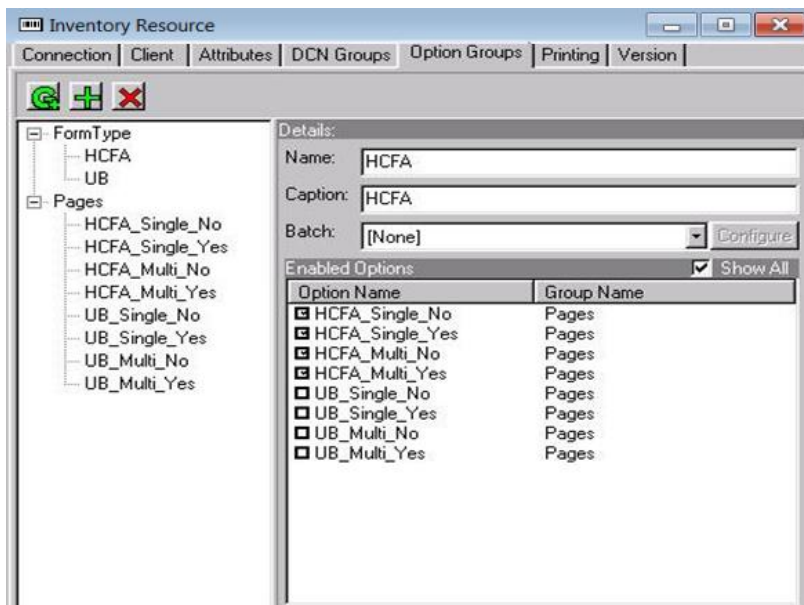
- Follow the instructions to [Create Pages and Variants](#) (and options).
- In the first group (Form Type), select the first option (HCFA for this example).
- In the Enabled Options section, select Show All if necessary to see a list of all options.

Note: The Show All setting shows all options from all successive groups, not other options within the current group.

- Select which options from the next group (Pages) should be enabled when the current option (HCFA) is selected by the user. For this example, check all HCFA options.

- Repeat the following steps for the second option in the FormType group (UB); enable all the UB options.
 - In the first group (Form Type), select the first option (HCFA).
 - In the Enabled Options section, select Show All if necessary to see a list of all options.
 - Select which options from the next group (Pages) should be enabled when the current option (HCFA) is selected by the user. For this example, check all HCFA options.

The Option Group should now appear as follows.



To Associate a Batch with an Option

In this example, each option listed in the Pages group is specific to only one Batch Type. If the system has additional levels of options, create another group with the appropriate options.

- Select the first option in the Pages group (HCFA_Single_No).
- Select the batch that is associated with this option in the Batch list (HN). Then select Configure to see batch-level options. (See [Batch Level Inventory Options](#)).
- Repeat for each option in the Pages group.

Batch Level Inventory Options

This Batch Level Inventory Options are accessed through the Option Groups tab in the Inventory configuration window. In the last group of options;

- Select the **Option** with which a batch is associated.
- Click the **Configure** button to open the batch settings for the selected batch.

Note: Unlike processes, resources (Inventory) cannot be configured by opening a database object and then double clicking an icon. All batch-level Inventory configuration options are accessed through the Inventory Resource, on the Option Groups tab.

General Tab

- Enter a caption, if different from the batch type name.
- Number of Batches – The default is the number of batches of this type to be created, unless changed by the operator.
 - Set appropriate minimum and maximum levels, (As a best practice, the minimum is normally set to 1).
- Number of Documents – The default batch size for this batch type. Typically, a default of 50, but the customer may have other requirements.
 - Set appropriate minimum and maximum levels, (As a best practice, the minimum is normally set to 1).
- Printing – These options are identical to those on the Printing tab of the Inventory Server configuration window, but for this example case they apply only to the selected batch. These settings control the printing of batch header sheets. If the print settings are changed, they override the system level settings. See Printing for additional information.

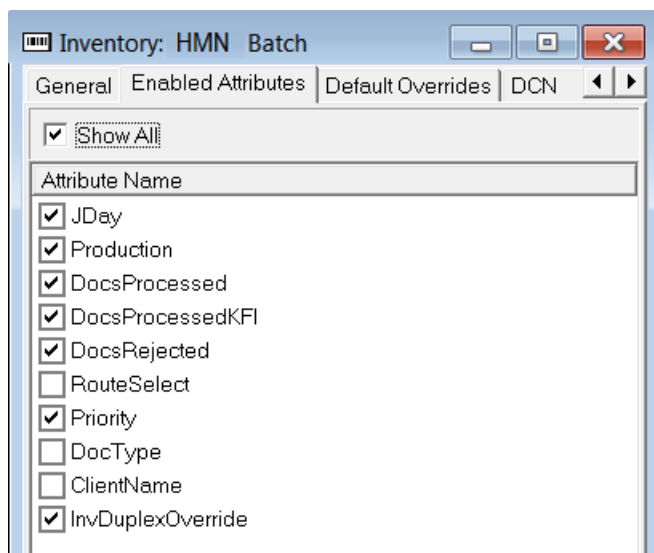
The screenshot shows a configuration window titled "Inventory: HMN Batch". It has four tabs: "General", "Enabled Attributes", "Default Overrides", and "DCN". The "General" tab is active. It contains the following fields:

- Caption:** A text box containing "HMN".
- Number of Batches:** A section with three input fields: "Default: *" (value 1), "Minimum:" (value 1), and "Maximum:" (value 10).
- Number of Documents:** A section with three input fields: "Default: *" (value 50), "Minimum:" (value 1), and "Maximum:" (value 100).

At the bottom of the window, there is a note: "(*) Blank default indicates user".

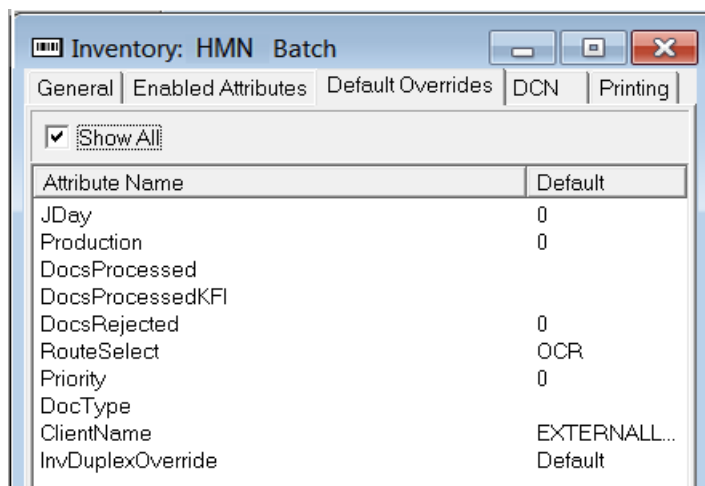
Enabled Attributes Tab

Select **Show All** to see a list of all user attributes. These are the attributes created on the Inventory Server Attributes tab, and assigned to selected batches. Enable or disable attributes as needed for the current batch.



Default Overrides Tab

This is a listing of the attribute override values for the current batch type.



To Set an Override Value

Warning: Make sure that any override value entered is valid, given the attribute type (string, signed integer, etc.). This functionality is considered an advanced feature, as the PCM performs minimal checking to determine if the override is legal.

- Click **Show All** to see the complete list of attributes (those that do not have an override value appear with the original default value).

- Right-click the **Attribute Name**, select **Edit**, and enter the override value for the batch.
- To Remove an Override Value, right-click the **Attribute Name** and select Remove. (If **Show All** is not enabled, the attribute disappears from the list; if **Show All** is enabled, the attribute appears with its original default value.)

DCN Tab

At the DCN tab, select from the **DCN Group** drop down the group for which the batch is to be assigned.

Note: This list contains the groups created at the Inventory DCN Groups tab

In the following figure, the batch type CMS_Y is assigned to the Default group. The prefix, length, and offset are plugged from the values you set when the group is created.

If this batch does not belong in any existing group, you can set the Prefix, Length, and Offset here. However, for the sake of consistency, as a best practice, it is best to return to the DCN Groups and create another group. Then, if other batch types are added that will follow the same DCN format, they need only be assigned to the group.

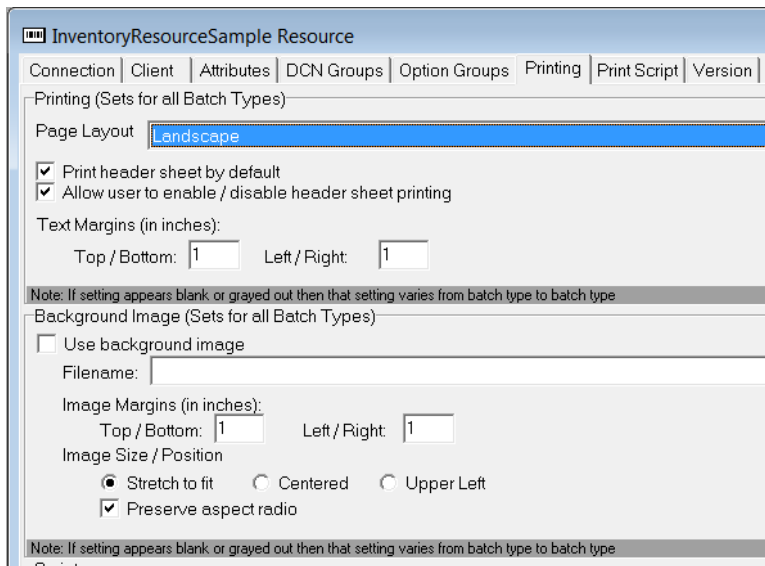
Important: DCN Groups (tab) is NOT a deprecated feature. The deprecated message displayed within the DCN Groups tab is erroneous. This message will not be displayed in future versions of FWEdit

The Print Option

The settings in the Printing tab control the default batch header sheet print options for all batches.

Note: As a best practice, when setting up a new system, it is helpful to disable the Print Header Sheets by Default setting to avoid printing unneeded header sheets during testing. (Retain the Allow User to Enable/Disable header sheet printing selection to header sheets may still be printed (if necessary)). Then enable Print Header Sheets by Default when the system is installed.

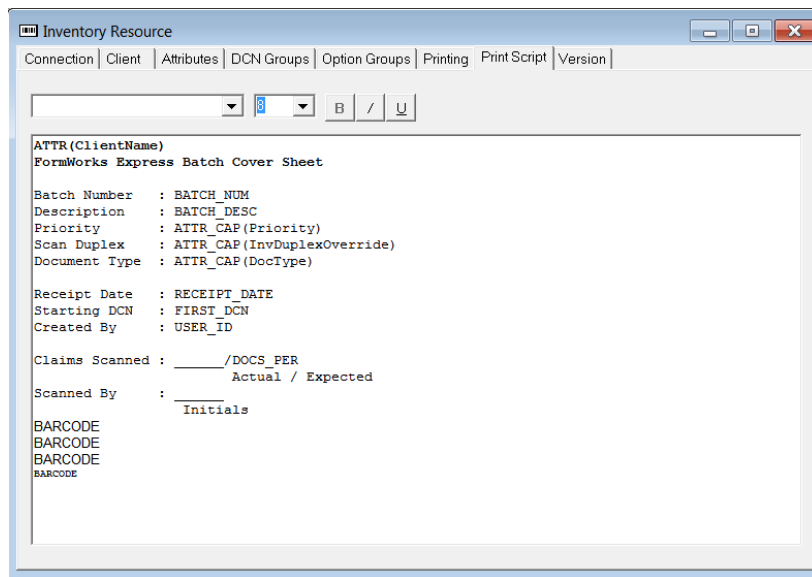
If different batch level print options are set (from the Option Groups configure button), the batch level setting overrides the system setting.



- Page Layout – Select either Landscape or Portrait.
- Print header sheet by default – Check to always print the header sheet.
- Allow user to enable/disable header sheet printing – Check to allow the user to turn on/off the printing of the header sheet.
- Top Margins – Set the value for which Top/Bottom and Left/Right margins should be set. As a best practice it is recommended not to set these values for less than .5 inches.
- Use background image – Check to allow a background image to be printed.
 - Filename – .TIFF format, enter the path and name of the file.
- Image Margins – Set the value for which Top/Bottom and Left/Right margins should be set. As a best practice it is recommended not to set these values for less than .5 inches and not to exceed the Page Layout margins.
- Image Size/Position – Select to offer the best presentation of the image.

Print Script

In the Print Script area, list the captions and attribute names for the items to be printed on the batch header sheets. When the header sheet is printed, the attribute names are replaced with the values for those attributes or with the associated value caption where specified and applicable.



Inventory Resource

Connection | Client | Attributes | DCN Groups | Option Groups | Printing | Print Script | Version

ATTR(ClientName)
FormWorks Express Batch Cover Sheet

Batch Number : BATCH_NUM
Description : BATCH_DESC
Priority : ATTR_CAP(Priority)
Scan Duplex : ATTR_CAP(InvDuplexOverride)
Document Type : ATTR_CAP(DocType)

Receipt Date : RECEIPT_DATE
Starting DCN : FIRST_DCN
Created By : USER_ID

Claims Scanned : ____/DOCS_PER
Actual / Expected

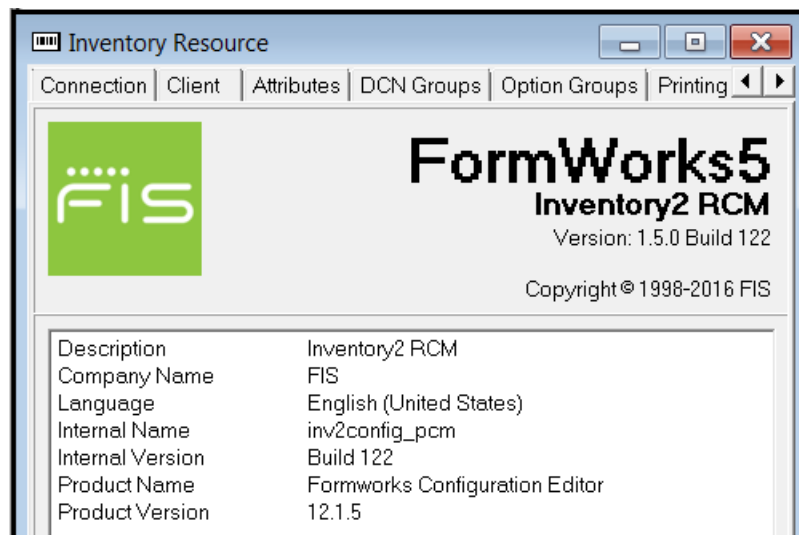
Scanned By : ____
Initials

BARCODE
BARCODE
BARCODE
BARCODE

Note: The attribute names in this window are case sensitive

Version

Within the window, displays to the user the current version and configuration information.



Inventory Resource

Connection | Client | Attributes | DCN Groups | Option Groups | Printing | Version

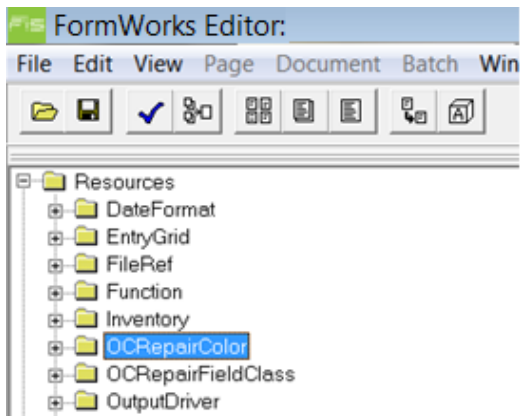
 **FormWorks5**
Inventory2 RCM
Version: 1.5.0 Build 122
Copyright © 1998-2016 FIS

Description	Inventory2 RCM
Company Name	FIS
Language	English (United States)
Internal Name	inv2config_pcm
Internal Version	Build 122
Product Name	Formworks Configuration Editor
Product Version	12.1.5

OCRepair Color

OCRepair workers use color coding to help operators determine what kind of data they are keying. These colors provide important clues about data when there is no other context, allowing an operator to differentiate between ID numbers, dollar amounts, dates, and other data.

To access the OCRepair Colors screen, Double-click the **OCRepairColor** group displayed in the tree.



The OCRepair Colors window displays (following figure). For the majority of systems, this configuration is acceptable for production. However, if colors do not display as expected on the customer's monitors, color definitions may be modified or new color definitions may be created.

Note: The additive color method is utilized.

 The screenshot shows the 'OCRepair Colors' window. It has a title bar with a pencil icon and the text 'OCRepair Colors'. Below the title bar are two buttons: a green plus sign and a red minus sign. The main area contains a table with 5 columns: Color, Red, Green, Blue, and Sample. The table lists 12 colors: Blue, Brown, Gray, Green, Olive, Orange, Pink, Purple, Red, and Teal. The 'Brown' row is highlighted. At the bottom of the window, a yellow banner contains the text 'Colors may differ on machines with different palette'.

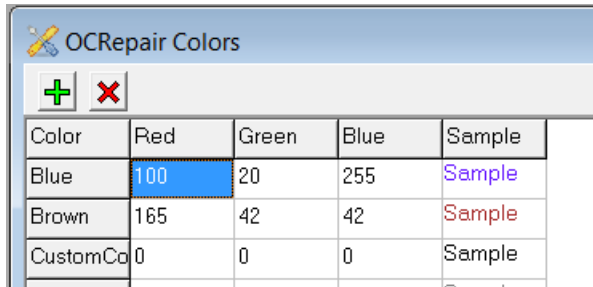
Color	Red	Green	Blue	Sample
Blue	100	20	255	Sample
Brown	165	42	42	Sample
Gray	128	128	128	Sample
Green	125	255	35	Sample
Olive	200	200	50	Sample
Orange	255	165	0	Sample
Pink	255	0	240	Sample
Purple	255	0	150	Sample
Red	255	100	50	Sample
Teal	2	200	175	Sample

Individual colors may be edited by selecting individual primary colors (three primary colors make-up each color) and editing the value.

Important: Once an individual primary color value has been modified, the only course to return the primary color to the original value is to manually reenter the original value. There is not a return to default function. As a best practice, it is recommended to take a screen capture of the OCRepair Colors screen so the original values may be reentered into the table if required.

Remove Color

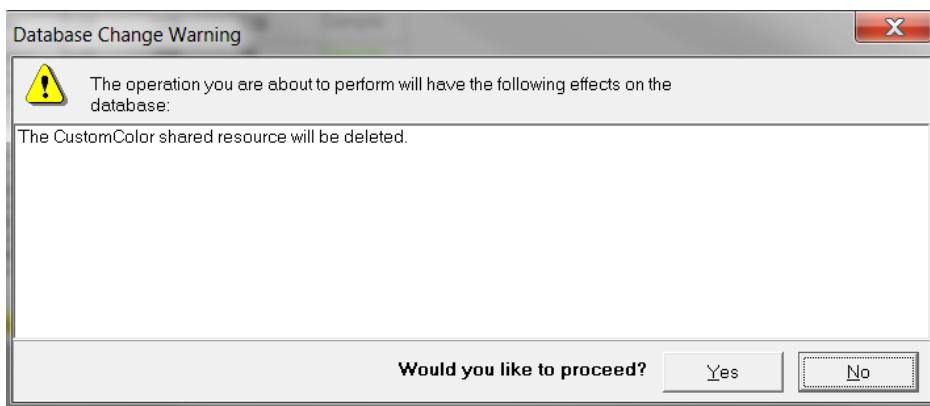
To remove a color from the listing within OCREpair Colors, place the cursor within one of the three primary color fields (Red/Green/Blue) adjacent to the color to be removed and select **Delete selected color** .



The image shows the 'OCRepair Colors' dialog box. It has a title bar with a wrench icon and the text 'OCRepair Colors'. Below the title bar are two buttons: a green plus sign and a red X. Below these buttons is a table with the following data:

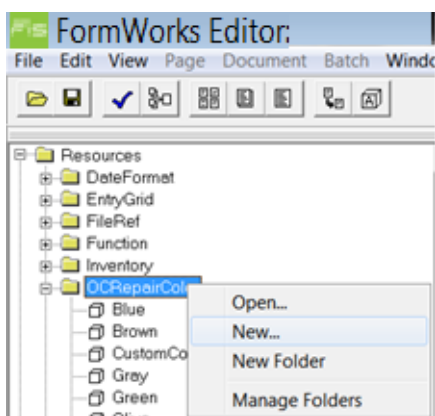
Color	Red	Green	Blue	Sample
Blue	100	20	255	
Brown	165	42	42	
CustomCo0	0	0	0	

The Database Change Warning window displays, select **Yes**  to delete the color.

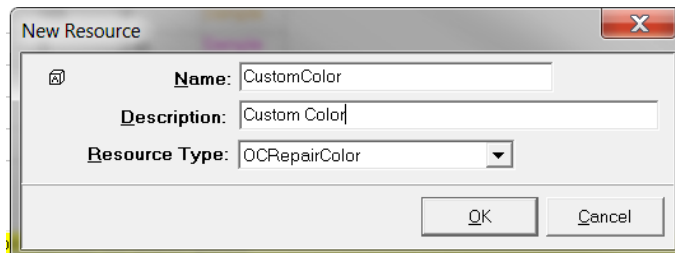


Add Custom Color

- Right-click **OCRepairColor** in the Resources list.
- Select **New**.



- The New Resource box is displayed.

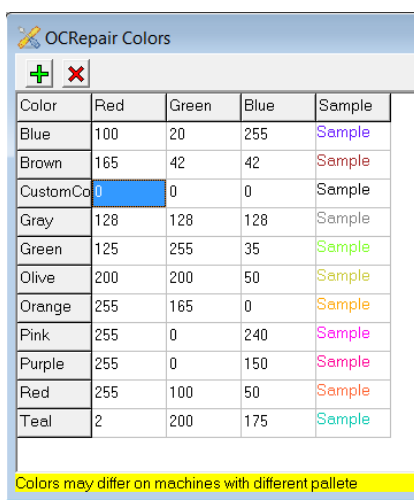


The 'New Resource' dialog box is shown. It has a title bar with a close button (X). Inside, there are three fields: 'Name' with the value 'CustomColor', 'Description' with the value 'Custom Color', and 'Resource Type' with a dropdown menu showing 'OCRepairColor'. At the bottom right are 'OK' and 'Cancel' buttons.

- Enter the name of the new Color in the **Name** field.
- Enter a brief description of the color in the **Description** field.
- Select the **Resource Type – OCRepairColor** (Group) from the drop down box.

Note: Resource Groups available may vary from system to system.

- Select OK, the newly created color (CustomColor in this example) is added to the OCRepair Colors listing.



The 'OCRepair Colors' window shows a table with columns: Color, Red, Green, Blue, and Sample. The 'CustomColor' row is highlighted. Below the table is a yellow warning bar.

Color	Red	Green	Blue	Sample
Blue	100	20	255	Sample
Brown	165	42	42	Sample
CustomColor	0	0	0	Sample
Gray	128	128	128	Sample
Green	125	255	35	Sample
Olive	200	200	50	Sample
Orange	255	165	0	Sample
Pink	255	0	240	Sample
Purple	255	0	150	Sample
Red	255	100	50	Sample
Teal	2	200	175	Sample

Colors may differ on machines with different palette

- For the newly created custom color, Enter a numerical value for each of the three primary colors.
- Select  to close the window and save changes.

Note: When creating a new color, be sure to update any field class resources that will utilize the new color.

OCRepairFieldClass

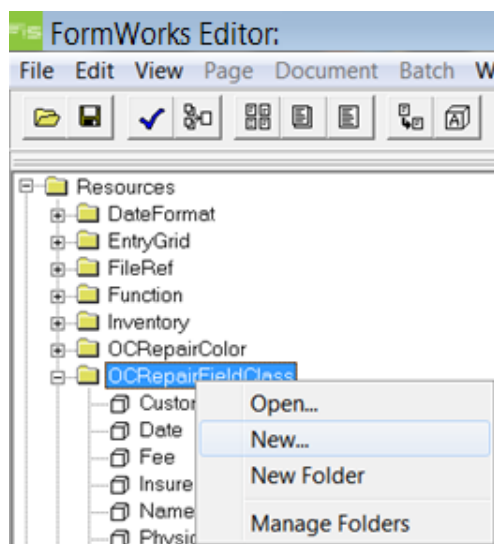
Use the Field Classes resource to create groups of fields, such as Date, Name, and Fee, and assign a color to each.

Note: For healthcare systems, there are standard color codes: Date = blue, Fee = Green, InsuredID = Red, Name = Olive, Unit = Orange, Procedure = Purple, Physician ID = Teal. All other fields are black.

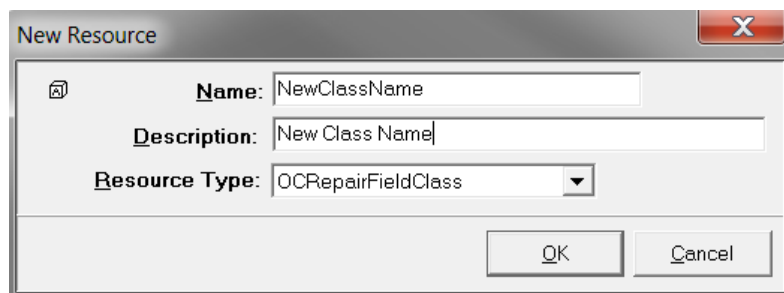
To Create a New Field Class

Important: As a best practice, for the creation of a new Field Class and attributes, consult a Sr. FormWorks System Engineer before attempting to configure.

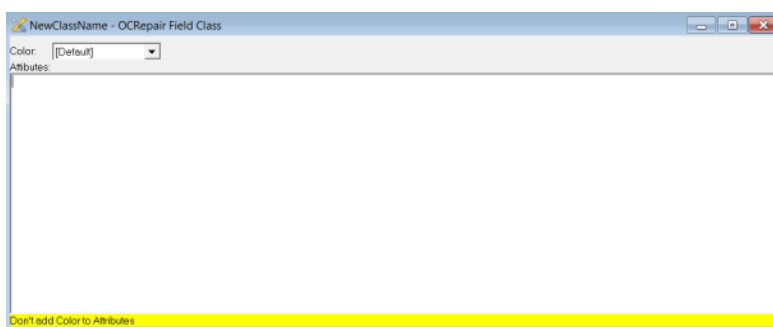
- Right-click **Resources** (or Double-click **OCRepairFieldClass** in the Resources list), select **New**.



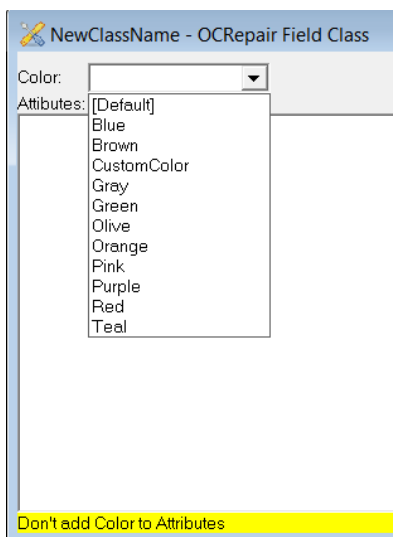
- In the New Resource dialog box, enter the new class **Name**, a **Description** and select **OCRepairFieldClass** as the Resource Type.



- Click **OK**, the OCRepair Field Class screen is displayed.



- Select a **Color** for this field class from the drop down color list.



- Enter any other attributes in the Attributes area.
- Select  to close the window and save changes.

Important: As a best practice, for the creation of a new Field Class and attributes, consult a Sr. FormWorks System Engineer before attempting to configure.

Output Driver Resource

An Output Driver is used to send formatted data that has made its way through the FormWorks pipeline to its final destination. This destination may be a file, fileref, or database, etc. Each destination requires a unique driver for one type of output (ODBC, fixed-length records, XML, etc.).

Note: The Output Driver and Destination are a unique one to one relationship.

In production, there is a need to transmit data to multiple destinations simultaneously; multiple output drivers may be utilized at the same time by the Store Worker.

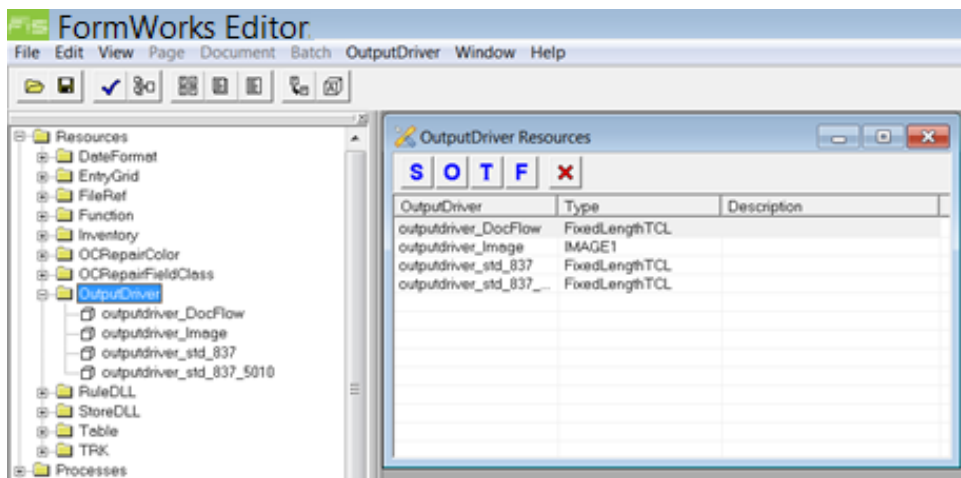
Note: The order in which the output drivers are listed in the Store Process (Store Configuration tab) is the order in which output is executed.

Configuration information is passed to the Output driver where the validity of the information received is confirmed against the information contained in the configuration. After the information has been processed, the driver cleans up after itself via the dispose function (within the driver).

Note: The driver is notified automatically whenever the Worker begins or ends the processing of a batch or document.

Add/Remove Output Drivers

- To access the Output Driver Resource screen, double-click the OutputDriver group displayed in the tree, the OutputDriver Resources screen is displayed.



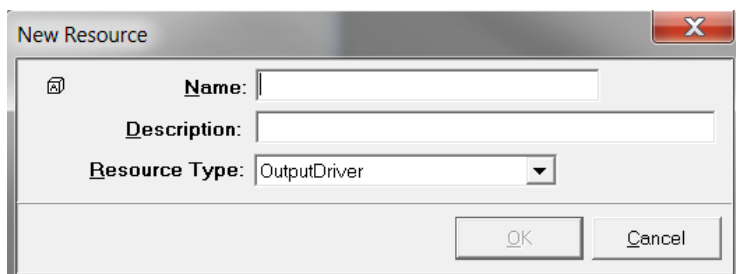
- Within the OutputDriver Resources screen are buttons, each providing a different function.



Button	Key Command	Function
S	Ctrl>F2	New Simple output driver Note: (This output driver does not have any configuration options).
O	Ctrl>F4	New ODBC1 output driver
T	Ctrl>F5	New TCL1 output driver
F	Ctrl>F6	New Fixed Length output driver

Button	Key Command	Function
		Delete output driver

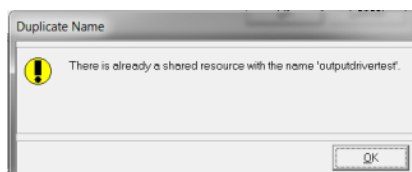
- Selection of any of the buttons responsible for the creation of new drivers (S, O, T, F) presents to the user the New Resource screen.



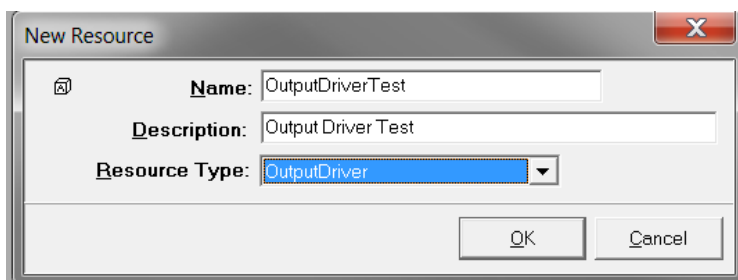
The 'New Resource' dialog box contains three input fields: 'Name', 'Description', and 'Resource Type'. The 'Resource Type' is a dropdown menu currently set to 'OutputDriver'. At the bottom right are 'OK' and 'Cancel' buttons.

- Enter the Name and Description of the new Output driver in the Name field.

Important: Output driver names previously used and deleted may not be reused. If an Output driver name previously used and deleted is entered, the system displays a Duplicate Name window.

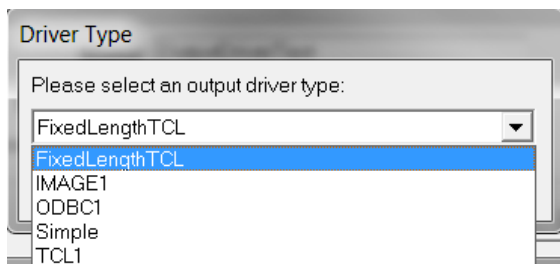


- From the Resource Type drop down, select OutputDriver
- Select OK, the newly created Output driver is added to the OutputDriver Resource listing.



The 'New Resource' dialog box is shown with the 'Name' field containing 'OutputDriverTest', the 'Description' field containing 'Output Driver Test', and the 'Resource Type' dropdown menu set to 'OutputDriver'. The 'OK' and 'Cancel' buttons are at the bottom right.

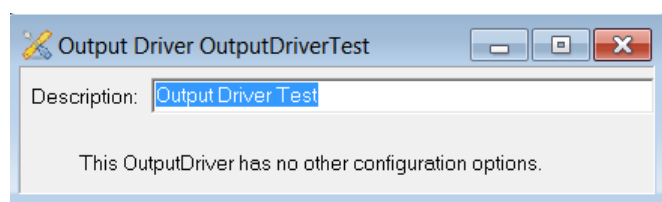
- The Driver Type screen is displayed. From the drop down, select the Output Driver type.



Note: Each output driver type, displays a different configuration window.

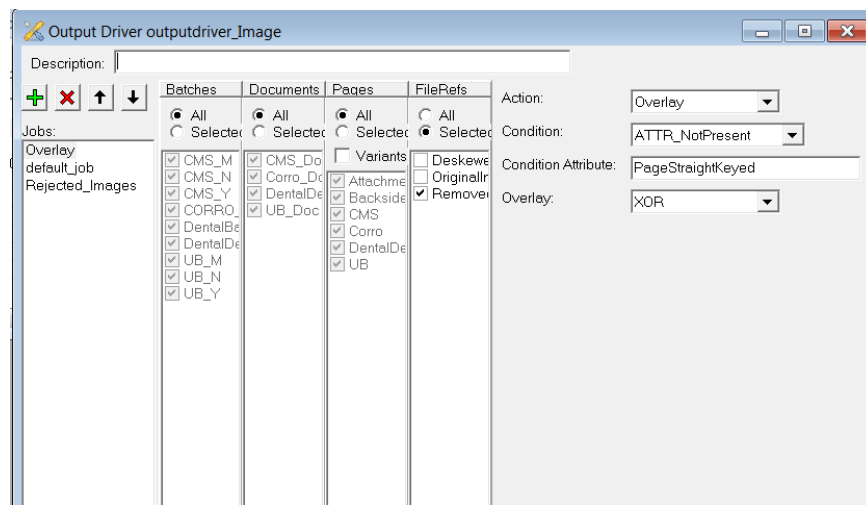
Simple Output Driver – Setup

By default, when the Simple Output driver is selected, the configuration is set, displaying the following screen.



The only configuration option available to the user for modification is the description.

Image Output Driver – Setup



- Jobs – List of jobs for which the driver will be applied.
- Batches – Batch listing for which the output driver is configured.
- Documents – A list of documents for which the driver configuration is to be applied.

- Pages – A list of page or page variants that will be included in the output for the document.
- FileRefs – A list of image filerefs that will be included in the output for a document.
- Action = Overlay – Combines the fileref image with the .FWD's blank image.

Action:
 Condition:
 Condition Attribute:
 Overlay:

- Condition – The action will be executed only if the condition is met.
 - None – No condition applied.
 - ATTR_Present –If ConditionAttr is present and non-zero, the action will be performed.
 - ATTR_NotPresent – If ConditionAttr is not present or is zero, the action will be done
 - Condition Attribute – Key the attribute to be used with the condition setting.
 - Overlay
 - OR – The image will be ORed with the variant image before being output.
 - XOR – The image will be XORed with the variant image before being output.
- Action = Set Tags – Assigns .TIFF tags for each image.

Action:
 Condition:
 Condition Attribute:

Tag	Value

- Condition – The action will be executed only if the condition is met.
 - None – No condition applied.
 - ATTR_Present –If ConditionAttr is present and non-zero, the action will be performed.

- ATTR_NotPresent – If ConditionAttr is not present or is zero, the action will be done
 - Condition Attribute – Key the attribute to be used with the condition setting.
 - Tag – The name of a .TIFF tag.
 - Value – An ATTR_SPrintfEx style string. This value will be put into the associated .TIFF tag.
- Action = Create Image – Creates the output image(s).

- Condition – The action will be executed only if the condition is met.
 - None – No condition applied.
 - ATTR_Present – If ConditionAttr is present and non-zero, the action will be performed.
 - ATTR_NotPresent – If ConditionAttr is not present or is zero, the action will be done
- Condition Attribute – Key the attribute to be used with the condition setting.
- Output Format – Select an option in which to output multipage documents. The available options are Tagged Image File Format (Tiff) and Portable Document Format (PDF). Tiff is the default output format.
- MultiPage – When checked, all images of a document will be combined into a single file. When unchecked, each image will be output separately.
- Base Dir – If not blank, the base directory name will be added to OutName in order to retrieve a full pathname.
- Output Name – This will be the filename of the output file. If the MultiTiff check box has been selected, the document attributes will be used. If the MultiTiff checkbox remains unselected, the page attributes will be used and the page attributes used should include an attribute that is different on each page.

- Rename – The output file will be renamed to the value entered after the file has been written to disk. If MultiTiff is checked, the document attributes will be used. If MultiTiff is unchecked, the page attributes will be used and the page attributes used should include an attribute that is different on each page.
- AttrToSaveOutName – The value of OutName will be saved to this attribute.
- File Operation
 - None – No fileref will be added.
 - Copy – A fileref will be added, and the output image will be copied to it.
 - Move – A fileref will be added, and the output image will be transferred into the fileref. The output image is assigned (belongs) to the work record and will follow the work record.
- File Ref Name – This is the name of an image fileref that the output image will be copied and moved to. If present it cannot be an empty string.
- File Op Condition
 - Always – The FileOperation will always be done.
 - ATTR_Present – If FileOpConditionAttr is present and non-zero, the FileOperation will be done.
 - ATTR_NotPresent – If FileOpConditionAttr is not present or is zero, the FileOperation will be done
- File Op Condition Attr – The attribute used with the FileOpCondition setting. This field is required if FileOpCondition is other than ALWAYS.
- Action = Add Page – Adds a page to the document.

Action: Add Page
 Condition: None
 Condition Attribute:
 PageID of page to add:
 Position to insert new page: Last

- Condition – The action will be executed only if the condition is met.
 - None – No condition applied.
 - ATTR_Present – If ConditionAttr is present and non-zero, the action will be performed.
 - ATTR_NotPresent – If ConditionAttr is not present or is zero, the action will be performed.

- Condition Attribute – Key the attribute to be used with the condition setting.
- PageID of page to add – The value added is the Page Variant name of the page to be added.
- Position to insert new page (Required when the PageID of page to add field is populated).
 - First – Adds a page to the beginning of the document.
 - Last – Adds a page to the end of the document.
- Action = Render – Render – Erases the image and then draws the OCR contents to the image.

Action:

 Condition:

 Condition Attribute:

 Render Process Name:

- Condition – The action will be executed only if the condition is met.
 - None – No condition applied.
 - ATTR_Present –If ConditionAttr is present and non-zero, the action will be performed.
 - ATTR_NotPresent – If ConditionAttr is not present or is zero, the action will be done
- Condition Attribute – Key the attribute to be used with the condition setting.
- Render Process Name – Name of the Render process to be utilized when rendering the image.

ODBC1 – Output Driver

- ODBC Tab

Output Driver OutputDriverSample

 Description:

 ODBC | Batch Settings | TemplateSet Settings

 Data source:

 Login:

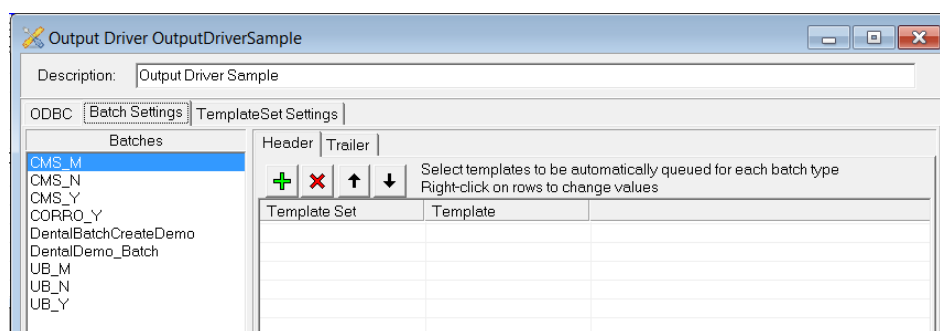
 Password:

 Timeout:

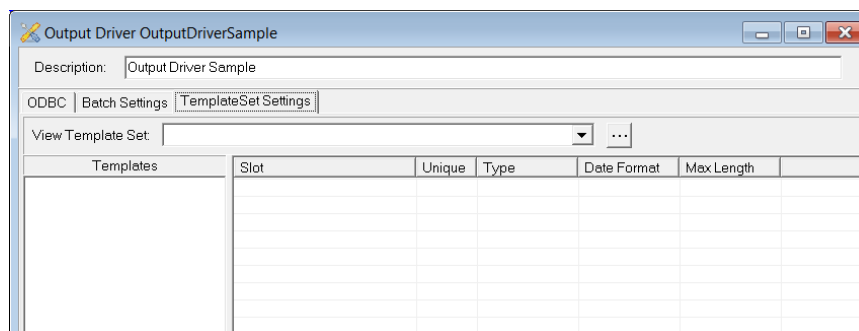
 DBMS:

 Date Format:

- Data Source – Name of the data source to be used. This field must be populated with a valid value.
- Login – The data source name (optional).
- Password – Data source password (optional).
- Timeout – The number of seconds the system it to try to long into the data source before timing out. While an optional field, if populated the value cannot be zero.
- DBMS – Name of the database software
 - Microsoft SQL Server
 - Microsoft Access
 - Oracle V.7.x or V.8.x
 - IBM DB2
- Date Format – Name of a Date Format resource, an optional field.
- Batch Tab – Name of the batch tab.



- Batches – Select the name of the applicable batch.
- Header/Trailer- Names of templates that will be output upon batch exit.
- TemplateSet Tab



- View Template Set – List of all names in the template (templates).
- Templates – Displays templates defined within the template set.
- Slot – List of all slot names in the template.
- Unique – Value is required.
 - 0= not a unique column.
 - 1= unique column.
- Type – Indicates to the SQL database format, optional.
 - 1 = String
 - 2 = Integer
 - 3 = Date
- Date Format – May be used to override the DateFormat assigned from the Main Configuration for a single column, optional.
- Max Length – Use to set the max length to place in the slot. It is only used for slots with the string type, or slots that have the undefined type and are set to String values in the rules.

TCL1 Output Driver

This driver will source a TCL file specified by the ScriptFile parameter and call procedures from that file for each event it receives.

Event	Procedure	Parameters
Batch Enter:	BatchEnterProc	
Batch Exit:	BatchExitProc	
Document Enter:	DocEnterProc	
Document Exit:	DocExitProc	
Output Template:	OutputTemplateProc	

- Description – Enter the description of the file.
- TCL Script File – Name of the TCL source file.
- Events
 - Procedure/Parameters (all Parameter fields are a string value and passed to the procedure).
 - Batch Enter – Name of the procedure to call on a Batch Enter event.
 - Batch Exit – Name of the procedure to call on a Batch Exit event.
 - Document Enter – Name of the procedure to call on a Document Enter event.
 - Document Exit – Name of the procedure to call on a Document Exit event.
 - Output Template – Name of the procedure to call on an Output Template event.

FixedLengthTCL Driver

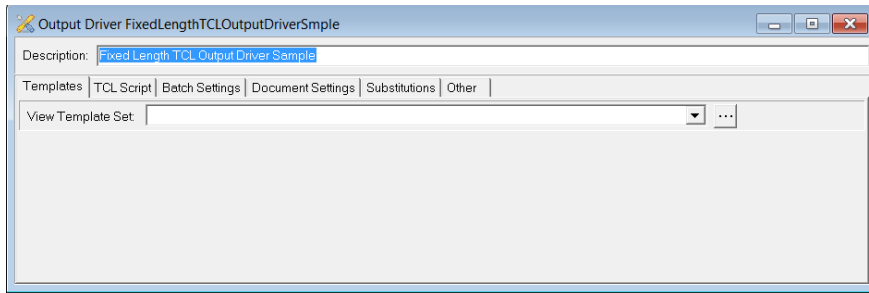
This driver will source a TCL file specified by the ScriptFile parameter and call procedures from that file for each event received.

On a BatchEnter event, if a template is specified for the batch type, a template with default values will be output by the driver.

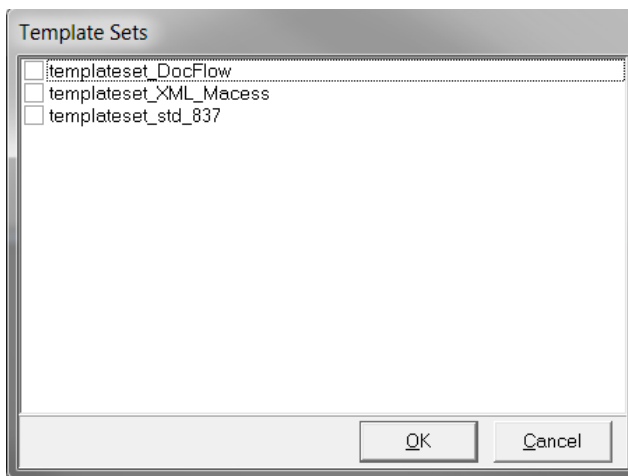
On BatchExit the template will be output again, with whatever information has been set by Store2 rules.

The template is cleared after output.

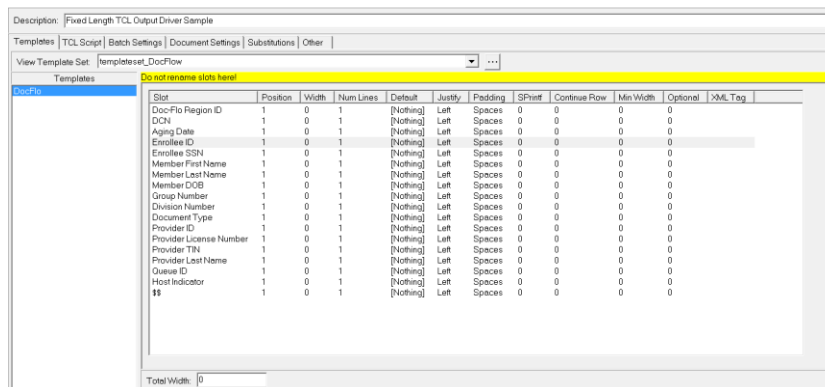
Templates Tab



- View Template Set – Select the ellipsis button  displaying the Template Sets window.



- From the Template Sets window select the correct **template** and press **OK**. Additional sections (Templates and Slot names are displayed).



Important: Do not rename slot fields from the slot list window. Only edit slot fields from the Edit List Item window (below).

- Double-click the Slot, displaying the Edit List Item window from which the individual Slot is defined. Modify and edit the values as required. Select Accept to save the changes.

Editing item 4 of 18.

Slot: Enrollee ID

Position: 1

Width: 0

Num Lines: 1

Default: [Nothing]

Justify: Left

Padding: Spaces

☐ SPrint

☐ Continue Row

Min Width: 0

☐ Optional

XML Tag:

Accept Cancel

- Slot – Name of the Slot.
- Position – Beginning position of the slot.
- Width – Number of spaces assigned for the slot.
- Num Line – Number of lines allowed for the slot.
- Default – Default action, Nothing is set by default.
 - Nothing
 - Linefeed
 - Unset
- Justify – Slot justification, one must be selected, left is selected by default.

- Left
- Right
- Padding – Select the type of padding to follow, by default Spaces is selected.
 - Spaces
 - Zeros
 - Spaces or Zeros
 - None

Insert/Append/Edit/Delete

To Insert, Append, Edit or Delete a slot, right-click the **Slot**, select the appropriate function from the list.

Slot	Position	Width	Num Lines	Default
Doc-Flo Region ID	1	0	1	[Noth
DCN	1	0	1	[Noth
Aging Date	1	0	1	[Noth
Enrollee ID	1	0	1	[Noth
Enrollee SS				[Noth
Member First				[Noth
Member Last				[Noth
Member DC				[Noth
Group Number				[Noth
Division Number				[Noth
Document Type				[Noth
Provider ID				[Noth
Provider License				[Noth
Provider TIN				[Noth
Provider Last Name	1	0	1	[Noth
Queue ID	1	0	1	[Noth
Host Indicator	1	0	1	[Noth
\$\$	1	0	1	[Noth

- Insert new item at top of list... – Inserts a new slot at the top of the current listing. When selected, displays the Edit List Item screen with the slot designated at Item 1.
- Append new item to end of list... – Adds a new slot at the end of the current listing. When selected, displays the Edit List Item screen with the slot designated as the last numbered item in the list.
- Insert new item before item #... – Inserts a new slot just before the slot selected. When selected, displays the Edit List Item screen with the slot designated in the position prior to the slot selected.
- Edit item #... – Displays the Edit List Item screen.
- Delete item # – Deletes the slot selected.

TCL Script Tab

Output Driver outputdriver_std_837_5010

Description:

Templates | **TCL Script** | Batch Settings | Document Settings | Substitutions | Other

TCL Script File:

Events	Procedure	Parameters
Batch Enter:		
Batch Exit:		
Document Enter:		
Document Exit:		
Output Template:		

- TCL Script File – Name of the TCL source file
 - Procedure/Parameters (all Parameter fields are a string value and passed to the procedure).
 - Batch Enter – Name of the procedure for a Batch Enter event.
 - Batch Exit – Name of the procedure for a Batch Exit event.
 - Document Enter – Name of the procedure for a Document Enter event.
 - Document Exit – Name of the procedure for a Document Exit event.
 - Output Template – Name of the procedure for an Output Template event.
- Batch Settings Tab

Output Driver outputdriver_std_837_5010

Description:

Templates | TCL Script | **Batch Settings** | Document Settings | Substitutions | Other

Batches

- CMS_M
- CMS_N
- CMS_Y
- CORRO_Y
- DentalDemo_Batch
- UB_M
- UB_N
- UB_Y

Events

Enter Procedure:

Enter Parameters:

Exit Procedure:

Exit Parameters:

Header | Trailer | Additional

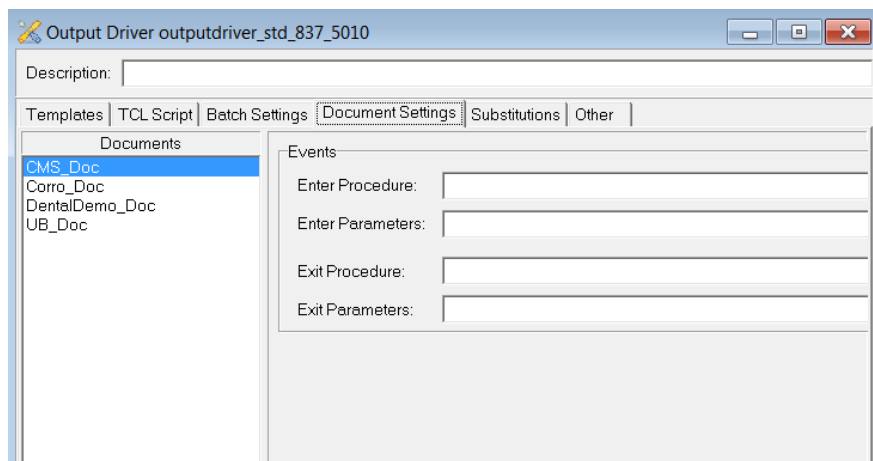
+ - ↑ ↓ Right-click on rows to change values

Template Set	Template
templateset_std_837	0000_ISA
templateset_std_837	0001_GS_5010

- Batches – Batches to select from for configuration.
- Events
 - Enter Procedure – Name of the procedure to call on a Batch Enter event. If not defined, the value of BatchEnterProc and BatchEnterParam from the TCL Script settings is used.
 - Enter Parameters – Parameters to pass to the Batch Enter procedure.
 - Exit Procedure – Name of the procedure to call on a Batch Exit event. If not defined, the value of Batch Exit Procedure and Batch Exit Parameter from the TCL Script settings is used.
 - Exit Parameters – Parameters to pass to the Batch Exit procedure.
- Header – Names of templates that will be output first for each batch or document for the selected batch type
- Trailer – Names of templates that will be output last for each batch or document for the selected batch type
- Additional – Names of templates that should be cleared at the beginning of every batch

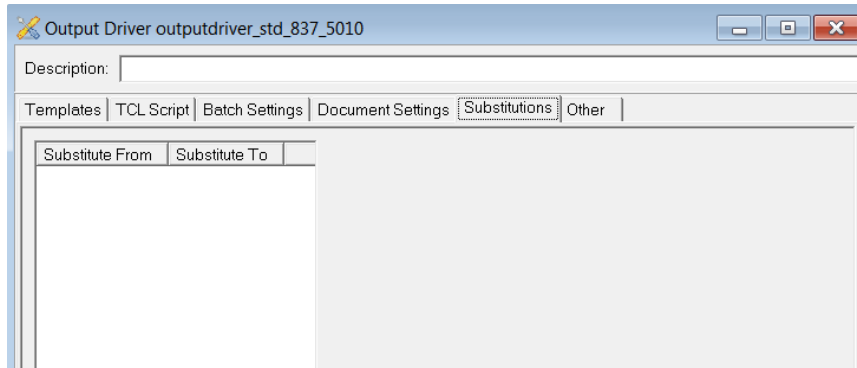
Note: Similar to how the Header and Trailer are configured. Select the TemplateSet, then the Template within that set.

- Document Settings Tab

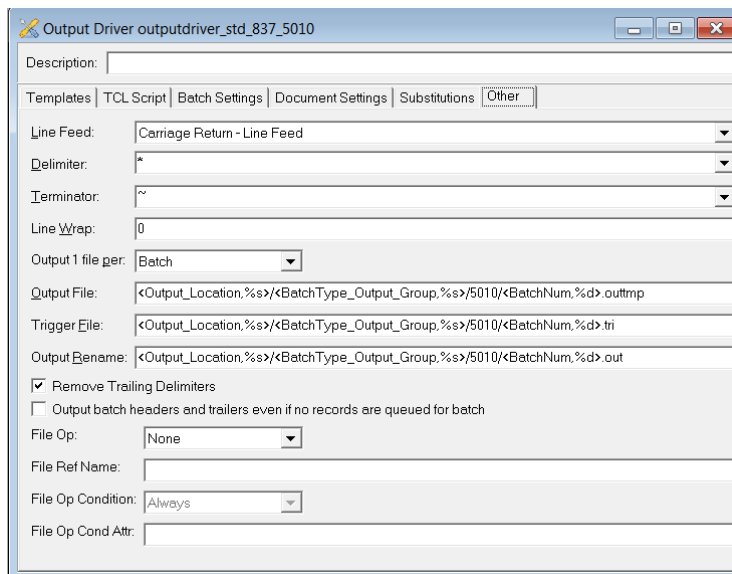


- Documents – Documents to select from for configuration.
- Events
 - Enter Procedure – Name of the procedure to call on a Document Enter event. If not defined, the value of DocEnterProc and DocEnterParam from the TCL Script settings is used.
 - Enter Parameters – Name of the procedure to call on a Document Exit event. If not defined, the value of DocExitProc and DocExitParam from the TCL Script settings is used.

- Exit Procedure – Parameters to pass to the Document Enter procedure.
- Exit Parameters – Parameters to pass to the Document Exit procedure.
- Substitutions Tab



- Substitute From – A list of strings that should be substituted (escaped), may not be zero, parallels Substitute to.
- Substitute To – The value defined in Substitute From is substituted by the value defined adjacent (to the right) of the value defined in Substitute To.
- Other Tab



- Line Feed
 - Line Feed – Move down to the next line without returning to the beginning of the line.
 - Carriage Return – Moves to the beginning of the line without advancing to the next line.

- Line Feed + Carriage Return – Populates (in order) both Line Feed and Carriage Return ASCII code.
- Carriage Return + Line Feed – As a best practice, most often used. Populates (in order) both Carriage Return and Line Feed ASCII code.

Note: The line feed setting needs to match the needs of the system for which being configured.

- Delimiter – Available for selection are:

- [Tab]
- [Space]
- [CR]
- [LF]
- [CRLF]

- Terminator – Available for selection are:

- ~ –

Note: the ~ is used as a terminator in 837 output.

- Line Feed –

- Line Wrap – If non-zero, extra Terminators will be inserted into template strings that are longer than this value.
- Output 1 file per – Select to which the Output file will be given.
 - Batch
 - Doc
- Output File – Assigns the path name for the output file to be saved to.
- Trigger File – Creates a new name of the output file after the initial creation.
- Output Rename – After the file has been created, the file is renamed to this value.
- Remove Trailing Delimiters (check box) – When selected, the contiguous Delimiters at the end of a formatted template string will be deleted.
- Output batch headers and trailers (check box) – Selected when outputting a file per batch, but all of the documents have been rejected at KE and none are to be output. The Header/Trailer records are still generated for the batch.
- File Op – May be used to add the output file to a fileref within the work record.
 - None – No action or effect.

- Copy – The output driver will make a copy of the output file produced and add it to a fileref in the work record.
- Move – The output driver will set the output file produced to a fileref in the work record. As such, when the work record is deleted, the output file is deleted with it.
- File Ref Name – Names the file reference that will contain the output file.
- File Op Condition – Dependent upon File Op being populated with a value of Copy or Move.
 - Always – No condition
 - ATTR Present – The output file will be added to the work record only if the FileOpConditionAttr is present.
 - ATTR Not Present – The output file will be added to the work record only if the FileOpConditionAttr is NOT present.
- File Op Cond Attr – Names the attribute that will be used by the File Op Condition.

Edit Output Drivers

To edit existing Output Drivers, double-click the driver, the appropriate configuration window displays.

Note: See Add/Remove Output Drivers for details regarding the configuration of each Output Driver type.

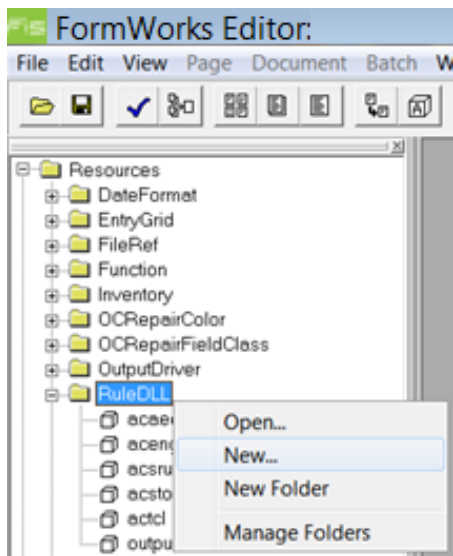
Rule DLL Resource

Rule DLLs contain the rules that are applied by AC and Store workers. Most systems use these DLLs:

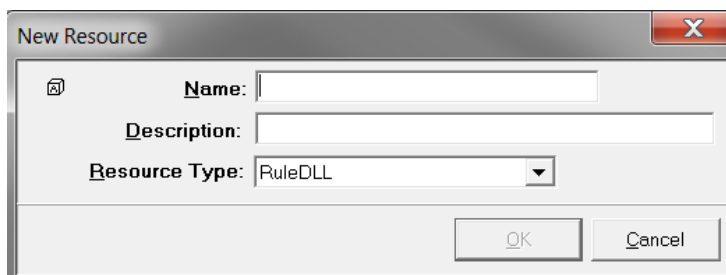
- acaeg
- acengine
- acsrules
- acstore
- actcl

To Create A Rule DLL Resource

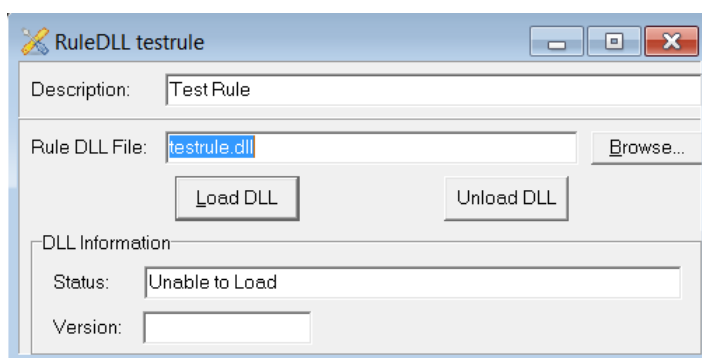
- Right-click Resources, Select New.



- In the New Resource dialog box, enter a Name for the resource and a Description. Select Rule DLL as the Resource Type. As a best practice, to easily identify the DLLs later, use the file-name as the resource name (i.e. acsrules).



- Select OK, the RuleDLL window opens



- If necessary, click Browse and select the appropriate Rule DLL file.

Important: The DLL should be in the bin folder and, as such, as a best practice Do NOT specify any path to the DLL here. In the event it is necessary to browse to a DLL, remove the path after selecting it.

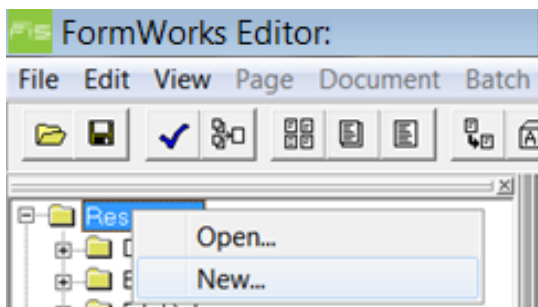
- Load DLL – Load the DLL now. The editor normally does this on FWD load too, but this would be used when changing a DLL to avoid having to reload the FWD.
- Unload DLL – Unload the one it already loaded from its memory. This is not typically used unless changing a DLL.
- Status – Success or failure of DLL loading.
- Version – Version information reported by the DLL.

Store DLL Resource

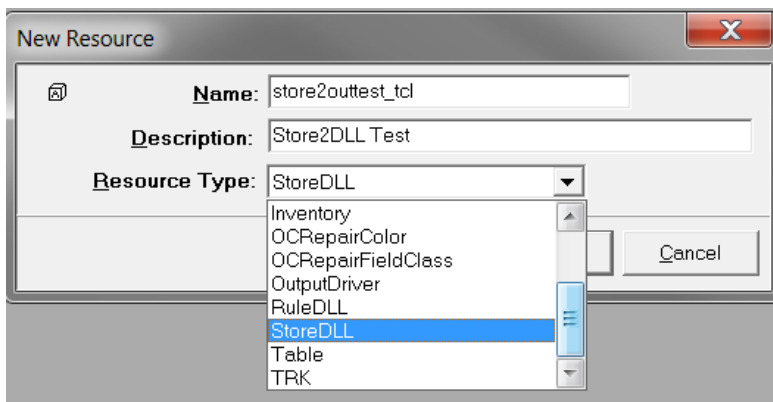
Store DLLs are the output driver DLLs used by the Store worker. Most systems use tcl.dll and store2out_image.dll. Also common is store2out_odbc.dll.

To Create A Store DLL Resource

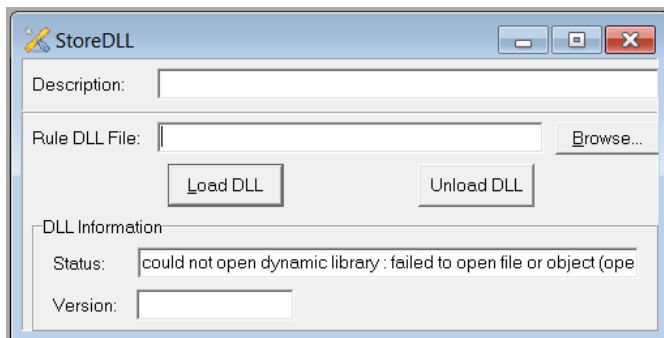
- Right-click **Resources**, select **New**.



- In the New Resource dialog box, enter a **Name** and **Description** for the resource, and select **Store DLL** as the Resource Type. As a best practice, to easily identify the DLLs later, use the file-name as the resource name (i.e. store2outtest_tcl).



- Select **OK**, the Store DLL window opens



- If required, click **Browse** and select the appropriate **Store DLL** file.

Important: The DLL should be in the bin folder and, as such, as a best practice Do NOT specify any path to the DLL here. In the event it is necessary to Browse to a DLL, remove the path after selecting it.

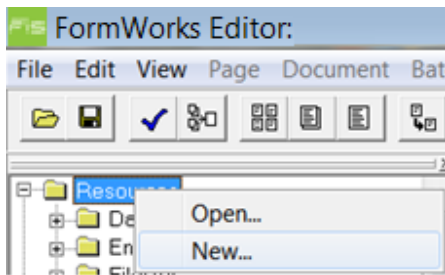
Table Resources

Character Sets

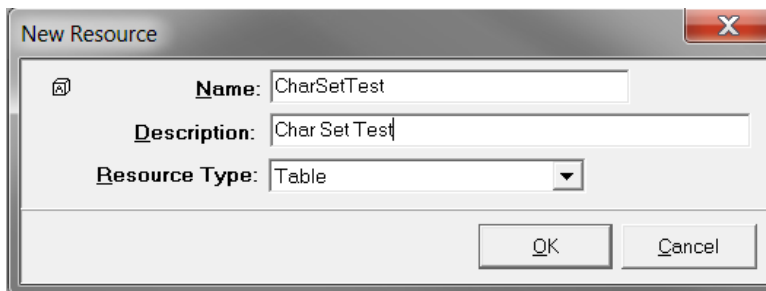
Character sets are used by AEG (OCR) in evaluating the characters read in a field and are also used by AC rules to filter out characters (punctuation, spaces) and to test the contents of a field.

To Create A Character Set

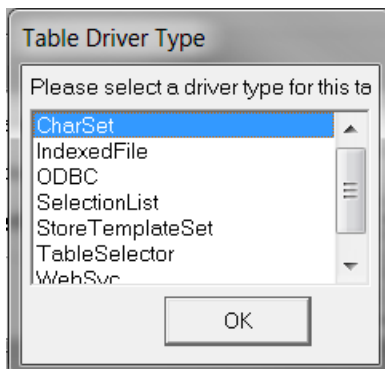
- Right-click Resources in the FWD tree (or Table in the resources list), select New.



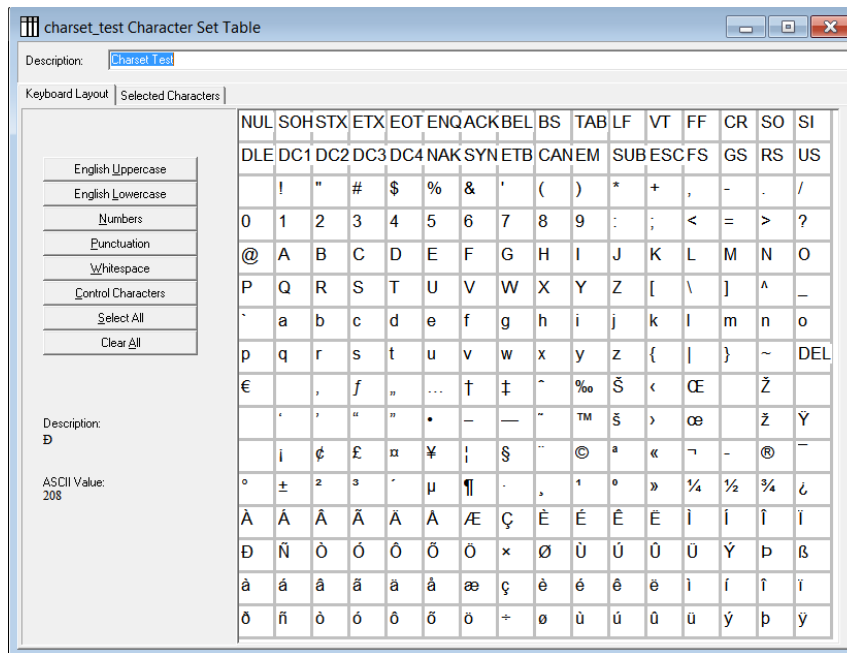
- In the New Resource dialog box, enter a Name and Description for the new resource, and select Table from the drop-down list of Resource Types. As a best practice, to differentiate character set tables from other kinds of table resources, use the naming convention “charset_name” to name the new table.



- From the list of table drivers, select CharSet, then select OK.

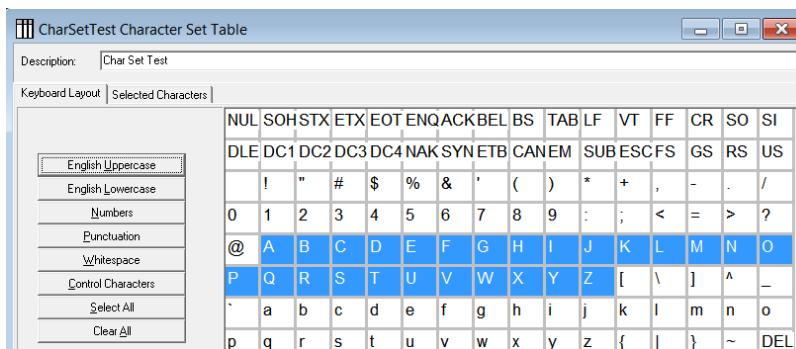


- The Character Set Table screen displays, showing a list of all available characters.



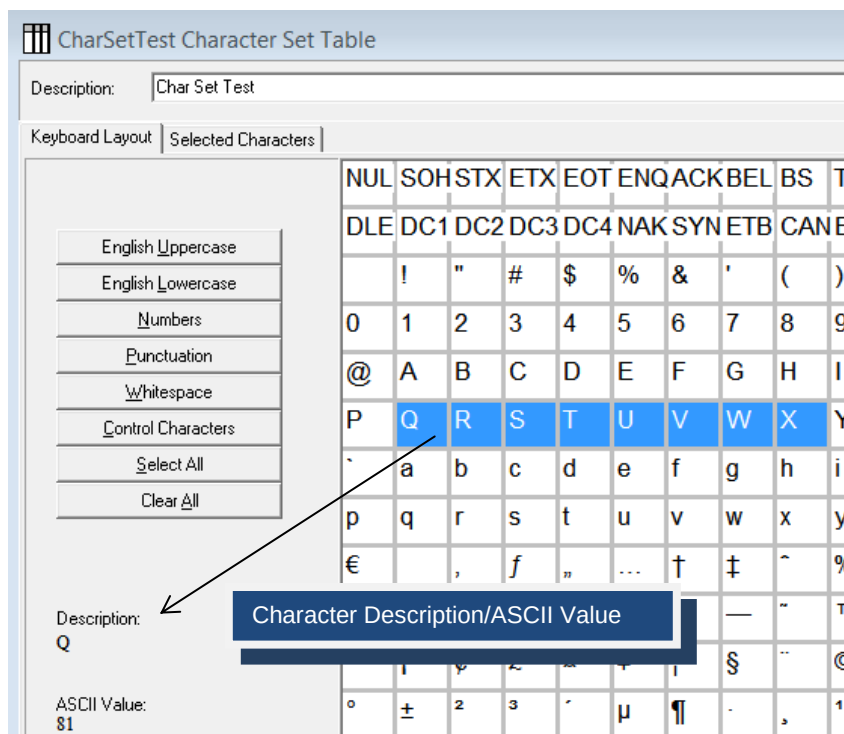
Keyboard Layout Tab

- Enter a Description of the new character set. The description should be detailed enough to communicate to various users the value of what is included without having to open the table.
- Use the buttons to the left of the displayed characters to select groups of characters (upper case letters, numbers, etc.). Additionally, an individual character may be selected (blue highlight) or cleared by selecting an individual character.



- To deselect characters, select Clear All.

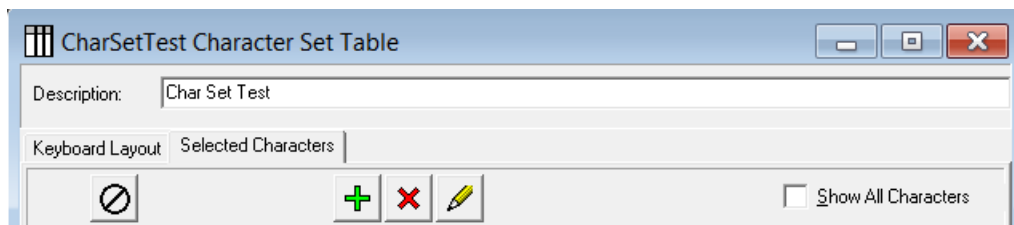
Notice that when the mouse hovers over an individual character, a description and the ASCII value for that character appear at the bottom left of the window.



Selected Characters Tab

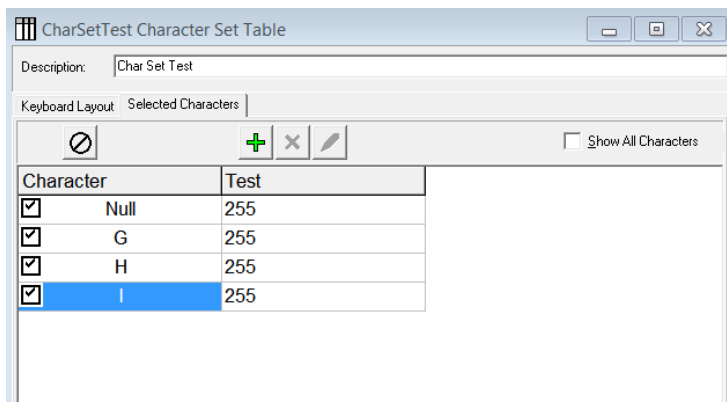
The **Selected Characters** tab allows the setting of a confidence level (Threshold) for individual characters. As a best practice, it is recommended to set confidence levels via OCRRepair.

- Enter a Description of the character set. The description should be detailed enough to communicate to various users the value of what is contained without having to open the table.

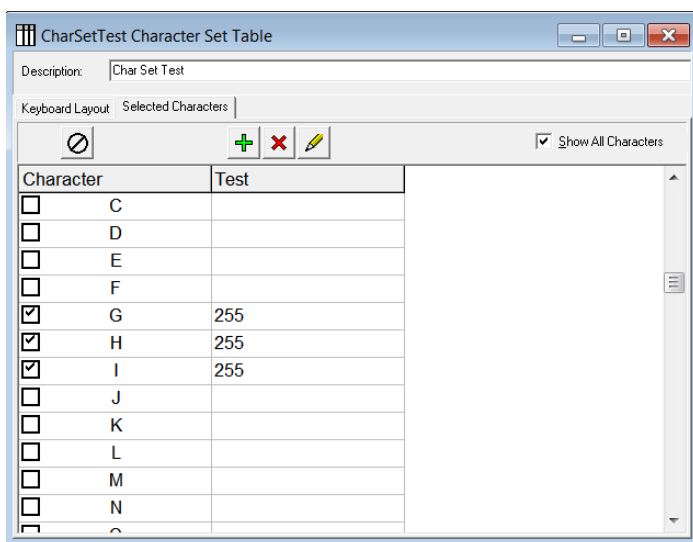


- Click the Selected Characters tab to view a list of all available characters.
 -  – Clear/Deselect all characters.
 -  – Add a Threshold column.
 -  – Delete Threshold column.

-  – Rename Threshold column.
- ☐ Show All Characters – Show All Characters.
- When unchecked, only the individual characters selected (checked) are presented.



- When checked, all characters (selected and unselected) are presented.



Note: Users processing non-English claims must select the relevant characters from the desired region from the keyboard layout tab. For example, to process Spanish claims, users must select Spanish characters. Refer to Region Settings for more information.

Add Threshold Column

This feature allows rules to use character confidences.

Important: As a best practice, consult a Sr. FormWorks System Engineer prior to the creation of threshold column or character confidences.

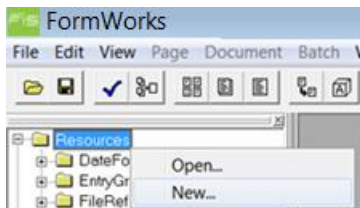
Indexed File Tables

Indexed File Tables are fixed-width tables normally provided by the customer and may be comprised of patients, providers, procedure codes, CSZ (city, state, zip) information.

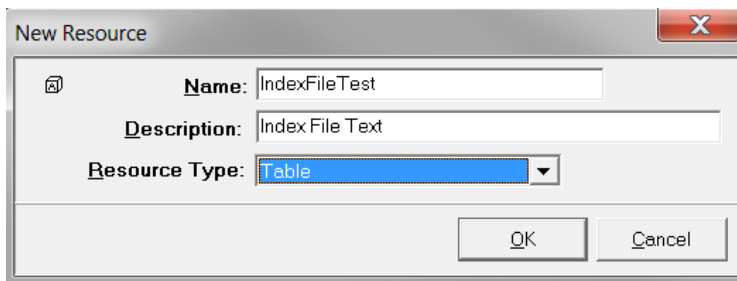
Note: For each indexed table, the locations of the table file and the index file must be specified.

To Create an Indexed File Table

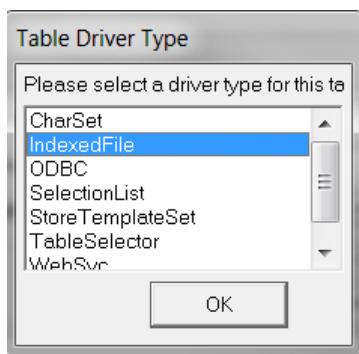
- Right-click **Resources** in the FWD tree (or Table in the Resources list), select **New**.



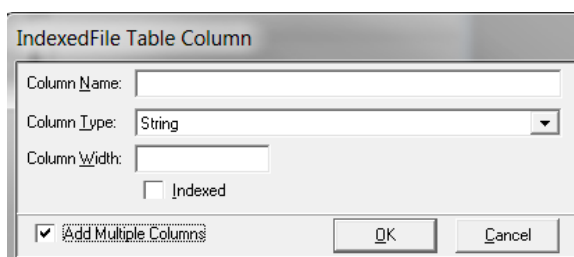
- In the New Resource dialog box, enter a **Name** and **Description** for the resource, next choose **Table** from the list of **Resource Types**, and select **OK**.



- From the list of Table Driver Types, select **IndexedFile**, select **OK**.



- The system displays the IndexedFile Table Column window.



- Column Name – Enter the Column Name.
 - Column Type – Select the Column Type
 - Column Width – Enter the number of characters (spaces) the column is to be allowed.
 - Indexed – Indexes speed searching. A column that will be used as the only exact match in a lookup rule needs to be indexed.
 - Add Multiple Columns – As a best practice, select the Add Multiple Columns check box when first setting up the table. This enables the Table Column dialog box to remain open after OK is selected, negating the need to repetitively select the Add a Column button.
- Select OK.

The system displays the Indexed File Table window. Specify the location (mapping) of both the **Table File Name** and the **Index File Name**. The table file must exist in the Table File Name location specified for FormWorks to load it; however, FormWorks creates the index file in the specified location. The graphic below illustrates a Table Resource for a patient table. As a best practice, use “../tables/<tablename>.tbl” and “../tables/<tablename>.idx”.

IndexFileTest Indexed File Table

Description: Index File Text

Table File Name: .../Tables/Patient.fil

Index File Name: .../Patient.ind

+ - ↑ ↓

Column	Type	Width	Indexed
Internal_ID	String	6	
Last_Name	String	20	
First_Name	String	15	
Middle_Initial	String	1	
Suffix	String	3	
Sex	String	1	
DOB	String	8	
Address	String	30	
City	String	20	
State	String	2	
Zip	String	5	
Zip_Extension	String	4	
Insured_ID	String	6	Yes
GroupName	String	30	
EffectiveDate	String	8	
CR/LF	String	2	

Total Width: 161

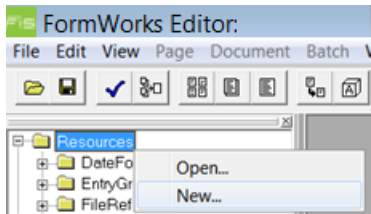
- Select the  button to enter a column into the table.
 - Column Name – Name of the column.
 - Column Type – Select a type (may vary dependent upon configuration) either String, Amount or Zip Code from the drop down box for this example.
 - Column Width – Enter the width of the column of the column.
 - Select Indexed (check box) if this column is indexed.
- A “newline” character at the end of each line must be specified in the table resource. This is typically a width of 2 for the “Carriage Return” and “Line Feed” characters but may be only 1 if just the “LineFeed” will be present.
- Total Width – The Total Width field at the bottom of the window automatically totals all of the Column width values entered.
- Click the Selected Characters tab to view a list of all available characters.

TRK Resources

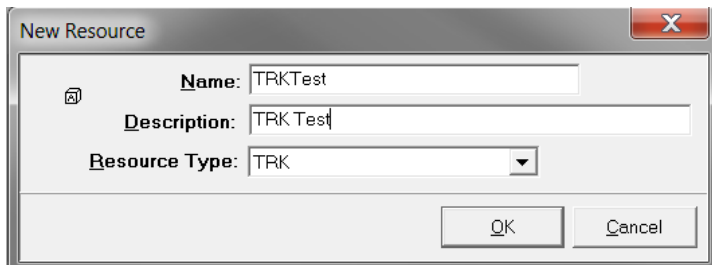
TRK Resources are used by the FormWorks Tracker to configure how FormWorks tracks changes within history files.

To Create TRK

- Right-click Resources in the FWD Tree (or select TRK in the Resources list), select New.

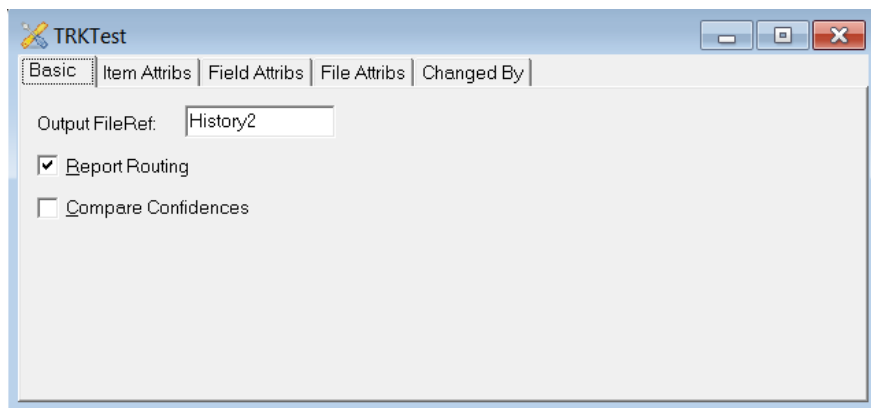


- In the New Resource dialog box, enter a Name and Description for the Resource, next select TRK from the list of Resource Types. Click OK.



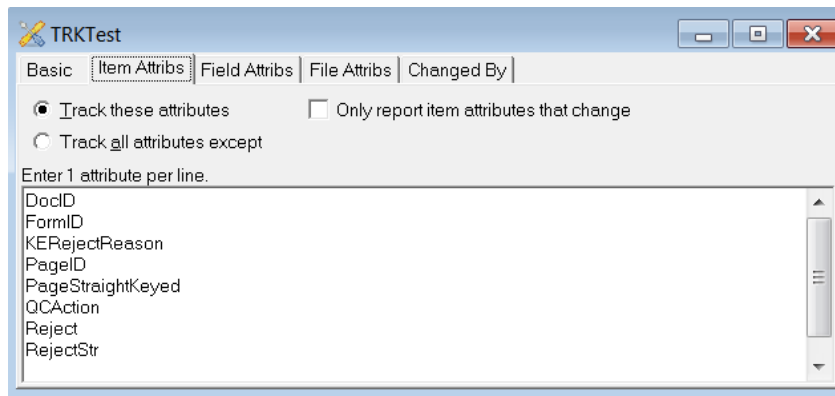
- The TRK window is displayed, which is comprised of five tabs.
 - Basic
 - Item Attribs
 - Filed Attribs
 - File Attribs
 - Changed By

Basic



- Output FileRef – User defined name of the Fileref. As a best practice, it is recommended the naming convention utilized should not exceed eight characters.
- Report Routing – If checked, creates route records (default value is checked).
- Compare Confidences – When checked, compares lines on a character by character basis (including the best guess confidence values).

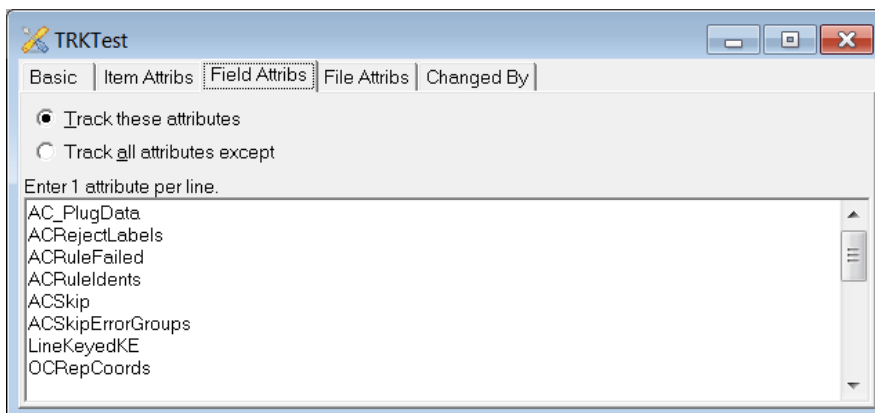
Item Attribs



- Track these attributes – Track the Item attributes listed in the window "Enter 1 attribute per line".
 - By default
 - FormID
 - KERectReason
 - PageID
 - PageStraightKeyed
 - QCAction
 - Reject
 - RejectStr
- Track all attributes except – Track all Item attributes except those listed in the window "Enter 1 attribute per line".
 - By default
 - WFFileBytes
 - WFFileHosts
 - WFWorkerHost

- WorkerInstance
 - WrkrInstance
 - FilePath
 - NewFileLoc
 - DCN
 - BatchNum
- Only report item attributes that change – When selected, TRK will not report item attributes that have not changed since the last time the history file was updated. The first time history is reported TRK will report the values of all attributes TRK is configured to track. After the initial report, TRK will only report on those attributes if there is a change.

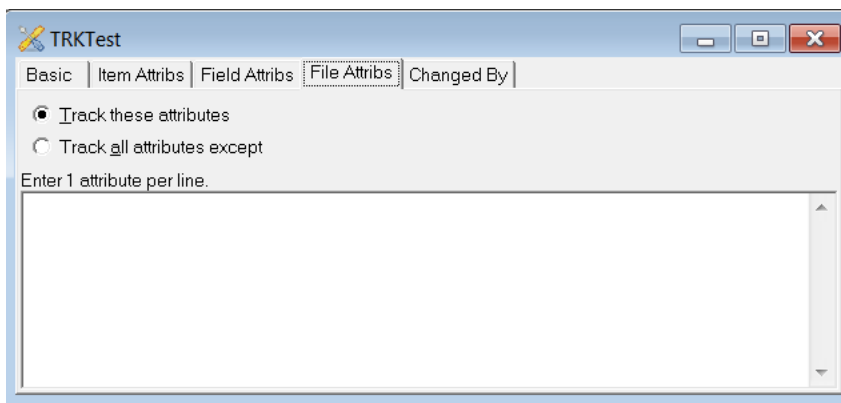
Field Attribs



- Track these attributes – Track the Field attributes listed in the window "Enter 1 attribute per line".
 - By default
 - OverrideValidations
 - OCRepFELines
 - ResolutionContext
 - ACRuleFailed
 - ACSkip
 - ACRejectLabels
 - ACSkipErrorGroups
 - ACRuleIdents

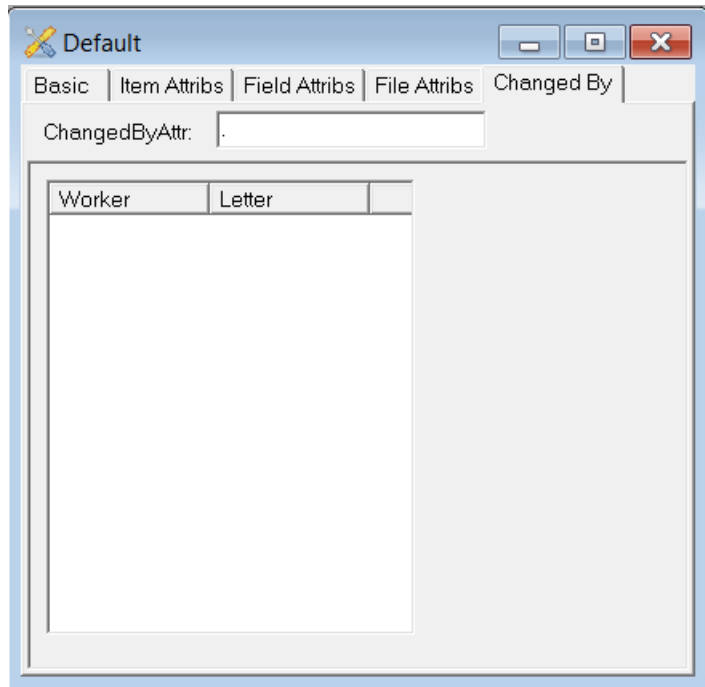
- Reject
 - RejectStr
 - LineKeyedKE
 - AC_PlugData
 - ACSkipStrings
 - LineSelectedKE
 - ACSkipFieldValues
- Track all attributes except – Track all Field attributes except those listed in the window "Enter 1 attribute per line".
 - No Field attributes defined by default.

File Attribs



- Track these attributes – Track the File attributes listed in the window "Enter 1 attribute per line".
 - No Field attributes defined by default.
- Track all attributes except – Track all Field attributes except those listed in the window "Enter 1 attribute per line".
 - No Field attributes defined by default.

Changed By

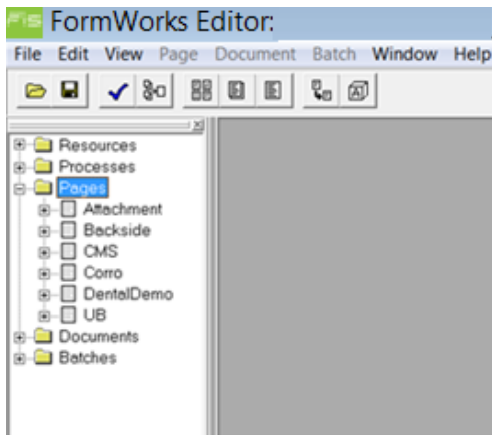


- **ChangedByAttr** – To modify, right-click in the ChangedByAttr field (blank space), then add or modify the entries. This applies to the work-record item attribute that tracks which workers changed an OCR file. When OCR files are compared, the tracker may optionally set attributes that provide some information about the changes. ChangedByAttr defines the name of the item-level attribute that the tracker will maintain. Each character position in the string corresponds to an element of the Letters column and is initialized to a period. If the corresponding worker changed something, the Letter will appear in the resulting string.

Note: If differences are detected between the OCR files and ChangedByAttr is set, then ChangedByLetters and ChangedByWorkers must also be set.

Pages

Each page in a FormWorks system consists of a field list (fields), one or more field variants, and processing steps and rules for the page. A page may have one or more variants but each individual variant contains a description that maps fields from the image to the field list.



Create Pages

There are two methods for creating a page in the FWD:

- Copy a page from an existing FWD.
- Create a new page from scratch.

Copy a Page

While it is always easiest to begin setting up a new system by copying an existing FWD that contains all page types needed, this may not always be possible. When copying data from an existing FWD is not possible, users may create a new blank FWD, and then copy selected objects into the new database.

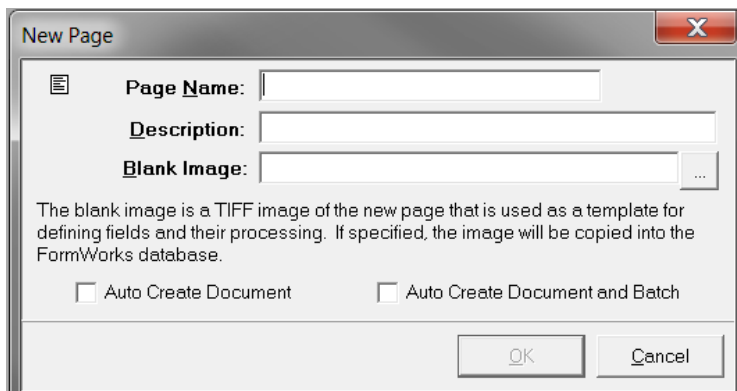
Warning: When copying an FWD Object, pay precise attention to names. The object copied will overwrite any existing object with the same name in the target database.

Create New Page

If there is not an existing page definition available that may be copied, added, and defined, a new page within FormWorks Editor must be created. To create a new page, first have a .TIFF image of the page, this image is to be the template.

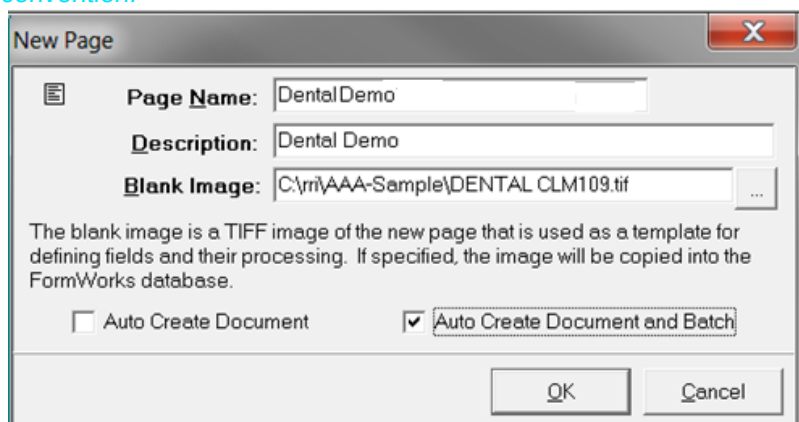
To Create A New Page

- Right-click Pages in the FWD tree and select New. The New Page dialog box opens.
- or
- From the File menu, select New + Page Type. The New Page Type dialog box opens.

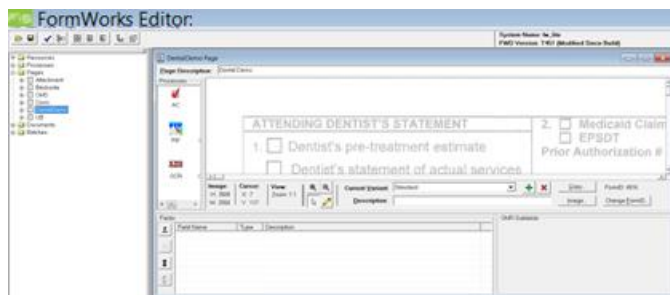


- Enter the Page Name.
 - Names utilized for Page/Document/Batch:
 - May be up to 31 characters.
 - Names may include only letters, numbers, or the underscore character.
 - All names must be unique.
 - Names must begin with a letter or the underscore character.
- Enter a **Page Description** (optional).
 - Click the browse button  to select a blank .TIFF image of the new page. (See [Selecting/Replacing Variant Image Templates](#) for information on selecting or replacing images.)
- Auto Create Document (check box) – Creates the document.
- Auto Create Document and Batch (check box) – Creates the document and batch.

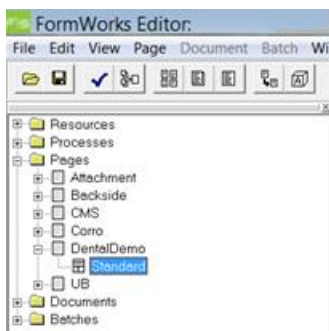
Note: As a best practice, it is recommended to create and apply a standardized naming convention.



- Select **OK**.
- Editor closes the dialog box and opens the specified .TIFF image for the new page.



- The page name, DentalDemo in this example, is listed in the Pages category in the FWD tree; the Editor assigns the variant name “Standard” by default.



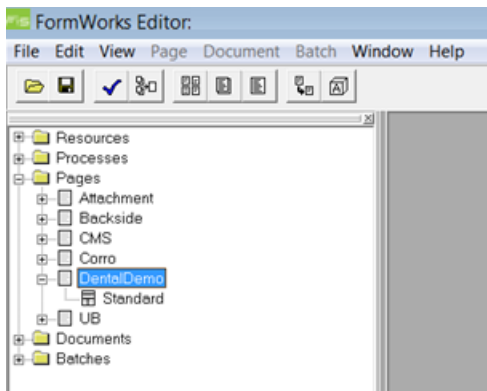
Fields may now be added as described in [Adding Fields To A Page](#) .

To add Processing Information, see [Configuring Page and Page Variant Processing](#).

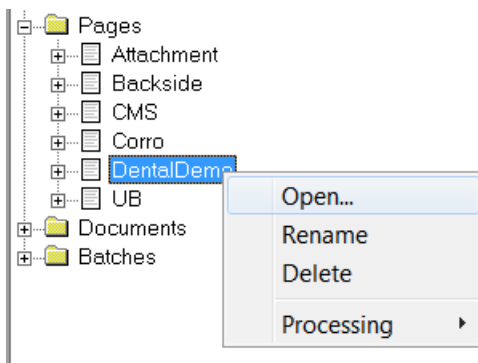
Create/Add a Graphical Variant to a Page

After the creation of a new page, a Graphical Variant for the Page may be created, if needed.

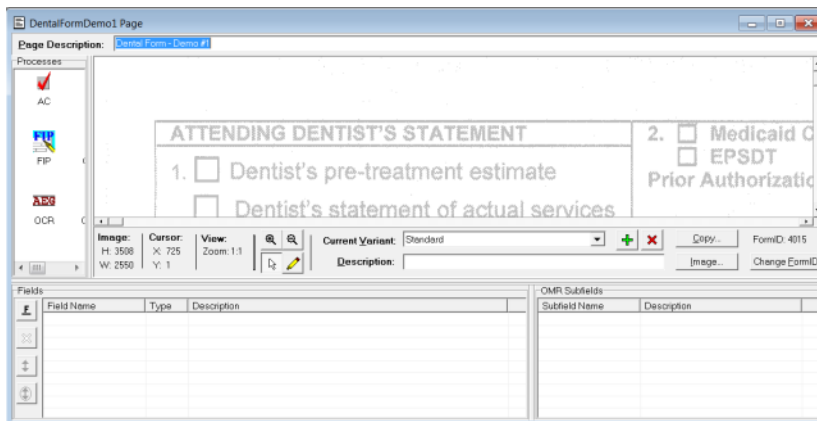
- Open the **Page** to which the variant is to be added. To do this, right-click the **Page** from the **Pages list** within the FWD tree.



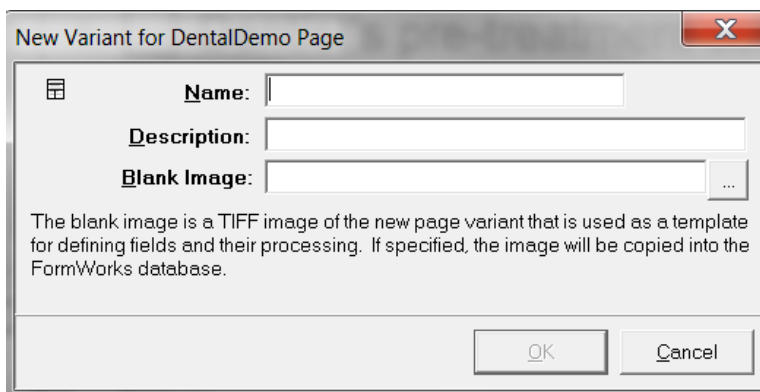
- From the FWD tree, select **Open**. The New Variant Type window opens, select **Open**.



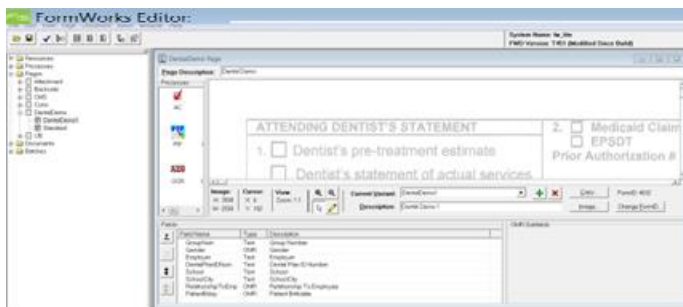
- The Page menu is displayed



- Select the **Create a new variant for this page** button , (or **CTRL + V**) displaying the New Variant dialog box.



- Enter a valid variant **Name**, **Description** and select the path (location) of the .TIFF image file in the **Blank Image** drop down box, select **OK**. The FWEditor adds the variant to the Graphic Variant tab of the page window.

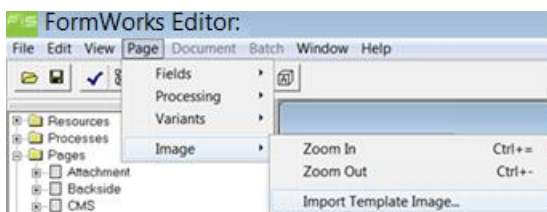


- Users may now create field variants (fields) on the image as needed.

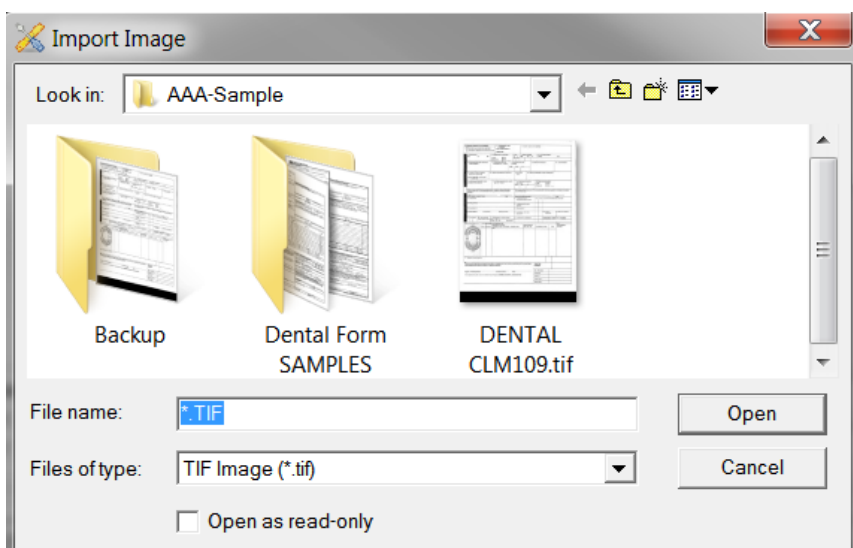
Select/Replace Variant Image Template

Be sure, when associating a graphic with a variant, to use a .TIFF image of a blank form

- Open the Page Variant to which the graphic is to be imported to.
- From the Variant menu, select Import Template Image.



- The Import Image window opens.



- Select the appropriate .TIFF file.
- Select Open. The Editor opens the blank image in the top portion of the Variant window.

Add Fields to a Page

Field Lists - Overview

Each page, including all its variants, has a single field list. Each field has:

- Field Name
- A bounding rectangle (for imaged fields)
- Text or OMR (Optical Mark Recognition)
- Description

A portion of a field list from a CMS-1500 form is provided in the example below.

Fields			
	Field Name	Type	Description
	ReceiptDate	Text	Date (MMDDYYYY) this batch
☒	_1_InsurancePlan	OMR	
☐	_1a_InsuredsIDNu...	Text	
☐	_2_PatientsName	Text	
☐	_2_First_Name	Text	
☐	_2_Middle_Initial	Text	
	MemberNumber	Text	

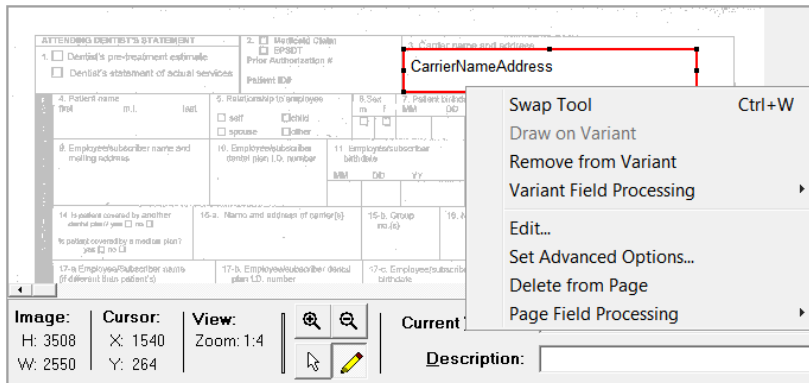
Additionally, all fields can be either:

- Imaged fields – Fields that actually appear on the form.

- Non-imaged fields – Do not appear on the form itself but are used throughout the system to hold information for data storage.

Shortcuts in the Field Lists

Once the fields have been created, users may right-click the field names in the list or on the page image (Configuration Window) to access a variety of options. The following example displays the menu after right-clicking a field on the image.



Text Fields

Text fields are fields that contain text data that may be read by the OCR engine and/or used by AC workers.

Note: Both imaged and non-imaged fields may be created using the procedure described below.

To Create a Text Field

- **Open the page** for which a field is being created. It does not matter which variant of the page is open, as all variants of a page share the same field list. However, the bounding rectangle for an imaged field is tied to a specific variant. See [Adding Page Variants](#), for additional information about variants.
- Select the Add Field button in the Field area in the lower portion of the window (shown at left). The New Field dialog box opens, as shown.

The 'New Field for Page Dental' dialog box is shown. It has a title bar with a close button. Inside, there are two text input fields: 'Name:' and 'Description:'. Below these is a 'Field Type' section with two radio buttons: 'Text Field' (which is selected) and 'QMR Field'. At the bottom left, there is a checkbox labeled 'Add Multiple Page Fields'. At the bottom right, there are 'Ok' and 'Cancel' buttons.

- Enter a new **Field Name**;

Field Naming Conventions

- Field Names may be up to 31 characters
- They may include only letters, numbers, and the underscore character
- They must begin with a letter or the underscore character; field names cannot begin with numbers
- Field Names must be unique only on a given page; different pages may use the same field names
- Enter a brief generic Description of the field, (recommended).
- Select Text Field in the Field Type area.
- Select OK.

Note: To create an imaged field, the user must define a bounding rectangle for the new field (see [Define Bounding Rectangles](#)).

Optical Mark Recognition Fields (OMR)

OMR, or Optical Mark Recognition fields are checkboxes, signatures, or other fields that simply need to be checked for a 'mark' inside the field. The data is not interpreted and the system merely judges if there is a mark or not inside the field. Every OMR field has two levels:

- The OMR field itself, which contains all the possible choices.
- The OMR subfields, which represent the individual checkboxes or options.

As an example, the OMR field Relationship to employee may contain (within) the subfields self, spouse, child and other.

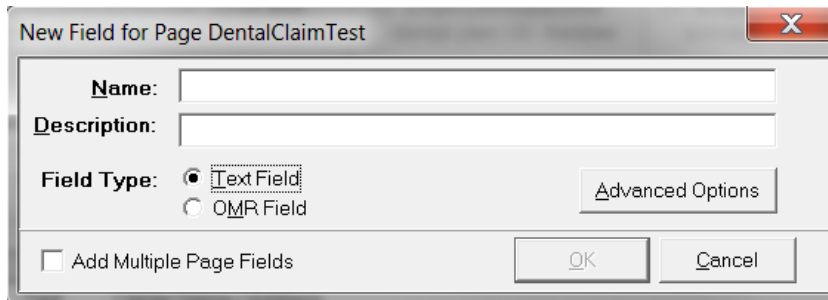
The screenshot shows a software interface for creating forms. On the left, there's a sidebar with icons for 'AC', 'FIP', and 'OCR'. The main area displays a form titled 'DentalDemo Page'. The form has several sections. One section is 'ATTENDING DENTIST'S STATEMENT' with checkboxes for 'Dentist's pre-treatment estimate' and 'Dentist's statement of actual services'. Another section is '2. Medicaid Claim' with checkboxes for 'EPBDT' and 'Prior Authorization #'. Below this is a 'Patient ID#' field. To the right is '3. Carrier name and address'. Further right is '6. Sex' with checkboxes for 'm' and 'f'. Below that is '7. Patient birthdate' with fields for 'MM', 'DD', and 'YY'. To the right of that is '8. If full time student' with checkboxes for 'school' and 'city'. Below these are fields for 'PatientDOB', 'PatientBda', 'School', and 'SchoolCity'. A callout box with the text 'OMR Subfield(s)' points to the 'Relationship to employee' field, which contains subfields: 'self', 'spouse', 'child', and 'other'.

Unlike text fields, OMR fields are read by FIP Worker, which analyzes only for the presence or absence of a mark in the box. This "mark" may take the form of a check, X or a line (either vertical, horizontal or diagonally) within the box.

To fully define a functional OMR field, the following components (described below) must be defined.

To Create an OMR Field

- Select the **Add Field button**  in the Field area in the lower portion of the window. The New Field dialog box opens, as shown.

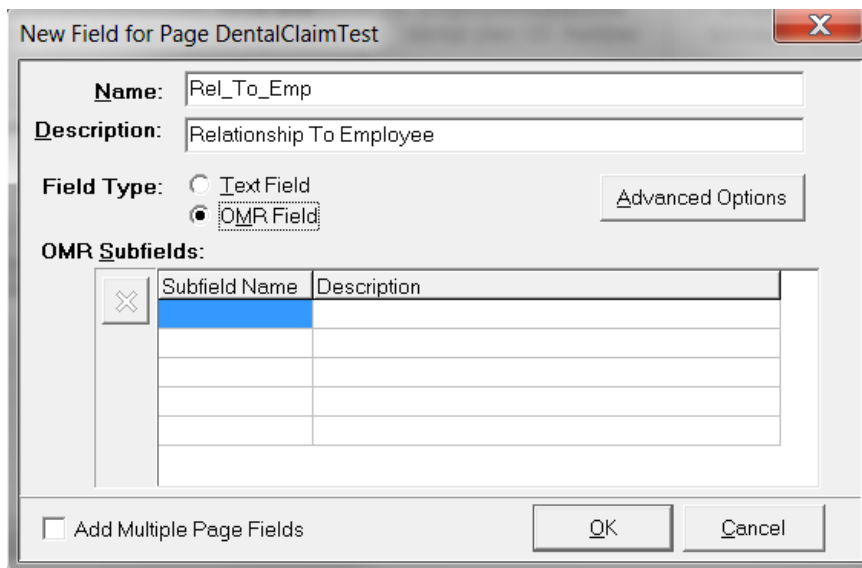


The dialog box titled "New Field for Page DentalClaimTest" contains the following fields and controls:

- Name:** A text input field.
- Description:** A text input field.
- Field Type:** Two radio buttons: ☒ Text Field and ☐ OMR Field.
- Advanced Options:** A button to the right of the Field Type section.
- Add Multiple Page Fields:** A checkbox at the bottom left.
- OK** and **Cancel** buttons at the bottom right.

Note: By default, the Text Field radio button is selected.

- Enter a Field Name for **the OMR field**; follow the [Field Naming Conventions](#) listed in To Create a Text Field.
- Enter a **Description** of the field, if desired.
- Select **OMR Field** in the Field Type area.
- The **New Field for Page** windows is displayed presenting **the OMR Subfields** area.



The dialog box is now showing the OMR Subfields area. The fields and controls are:

- Name:** Rel_To_Emp
- Description:** Relationship To Employee
- Field Type:** ☐ Text Field and ☒ OMR Field.
- Advanced Options:** A button to the right of the Field Type section.
- OMR Subfields:** A table with two columns: Subfield Name and Description. The first row is highlighted in blue.
- Add Multiple Page Fields:** A checkbox at the bottom left.
- OK** and **Cancel** buttons at the bottom right.

Subfield Name	Description

- Enter **each Subfield Name**, with a brief Description. Use the Tab key or the arrow keys to move to the next field or line.

New Field for Page DentalClaimTest

Name: Rel_To_Emp

Description: Relationship To Employee

Field Type: ☐ Text Field ☒ OMR Field Advanced Options

OMR Subfields:

Subfield Name	Description
Self	Self
Spouse	Spouse
Child	Child
Other	Not Self, Spouse or Child

☐ Add Multiple Page Fields OK Cancel

- Select **OK**.
- The OMR field is created and displayed along with the defined OMR Subfields

Fields

Field Name	Type	Description
CarrierNameAddress	Text	Carrier Name / Address
Rel_To_Emp	OMR	Relationship To Employee

OMR Subfields

Subfield Name	Description
Self	Self
Spouse	Spouse
Child	Child
Other	Not Self, Spouse or Child

- Next, define bounding rectangles for both the OMR field as a whole, and for each subfield. See [Defining Bounding Rectangles](#).

Important: The FormWorks Editor currently does not issue a warning if an OMR field is created and fails to generate subfields (or their bounding rectangles). It is the users' responsibility to properly create all necessary subfields and bounding rectangles in order for the system to properly read the data on the form.

Advanced Options

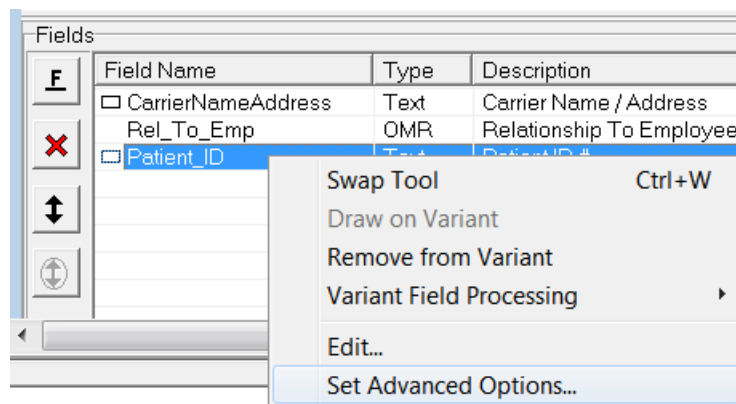
Dynamic Field Creation

Dynamic field creation takes place when a field must be created by the system on-the-fly. This may be necessary when a field is added to a page. In this scenario, work entered into the system prior to the newly added field would in all probability return an error as the newly added field would not be recognized.

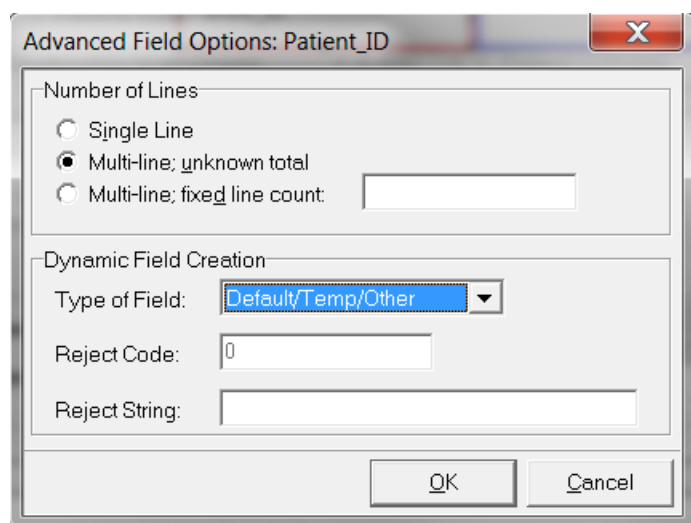
By default, FWEditor automatically assigns a newly created field as a "Default/Temp/Other" field type. For these types of fields, AC will automatically create the field if the field did not previously exist, and no reject code is set. In this instance it is necessary to force the field to be keyed or populated at DV by utilizing the Advanced Options to utilize the preferred settings.

Set Advanced Field Options

To configure **Advanced Field Options**, Right-click the **Field Name** in the field list. Select **Set Advanced Options** from the menu. The **Advanced Options** dialog box opens.



After selection of **Set Advanced Options**, the Advanced Field Options window is displayed (note, the field name selected extends past the name of the window).



The dialog box titled "Advanced Field Options: Patient_ID" contains two main sections. The first section, "Number of Lines", has three radio button options: "Single Line", "Multi-line; unknown total" (which is selected), and "Multi-line; fixed line count:" followed by an empty text input field. The second section, "Dynamic Field Creation", includes a "Type of Field:" dropdown menu currently set to "Default/Temp/Other", a "Reject Code:" text input field containing the value "0", and a "Reject String:" text input field. At the bottom right are "OK" and "Cancel" buttons.

To configure advanced field options:

- Select the **Type of Field**.
 - Default/Temp/Other – Normal fields. Initial reject code is 0; with an empty reject string (no reject is set).
 - Keyed – Editor automatically sets the reject code to 1, and the reject string “Enter data”. The page will be forced to KE, and this field shown to the keyer.
 - Populated at DV – Editor automatically sets the reject code to 2, and the reject string
 - “Return to DV”. The page is sent to DV for parsing, table lookup, or other action to populate this field.
 - Custom – Set a user assigned reject code (other than 0, 1, or 2) and reject string.
- Select **OK** to save changes

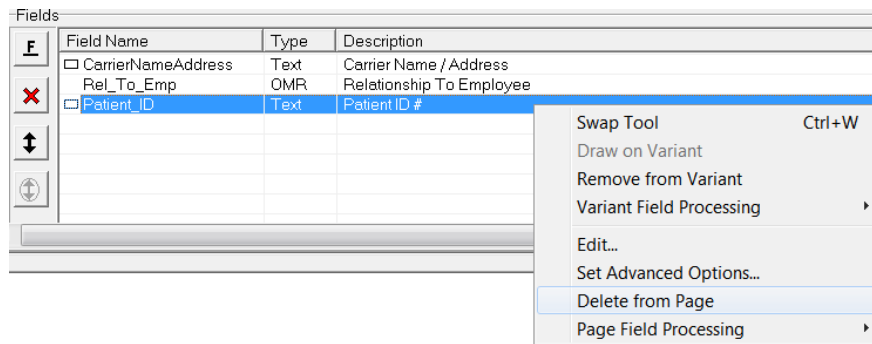
Deleting a Field

To delete a field from a page

- Select the Field Name to be deleted.

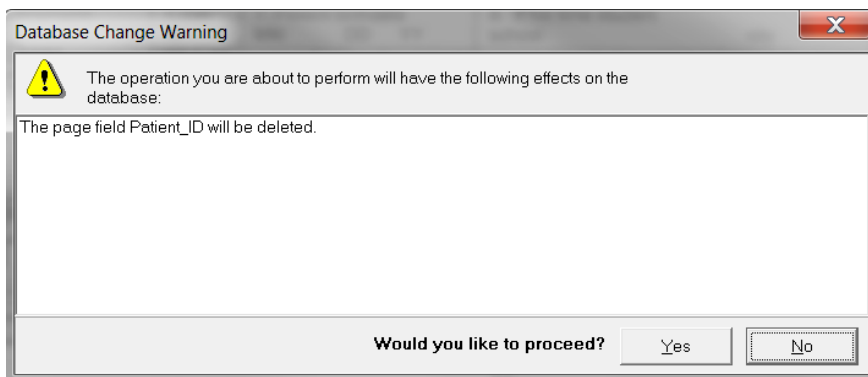
Fields			
	Field Name	Type	Description
☐	CarrierNameAddress	Text	Carrier Name / Address
☐	Rel_To_Emp	OMR	Relationship To Employee
☒	Patient_ID	Text	Patient ID #

- Right-click the field and select Delete from Page.



Important: DELETE on the keyboard removes the field only from the current variant; it does not delete the field. To CORRECTLY delete the field from the page, the Delete from Page selection MUST be used.

- Upon selection of Delete from Page, the Database Change Warning window is displayed.

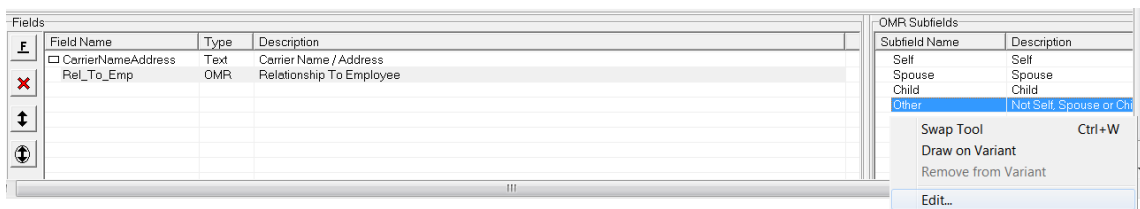


- Select Yes to delete the field from the page.

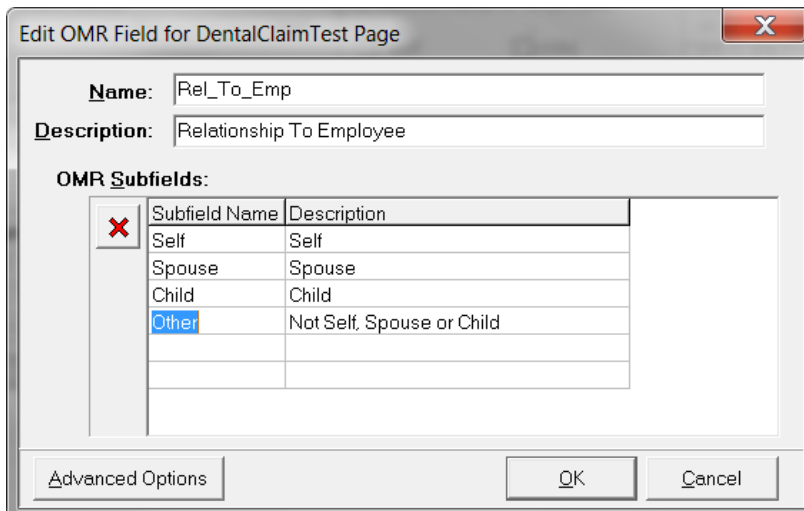
Important: Once Yes has been selected neither the field nor the changes to the database may be recovered. Also, when an OMR field is deleted, all subfields are deleted in the same operation.

To Delete an OMR Subfield from a Page

- Right-click the **Subfield name** and select **Edit** from the menu.



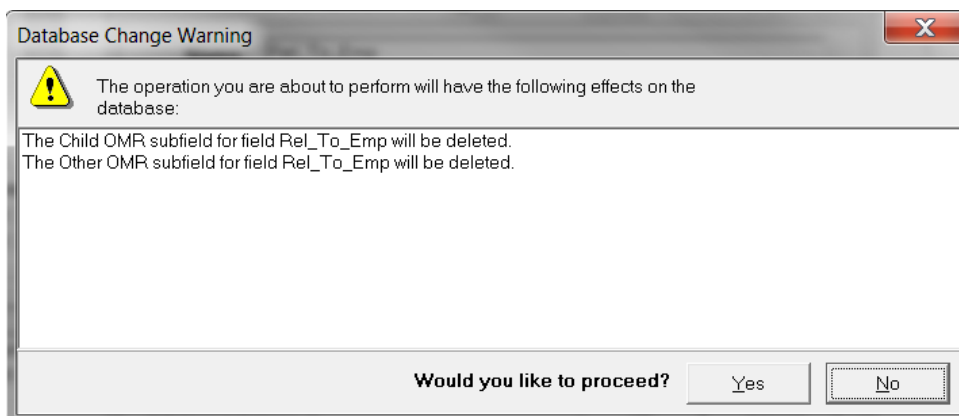
- Displayed is the **Edit OMR Field** screen.



- In the **Edit OMR Field** window, select the **Subfield** to be deleted, select the **Delete** button .

Note: Pressing the delete key will delete the field data. But, each field (Subfield Name/Description) must be deleted individually.

- Click OK. A Database Change Warning window is displayed communicating to the user that the operation about to be performed will have the following effects. The system prompts the user with the question; Would you like to proceed?

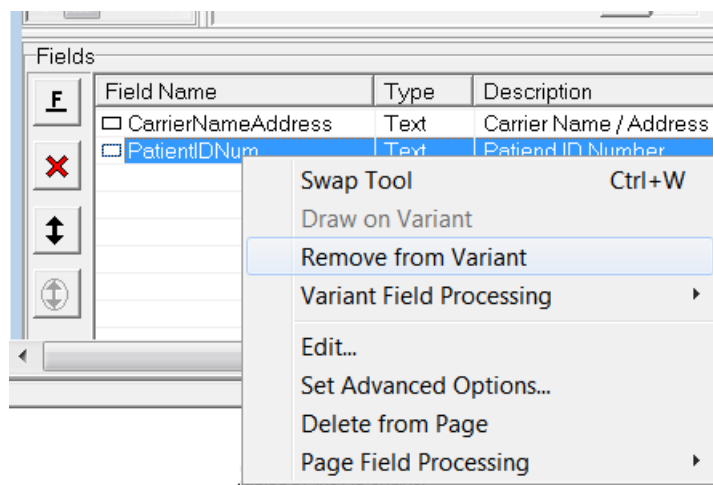


- Select Yes to complete the OMR Field deletion.

To Remove a Field from a Variant

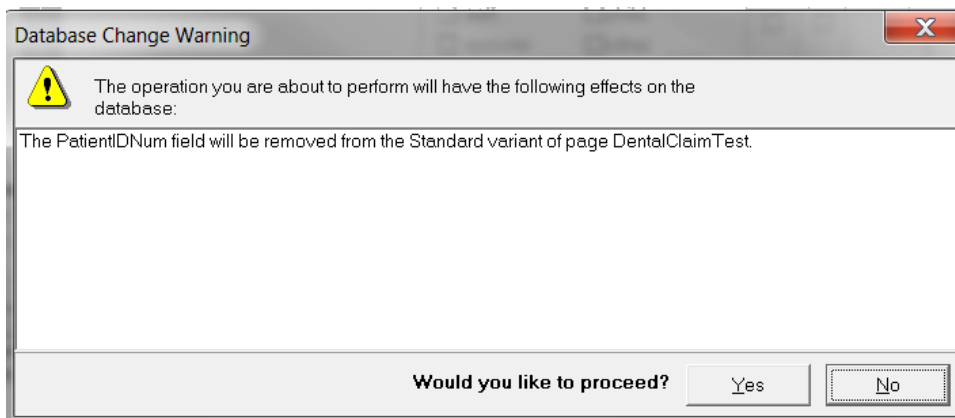
Removing a field from a variant does not affect the page's field list, or the mapping for the field on other variants. This procedure only removes the bounding rectangle for the field on the current variant.

- Right-click the Field Name in the Fields list select Remove Field from Variant.

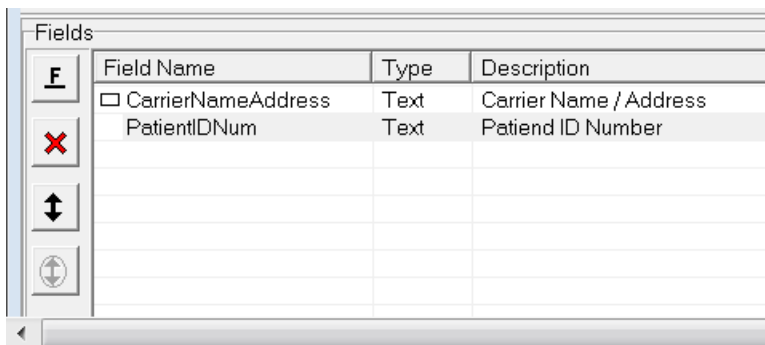


Important: There are separate options for removing the field from the current variant and deleting the field from the page. Deleting a field removes it from the page and all variants of the page.

- Read the message in the Database Change Warning box. Verify that that the field is being removed from the current variant only (not the page). Select Yes to proceed with the deletion.



After removing a field from a variant, the field name still appears in the field list (because it is still a field on this page), but the imaged field indicator (☐) disappears.



To Remove a Subfield from a Variant

Subfields are to be removed in the same manner as fields.

Note: The DELETE key on the keyboard may not be used.

Right-click the **Subfield** and select **Remove** from variant. This removes the subfield from the current variant only.

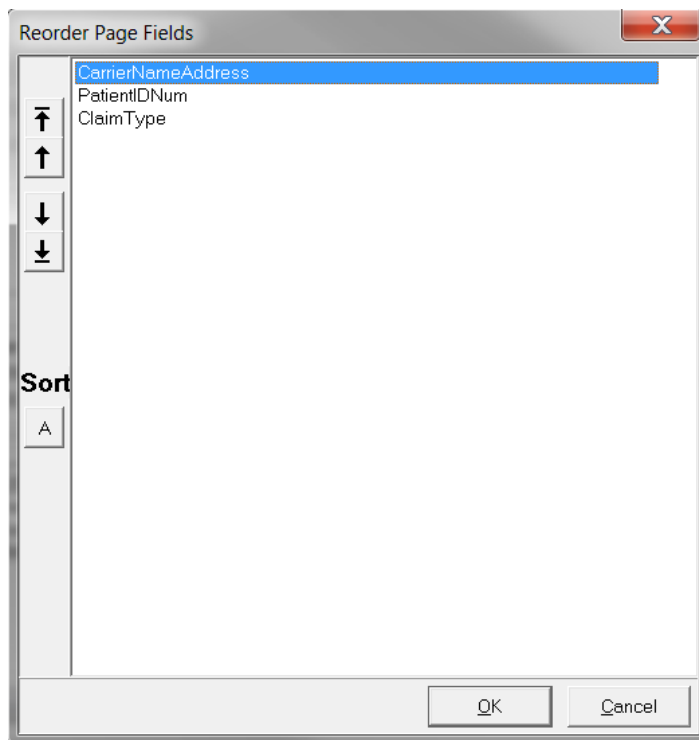
Reordering Fields and Subfields

Once the fields have been created, rearrange the order in which they appear in the field or sub-field list by utilizing the Reorder Fields or Reorder Subfields icon(s).

-  **Reorder Fields** – Selecting the Reorder Fields displays to the user the **Reorder Page Fields** window. Select the field or fields (Ctrl Click) for which the change of order is to occur.
 -  – Moves the selected field(s) to the top of the list.
 -  – Move the selected field(s) up one line.

-  – Moves the selected field(s) down one line.
-  – Moves the selected field(s) to the bottom of the list.
-  – Sort the entire list alphabetically.

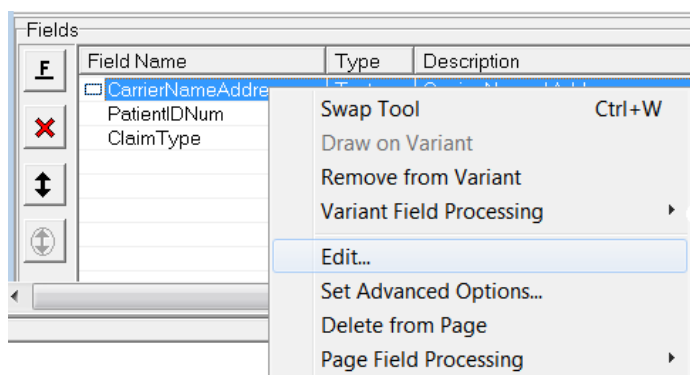
Note: Once the list has been sorted alphabetically, to resort the user must utilize one of the arrow keys. Numbers are sorted as text strings.



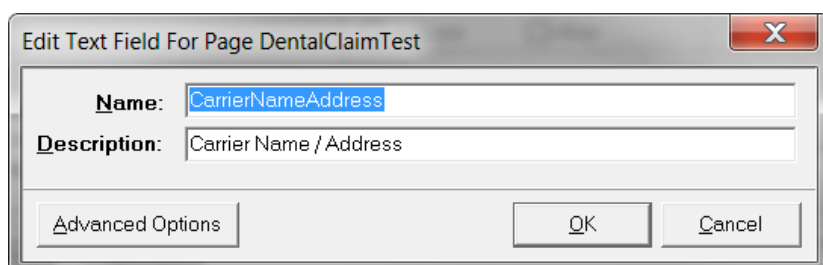
- Select **OK** to save the new order.

Rename a Field

To change the name or description of a field, right-click the **Field Name** and select **Edit** from the menu (or press **CTRL+E**).



The Edit Field window opens.



Modify the field Name and Description in accordance with standard [Field Naming Conventions](#).

WARNING: Modification of lines or field types within the Advanced Filed Options (for fields already in production) may render field data inoperable.

Define Bounding Rectangles

Every imaged field must have a bounding rectangle which defines where a particular field is located on the current variant of a form.

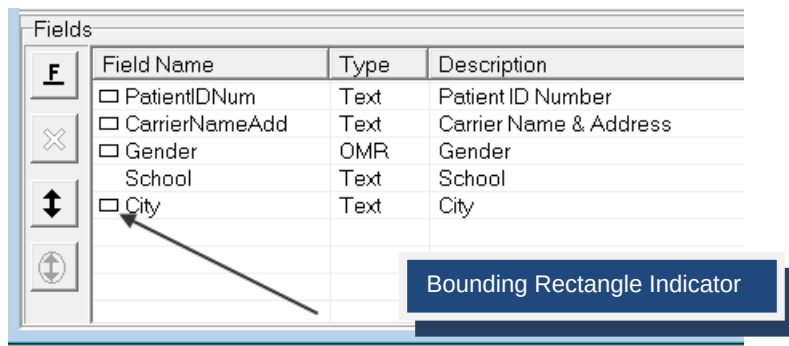
For OMR fields, first define one bounding rectangle for the field as a whole, and then separate rectangles for each subfield checkbox.

Note: Field bounding rectangles must be defined on each variant. See [Adding Page Variants](#), for additional information.

To Define Bounding Rectangles

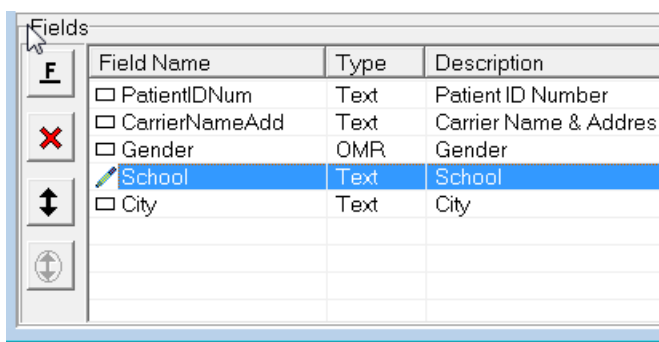
There are two methods in which users may go about activating the drawing tool used to create bounding rectangles. The behavior of the tool and the options presented will vary, depending on which method is utilized.

Note: A small rectangle appears next to the field name in the field list after the bounding rectangle is defined.

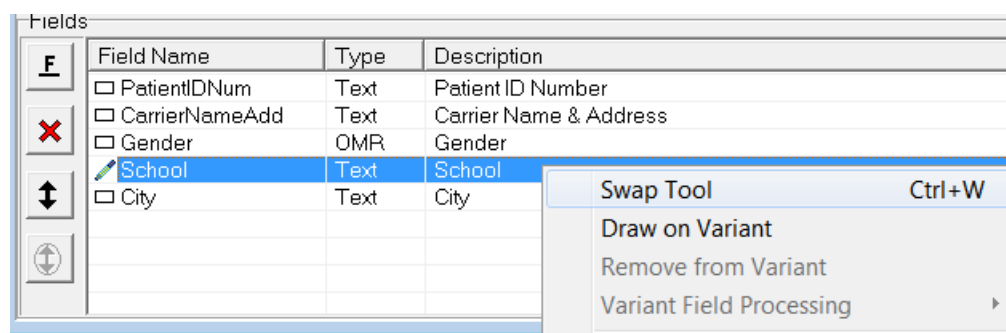


Method 1: Double-Click Field Name

- Double-click a field name in the field list (or subfield name if a bounded rectangle for an OMR field has already not been created, (example, "School" in the screen shot above)). To the left of the field name the Pointer is displayed.



Note: To deactivate the Pointer, select Ctrl W or right click on the field and click Swap Tool.

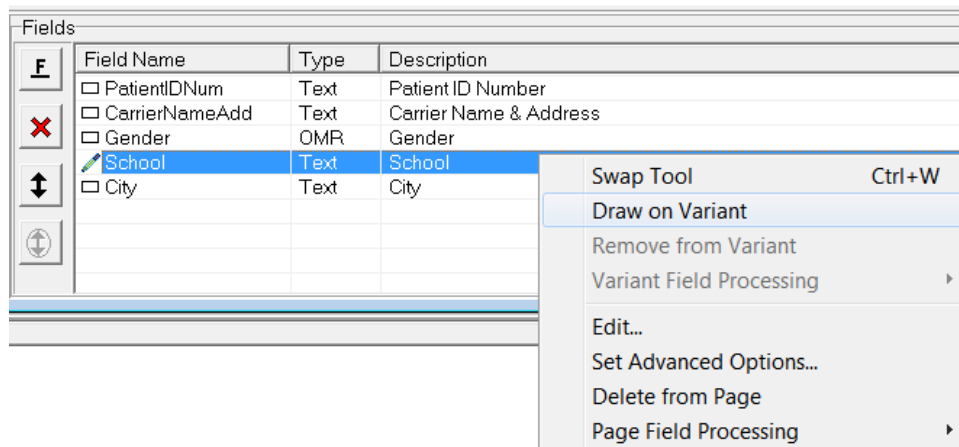


- Within the Configuration window, the mouse pointer becomes a plus sign .
- Click and drag a rectangle around the area on the form image that corresponds to the field.

Note: If drawing an OMR subfield, the rectangle drawn may resize, or “snap” to the box on the image. See [Enabling/Disabling Snap Feature](#).

- The mouse pointer returns to the arrow shape automatically after the bounding rectangle has been drawn.

*Note: An alternate method to draw the field is to right-click the **Field Name**, and select **Draw on Variant**.*



Method 2: The Drawing Tool

- Click the drawing tool button . Within the Configuration window, the mouse pointer changes from a pointer to a plus sign $+$.
- Click and drag a rectangle around the area on the form image that corresponds to a field.
- Verify that the correct field name is selected in the Choose Field dialog box, and select OK. If the field name has not been created, select the Add Field button , and enter the Field Name and field type.



- Press Esc to deactivate the drawing tool, and return to the arrow pointer.

The bounding rectangles in the Editor window are color coded as follows:

- Blue – Text field
- Green – OMR field
- Burgundy – OMR subfield

- Red – currently selected field or subfield

Enable/Disable Snap Feature

“Snap” allows users to draw bounding rectangles somewhat larger than the box on the image. When the mouse is released, the rectangle automatically resizes to the image.

To disable Snap, select the **Drawing Tool button**  and drag a box on a subfield in the form. The **Choose Field** dialog box opens. Click **Snap Checkbox to Image** to unmark the option.

Note: When a page is initially opened, the Snap feature is by default enabled. However, when Snap is disabled the functionality (Snap Checkbox to Image) will be automatically enabled the next time the page is opened.

Tips for Defining Bounding Rectangles

- OMR Subfields – Make the bounding rectangle somewhat larger than the actual checkbox on the image to capture marks that fall outside the box.
- Don't overlap fields – Overlapped bounding rectangles may result in the same character or characters being read in two different fields.
- Don't leave gaps between adjacent fields – A character that falls exactly on the boundary between two fields will generally be picked up in one field or the other. If there is a gap, the character may not be read at all. Use the magnifier button to zoom in for a closer review.
- The following is a magnified view of properly drawn bounding rectangles. Note that rectangles touch surrounding fields.



Add Page Variants

A page variant is a form similar to the standard variant previously defined but is graphically different.

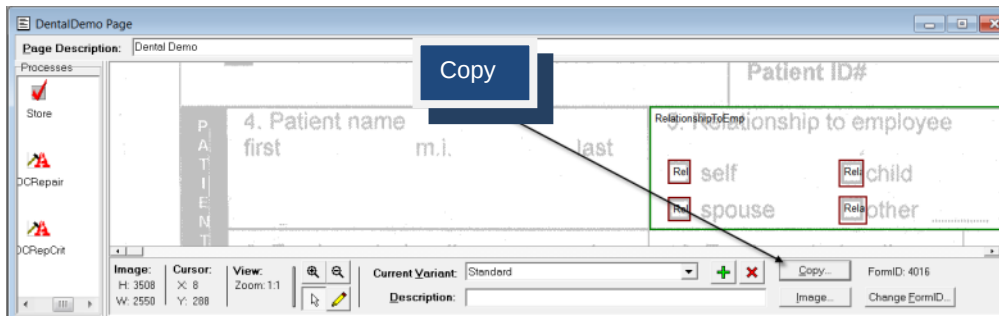
All variants of a given page have the same fields, and more importantly, the same page processing rules. If necessary, users may set variant field level processing configurations for FIP and OCR to override the page settings.

Often Variants are versions of a form designed or printed by different people or printers, resulting in forms containing the same information, but using alternate layouts. To handle such situations, it is often best to copy an existing variant with a similar layout; then modify bounding rectangles for fields that are in different locations.

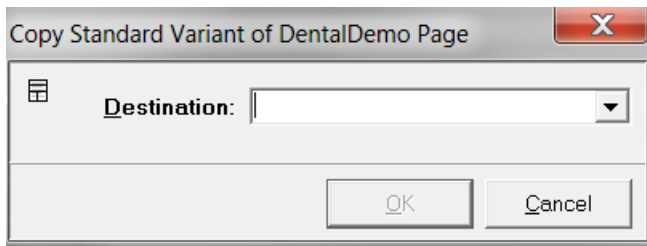
For a variant with a substantially different layout, a best practice is to first create a new variant and then set new bounding rectangles for each field.

To Copy an Existing Variant

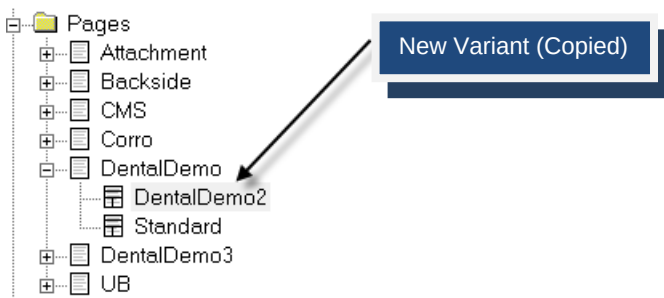
- Open the variant to be duplicated.
- In the image template window, select the Copy button (or press CTRL + C).



- Enter the destination (name of the new variant) in the Copy Standard Variant dialog box, and select OK.



- The newly created variant is placed in the directory (Pages) of which the original variant resides. The copy is an exact duplicate of the original, including the bounding rectangles for all imaged fields, and all processing specifications.



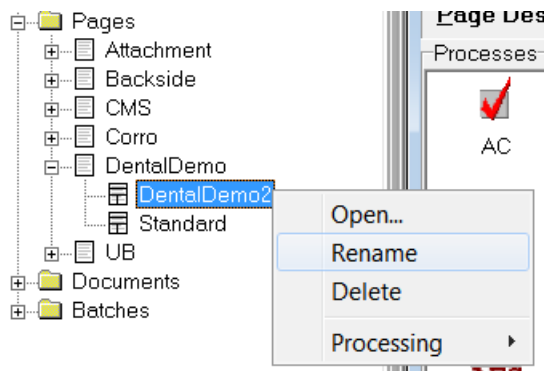
Rename Pages and Variants

By default, the first variant of a page is given the name Standard. Additional variants are named by the user when they are created.

To Rename a Page or Variant:

- Right-click the **Page Name** or **Variant Name** in the FWD tree.

- Select **Rename** from the menu.

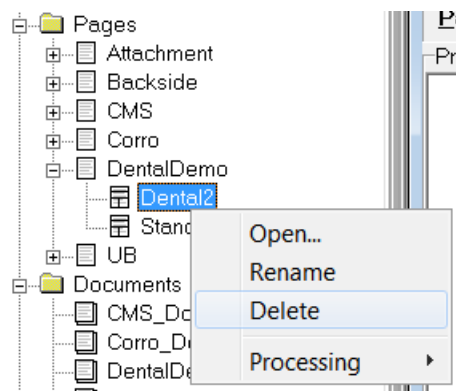


- Type the **new name** for the page or variant, **press Enter** to save the new name.

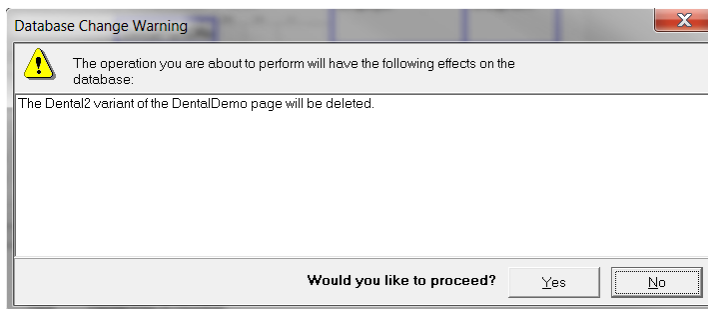
Delete Pages and Variants

To Delete a Page:

- Right-click the **Page Name** in the FWD tree, then select **Delete** from the menu.



- The **Database Change Warning** window is displayed listing the effects of the deletion. To continue the delete a page process, select **Yes**.

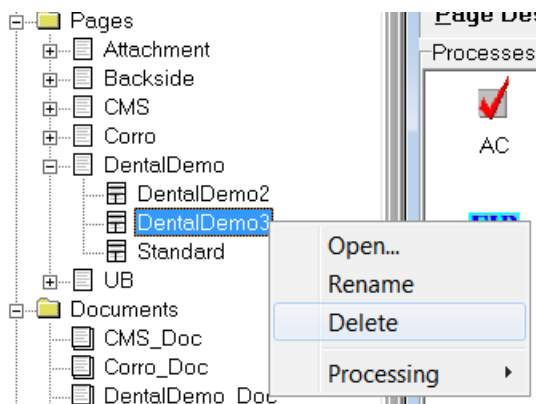


*Note: Users may also select the page name and press **DELETE**.*

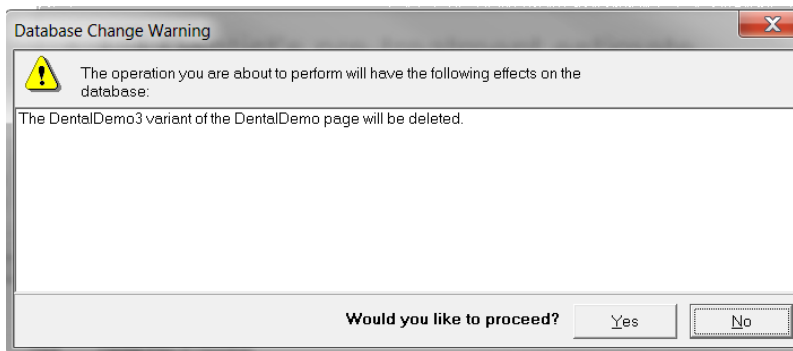
Warning: When a page is deleted, all of the variants and fields are deleted in the same operation.

To Delete a Variant:

- Right-click the variant name in the FWD tree.



- Select **Delete** from the menu.
- The **Database Change Warning** window is displayed listing the effects of the deletion. To continue the Delete A Page process, select **Yes**.

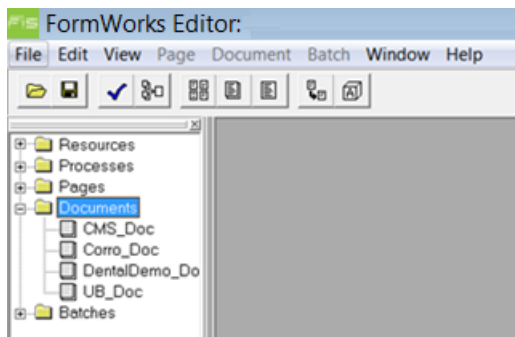


Working with Documents

A FormWorks document is composed of:

- A document field list.
- Processing steps and rules for the document.
- May contain additional fields at the Document level.

Note: Different documents may share one or more pages.



Use the document level Collator processing configuration to specify the minimum and maximum numbers of each page allowed in the document.

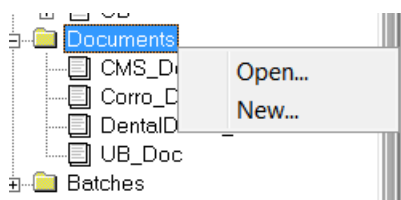
Create a New Document

When creating a new document, list all possible pages that could be included. Be sure to include pages such as Attachments and Separators, even though not every document of this type will contain every page.

Note: When creating a new Page (Variant) if the Auto Create Document and Batch option is selected when the Page was created, the Document is created automatically.

To Create a New Document

- Right-click Documents in the FWD tree and select New from the menu.



- The New Document window opens.

Page	Description
☐ Attachment	
☐ Backside	
☐ CMS	
☐ Corro	
☐ DentalDemo	Dental Demo
☐ UB	

- Enter a **Name** and Description for the document; ([Field Naming Conventions](#) apply).
- From the page list (Pages in Document); check each **Page** that may be in the new document type. To remove a page from the document, select the **Page** again to clear the checkbox.

Page	Description
☐ Attachment	
☐ Backside	
☐ CMS	
☐ Corro	
☒ DentalDemo	Dental Demo
☐ UB	

- Click **OK**; the new document appears in the FWD tree.

Add Document Level Fields

Document level fields are simply fields that are attached to the document as a whole, rather than to a specific page of a document. Document level fields are all non-imaged, since they may occur anywhere within the document.

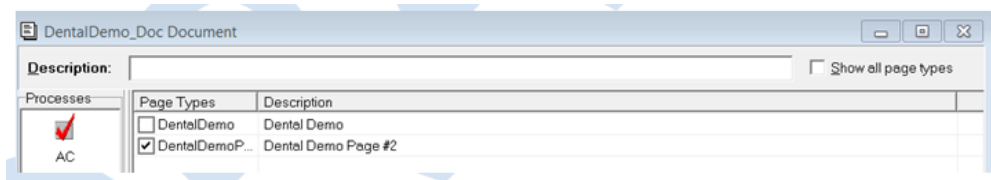
The procedure for adding fields at the document level is identical to adding page level fields. See [Adding Fields to a Page](#), for additional information.

Add and Remove Document Pages

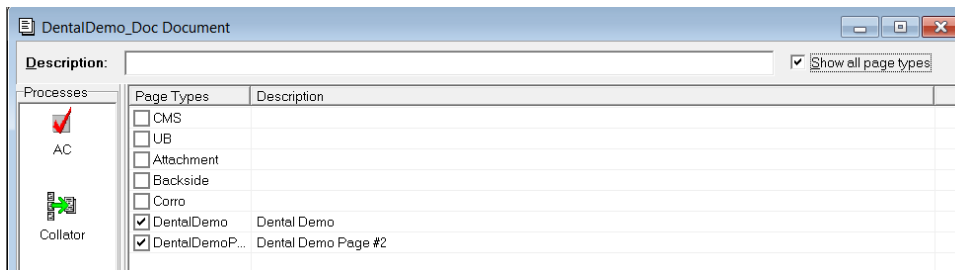
When a document has been created and Document Pages require removal, the user selects the page(s) that are to be removed from the document.

Note: Individual Pages may (at a later time) be added to or withdrawn from documents (unless the Page has been deleted).

- Double-click **the Document Name** in the FWD tree to open the document. Use the checkboxes next to the page names to exclude pages.



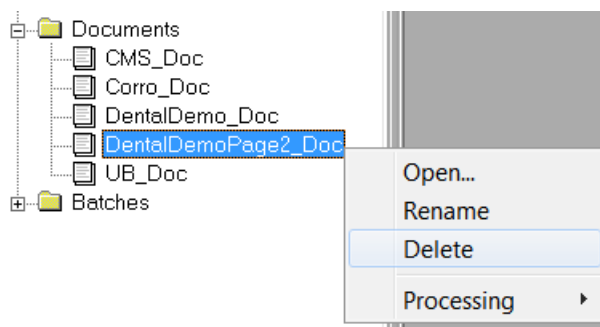
*Note: If the desired page name does not appear, select the **Show All Page Types** checkbox.*



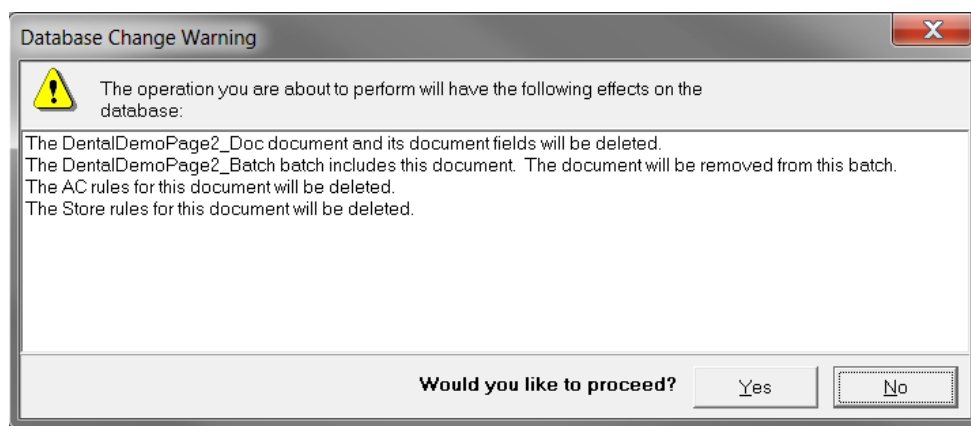
Delete a Document

Important: A document that is currently in use in an active system should not be deleted.

- In the FWD tree, right-click the Document Name to be deleted.
- Select Delete from the menu.



- A Database Change Warning appears. Read the list of effects of this change before proceeding with the deletion.

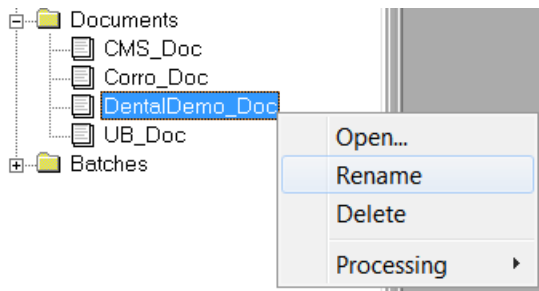


- Select Yes to proceed to delete the Document.

Rename A Document

Important: A document that is currently in use in an active system should not be renamed.

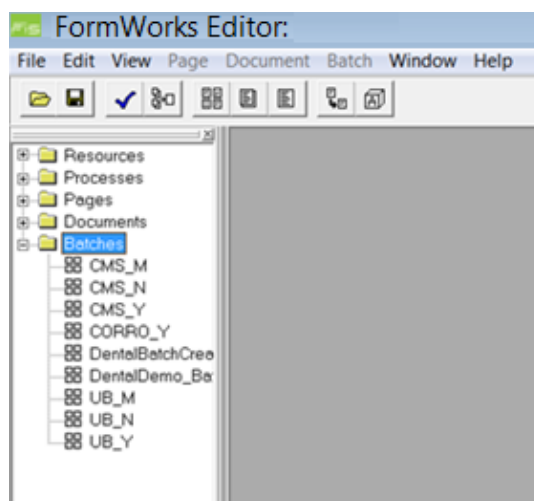
Like most other objects (Pages), FormWorks Documents may be renamed at any time. Simply right-click the **Document Name** and select **Rename** from the menu. Then type the **New Name**.



Batches

A FormWorks Batch consists of one or more document types and processing rules for the Batch. An operator enters a batch into the system using an Inventory Client to specify various options, and to enter values for batch attributes. These options and attributes are configured through the Inventory Resource PCM. (See [Creating/Configuring an Inventory Resource](#), for additional information).

Note: Batches must be created before the Resources – Inventory may be fully configured. Batch level Inventory options are accessed through the Inventory resource window (unlike processes, which can be accessed from the batch window).



Collator, FIP, and Scan processes have batch level configurations. For additional information see:

- [Collator: Batch Processing](#),
- [FIP: Batch Processing](#)
- [Scan: Batch Processing](#)

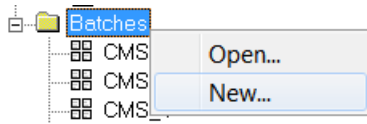
Create a New Batch

When creating a new batch, list all documents that could be included in the batch. Make sure that each document definition contains all of the pages that may be included.

Note: When creating a new Batch if the Auto Create Document and Batch option was selected when the Page was created, the Batch is created automatically.

To Create a New Batch

- Right-click **Batches** in the FWD tree and select **New** from the menu.



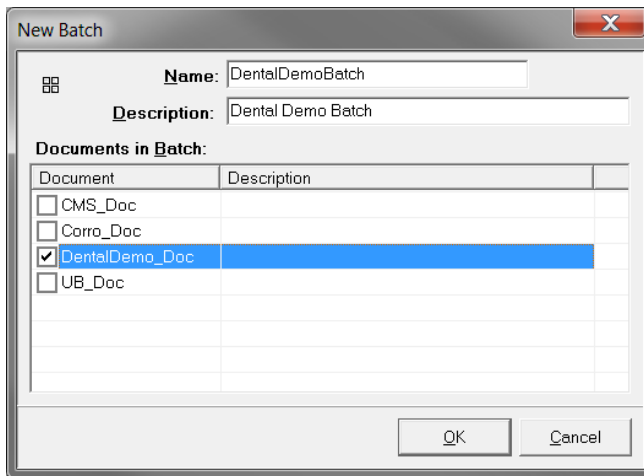
- The **New Batch** window opens.

A screenshot of the 'New Batch' dialog box. The dialog has a title bar with 'New Batch' and a close button. Inside, there are two text input fields: 'Name:' and 'Description:'. Below these is a section titled 'Documents in Batch:' which contains a table with two columns: 'Document' and 'Description'. The table has four rows, each with a checkbox in the 'Document' column and an empty 'Description' cell. The checkboxes are for 'CMS_Doc', 'Corro_Doc', 'DentalDemo_Doc', and 'UB_Doc'. At the bottom of the dialog are 'OK' and 'Cancel' buttons.

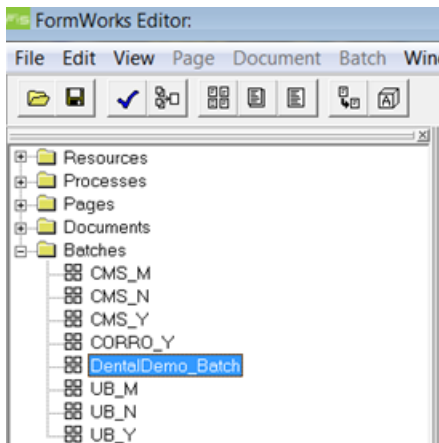
Document	Description
☐ CMS_Doc	
☐ Corro_Doc	
☐ DentalDemo_Doc	
☐ UB_Doc	

- Enter a **Name** and a **Description** for the batch.
- Select the **Documents** that may be included in this batch type.

Note: Documents may still be added to batches (and removed from them) later.



- Select OK, the newly created Batch is added to Batches



Adding and Removing Documents

To open the batch, double-click the **Batch Name** in the FWD tree. Use the check-boxes next to the document names to include or exclude documents.

Note: If the desired document name does not appear, select the Show All Document Types checkbox. Select the needed documents via the check box.

Document	Description
☐ CMS_Doc	
☐ Corro_Doc	
☒ DentalDemo_Doc	
☐ UB_Doc	

Close the window  to save the changes.

Processes

FWEditor provides an environment where the user may define the processing steps needed for a specific system as well as how each processing step should work in a given system. Process information is stored in and managed by Process Configuration Modules (PCM) that are dynamically loaded by FWEditor.

Each processing step (often also referred to as a Worker) is a distinct stage in which some operation is performed on the data. A given FormWorks database may contain processes for Inventory, Scan, FIP, OCR, Collator, AC, OCRepair, Store, and other processes. There can be more than one process of a given type. Each process has a system level configuration as well as one or more of the following levels: page, page variant, document, and batch. For example, Collator is configured at the document and batch levels; FIP operates on page, page variant, and batch level data.

It is recommended users create the processing step first, then set the appropriate page, document, and batch level process configurations. See [Configuring Page and Page Variant Processing](#), and [Configuring Document and Batch Processing](#) for more information.

Important: For configuration of Process windows within the Process module, there is NOT a Save/OK button. Any changes made to any of the Processes windows are by default saved when the user closes the window utilizing the close window function .

Users are advised to use caution when making changes to all screens with the Processes module.

Create A New Processing Step

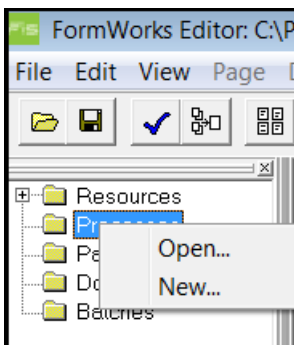
Use the Editor to define the processing steps for data. Each processing step has a base processing type.

Note: There may be multiple steps using a single type.

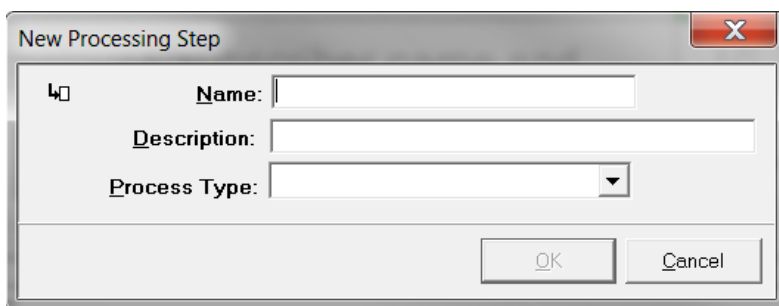
For example, the system may have the processing steps OCRepCrit, OCRepNum, and OCRepair. These steps are all based on the processing type OCRepair; one PCM maintains the configuration for the processing type.

To Create A New Processing Step

- Right-click Processes in the FWD tree and select New from the menu.



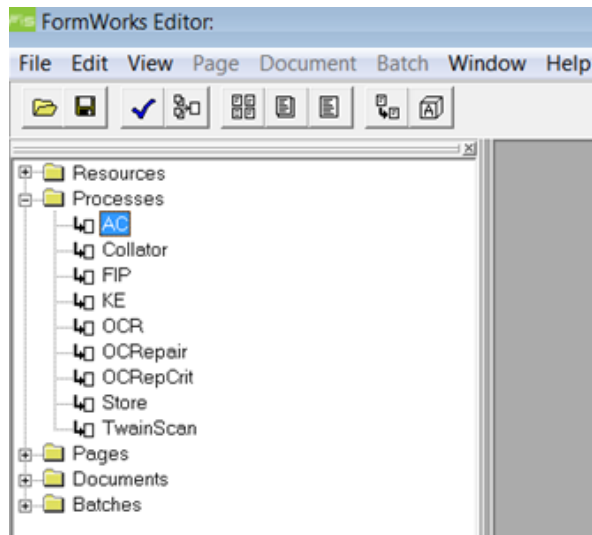
- In the New Process Step window, enter a Name, Description and select a Process Type from the Process Type drop down box.



- The FormWorks Express build is comprised of the following processes.
 - AC
 - AEG
 - Collator
 - FIP
 - GenPickup2
 - Inventory
 - KE
 - OCRepair3
 - Render
 - Store
 - TWAINSCAN
 -
- Click **OK**. The newly created processing step appears in the list of processes displayed in the FWD tree.

Configure A Process

- Double-click the **Process** to be configured from the tree (Processes) to open the configuration dialog box.



The options presented in the configuration dialog boxes are different for each process type. Each of the process type's native to FormWorks Express is detailed in the following sections. For more information on specific processes, refer to the documentation specific to the process (PCM documentation).

AC Process

AC captures, corrects, and validates OCR data using the AC Engine. Generally, the AC Worker is located in at least two places in workflow: before OCRRepair and before KE. In many systems, AC runs before and after OCRRepair Critical (OCRRepCrit), and before KE. AC identifies (potentially) erroneous characters which are sent to OCRRepair to be repaired by an operator. Most commonly, many systems define "AC I" before OCRRepair Critical, and "AC II".

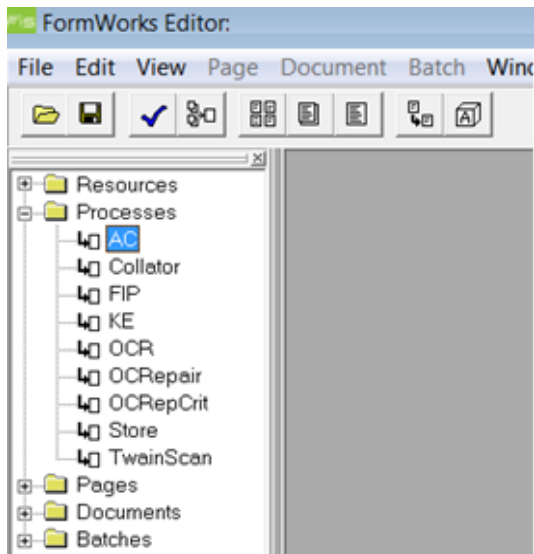
The worker prior to KE is usually known as DV (Data Validation) and is used primarily to validate and format OCR data on a field-by-field basis. DV identifies invalid fields to be sent to a KE operator for review and correction.

The validations and formatting performed by AC workers are accomplished through the application of rules. Within most processing environment, rules are most often referred to as functions, which are contained in several rule DLLs.

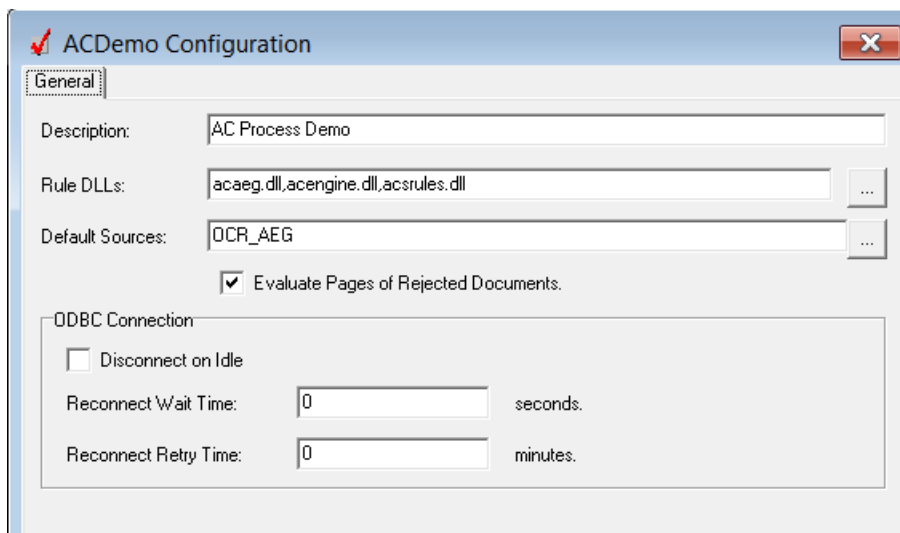
AC configuration options are available at the page and document levels. See [AC: Page Processing](#), and [AC: Document Processing](#), for information on rule configuration.

In the Editor, open the AC Configuration window to specify which rule DLLs and source file being used.

- To open the configuration window, double-click on AC in the Processes list in the FWD tree.



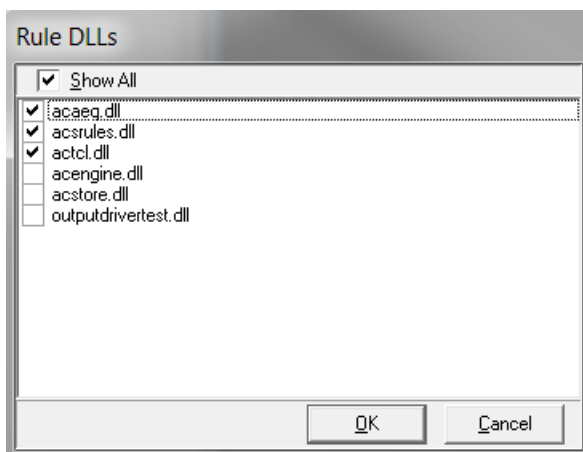
- The AC Configuration window is displayed (ACDemo as our example).



Important: The AC configuration window within the Process module DOES NOT have a Save/OK button. Any changes made to any of the Processes windows are by default saved when the user closes the window utilizing the close window function .

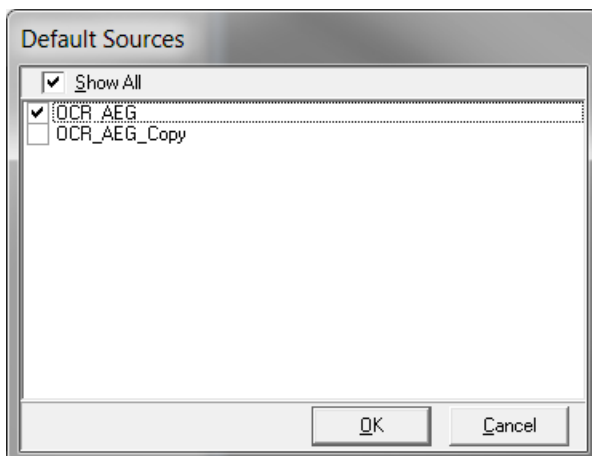
General

- Description – A brief description of the configuration.
- Rule DLLs – Use the browse button  to view a selection list of all available rule DLLs (Resources -> RuleDLL). For our example, DLLs acaeg.dll, acsrules.dll, actcl.dll have been selected. Select OK upon completion of the selection(s).



Note: As a best practice, do not select the acengine.dll here.

- Default Sources – Specify the fileref of the OCR file (OCR_AEG), select OK upon completion of the selection(s).



- Evaluate Pages of Rejected Documents – If selected, AC rules are run on rejected documents.

Note: As a best practice this feature is most often checked.

ODBC Connection

- Disconnect on Idle – If checked, the system will disconnect from the ODBC connection when not in use.
 - Reconnect Wait Time (seconds) – Number of seconds until the system waits before trying to automatically reconnect (maximum allowable entry 10 digits).
 - Reconnect Retry Time (minutes) – Number of minutes until the system waits before trying to automatically reconnect (maximum allowable entry 10 digits).

Collator Process

The Collator (Collator System) assembles pages into documents and assigns a document ID.

Typically, the Collator determines which pages belong together through one of several collation techniques (described below). When a collation technique is selected for the Collator process, it is a system-level setting used as a default. At the batch level, users may specify a different technique if necessary. See [Collator: Batch Processing](#) for more information.

As a best practice, select a collation technique that is reliable (identifies all pages as part of a document).

Collation Techniques

- Fixed Number Of Pages Per Document – Uncollated pages are moved into documents consisting of a fixed number of pages. If there are not enough pages to make an entire document, the pages available are placed into a document, but no error is set. This technique should only be used when all documents in the batch contain "n" pages, and are sorted correctly.
- Attributes Change Values – Collator compares the values of an attribute or set of attributes, usually DCN. The first uncollated page is assumed to be the first page of a document; a new document is created when the value of the specified attribute(s) changes. All pages with the same value for all specified attributes are moved into a document. The separator page (if there is one) can be deleted based on the attribute that changed.

Note: The Fixed Number of Pages Per Document or Attributes Change Values technique never results in uncollated pages.

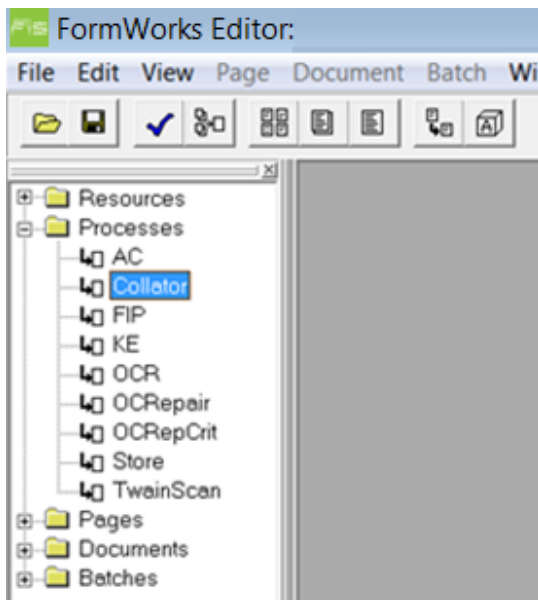
- Attributes Present – Collator looks for the presence of one or more specified attributes to determine if a page is a separator page, and creates a new document each time a separator is found. Each successive page not having the specified attribute(s) is moved into the new document. The separator page can be moved into the new document or deleted based on the attribute found.
- PageIDs Used As Separators – Collator is configured with a set of PageIDs that are separator sheets. When Collator finds a page with a separator PageID, a new document is started. All successive pages that are not separators are moved into the document.

Note: PageIDs used as Separators are configurable as a Batch-Level option only.

Important: The FormWorks 5.0 systems currently cannot processes uncollated pages; uncollated pages are ignored by the system. Any Uncollated pages that occur before the first separator sheet in the work record will remain uncollated.

To Configure the Collator Process

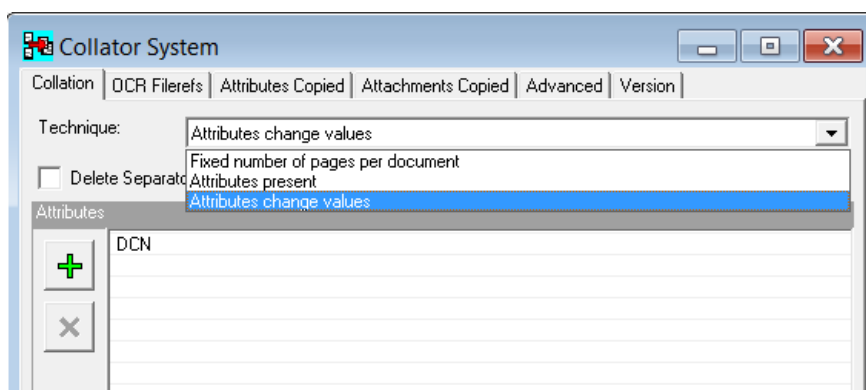
- Double-click Collator in the processes list to open the configuration window.



- The Collator System window is presented.

Collator System – Collation Tab

- Select the System Collation method from the Technique drop down box (see [Collation Techniques](#) for details).

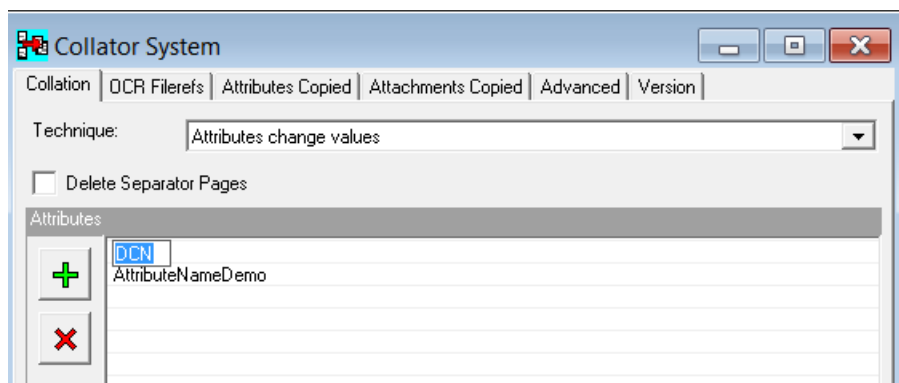


Note: The majority of FormWorks systems process utilizing the “Attributes change values” technique based on change in the DCN.

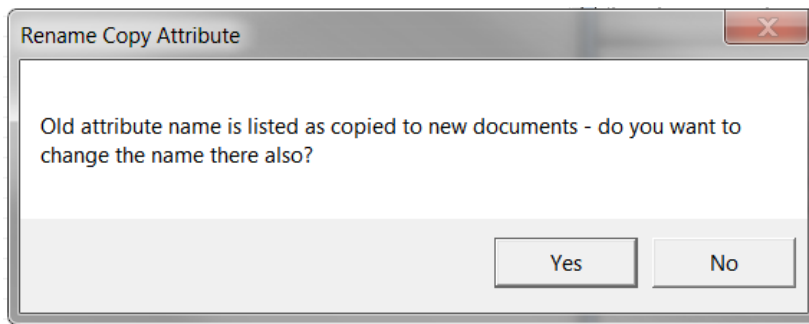
- Delete Separator Pages – Select the check box to direct the system to delete all separator pages.
- For attribute-based techniques, use the Add New Attribute button  to add the appropriate attribute(s) to the list.
- Select the Delete Attributes  button to delete an attribute from the list.

Note: Attribute names may only contain letters and numbers without any spaces or punctuation. Attributes may not start with numbers.

If required, to rename an Attribute, place the mouse over the **Attribute** and select, **pause** and select again. The Attribute name window opens, modify the name as required.

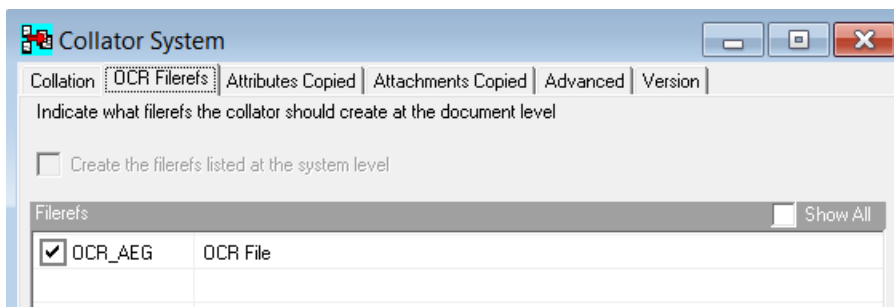


The system may display to the user the Rename Copy Attribute Window. Select **Yes** or **No** appropriately.



Collator System - OCR Filerefs Tab

- Click the OCR Filerefs tab. If most or all of the batch types in the system will have the same OCR filerefs, as a best practice it is recommended to specify the fileref within the OCR Filerefs tab (instead of at the batch level). In using this methodology, should changes be required at a future time, the change will only need to be made once.
- Select the name of the OCR file to be created (usually OCR_AEG).



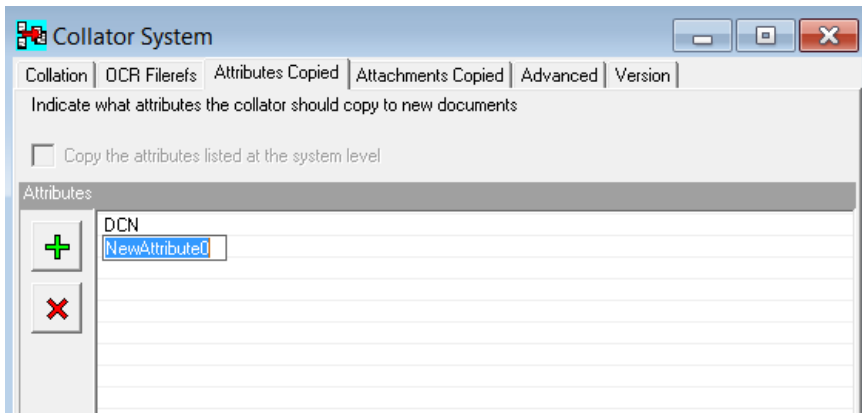
- When specifying the OCR fileref at the system level, it is recommended to use the same fileref for all batches.
- At the batch level, check the Create the filerefs listed at the system level checkbox. [See Collator: Batch Processing](#) for more information about batch level settings.

Collator System – Attributes Copied Tab

The collator makes a multi-level copy of attachment page items by making copies of all the filerefs with new file referents). When copying attributes into the new fileref the collator retains the original page item's attribute values with two exceptions.

- All attributes (configured to copy) that are copied into new document items are overwritten with the values from the new document item
- The collator will add a Collator_Attachment_Copy signed long attribute to the new copy of the page item (with value of one) to indicate the origin of the new page item.

For example, during the collation pass the collator is configured to copy the DCN attribute from separator pages to the document item of new documents. When the collator makes copies of attachment pages then the DCN attribute from the new attachment page's destination document is also copied to the new page item.

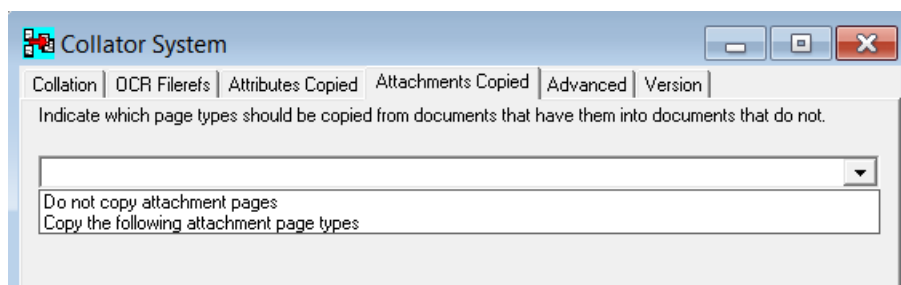


- Use the Add New Attribute button  to add additional attributes into the Attributes window.
- Use the Delete Attribute button  to delete an attribute from the Attributes window.

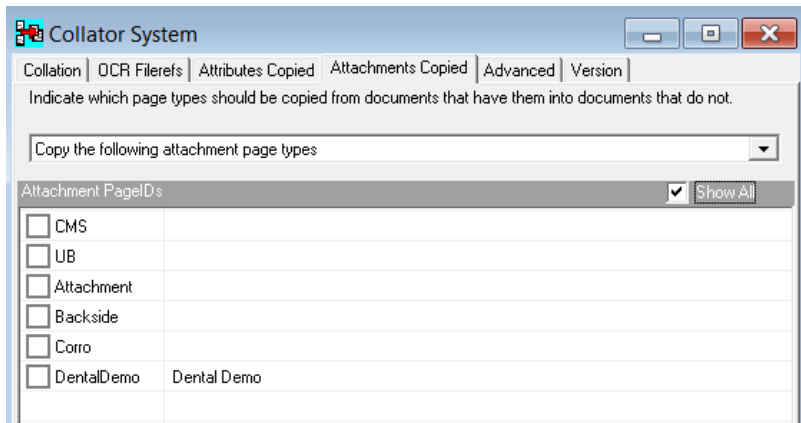
Collator System – Attachments Copied Tab

The collator loops through the documents and copies attachment pages from documents (which have attachment pages) into documents that do not. The Collator then checks each document, if the document does not contain any attachment pages, the attachment pages from the next document (with attachments) are copied into that document.

So, if an attachment page was copied from document one to document three, then the new copy off the attachment page in document three would have the same DCN has listed in document three's document item.

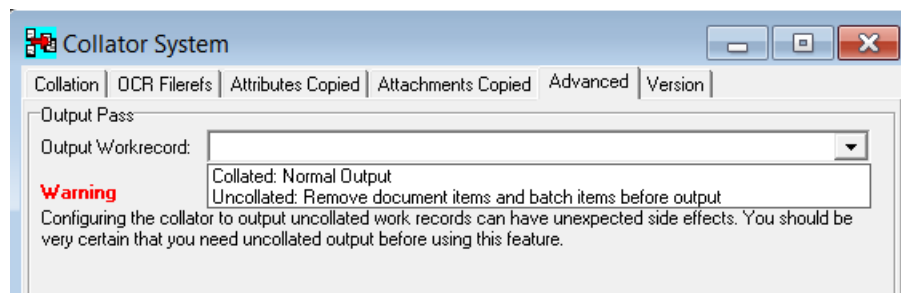


- Select (as applicable) from the drop-down box.
 - Do not copy attachment pages
 - Copy the following attachment page types – when selected the Attachment PageID window displays. Select the Show All checkbox to display a list of Attachment PageIDs from which to select.



Collator System - Advanced Tab

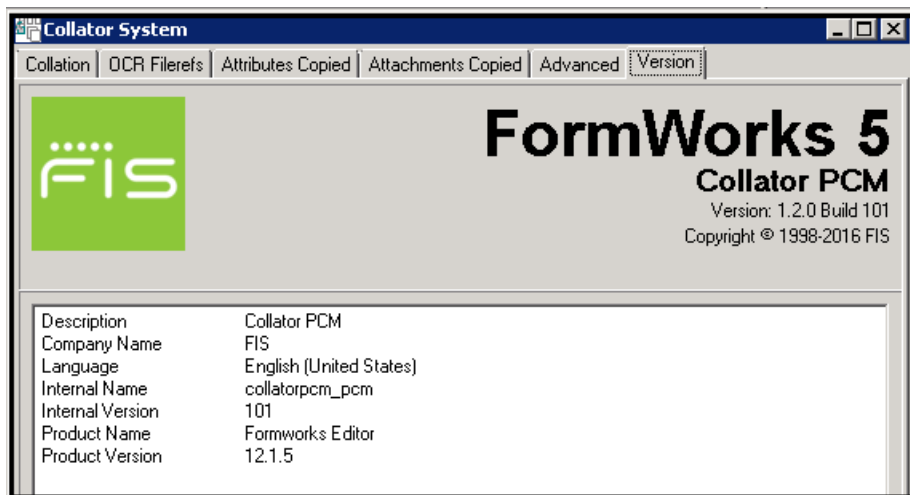
From the Advanced Tab, configure the output work record for normal output or uncollated output.



Note: The Document ID pass is as a best practice, generally skipped if the Output Pass is configured for Uncollated output.

Collator System - Version Tab

Within the window, displays to the user the current version and configuration information.

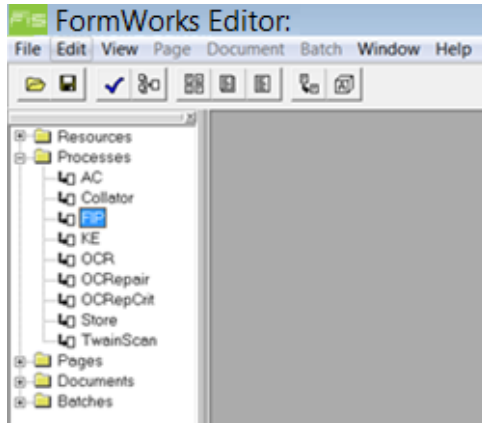


FIP Process

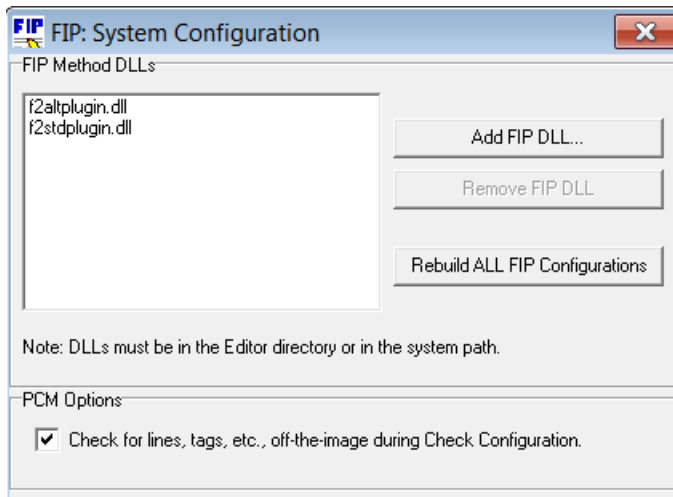
FIP (Forms Image Processing) is the FormWorks component that identifies forms and assigns PageIDs. FIP also performs mark recognition (reading of checkbox fields), removal (form instructions, lines), creates the OCR Files, and deskews and cleans up scanned images.

As a best practice, the default DLLs, f2stdplugin.dll and f2altplugin.dll are recommended for most systems.

- Open the FIP: System Configuration window by double-clicking the FIP in the Processes list.



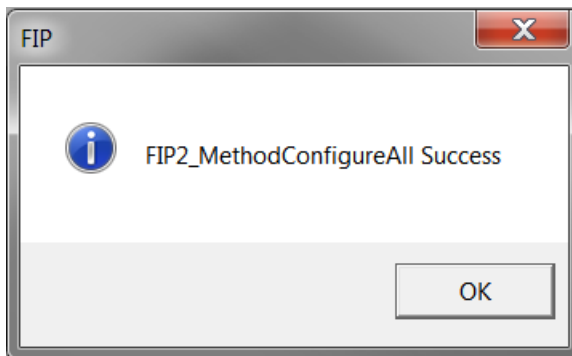
- The FIP: System Configuration window is presented.



- Add FIP DLL – Select to display a drive mapping window from which FIP DLLs may be added.
- Remove FIP DLL – Select the DLL to be removed upon which the Remove FIP DLL button will become active. Next, select Remove FIP DLL.

Note: DLLs must be in the FWEditor director or in the system path.

- Rebuild ALL FIP Configurations – Select to rebuild all FIP Configurations listed in the FIP Method DLL window. As a best practice, select Rebuild ALL FIP Configurations after the addition or deletion of one or more DLLs. After selection of Rebuild All FIP Configurations, if the rebuild is successful the system presents to the user a ConfigureAll Success message.



Note: If the ...ConfigureAll Success message is not displayed, contact a FIS FormWorks Support.

- PCM Options – Selecting the check-box instructs the system to inspect for any lines, tags etc. that may be present away from the defined image area during the Check Configuration process.

Registration Worker

The ability to process higher resolution dropout images is available in FormWorks. Depending on the scanner or image source, higher resolution images may increase text readability and in some cases, can result in increased accuracy of the OCR process for forms with small text. This reduces keying effort and time. In addition, the Imaging worker improves dropout registration, and is also capable of detecting and fixing images that are stretched or expanded during the scanning process. File sizes are larger due to increased image resolution, so more disk space may be needed during processing and for archival purposes. Copy time may also increase. Contact your FIS representative to reconfigure and add this worker to your FormWorks system.

*Note: You need to manually select the **Identity** option in the **Methods** field to be able to process the 300 dpi images. This option is available in the FIP page setting for the Variant as illustrated in the screenshot below.*

FIP: Variant Standard of Page CMS

PageID
 Confidence: 45
 Orientation: ☒ Upright ☐ Upside Down
☒ Process as a Dropout

Registration
 Methods: ☒ IDENTITY ☐ MANUAL ☐ AUTO ☐ ALTAUTO ☐ SCALEAUTO

Removal
 Confidence: 80
 Methods: ☒ STANDARD

Deskew
 Methods: ☒ STANDARD ☐ EXTRACTSTANDARD

Filter
 Methods: ☒ STANDARD

QMR
 Methods: ☒ CHECKSTD ☒ SIGSTD ☐ CHECKALT

Blank Detection
☐ Detect Whether this Variant is Blank
 Full Page Blank Conf:
 Key From Screen Conf:

Inventory Process (Optional)

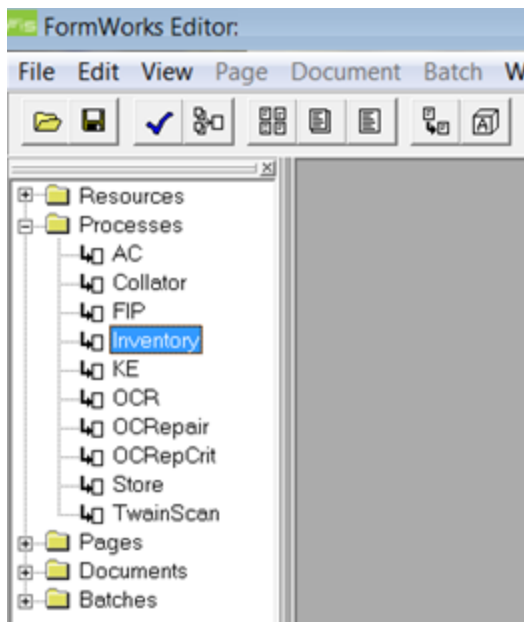
Note: Inventory Process (Optional) is generally utilized on FormWorks legacy systems.

Inventory is an optional FormWorks subsystem which is meant to provide an independent system to track what work should be in the system, in effect providing a double-check of processing. If a batch should be lost due to a system error, the inventory reports provide information as to what batch was lost.

Additionally, Inventory Worker enters batches into a FormWorks system, and is used to manually retire batches when necessary. Inventory is composed of an Inventory Server, an Inventory Client, and a multirole Inventory BWS Worker that usually runs in *Inventory In* and *Inventory Out* capacities. TWAINScan can perform Inventory In duties, and often the Inventory Out function is handled by Store. Additionally the MWM is also involved when a batch is deleted from Navigator

IMPORTANT An Inventory Resource must be configured before configuring the Inventory Worker (See [Creating/Configuring an Inventory Resource](#)).

- Open the Inventory Worker configuration window by double-clicking the Inventory in the Processes list.



- The Inventory Worker window is displayed to the user.

Inventory Worker – General Tab

- Setup to login to the SQL Server database if using SQL server authentication.

- User ID – SQL server user id.
- Password – Password for the SQL server user id.
- Server – SQL server name.
- Worker Type
 - Input Worker – Workers capable of performing batch create and enter operations. Also, TWAINScan and the Inventory Worker are considered Input Workers when operating in CreateAndEnter mode (sometimes referred to as InvIn (Inventory In)).

Note: When Worker Type equals Input Worker, only the BatchNum, Page count and Allow Mismatch attributes are definable.

- Output Worker – Workers that can perform batch retirement. Also included as an Output Worker are Store and Inventory workers when running as Retire or InvOut (Inventory Out).

Note: When Worker Type equals Output Worker, only the BatchNum, Batch Status and Batch Note attributes are definable.

- Optional
 - BatchNum Attribute – The name of the work record attribute containing the inventory batch number (usually BatchNum). This attribute MUST be present in the work record.

- Page Count Attribute – A numeric attribute designed to inform the worker the maximum number of documents for which the batch is to be created. This number must be at least equal to the number of documents that are actually present in the work record.

Note: The "Page" part of the name is historical, dating back to when Formworks handled single-page forms only.

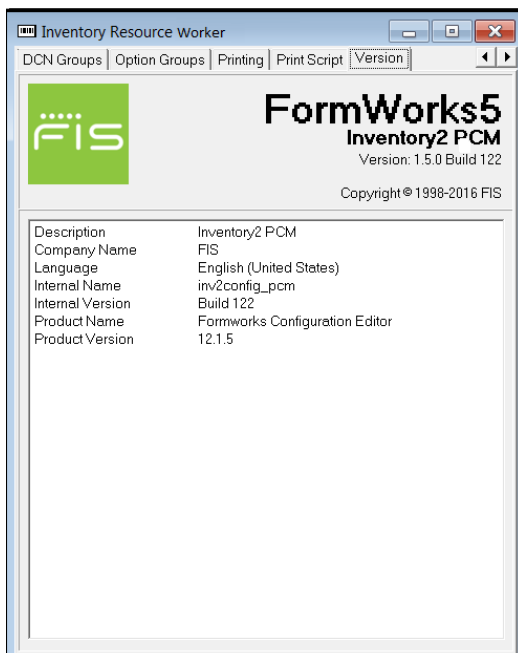
- Allow Mismatch Attribute – Designed to inform the Worker whether a mismatch between the expected page count (i.e. Page count) and the actual page count (i.e. EnteredPageCount) is to be allowed.

Note: Used primarily in cases where Batch-Create and Batch-Enter are performed by separate workers. As a best practice, systems are not configured in this manner.

- Batch Status Attribute – The name of the attribute used as the batch retire note (usually StoreInvRetireString). If no attribute is listed, the note defaults to an empty string.
- Batch Note Attribute – Batch Note Attribute – The name of the attribute used as the batch retire note (usually StoreInvRetireString). If no attribute is listed, the note defaults to an empty string.

Inventory Worker – Version Tab

Within the window, displays to the user the current version and configuration information.



OCR Process

OCR is the stage at which data on forms is read by the OCR engine (usually AEG). The OCR worker writes the data to an output file and then the file is modified by other workers as the data is changed at subsequent processing stages.

Note: Detailed OCR configuration options are accessible at the Page and Field variant levels via the [OCR Variant Field Processing](#) dialog box.

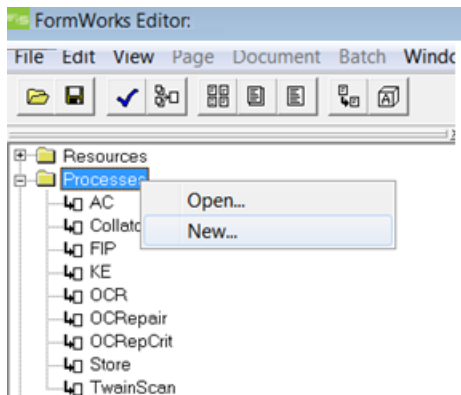
FormWorks utilizes Optical Character Recognition (OCR) which integrates with the Open Text 6.0 engine. This integration provides the following benefits.

- Reads and processes data much faster with improved accuracy and character confidence over previous OCR engines.
- The ability to support Western European languages which are supported by Windows 1252-character set. This includes Spanish, French, and many other common languages.

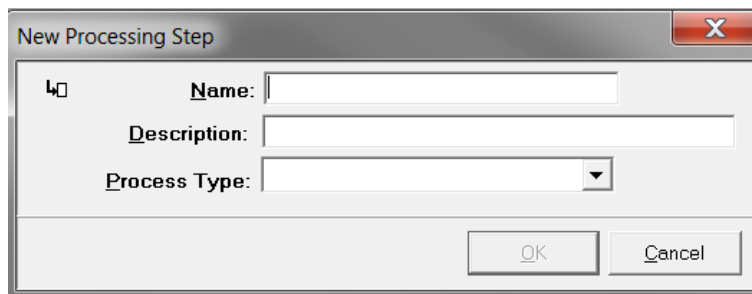
OCR: Create

Creating an OCR, OCRepair or OCRepCrit (or additional OCR Processes) begins as follows.

- Right-click Processes, Select New.



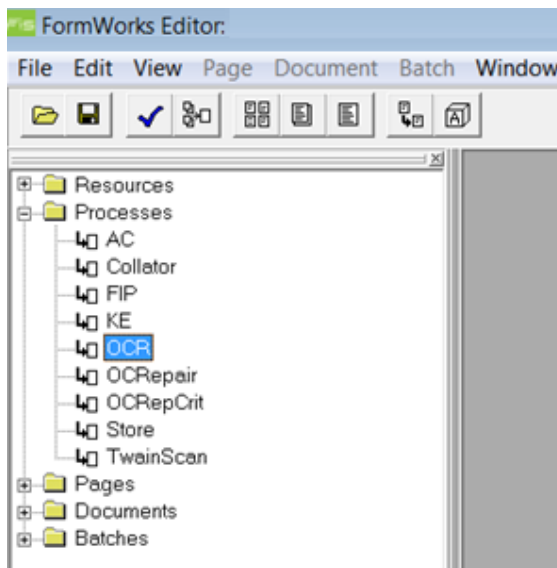
- The New Processing Step window displays.



- Name – Enter the Name of the process.
- Description – Enter the Description of the process.
- Process Type – Applicable to an OCR Process, are the selections AEG or OCRepair3.
- Select OK, the newly created Process displays in the Processes tab of the FWEdit Tree.

OCR System Configuration

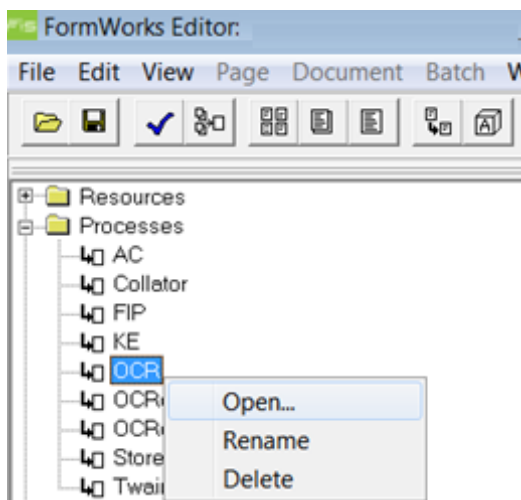
- Open the OCR process configuration window by double-clicking OCR in the Processes list.



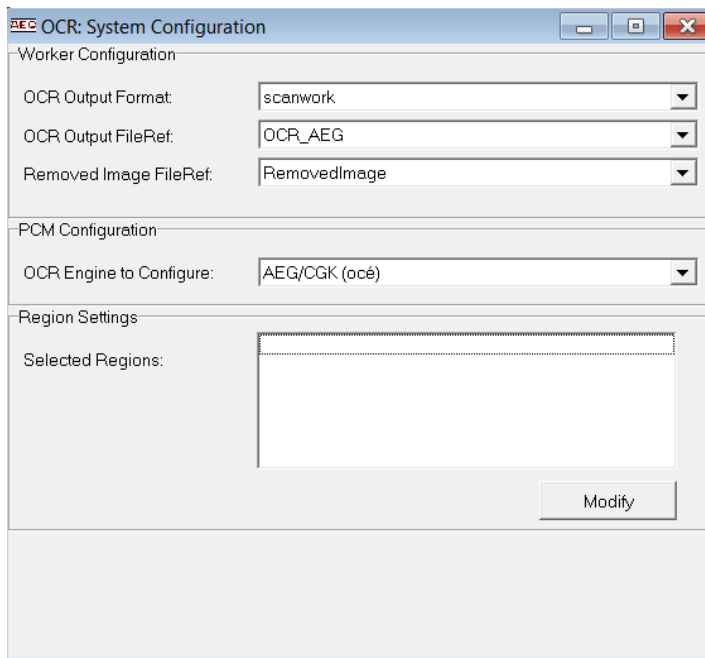
- Upon selection of OCR, the OCR: System Configuration window is presented.

Note: From the Worker, PCM Configuration, and Region Setting sections of the OCR System Configuration window, select the appropriate configuration for each OCR Process.

- Double-click (or Right-click > Open) OCR.



- The OCR System Configuration window displays.



- Configure the Worker, PCM and Regions Settings sections as follows.
- OCR Output Format – Select one of the available formats.

Define the Output Format and FileRef in the Processes OCR System Configuration window. File references must be defined in the Resources Filerefs list before being selected.

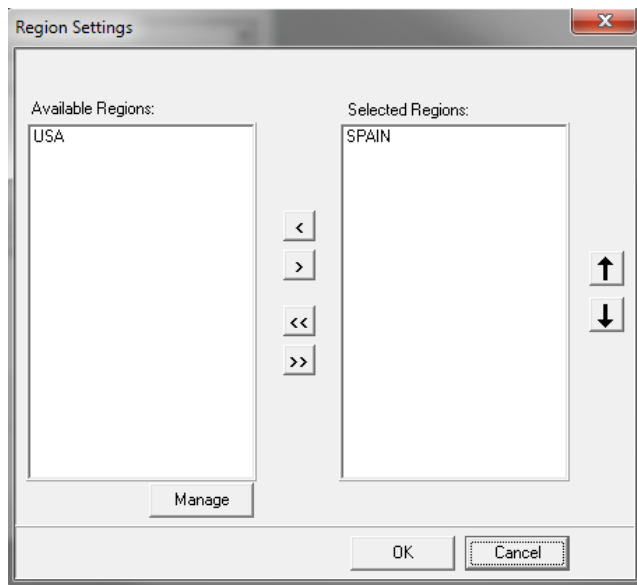
Note: As a best practice, the rri5 format should be used.

- | | | |
|---------|--------|---------|
| • rri3 | • rri5 | • rriez |
| • rri3u | • pda | • |
| • rri4 | • aeg | • |
- OCR Output FileRef
 - OCR_AEG (most frequently utilized) – This is the FileRef to be used.
 - OCR_AEG_Copy – This is for Store only; do not use for OCR.
 - Removed Image Fileref

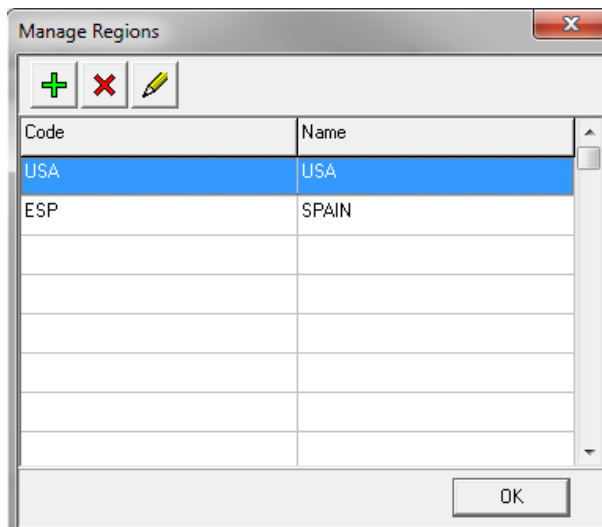
- RemovedImage (most often used) – This is the image that has been prepared for OCR and should be used here.
- DeskewedImage – This image is typically not for OCR.
- OriginalImage – This image is typically not for OCR.
- OCR Engine to Configure
 - AEG/CGK (oce') (most often selected)
 - FieldScript (ParaScript) – Not Applicable

Important: The FieldScript option is for testing purposes only and should not be selected. Future versions of FWEdit will not reflect the FieldScript option. Users are instructed to always utilize the AEG/CGK (oce') option.

- Selected Regions – Selected Regions displays a list of regions that will be used by Page and Variant field OCR configuration settings. Refer to [OCR: Variant Field Processing and OCR: Page Field Processing](#) for more information.
- Modify – Select the Modify button to open the Region Settings window of the OCR System Configuration from which configurations to individual regions may be made. Refer to [Selecting Regions](#) for more information.
 - Regions Settings Window – Displayed are the Available Regions and Selected Regions.

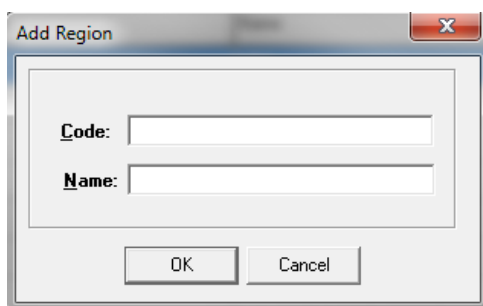


- If the desired region is not displayed, select Manage.
- The Manage Regions window is displayed.



From Manage Regions, the user may:

-  – Add Region, selecting displays the Add Region window.


 A screenshot of a Windows-style dialog box titled "Add Region". It contains two text input fields: "Code:" and "Name:". Below the fields are two buttons: "OK" and "Cancel". The dialog box has a standard Windows title bar with a close button (X) in the top right corner.

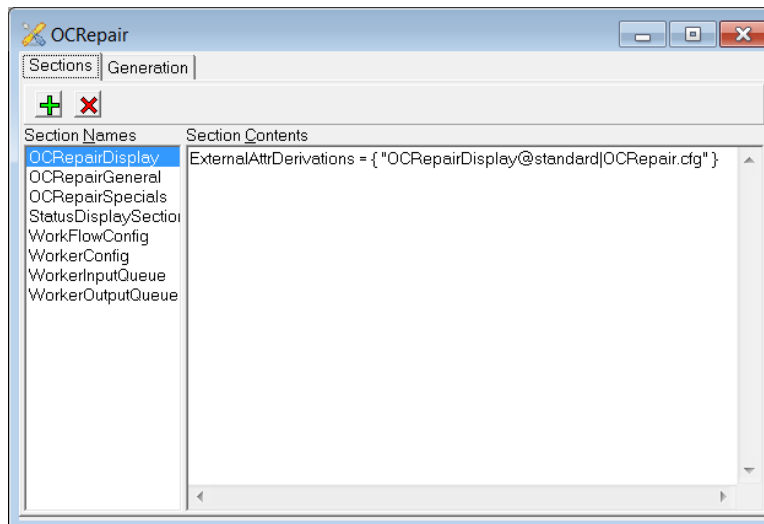
- Code – User defined, unique (3 letter) word to define the region. For example, Code USA for USA or ESP for Spain.
 - Name – Name of the region.
 - Select OK to save.
 -  – Delete Region – To delete a Region, select the Region and select.
 -  – Edit Region – Displays the Add Region window from which only the Name may be edited.
- Note: Once named, the Code field fixed and may not be edited.*
- Close the OCR Configuration Window by Selecting the Close Window icon .

Warning: Changes to  the OCR Configuration Window are saved automatically when the Close Window icon is selected

OCRepair/OCRepCrit – Configuration File

An OCRepair Process must be created and an OCRepair System Level Configuration built for each OCRepair stage, (OCRepair, OCRepCrit, OCRepNum etc.) in the system.

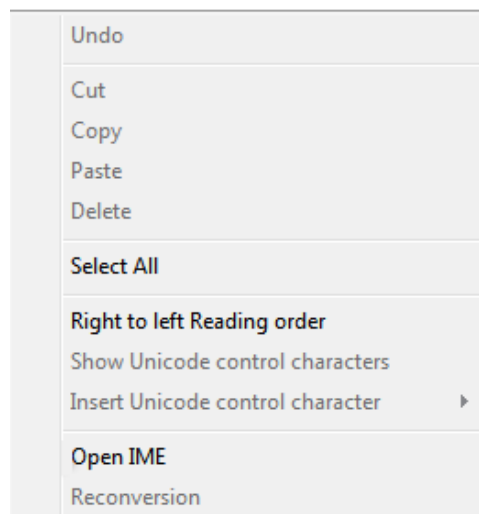
Note: The number OCRepair Configurations and Type within the Process module are determined by individual customer requirements. See [OCR Create](#) regarding the creation of OCR Process.



- Select a Section Name (Sections tab, left column) to view and edit the Section contents. Once the Page level configurations have been completed, return to the system level to generate a [Configuration File](#) (Generation tab).

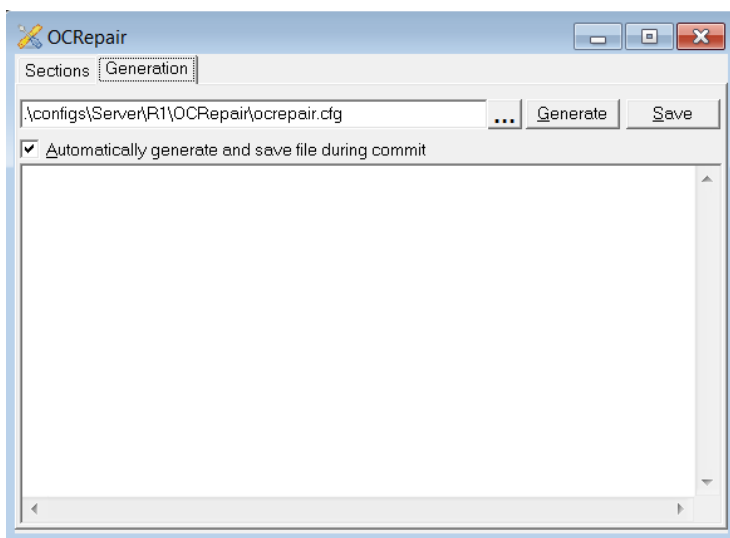
Note: If the user right-clicks within the Section Contents window, displayed to the user is a window from which the following may be selected.

- Select All – Selects all within the Section Contents window.
- Right to left Reading order – Modifies the reading order of the content
- Open IME – Specifically for Windows 7 input. Not a FWEdit function. Do Not Use.



Generate An OCRepair/OCRepCrit Configuration File

- From the OCRepair, OCRepCrit configuration window, select the Generation tab.



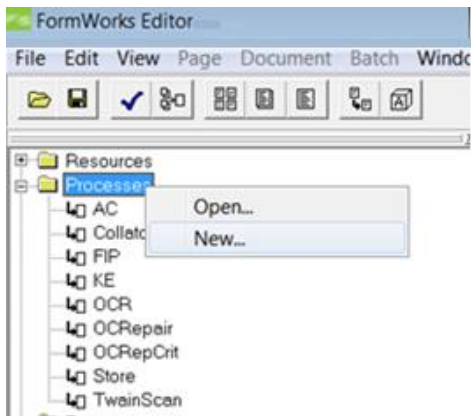
- Automatically generate and save file during commit – When checked the config file is generated and saved every time the CFD is built.
- Select Generate to view the configuration file resulting from the current system and page configuration.
- If path and file name are not displayed at the top of the window, select the Browse button  to select the location and name.
- Click Save.

Store Process

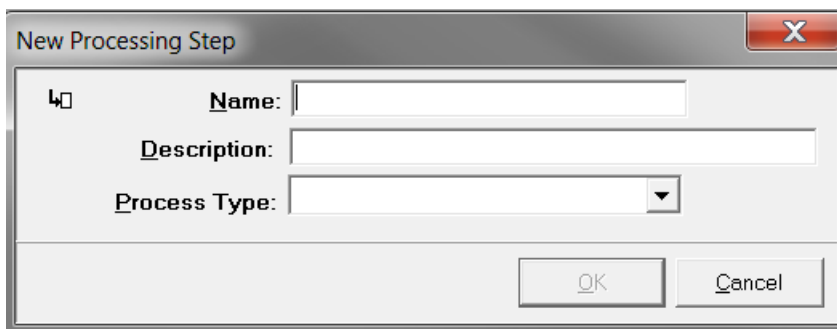
Store compiles all data into output in your specified format as well as handling image output and retires batches from Inventory.

Create Store Process

- Right-click Processes, Select New.



- The New Processing Step window displays.



- Name – Enter the Name of the process (typically Store).
- Description – Enter the Description of the process.
- Process Type – Select Store from the Process Type drop down box.
- Select OK, the Store Configuration window displays.

Store Configuration – General Tab

Select the rule DLLs being used (usually acsrules.dll, acstore.dll, actcl.dll).

Important: Do not select acengine.dll .

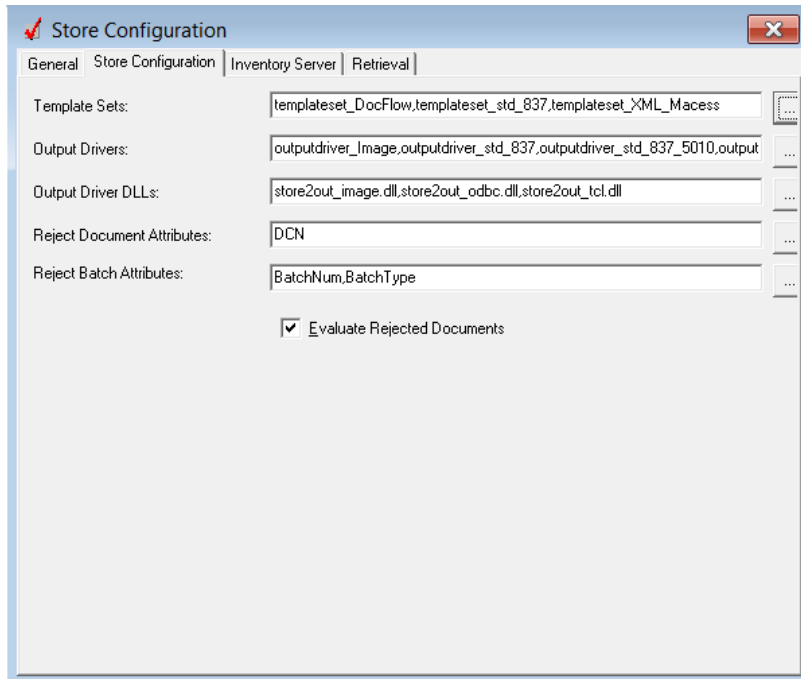
- Description – (Optional) A brief description of the Store process.
- Rule DLLs – From the drop-down box, select the Rule DLLs available (usually acsrules.dll, acstore.dll, actcl.dll). Rule DLLs are built and defined in Resources > RuleDLL within the FWTree.

Important: Do not select acengine.dll .

- Default Sources – The OCR fileref to use by default. Filerefs are built and defined in Resources > FileRef within the FWTree.

ODBC Connection

- Disconnect on Idle – If checked, the system will disconnect from the ODBC connection when not in use.
 - Reconnect Wait Time (seconds) – Number of seconds until the system waits before trying to automatically reconnect (maximum allowable entry 10 digits).
 - Reconnect Retry Time (minutes) – Number of minutes until the system waits before trying to automatically reconnect (maximum allowable entry 10 digits).

Store Configuration – Store Configuration Tab

- Template Sets – Select the **Template Sets** to be used by the Store worker.
- Output Drivers – Select **Output Drivers** used by Store.
- Output Driver DLLs – Typically store2out_tcl.dll, store2out_odbc.dll, and store2out_image.dll.
- Reject Document Attributes – Usually DCN.
- Reject Batch Attributes – Usually BatchNum, BatchType.
- Evaluate Rejected Documents – Controls whether rejected documents will be processed by Store at all. Typically this would be checked and Store rules would determine what output to generate for rejected documents.

Note: As a best practice, it is recommended to select all of the available Template Sets and Output Drivers unless some are known to not be needed.

Store Configuration – Inventory Server Tab

Un-check the Use Inventory checkbox if the Store process does not need to interface with a Formworks Inventory Server.

Store Configuration

General | **Store Configuration** | Inventory Server | Retrieval

☒ Use Inventory

Server Name: Inventory

User Name: Store2

Password:

Batch Status Attribute: StoreInvRetireStatus

Batch Note Attribute: StoreInvRetireString

Batch Number Attribute: BatchNum

Sub-Batch Number Attribute: SubBatchNum

☒ Allow Retired Batches to be re-Stored.

Retire Attributes:

- ParentBatchNum
- ClaimType
- DuplexScan
- DocFlow

- Server Name, User Name, Password – Select the Inventory resource and enter the user name and password to be used by Store to access it.
- Batch Status Attribute – The name of the attribute used as the batch retire status (usually StoreInvRetireStatus). If no attribute is listed, the status defaults to a value of zero.
- Batch Note Attribute – The name of the attribute used as the batch retire note (usually StoreInvRetireString). If no attribute is listed, the note defaults to an empty string.
- Batch Number Attribute – The name of the work record attribute containing the inventory batch number (usually BatchNum). This attribute MUST be present in the work record.
- Sub Batch Number Attribute – The name of the work record attribute containing the inventory sub-batch number.
- Retire Attributes – List any attributes whose values that are to be recorded in the Inventory Server during batch retire.

Store Configuration - Retrieval Tab

Important: Select Use Retrieval ONLY if the system is using FIS's Formworks retrieval system.

Note: As a best practice, contact your FIS technical representative prior to execution.

The screenshot shows the 'Store Configuration' dialog box with the 'Retrieval' tab selected. The 'Use Retrieval' checkbox is checked. The 'DSN' field is empty, and the 'Schema' field is empty. The 'DBMS Type' is set to 'Microsoft SQL Server'. The 'Image Server' field is empty. The 'User Name' is 'Store2', and the 'Password' field is empty. The 'Image FileRef' field is empty, and the 'OCR FileRef' field is empty. The 'Base Path' field is empty. The 'Data Order' is set to 1: Template, 2: Driver, and 3: OCR Files. Below these fields is a table with columns: Caption, Field, Attribute, and Batch Types. The table is empty. Below the table are buttons for 'Add', 'Edit', 'Delete', and two arrow buttons (up and down). Below these buttons is a table with columns: Caption Order and Query. The table is empty. To the right of the table are fields for 'DCN Attribute' and 'Column Width' (set to 255). At the bottom are buttons for 'Rename' and 'Query'.

Store Configuration

General | Store Configuration | Inventory Server | **Retrieval**

☒ Use Retrieval

DSN: Schema:

DBMS Type: Image Server:

User Name: Password:

Image FileRef: OCR FileRef:

Base Path:

Data Order: 1: 2: 3:

Caption	Field	Attribute	Batch Types
---------	-------	-----------	-------------

Add Edit Delete ↑ ↓

Caption Order	Query
---------------	-------

DCN Attribute:

Column Width:

↑ ↓ Rename ☐ Query

Store Configuration

General | Store Configuration | Inventory Server | **Retrieval**

☒ Use Retrieval

DSN: Schema:
DBMS Type: Microsoft SQL Server Image Server:
User Name: Store2 Password:
Image FileRef: QCR FileRef:
Base Path:
Data Oder: 1: Template 2: Driver 3: OCR Files

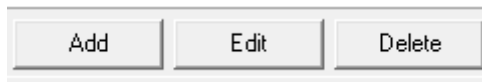
Caption	Field	Attribute	Batch Types

Caption Order	Query

DCN Atttribute:
Column Width:

- Use Retrieval – Activates the Use Retrieval configuration command(s).
- DSN – Doman Service Name defined.
- DBMS Type – Database Management System type defined.
- User Name – Enter the user name Retrieval will use to access the database.
- Image Fileref – RemovedImage
- Base Path – See the system engineer or contact your FIS technical representative.
- Schema – See the system engineer or contact your FIS technical representative.
- Image Server – See the system engineer or contact your FIS technical representative.

- Password – Enter the password Retrieval will use to access the database.
- OCR Fileref – OCR_AEG
- Data Order –
 - 1 – Selected item appears 1st.
 - 2 – Selected item appears 2nd.
 - 3 – Selected item appears 3rd.
- Caption – Populated from the previous caption entries.
- Field – The field that appears in the client main window.
- Attribute – Attributes that may be entered at an Inventory Client Worker by an operator.
- Batch Types –



- Add – Add Retrieval 2 Field/Attribute.
- Edit – Edit Retrieval 2 Field/Attribute.
- Delete – Delete Retrieval 2 Field/Attribute.
-   – After making a Field selection, moves the cursor up or down.
- Caption Order – Displays caption order.
- Query – Displays query.
-   – After making a Field selection, moves the cursor up or down.
- – Rename – Rename Caption (updates Caption field above Caption Order).
- Query – Check to make query active
- DCN Attribute – A unique number assigned to each document in a system, usually imprinted on the form as it's scanned (Document Control Number).
- Column Width – Enter the number of characters (spaces) the column is to be allowed.

TwainScan Process

The FormWorks TWAINScan application allows the operator to scan batches into the system using the specifications set in Inventory Client. The TWAINScan application traditionally picks up the barcode from the

batch cover sheet, the retrieves the batch information from the inventory server. Based upon the batch type, the TWAINScan application will initialize the scanner with the appropriate settings for the batch.

TWAINScan provides both a system level configuration and an individual batch level configuration.

At the system and process level, TWAINScan communicates to the scanning application where to locate the Inventory Server used to retrieve batch information, enter batches, and the fileref for scanned images. All other configuration for TWAINScan can be done either at the process- or batch-level. Any setting at the process level will be used for all batches unless individual settings are modified for a batch.

Note: The batch-level settings do not show modifications made at the process level, though they are indeed inherited when they are set to use process default (grayed out). Also, Fileref is configured within the Resources tab.

Custom batch level settings are configured through the TwainScan Batch Configuration window.

- TWAINScan is configured at the system level within the [TwainScan Process Configuration](#) window.
- TWAINScan is configured at the batch level within the [TWAINScan Batch Configuration](#) window.

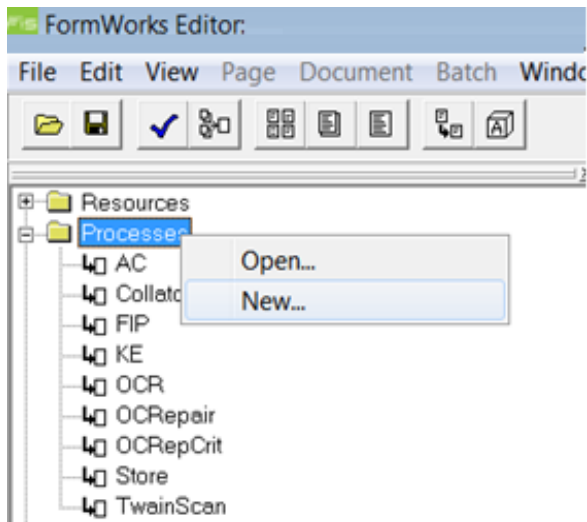
Important: Users must set up FileRef and Inventory resources before configuring TwainScan.

- Fileref Name – Specify the fileref that scanned images will be saved to (usually OriginalImage)
- Device Type or Device Alias – Select Device Type to specify the scanner by type (scanner or disk) in the drop down list. Select Device Alias to specify the scanner by name. Type in the source name in the box to the right.
- Inventory Server – Select the name of the Inventory resource.
- User Name and Password – The name and password Scan used to log into the Inventory server. If the Inventory server uses an authentication database, such as auth.db, the password must be entered.

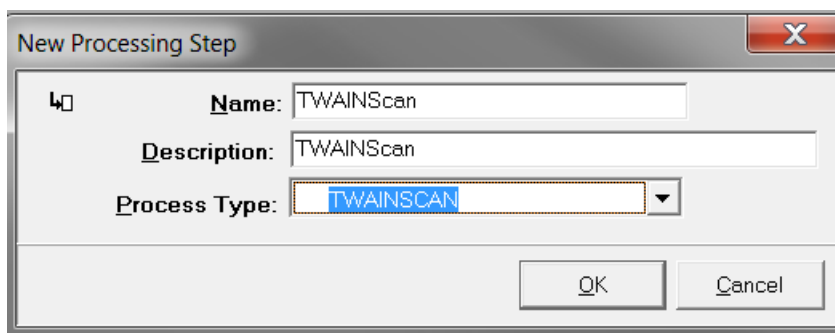
TWAINScan Process Configuration

Create TWAINScan Process

- Right-click Processes, Select New.



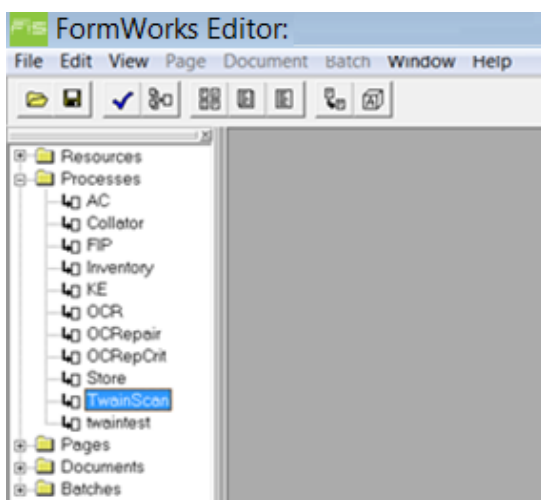
- The New Processing Step window displays.



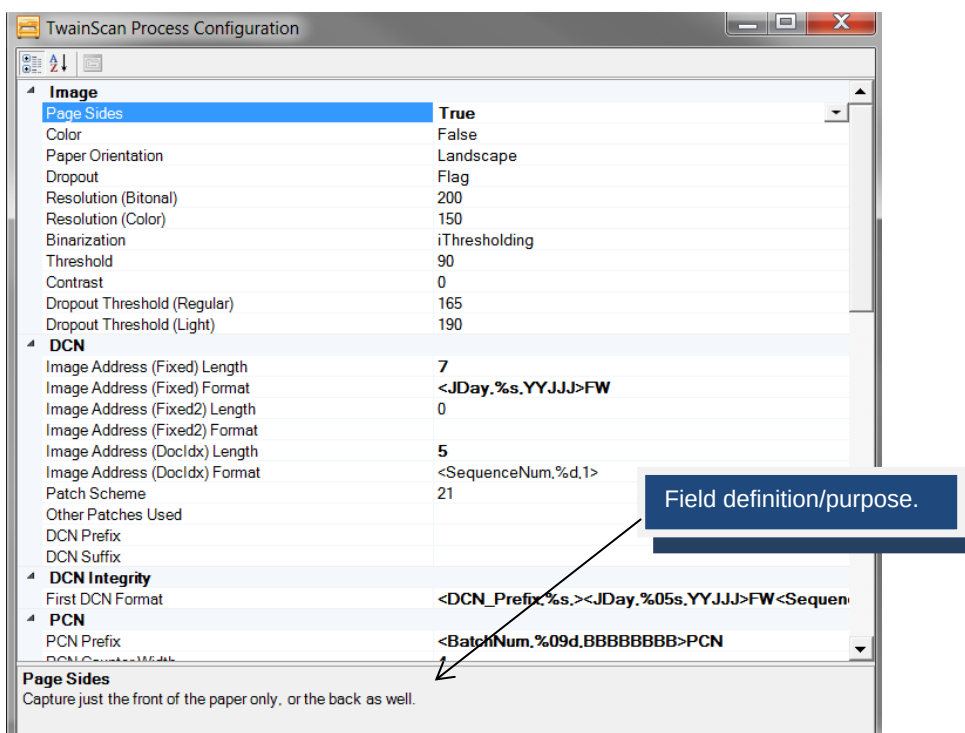
- Name – Enter the **Name** of the process.
- Description – Enter the **Description** of the process.
- Process Type – Select **TWAINSCAN** from the Process Type drop down box.
- Select **OK**, TWAINScan within the FWEdit Tree.

TWAINScan – Process Configuration

- From Processes, double-click on TwainScan,



- The TWAINScan Process Configuration window displays.



Note: For each field, when selected, the window displays (at the bottom) a brief definition of the function or purpose of the field.

- From within the TWAINScan Process Configuration window, displayed to the user for configuration are multiple categories each listing individual items for configuration.

- Image
 - DCN
 - DCN Integrity
 - PCN
 - Page ID
 - Batch Integrity
 - Printer (Ink)
 - Printer (Electronic)
 - Scanner
 - Workflow
 - Extension
- Fields within the TwainScan Process Configuration window may be sorted using the following functions

- 

Categorized – Sort by category.
 - 

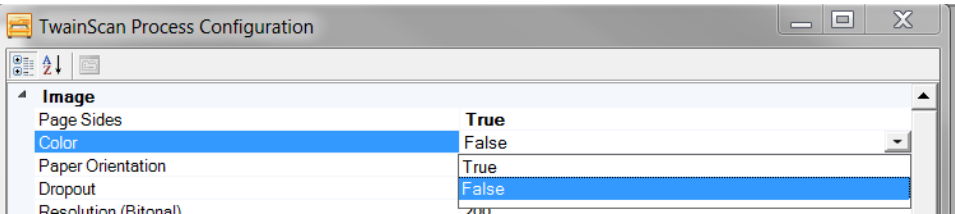
The user configures setting by the item.

The system to the user right drop down which a may be and
 - 

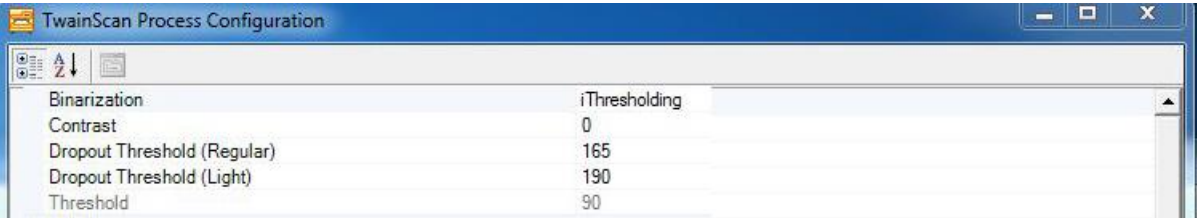
Property Pages – Right-click to display:

 - Reset – Resets the selected item back to its default setting.
 - Reset All – Resets all items to their default settings.

Warning: Once selected, Reset or Reset All may not be reversed unless the item(s) are manually reentered.
- each selecting individual presents in the column a box from value selected entered.



Or, the system allows the user to place the cursor in the right column and input the appropriate value.



TWAINScan Process Configuration

Option Group	Option	Option Values (if enumerated)	Description	Attribute Default	Attribute Values	Notes
Image	Page Sides	Front, Front and Back	Capture just the front of the paper only, or the back as well.	"False"	"False" (Front), "True" (Front and Back)	
	Color	Bitonal, Bitonal and Color	Capture bitonal images, or color as well. Color is generally bigger and slower.	"False"	"False" (Bitonal), "True" (Bitonal and Color)	
	Paper Orientation	Landscape (Top to the Right), Portrait, Landscape (Top to the Left)	Landscape is the fastest way to get paper through the scanner. Top to the right is most compatible with printers.	"Landscape"	"Landscape" (Top to the Right), "Portrait" (Portrait), "LandscapeTopToLeft" (Top to the Left)	
	Dropout	Flag, Dropout, Non-dropout	Drop out (red) from image electronically.	"Flag"	"Flag" (Flag), "True" (Dropout), "False" (Non-dropout)	
	Resolution (Bitonal)	200, 300	Resolution of black and white images.	200	200, 300	Int32
	Resolution (Color)	100, 150, 200	Resolution of color images	150	100, 150, 200	Int32

Option Group	Option	Option Values (if enumerated)	Description	Attribute Default	Attribute Values	Notes
	Binarization	iThresholding, Adaptive Thresholding	iThresholding automatically determines the best threshold to separate text from background. Adaptive Thresholding allows manually setting the threshold.	"iThresholding"	"iThresholding" (for iThresholding), "Adaptive" (for Adaptive Thresholding)	String
	Threshold		If Adaptive Thresholding is selected, this determines the pixel gray color that differentiates text from background.	90	0 to 255	Int32
	Contrast		Contrast is a smoothing factor.	0	(-1000) to 1000	Int32
	Dropout Threshold (Regular)		Dropout filter threshold (regular) determines the amount of red that must be present in order to be removed.	165	0 to 255	
	Dropout Threshold (Light)		Dropout filter threshold (light) determines the amount of red that must be present in order to be removed.	190	0 to 255	
DCN	Image Address (Fixed) Length		Length of the fixed part of the image address.	5	1 to 9	

Option Group	Option	Option Values (if enumerated)	Description	Attribute Default	Attribute Values	Notes
	Image Address (Fixed) Format		Format string (ATTR_SPrintf) of the fixed part of the image address.	"<JDay,%s,YYJJ>"	string, non-blank	
	Image Address (Fixed2) Length		Length of the fixed2 part of the image address	0	0 to 9	
	Image Address (Fixed2) Format		Format string (ATTR_SPrintf) of the fixed2 part of the image address.	""	string, can be blank	
	Image Address (DocIdx) Length		Length of the document index part of the image address.	7	1 to 9	
	Image Address (DocIdx) Format		Format string (ATTR_SPrintf) of the document index part of the image address.	"<SequenceNum,%d,1>"	string, non-blank	
	Patch Scheme	Patches 2 and 1, Patches 3 and 2	Patch 2 starts a doc and Patch 1 starts attachments (recommended), or Patch 3 starts a doc and Patch 2 starts attachments.	"21"	"21" (Patches 2 and 1), "32" (Patches 3 and 2)	
	Other Patches Used		Other patches that should be recognized (used for super batching)	""	string with any values: 12345T	

Option Group	Option	Option Values (if enumerated)	Description	Attribute Default	Attribute Values	Notes
	DCN Prefix		DCN Prefix is a static prefix that is printed and added to the beginning of the DCN, but is not part of the fixed part of the image address.	""	string	
	DCN Suffix		DCN Suffix is a static suffix that is printed and added to the end of the DCN, but is not part of the fixed part of the image address.	""	string	
DCN Integrity	First DCN Format		Format string (ATTR_SPrintf) of the first DCN of the batch. This makes sure the image address has been set correctly in the scanner.	"<DCN_Prefix,%s,><JDay,%05s,YYJJ><SequenceNum,%07d,0000001><DCN_Suffix,%s,>"	string	
PCN	PCN Prefix		PCN Prefix is a static prefix that is printed and added to the beginning of the Page Control Number.	"<BatchNum,%08d,BBBBBBBB>PCN"	string	
	PCN Counter Width		Number of digits in the counter used in the Page Control Number.	5	0 to 9	

Option Group	Option	Option Values (if enumerated)	Description	Attribute Default	Attribute Values	Notes
	PCN Format		Format string (ATTR_SPrintf) of the Page Control Number.	"<PCN_Prefix><TWEI_TS_COUNTER,%0<CounterWidth>d,001>"	string	
PageID	Page 1 Front ID		Page.Variant ID of the main document page, commonly "HCFA05.Standard", "UB04.Standard", etc.	UNKNOWN	string; select from Page.Variant; allow free form	
	Page 1 Back ID		Page.Variant ID of the main document page backside if scanning both sides.	Attachment.Standard	string; select from Page.Variant; allow free form	
	Page 2 Front ID		Page.Variant ID of the second document page if needed.	Attachment.Standard	string; select from Page.Variant; allow free form	
	Page 2 Back ID		Page.Variant ID of the second document page if needed and scanning both sides.	Attachment.Standard	string; select from Page.Variant; allow free form	
	Page 3 Front ID		Page.Variant ID of the third document page if needed.	Attachment.Standard	string; select from Page.Variant; allow free form	

Option Group	Option	Option Values (if enumerated)	Description	Attribute Default	Attribute Values	Notes
	Page 3 Back ID		Page.Variant ID of the third document page if needed and scanning both sides.	Attachme nt.Standar d	string; select from Page.Variant; allow free form	
	Page 4 Front ID		Page.Variant ID of the fourth document page if needed.	Attachme nt.Standar d	string; select from Page.Variant; allow free form	
	Page 4 Back ID		Page.Variant ID of the fourth document page if needed and scanning both sides.	Attachme nt.Standar d	string; select from Page.Variant; allow free form	
	Page 5 Front ID		Page.Variant ID of the fifth document page if needed.	Attachme nt.Standar d	string; select from Page.Variant; allow free form	
	Page 5 Back ID		Page.Variant ID of the fifth document page if needed and scanning both sides.	Attachme nt.Standar d	string; select from Page.Variant; allow free form	
Batch Integrity	Action when Too Many Docs	Error, Warn	When there are more documents scanned than indicated in Inventory, display either an error (unoverridable) message (recommended), or a warning (overridable) message.	"Error"	"Warn", "Error"	

Option Group	Option	Option Values (if enumerated)	Description	Attribute Default	Attribute Values	Notes
	Action when Too Few Docs	Warn, Error	When there are more documents scanned than indicated in Inventory, display either an error (unoverridable) message, or a warning (overridable) message (recommended).	"Warn"	"Warn", "Error"	
	Ultrasonic Double-feed Detection	Disabled, Low, Medium, High	Ultrasonic double-feed detection detects if two pieces of paper feed together.	"Disabled"	"Disabled", "Low", "Medium", "High"	
Printer (ink)	Printer	Flag, Print, Don't Print	This controls whether the physical printer is enabled, disabled, or respects the GUI flag (recommended). If it is specifically set here, the GUI flag is ignored. Global GUI flags defaults can be changed in the config file.	"Flag"	"True" (Print), "False" (Don't Print), "Flag" (Flag)	
	Printer Offset		Printer offset specifies how far off the leading edge of the paper to start printing.	4.5	0.0-12.0	

Option Group	Option	Option Values (if enumerated)	Description	Attribute Default	Attribute Values	Notes
		(hidden)		5400	(int)(TSPCM_PrinterOffsetIn * 1200)	This is a convenience attribute for TWAIN Scan
	Image Address Prefix		Image Address Prefix is a static prefix that is printed before, but not part of, the DCN	""	string, 9 char max	
	Image Address Suffix		Image Address Suffix is a static suffix that is printed after, but not part of, the DCN	""	string, 9 char max	
Printer (electronic)	ePrinter	Flag, ePrint, Don't ePrint	This controls whether the electronic imprinter (image only) is enabled, disabled, or respects the GUI flag (recommended). If it is specifically set here, the GUI flag is ignored. Global GUI flag defaults can be changed in the config file.	"Flag"	"True" (ePrint), "False" (Don't ePrint), "Flag" (Flag)	
	ePrinter Format		Format string (ATTR_SPrintf) to be electronically imprinted on the image.	"<TWEI_TS_DCN>"	string	

Option Group	Option	Option Values (if enumerated)	Description	Attribute Default	Attribute Values	Notes
	ePrinter X-offset		X offset (in inches) to start imprinting. -1 centers the text on the image.	-1	(-1) or 0.0 to 12.0	
	ePrinter Y-offset		Y offset (in inches) to start imprinting. Negative numbers mean offset off the bottom of the image.	0.3	(-20.0) to 20.0	negative numbers mean offsets off the bottom of the image
	ePrinter Font Size		Font size of imprinted text (default 36)	36	greater than 8	
	ePrinter Font		Name of font style for imprinted text. Blank defaults to the system font.	""	string, empty, haven't tested allowed values	
	ePrinter Opacity	"Transparent", "Opaque"	Background opacity of the imprinted text. If set to transparent, the text is XORed onto the image. If Opaque, the area around the text is erased for legibility.	"TRANSPARENT"	"TRANSPARENT", "OPAQUE"	

Option Group	Option	Option Values (if enumerated)	Description	Attribute Default	Attribute Values	Notes
Scanner	Paper Feeder Level	100, 250, 500	Paper Feeder Level controls how many pages to allow on the feeder at startup. This affects how far down the feeder carriage is at startup.	500	100, 250, 500	
	Transport Timeout (Seconds)		Time (in seconds) after which the transport will automatically timeout and return control back to the application.	10	0 to 60	
	Flatbed Attachment	Not Present, Use Flatbed Attached	Controls whether an attached flatbed will be used. (i2xx)	"False"	"True" (Use Flatbed Attached), "False" (Not Present)	
	Scanner Mode		Specifies a pre-configured scanner mode when using a Kodak Digital Science of imageLink scanner (i55xx, 75xx, 95xx).	4	0 to 18	

Option Group	Option	Option Values (if enumerated)	Description	Attribute Default	Attribute Values	Notes
Workflow	Calculated Work Record Attributes		Attributes that are calculated set into the record level of the work record.	"DocCount", "<TWAINSCAN_DocCount>", "JDay", "<JDay, %s,00000>", "Production", "<Production,%s,0>", "RouteSelect", "<RouteSelect>"	list of strings	

Option Group	Option	Option Values (if enumerated)	Description	Attribute Default	Attribute Values	Notes
	Calculated Item Attributes		Attributes that are calculated and set into each item of the work record.	DCN, "<TWEI _TS_DCN >," "Patch", "<TWEI _TS_PAT CHCODE >," "PageSide ", "<TWEI _HDR_PA GESIDE, %1d,0>," ", "PageSide FileSize", "<TWEI _HDR_PA GESIDE, %1d,0>," "<TWEI _TS_FILE SIZE>," "BarCode" , "<TWEI _TS_BAR CODE>," ", "PCN", "<PCN_F ormatStr, %s,>"	list of strings	

Option Group	Option	Option Values (if enumerated)	Description	Attribute Default	Attribute Values	Notes
	Page Deletion Regular Expressions		Regular expressions that cause page deletion.	Patch, "[<DocSeparator_Patch,%s,3><IntraDocSeparator_Patch,%s,2>]", "PageID", "^Separator.Back\$ ^Backside.Standard\$"	list of strings	
Extension	Extra Attribute Group Name		If specified and this group is present in the config, it is loaded. This is intended to load extra attributes not available in the PCM.	""	string, valid attr name (32 chars max, etc...)	
	Extra Capability Group Name		If specified and this group is present in the config, it is loaded. This is intended to load extra capabilities not available in the PCM.	""	string, valid attr name (32 chars max, etc...)	
		(hidden)			000-999, "def" is reserved for the config file	"def" is reserved for config file, B### comes from this value

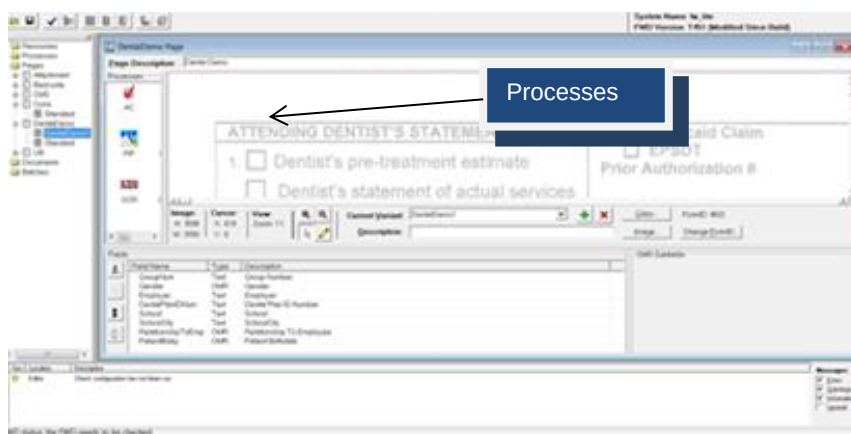
Warning: Changes to the TwainScan Process Configuration Window are saved automatically when the Close Window icon is selected . A best practice is to note the field selections in the event the original settings are to be restored. Additional option groups may be available. Contact your FIS Technical representative for details.

Configuration Page and Page Variant Processing

Open a Configuration Window (Page Level)

When a page name is double-clicked, its image template opens. To the left of the image is a Processes panel, providing quick access to each process that operates at the page level.

Alternate method: right-click the page name in the FWD tree, then select Processing > Process Name. The same shortcut is in place for documents and batches. The object does not have to be open to access the processes with this method.



Double-click a **Processing** icon (within the Processes window pane) to open a configuration dialog box.

AC: Page Processing

AC and DV workers (and KE and KFI) apply rules to evaluate data in a FormWorks system. The differences between AC and DV are in which rules are applied, and where each worker appears in the workflow.

Many rules use Resource objects, such as Tables, Functions, and Date Formats. These resources must exist before they may be used in rules. However, if a resource is added, deleted, or renamed, all open rule windows are refreshed to accommodate these changes.

To View Rules for a Page

If the page image template is open, double-click **AC** in the processes panel. If the page is not open, right-click the **Page Name** (not the page variant) in the FWD tree, select **Processing**, then select **AC**.

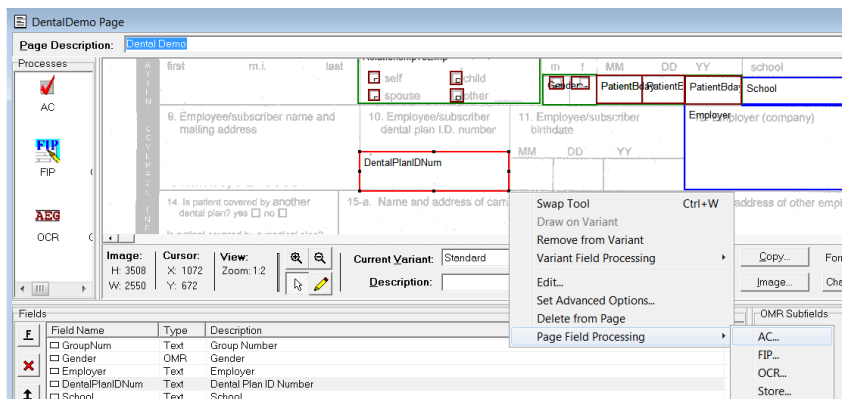
Before working with AC rules, become familiar with the following terms:

- **Function** – An operation that is called from a rule list, and typically tests or modifies field contents. All functions return a result and have a fixed number of status results. Most functions also have parameters, in the form of field lists and other configuration information. Functions are divided into categories, or types, described in [AC Function](#) Types, below.
- **Rule** – A function that has all configuration information specified, including the fields to be processed.
- **Rule List (and Sub-List)** – An ordered collection of rules to evaluate, along with an action for each possible status result of the function.
- **Parent Rule** – A rule that contains one or more sub-lists of actions.
- **Action** – A rule list that executes when the evaluation of a rule arrives at a particular status.

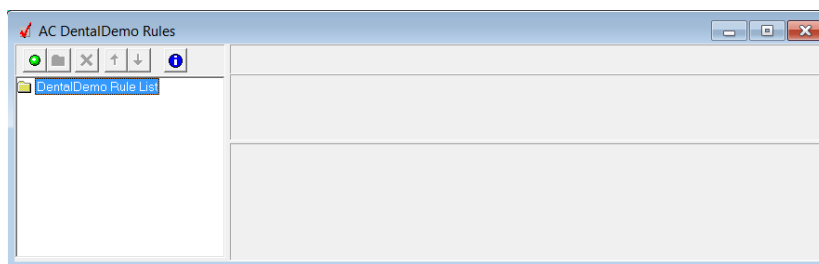
To View Rules that Reference a Specific Field

- Open the **Page**. Right click the desired **field** in the page image. Select **Page Field Processing > AC**.

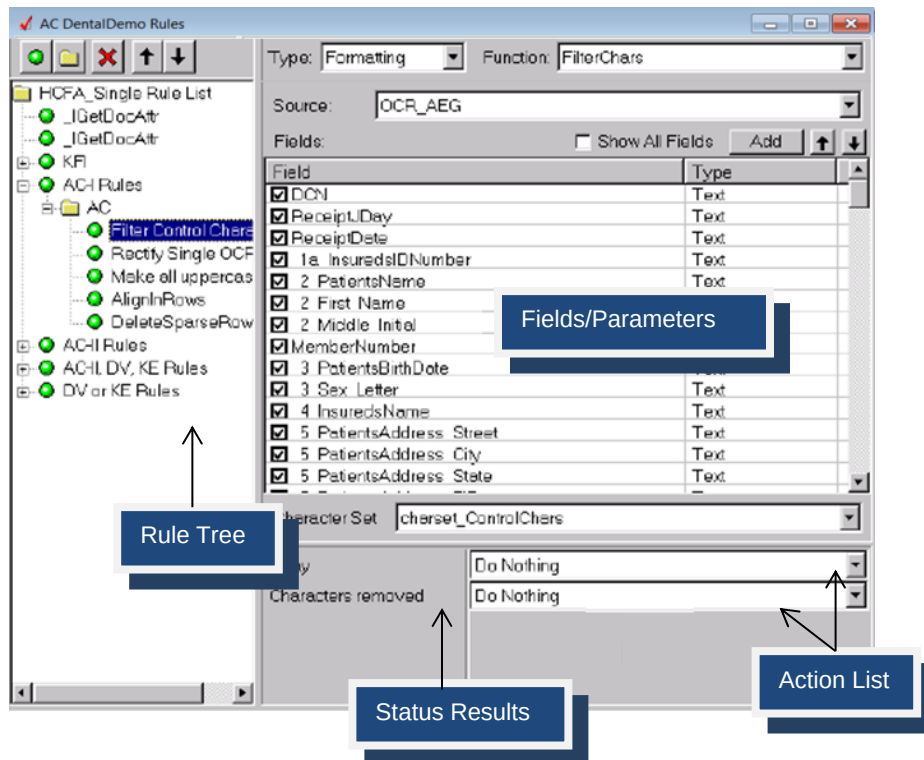
Note: More commonly, if the field appears in one or more rules, the contextual menus will appear structured similar to Page Field Processing -> AC -> (list of rules).



- The AC Rules page is displayed.



Warning: Changes to the Rules are saved automatically when the Close Window icon is selected. A best practice is to note the field selections in the event the original settings are to be restored.

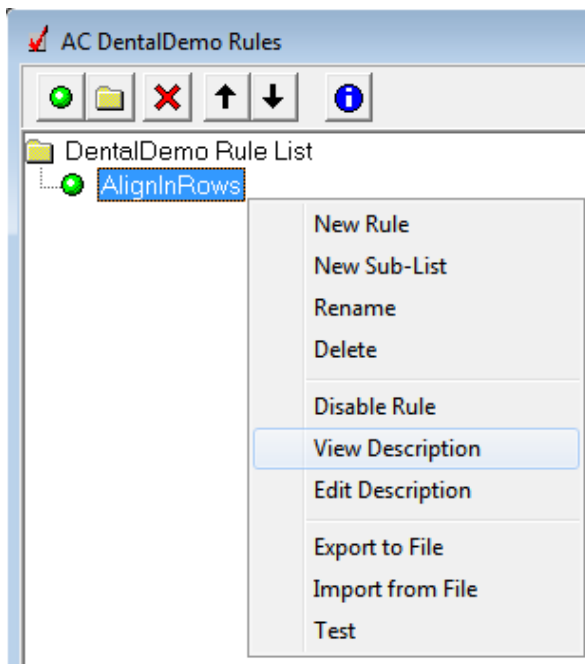


To View Information about a Rule

For many rules, users may view a description of the function and the parameters that have been set.

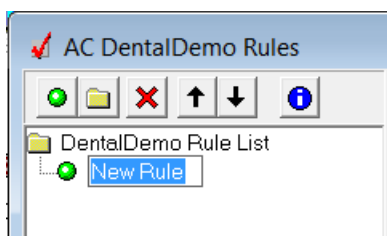
Select the rule, then select **Rule > Rule Description**.

Note: A description is only available if the creator of the rule has written and inserted a rule description.

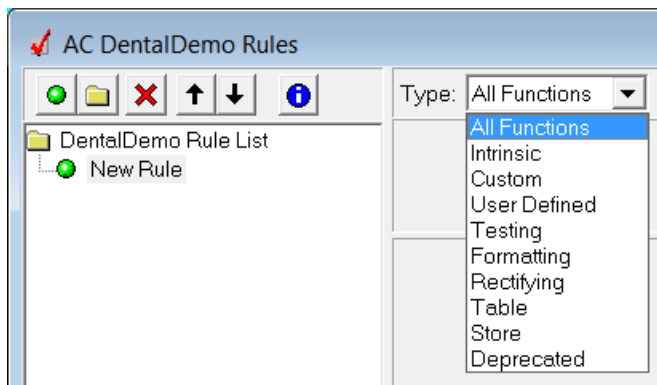


Adding New Rules

- In the rule list, select the line at which the new rule is to be added. (The new rule is inserted immediately below the current line; if the rule list name is selected, the new rule becomes the first item in the rule list).
- Select the **Create a new rule**  (or press **Ctrl-U**), and enter the **Name** of the new rule.



- This adds a rule node to the tree, with a default rule name, and selects the node.
- Enter a **Name** for the rule (or allow Editor to fill in the name when a Function is chosen).
- Select a **Function Type** from the drop-down list.



- Specify field names and parameters as needed.

Most functions have a customized interface; as such users see only the appropriate options for specifying fields and other parameters. However, if no customized interface is available, a default interface opens. The default interface (most often,) contains two tabs: Fields and Attributes.

Note: Some Functions when selected present to the user only fields (i.e. Attribute Name, etc.).

- For each possible status result, specify an action from the drop-down list. For most functions, the default options are Do Nothing, Reject, and (New sub-rule).
 - When the Reject action is selected, the Editor creates a new sub-list called Reject: reason, where reason is the name of the status result, and adds a Reject rule to the sub-list.
 - If a (New sub-rule) is selected from the actions list, Editor creates a new sub-list folder icon named for the status result you selected, with a new rule in the folder.
- Click on another rule in the rule tree to save the current rule (and display the configuration for another rule).

Note: To change an existing rule, follow steps above. Then right-click the rule and select Rename; enter a descriptive name for the modified rule.

AC Function Types

The list below contains general descriptions of the rule categories. Additional information about specific functions may be found within the FormWorks AC Functions document.

- **Intrinsic** – An intrinsic function is built into the AC engine (as opposed to being contained in a separate rule DLL). Intrinsic functions perform specialized tasks, such as calling other functions or interacting with the work record.
- **Custom (Tcl Procedure)** – Used to write new rules for specialized customer needs. AC/Tcl functions incorporate Tcl scripts into them. These functions are not implemented in the Standard AC Plugin DLL (acsrules.dll), but in their own custom DLL (actcl.dll). Please refer to the internal website for additional information.
- **User-Defined** – These are functions that call other functions, and are typically used for complex rules. The user creates the function implementation as a Function resource in the Editor, defining named field

list parameters and status results. Generally, a user-defined function is a rule list, in which the rules refer to the named field list parameters rather than any specific fields. These field lists are expanded against the list of fields passed to the user-defined function when it is called by an AC rule.

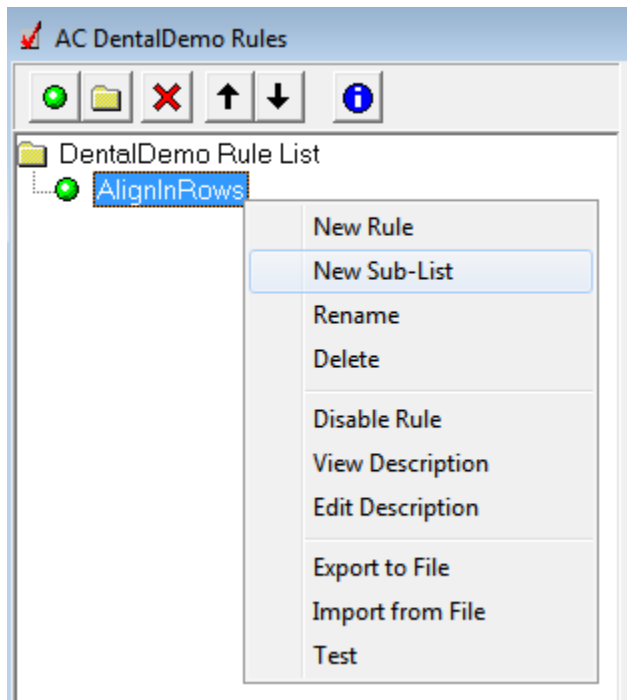
- **Testing** – Checks the contents of a field, without changing it, and then returns a status. Contained in acsrules.dll.
- **Formatting** – Changes the representation of information in a field. Formatting functions generally do not reject fields or return status. Contained in acsrules.dll.
- **Rectifying** – Functions that examine confidence levels to determine whether characters should be accepted or repaired. Contained in acsrules.dll.
- **Table** – Functions for checking selection lists and plugging values from tables. Contained in acsrules.dll.
- **Store** – Store-specific functions used to facilitate Store template population and formatting requirements. Contained in acstore.dll.
- **Deprecated** – These functions are older versions of functions found in other categories. As a best practice, they should generally not be used and are only present for backward compatibility with existing rules. When the Editor finds a rule using a deprecated function, it issues a warning. The user should replace any deprecated functions whenever practical.

Note: The AC configuration window for page field rules is shown in [Configuring Page and Page Variant Processing](#). Windows for document level fields and Store rules are identical and use the same procedures. Functions are further documented with the [AC Functions document](#).

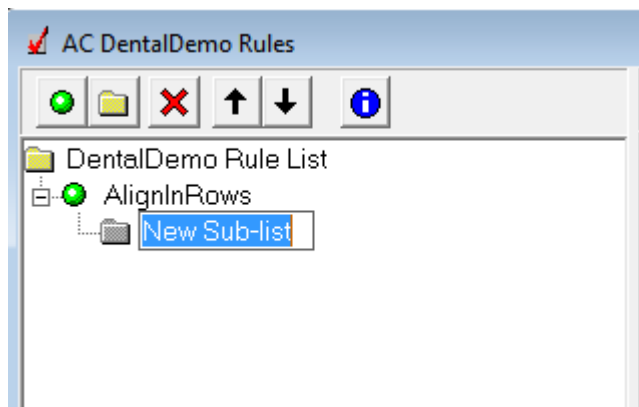
Adding New Sub-lists

To Add A New Sub-List

- In the rule list, select the line at which the new sub-list is to be added. (The sub-list is inserted immediately below the current line; if the rule list name is selected, the new rule becomes the first item in the rule list.)
- Click on the **Create a new sub-list button** (or press Ctrl-L).



This creates a new sub-list node, with a sub-list name, under the parent's rule node. The new sub-list is added to the parent's action list when the parent is selected.



- Create the desired rule(s) for the new rule sub-list.

Note: The new sub-list folder icon is shown in gray until this action is selected in the action list of a status result in the parent rule. CTRL-L Rule > New Sublist

Deletion of Rules (or Sub-Lists)

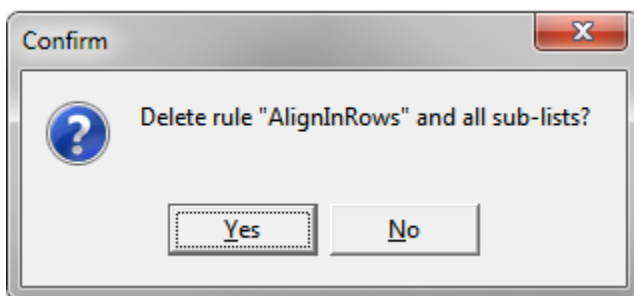
Deleting a rule or sub-list is a permanent action. Once a rule or sub-list has been deleted it cannot be restored. As a best practice, if the rule or sub-list may be needed later, disable it instead of deleting it.

To Delete a Rule

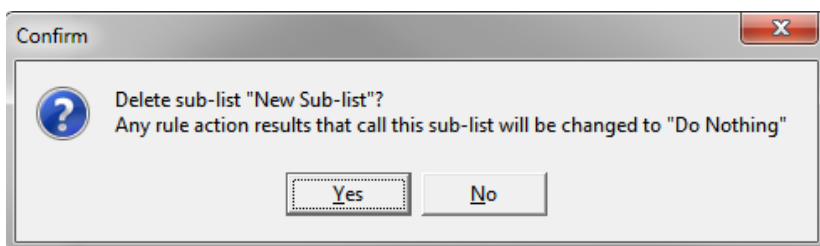
- Select the **rule or sub-list to be deleted**.
- Click on the **Delete selected rule or sub-list**  button (or press Delete).

A Confirm Delete dialog box opens. Be sure to read the message before deleting the rule or sub-list. Note that when a sub-list is deleted, the corresponding action is deleted from the parent rule, and all of the sub-list rules are deleted.

For delete rule, the following confirmation message is displayed.



For delete sub-list, the following confirmation message is displayed.



Disable a Rule

Disabling a rule causes the AC engine to ignore the rule during evaluation. When (or if) the rule is needed later, just remove the disable setting using the same steps.

To Disable/Enable a Rule

- Select the rule.
- **Select Rule > Disable** Rule (also available on the right-click menu, or **CTRL-BKSP**).

Move a Rule

To Move a Rule or Sub-List

- Select the rule or sub-list.
- Click the up or down arrow button to move the rule to the desired position. Drag-and-drop moving and copying is also supported for rules and sub-lists.

*Note: CTRL- , CTRL-
Rule > MoveUp
Rule > MoveDown
Drag*

Drag This Item	Drag To	Result
Rule	Rule	Rule moved/copied <i>above</i> target rule. *
Rule	Sub-list	Rule moved/copied to the end of target sub-list.
Sub-list	Rule	Sub-list moved/copied to the end of all sub-lists for the rule.

* Generally, any item dropped within the same window is moved; an item dropped into a different window is copied. However, users may press **CTRL** to force a copy, or press **SHIFT** to force a move.

Replace Fields, Tables, or Sources (Filerefs) in Rules

FormWorks Editor provides a search and replace feature to quickly change field names, tables or sources used in multiple rules. This feature can be used on just a selected rule, a sub-list, or on an entire rule tree. Use this feature to update rules when the name of a field or resource has changed.

- Select the **rule or sub-list** to be replaced.
- Select **Rule > Replace (press Ctrl-P)**. The Replace dialog box opens
- Select the type of item to be replaced (field, table, or source).
- Specify the name of the item to find, and what to replace it with.
- Pay particular attention to the scope of the replacement. Notice that “Entire rule tree” is selected as the default.

Unused Functions/Options

As of this writing, some functions appearing on the menu are not enabled, or are likely to undergo significant changes.

Validate (on right-click menu): nothing is attached to this command; it appears disabled on the menu.

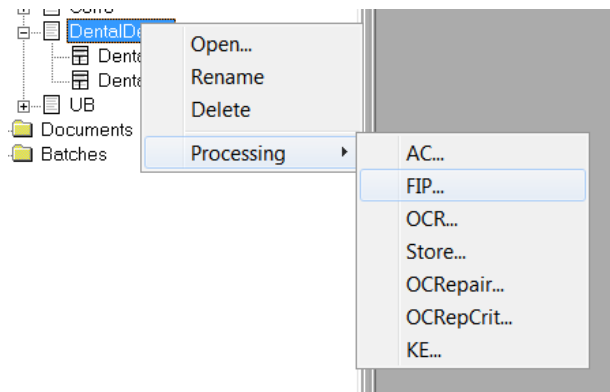
Debug (on Rule menu): these functions are not fully implemented.

FIP: Page Field Processing

To Open The FIP: Page Field Window

If the page is open, double-click **FIP** in the Processes panel

If the page is not open, right-click the **Page Name** (not the variant name), select **Processing > FIP**.



Depending on which field is selected in the Field list, the center section of the dialog box displays either Text Field options or OMR Field options.

FIP – Text Field

- Field – Drop down box, select the field for configuration.

Filter

- Filter
 - Custom – Not Applicable

Note: The customer option is currently not available (implemented).
 - None – No filter applied.
 - Despeckle – Removes noise from images.
 - Preview – Displays the [Filter Preview](#) window.

Text Field

- Field Type
 - Regular – Text/character field type.
 - Barcode – Barcode field type.

Important: Contact your FIS technical representative regarding Barcode configuration and support.

The FormWorks system has been enhanced to include a new standard Barcode worker that is capable of reading both one-dimensional (1D) and two-dimensional (2D) barcodes.

Barcodes provide an accurate, efficient way of labeling and organizing data, and allow for its easier retrieval. The characteristics of both barcode types are as below:

- 1D barcodes can store up to 30 alphanumeric characters, and are also known as linear barcodes.
- 2D barcodes are generated in various geometric shapes and can store large amount of data in a very small area (up to 7089 characters). It supports numbers, characters, pictures, voice and other binary information.

The barcode worker supports DataMatrix, PDF417, and QR types of 2D barcodes, in addition to most linear barcode types.

Users can define the maximum number of characters that can be read from the barcode using the 'MaxBarcodeLength' attribute in the Barcode.cfg file. Maximum barcode length that is allowed is 128. Value assigned to the MaxBarcodeLength attribute should always be greater than the MinBarcodeLength value.

Note - If the Barcode worker is configured for OCR Field output, the barcode geometry will be saved to the OCR file. Barcode geometry is saved in the OCR format: length, X, Y, W, and H where length = number of characters in the Barcode, and X, Y, W, and H are location pixels.

Licenses required

The barcode worker uses the ClearImage Barcode Recognition Engine developed by Inlite Research Inc. Depending on the types of barcodes to be read, runtime licenses for one or more of the following products are required:

- ClearImage Barcode 1D Basic or Pro
- ClearImage DataMatrix
- ClearImage PDF417
- ClearImage QR

Usage

The FormWorks system will read the barcodes printed anywhere in a document, and return the encoded data as part of the document that can be used for further processing. Typical uses of this worker are:

- Survey identification
- Member identification
- Form identification
- X-ray attachment identification

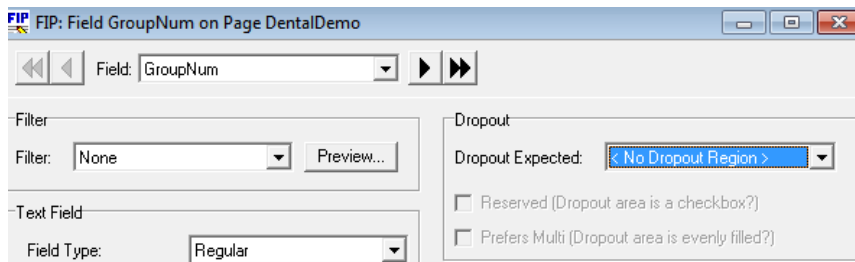
The barcode worker supports both collated and uncollated work records. There are two ways in which the barcodes are saved to the work records.

- As attributes at the item level (page item level, if the work record is collated).
- As field values in OCR files to be accessed and used by the AC rules engine.

Important: Contact your FIS technical representative regarding Barcode configuration and support.

Dropout

Note: As a best practice, do not use field level dropout settings. The Dropout Expected should generally be defined as "No Dropout Region". Instead, as a best practice utilize FIP Variant Image Layout.



- Dropout Expected
 - No Dropout Region – Field does not lie within a dropout region and should not be used for form registration or alignment.
 - May Be Empty – Field may be useful for registering the form. Two or more such fields can be grouped into a single dropout region, such as the Insured Name, Address, City, State and ZIP fields on a HCFA.
 - Will Not Be Empty – These are fields that are required on the form. FIP can reliably use these areas in matching data on a scanned image with an area known to contain data in the layout.

- Will Be Empty – Parts of the form that should never contain data. Areas may be defined at the sides of the form as “Will be Empty” so that if FIP sees data in the areas defined, it will shift the image to avoid having data there.
- Reserved – (Dropout area is a checkbox?) – If the field contains one or more checkboxes, select **Reserved** (Dropout area is a checkbox).
- Prefers Multi (Dropout area is evenly filled?) – If the field is an evenly filled area, select **Prefers Multi** (Dropout area is evenly filled).

Note: As a best practice, it is recommended not to use field level dropout setting but allow them to remain defined as a No Dropout Region.

FIP – OMR Field

- Field – Drop down box, select the field for configuration.

Filter

- Filter
 - Custom – Allows the user to customize settings.
 - None – No filter applied.
 - Despeckle – Removes noise from images.
 - Preview – Displays the [Filter Preview](#) window.

OMR Field

- Field Type
 - Checkbox – Standard checkbox.
 - Signature – Used for signature detection

- Check Type
 - Box – For boxes with box outline.
 - Fill-in – For boxes in which the user completely fills the box (solid fill).
 - Dropout – for boxes in which the checkbox itself drops out.
 - Letter – for boxes in which a letter (like M or F) appears in the box.
- Signature Detect Threshold – May be populated if Signature is selected in the Field Type box.
- Use Aura: The aura is an area around the outside of the checkbox (or signature box). Use this setting to slightly expand the area in which FIP looks for a mark or signature.
- Thick Checkbox – Use this setting if the lines of the checkbox itself are especially heavy.
- Letter in Checkbox – Use for checkboxes that contain a letter (like M or F).

Warning: The configuration windows for Variant Field Processing and Page Field Processing are very easy to confuse.

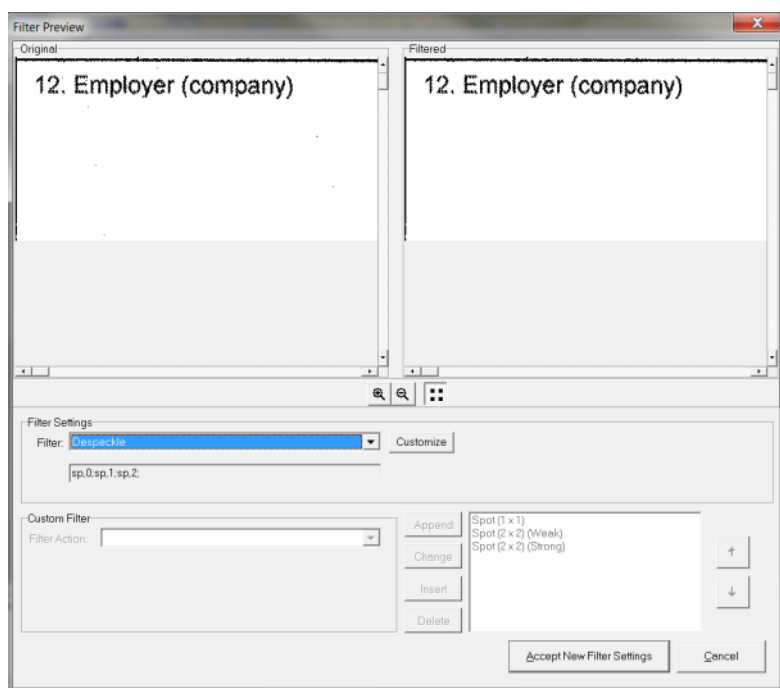
At the variant level, changes may only be made after selecting Override page settings. If options are set for a field at both the page and variant levels, Editor uses the variant level setting.

Generally, changes should always be made at the page level, so that all variants are affected (including variants that might be added at a later date).

Filter

Use [Filter Regions](#) to specify small image cleanups where necessary. A variety of cleanup options, such as despeckle, erosion, and dilation are available to enhance image quality in the filter area.

If you specify a filter at the page level, the filter applies to all variants of the page. The filter region corresponds to the field's size and location on the current variant (as listed in the current variant box on the page window).



To Filter a Field Area

- Select the desired field from the field list.
- Select Despeckle or Custom from the filter drop down.

If Custom is selected, the Filter Preview dialog box opens

Custom Filter Area

- Select a **Filter Action** and **Filter Mode** from the drop-down lists.
- Click **Append** to add this filter to the list at the right side of the window. The preview area now shows the original image on the left, the filtered image on the right.
- Select and append any **additional filter actions**.
- Click **Accept New Filter Settings** to retain changes.

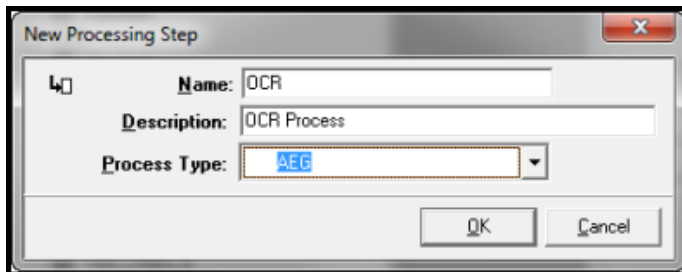
Warning: Do NOT set filters at both the page and variant levels. If both are set, the filter is applied twice, usually resulting in poor image quality. See [Filter Regions](#) for more information on setting filters at the variant level.

Configure Regions for the Optical Character Recognition (OCR) Worker

Optical Character Recognition (OCR) worker v. 6.0 and newer allows users to process documents of English and Western European languages that are supported by the Windows-1252 character encoding. This section provides information about various steps required to configure region settings like creating an OCR process, adding regions and selecting regions for document processing.

Create an OCR Process

- In FWEditor, right-click Processes, and then select **New**.



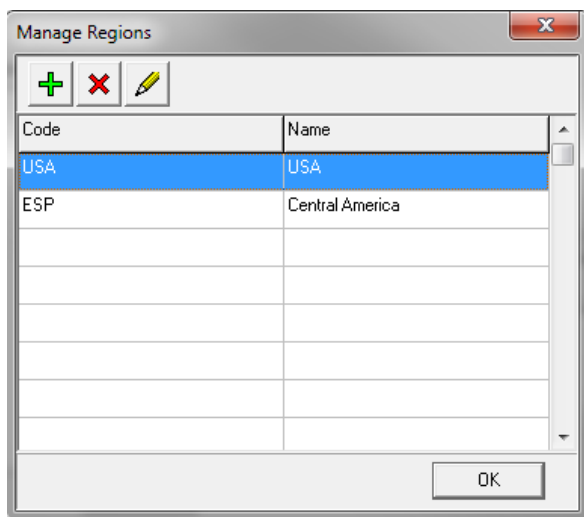
- Enter the process name and a corresponding description.
- Select **AEG** as the **Process Type**.
- Click the **OK** button to save the process.

Manage Regions - OCR

The Manage Regions window can be used to create, modify and delete regions. Only regions that are created and displayed in this window can be configured in the **Region Settings** window. Refer to the [Selecting Regions](#) section for more information.

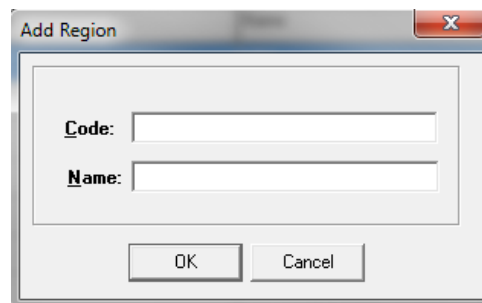
- Right-click **Processes** -> **OCR**, select Open.
- Click the **Modify** button in the **Region Settings** panel. The Region Settings window is displayed.
- Select **Manage**.
 - The Manage Regions window is displayed.

Note: As a best practice, it is strongly recommended that users configure the Name as USA for English and CENTRAL_AMERICA for Latin American documents.



From Manage Regions, the user may:

-  – Add Region, selecting displays the Add Region window.



- Code – User defined, unique (3 letter) word to define the region. For example, Code USA for USA or ESP for Spain.
- Name – Name of the region.
- Select OK to save.
-  – Delete Region – To delete a Region, select the Region and select.

Note: A region cannot be deleted if it has already been moved to the Selected Regions list in the Region Settings window.

-  – Edit Region – Displays the Add Region window from which only the Name

may be edited.

Note: Once named, the Code field fixed and may not be edited.

- Close the OCR Configuration Window by Selecting the Close Window icon .

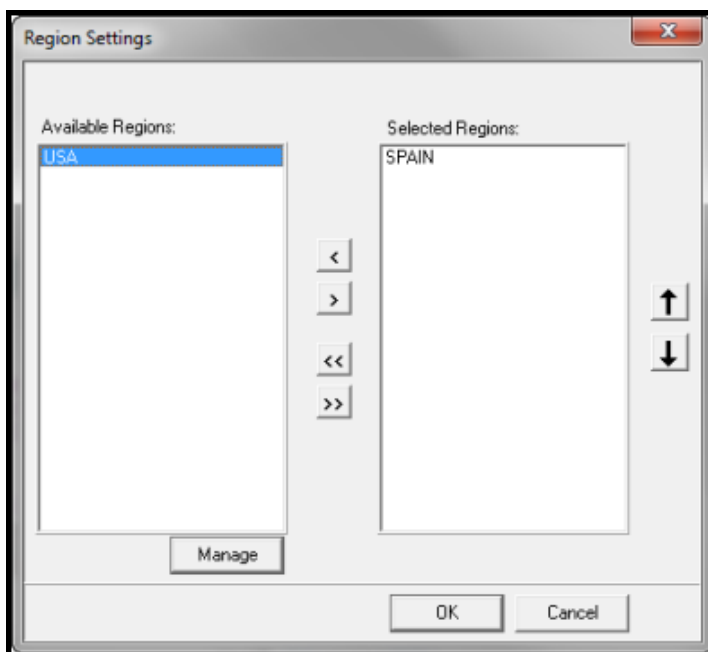
Warning: Changes to the OCR Configuration Window are saved automatically when the Close Window icon is selected.

Selection of Regions

Use the **Region Settings** window to configure regions for the OCR worker. All regions that were created in the **Manage Regions** window are displayed in the **Available Regions** list. The user can move the regions from the available to the selected list to process documents of those regions.

Note: Contact a FIS representative regarding system configuration for 1I8N (Internationalization) enhancements.

- Open the **Region Settings** window by clicking the **Modify** button in the **OCR: System Configuration** window. Refer to the [OCR Process](#) section for more information.



- Move a region from **Available Regions** to the Selected Regions list. Refer to [Managing Regions - OCR](#) for information on creating regions. For example, if Spanish claims are to be processed, then Spain should be moved to the **Selected Regions** list.

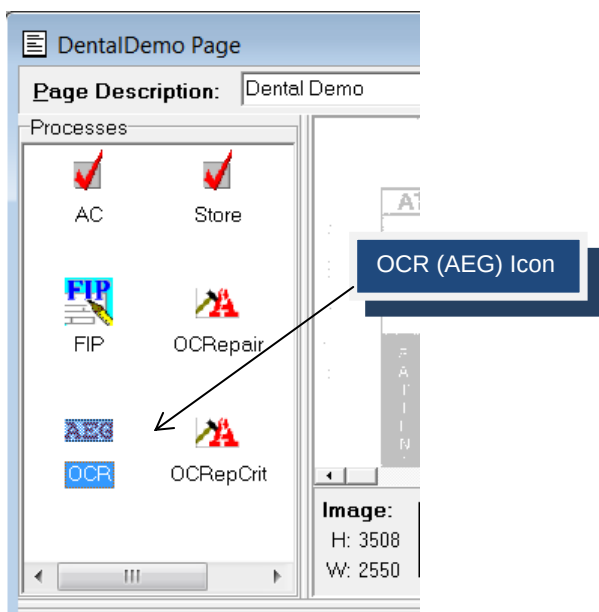
- Click the **Manage** button to open the **Manage Regions** window. This button is available in this window only if it is opened from the System Configuration level (OCR System Configuration).

OCR: Page Field Processing

To improve OCR results, first make sure that the bounding rectangles have been defined correctly for all imaged fields on your form. Check that the bounding boxes do not overlap, and that there are no gaps between adjacent fields. See [Defining Bounding Rectangles](#), for more information.

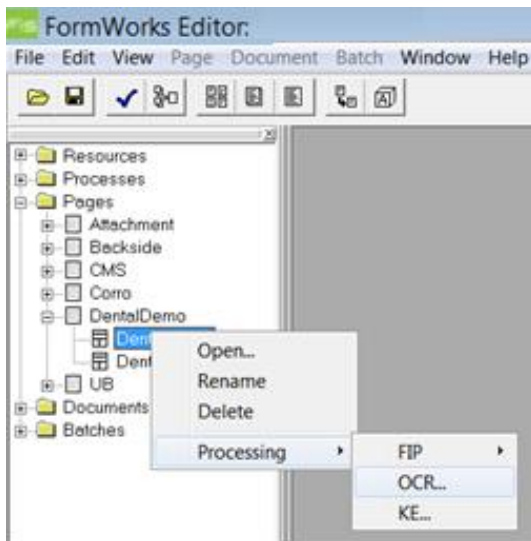
To Open the OCR Page Field Window

- If the page is open, double-click the **OCR (AEG)** icon in the Processes panel.

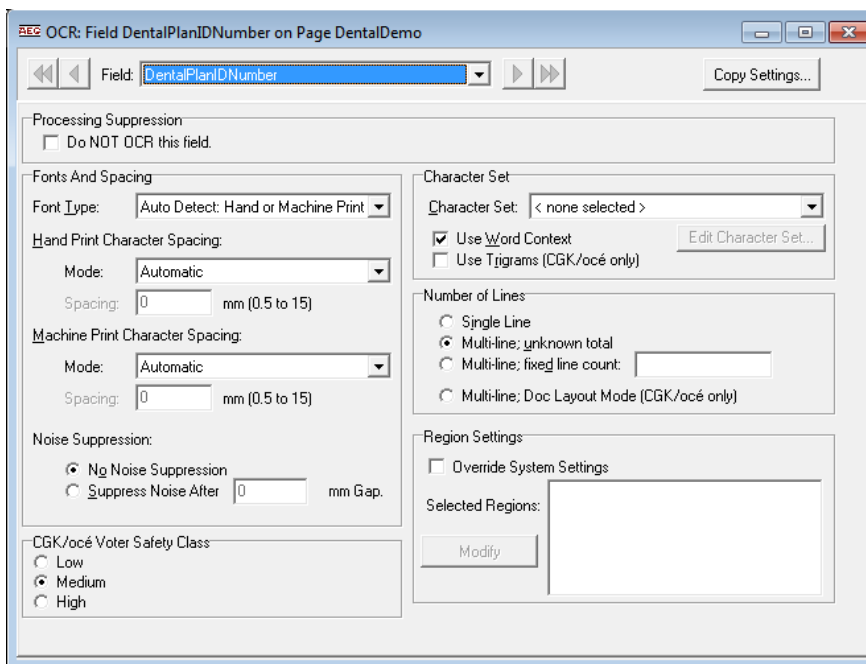


or

- Right-click the **Page Name** (not the variant), select **Processing > OCR**.



- The OCR:Field on Page screen is displayed.



- Within the OCR:Field on Page screen (above) the following settings from which the user may define the best Page Field Processing configuration are provided.

Field/Copy Settings

Field

From the Field drop down, select the field to be configured

Important: The OCR Field Page Settings window is almost identical to the OCR Field on Variant window.

The [OCR Field Page](#) window does not contain the three Override Page Settings checkboxes which are present in the OCR Field on Variant window.

(See [OCR: Variant Field Processing](#) for details regarding remaining configuration options).

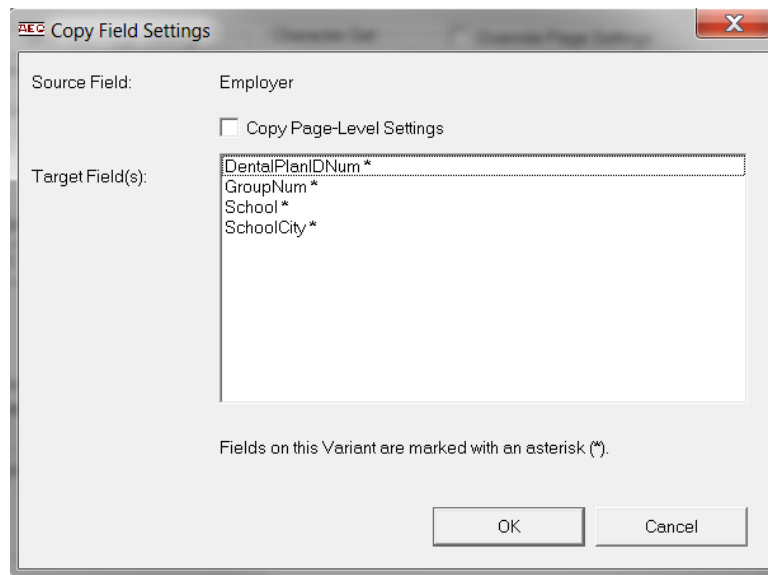
The screenshot shows the 'OCR: Field DentalPlanIDNumber on Page DentalDemo' window. It features a 'Field' dropdown set to 'DentalPlanIDNumber' and a 'Copy Settings...' button. The settings are organized into several sections:

- Processing Suppression:** A checkbox for 'Do NOT OCR this field'.
- Fonts And Spacing:**
 - Font Type:** A dropdown menu set to 'Auto Detect: Hand or Machine Print'.
 - Hand Print Character Spacing:**
 - Mode:** A dropdown menu set to 'Automatic'.
 - Spacing:** A text input field set to '0' mm (0.5 to 15).
 - Machine Print Character Spacing:**
 - Mode:** A dropdown menu set to 'Automatic'.
 - Spacing:** A text input field set to '0' mm (0.5 to 15).
- Noise Suppression:**
 - No Noise Suppression:** A radio button that is selected.
 - Suppress Noise After:** A text input field set to '0' mm Gap.
- CGK/océ Voter Safety Class:**
 - Low:** A radio button.
 - Medium:** A radio button that is selected.
 - High:** A radio button.
- Character Set:**
 - Character Set:** A dropdown menu set to '< none selected >'.
 - Use Word Context:** A checked checkbox.
 - Use Trigrams (CGK/océ only):** An unchecked checkbox.
 - Edit Character Set...** A button.
- Number of Lines:**
 - Single Line:** A radio button.
 - Multi-line; unknown total:** A radio button that is selected.
 - Multi-line; fixed line count:** A text input field.
 - Multi-line; Doc Layout Mode (CGK/océ only):** A radio button.
- Region Settings:**
 - Override System Settings:** An unchecked checkbox.
 - Selected Regions:** A text area.
 - Modify:** A button.

Copy Settings (Field Level/Page Level)

The copy settings feature provides an easy way to configure fields with the same settings.

- Select the **Copy Settings button**, the Copy Field Settings window is displayed.



- Select each field that is to be configured identically to the source field (listed at the top of the dialog box). Hold the **CTRL + click** each field individually with the mouse to select multiple fields.

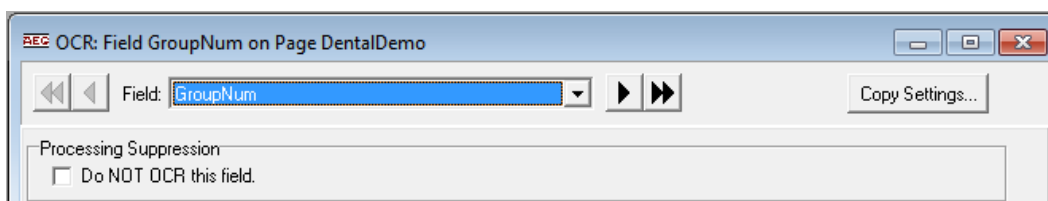
Note: To copy the Page Level Settings instead of the Field Settings, select the Page Level Settings check box.

- Select **OK** to copy the fields selected.

OCR Field Page Configuration

Processing Suppression

Select this option for any field in which the data will not be read by the OCR engine. This may include DCN, values plugged from a table lookup, fields to hold parsed names, temporary fields, and so on. When the **Override Page Settings** and **Do NOT OCR this field** checkboxes are selected, all other options are disabled.



Note: The OCR Field on Page window does not contain the Override Page Settings checkboxes.

Fonts and Spacing

The screenshot shows a dialog box titled "Fonts And Spacing". It has four main sections:

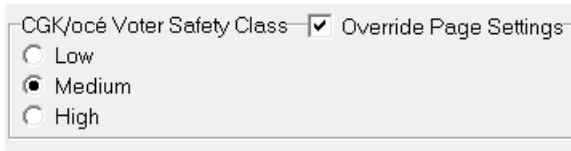
- Font Type:** A dropdown menu set to "Auto Detect: Hand or Machine Print".
- Hand Print Character Spacing:**
 - Mode:** A dropdown menu set to "Automatic".
 - Spacing:** A text input field set to "0", followed by "mm (0.5 to 15)".
- Machine Print Character Spacing:**
 - Mode:** A dropdown menu set to "Automatic".
 - Spacing:** A text input field set to "0", followed by "mm (0.5 to 15)".
- Noise Suppression:**
 - ☒ **No Noise Suppression**
 - ☐ **Suppress Noise After** [0] mm Gap.

- **Override Page Settings (Check box)** – Select to enable the fields and drop down boxes within the Fonts And Spacing grouping.
 - Note: Within the OCR Field on Page window, all fields and drop down boxes within Fonts And Spacing are by default enabled.*
 - **Font Type** – Specify whether data in the selected field is hand print, machine, mixed, etc. As a best practice, for most forms, use the default setting; Auto Detect Hand or Machine Print.
 - **Hand Print Character Spacing** – As a best practice, for the majority of forms, selects the default mode; Automatic. However, if a form contains constrained data, select Fixed Known and define the Spacing (the center to center distance between hand print characters) value in mm.
- **Machine Print Character Spacing** – Defines how a hand print will be spaced on the field. For most forms, use the default mode, Automatic. If a form contains constrained data, select Fixed Known, and define the Spacing (the center to center distance between hand print characters) value in mm.
- **Noise Suppression** – Instructs how to handle data after blank areas within the field. Noise suppression causes OCR to “ignore” characters in the right side of a field box. Use this only if it known that extraneous data is present (like a stamp), in a given field. By default, the No Noise Suppression option is selected.
 - Select Suppress After <value> mm Gap; to instruct the system to ignore any characters after the specified blank gap distance in millimeters.

Note: If characters are consistently lost from the right side of a field, noise suppression may have inadvertently been set.

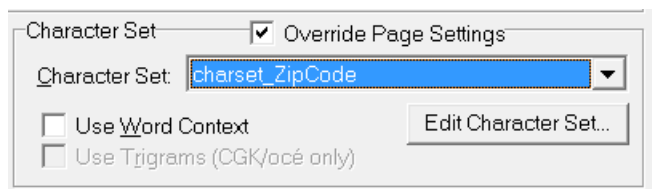
CGK/océ Voter Safety Class

CGK/océ Voter Safety Class – The CGK engine uses a multi-pass OCR algorithm with voting to determine which pass is applied. The level of how much CGK trusts its determination of the pass utilized may be adjusted with this setting.



- **Override Page Settings (Check box)** – Select to enable the fields and drop down boxes within the CGK/océ Voter Safety Class grouping.
 - To utilize the CGK/océ Voter Safety Class, select **Override Page Settings** adjacent to CGK/océ Voter Safety Class.
 - **Low** – For fields that will have secondary validation, such as ZIP codes or NPI, as a best practice set voter safety class to LOW. Selecting LOW will result in less keying (allowing CGK to be more confident with results that are less restrictive). In this scenario, the job of validating the ZIP code or NPI falls onto AC rules.
 - **Medium** – As a best practice, the best selection.
 - **High** – For fields that have no secondary validation, however the result will most often be additional OCREpair keying. This selection will provide the most accuracy possible.

Character Set



- **Override Page Settings (Check box)** – Select to enable the fields and drop down boxes within the Character Set grouping.
 - **Character Set** – Select a Character Set appropriate for the field, from the Character Set drop down box form which a listing of all possible characters that may be found in this field. As a best practice, select a character set that accurately reflects the expected contents of the field and always select a Character Set for any field that is configured to be OCR'd.

Important: It is not recommended to accept the default of <none selected> as this may contribute to anomalies within the .CFD file.

- **Edit Character Set** – Select Character Set to edit the set of characters. [See Table Resources – Characters Sets](#) for information on editing [Character Sets](#).
- **Use Word Context** – Select Use **Word Context** if characters in the field follow a consistent pattern that can be used to determine what a character is. As a best practice, if a field will probably contain groups of characters separated by spaces, and those groups of characters are each entirely alpha or

entirely numeric, turn word context on. Consequently, Word Context should not be select for fields that are known to contain words with mixed alpha and numeric characters (account numbers, diagnostics, procedures, modifiers, etc.).

- Use Trigrams (CGK/océ) – A trigram is a set of three letters that is common within the English language ("ing", "pre" etc.). As a best practice, when using Word Context, select Use **Trigrams (CGK/océ)**.

Number of Lines

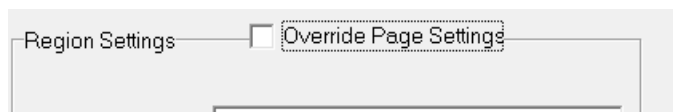
- Single: Fields in which there will be only one line of data, i.e. ID fields, DOB and other dates, most name fields.
- Multi-line; unknown total: This setting is used in grid fields, such as the detail lines in a HCFA and UB claim forms. These fields can contain a variable number of lines.
- Multi-line; fixed line count: Use only for fields in which the number of lines is always the same.

Note: The Number of Lines options can also be accessed by right-clicking a field on a page or in the field list. Select Set Advanced Options.

Regions Settings

The Regions Settings section is compatible with OCR Worker versions 6.0 and above.

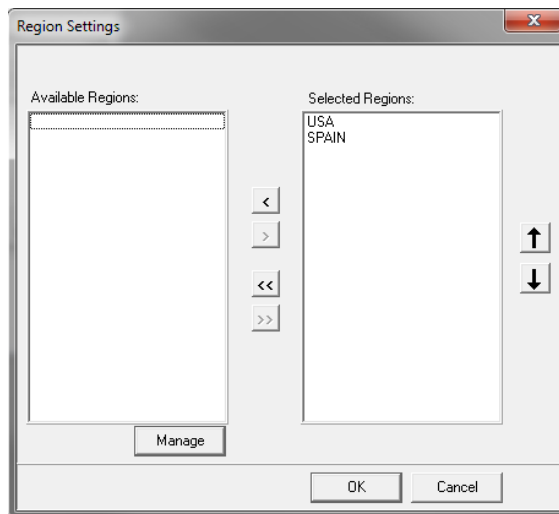
- The **Selected Regions** displays all regions assigned to the field.
- Select the **Override System Settings** checkbox to select the option to override regions configured at the System level.



The screenshot shows a form with a label 'Region Settings' followed by a checkbox labeled 'Override Page Settings'. The checkbox is currently unchecked. Below the checkbox is a horizontal line, likely representing the 'Selected Regions' list mentioned in the text.

Note: Leave this checkbox unmarked to assign the regions selected at the system level to the field. If checked, System Configuration must have regions defined for Page to override.

- After selecting the **Override System Settings** checkbox, select the **Modify** button to open the **Region Settings** window. Modify button is available only if the **Override System Settings** checkbox is selected.



- In the **Region Settings** window, move the regions to configure from **Available Regions** to the **Selected Regions** list.

Important: The regions configured here will override the regions configured at the system configuration level.

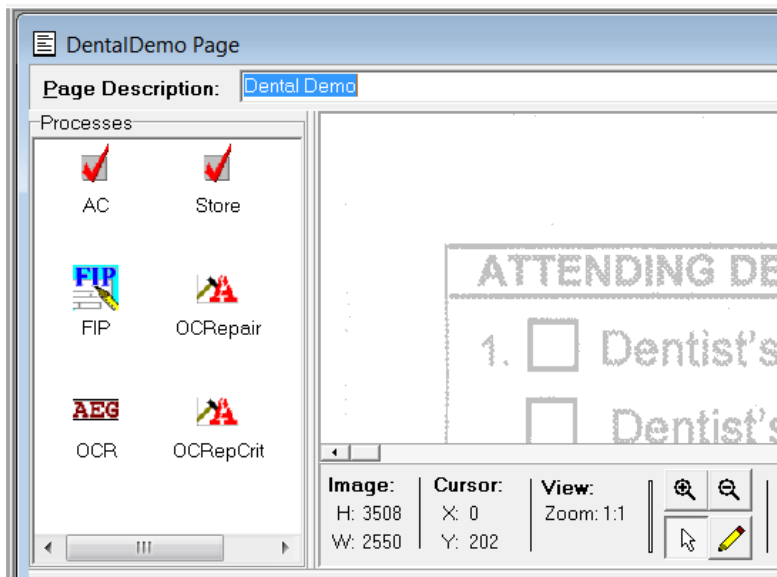
- Click **OK** to save the configuration and return to the **OCR Page Field** window.

OCRepair - Page Processing

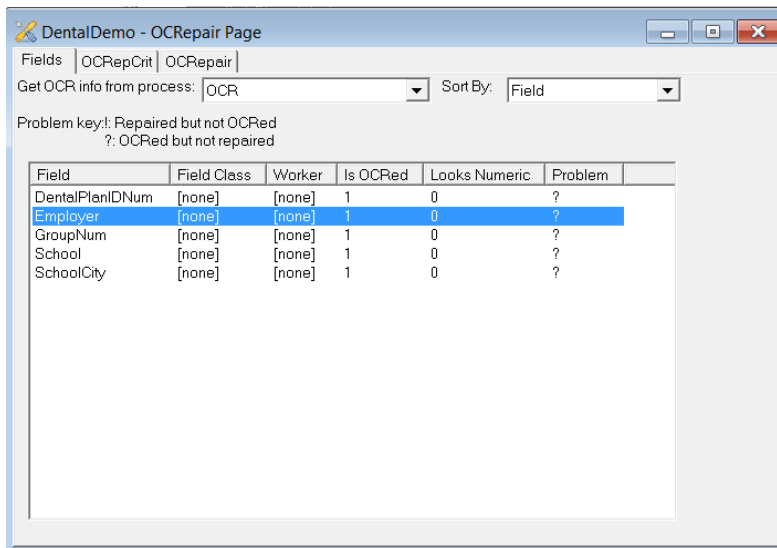
At the page level, users must select which fields will be processed by each OCRepair step (OCRepair/OCRepCrit). Colors and field classes control the display color for groups of fields; they are configured as resources, see [OCRepair Color](#), and [OCRepair Field Class](#).

To Open the OCRepair Page Configuration Window

- If the page is open, double-click any **OCRepair icon** in the Processes panel, or right-click the Page Name (not the variant).



- Select **Processing > OCRepair** or **OCRepCrit**.
- The OCRepair page window is shown below.



To Configure OCRepair Steps for the Page

Each field that is to appear at an OCRepair/OCRepCrit step requires that the Field Class to which the field belongs be configured.

To configure, double-click the **Field**, from the **OCRepair Page** window, the **Edit List Item** window is displayed.

To navigate the Edit List Item window (change fields) use the arrow buttons to forward or reverse through the list of Fields.

- Field – Select the **Field** to be configured.
- Field Class – Select from the **Field Class** drop down a **Field Class** (defined within Resources -> OCRepairFieldClass) ONLY for the appropriate OCRepair stage.
- Worker – From the Worker drop down, select the applicable **Worker**.
- IS OCRed – Checkbox, select if the field is to have the OCR process applied
- Looks Numeric – Check

When the Page Level processing configuration has been completed, return to the system level configuration window to generate and save a configuration file.

Store: Page Processing

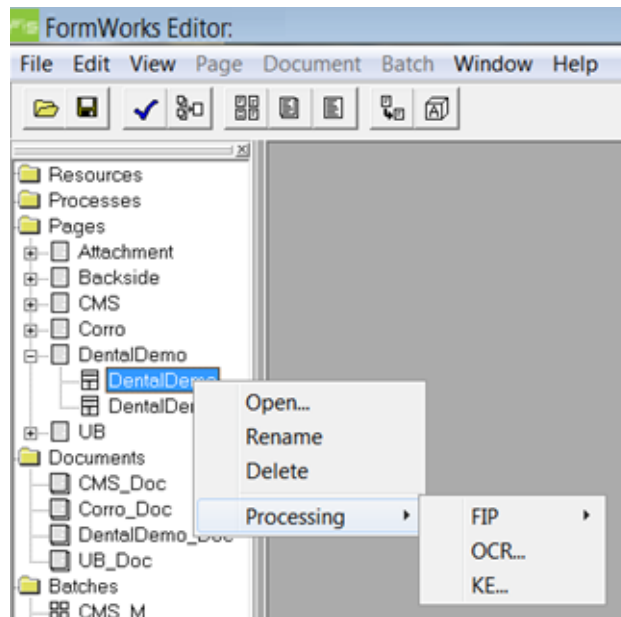
The interface and procedures for configuring rules for Store are identical to those for configuring rules for AC workers. Please refer to [AC: Page Processing](#) for additional information.

Page Variant Processing

FIP and OCR processes operate on the Page Variant (field level) as well as the Page Level.

To Set Variant Level Processing Options

- Right-click the **Page** name (Page Variant) in the FWD tree.
- Select **Processing**, both FIP and OCR are available on the submenu.



Important: Some configuration options can be set at both the Variant field and Page field levels. FormWorks Editor currently does not issue a warning when attributes at both levels are set.

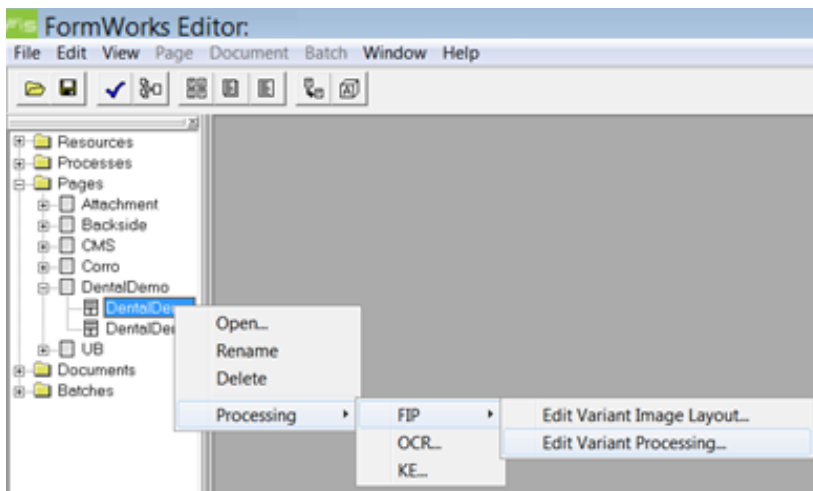
The variant level setting always takes precedence over the page level setting.

For example, users may select an OCR character set for the Name field at the Variant level. If later a different character set for Name at the page level is selected, it will be ignored, and the character set selected for the Variant is used.

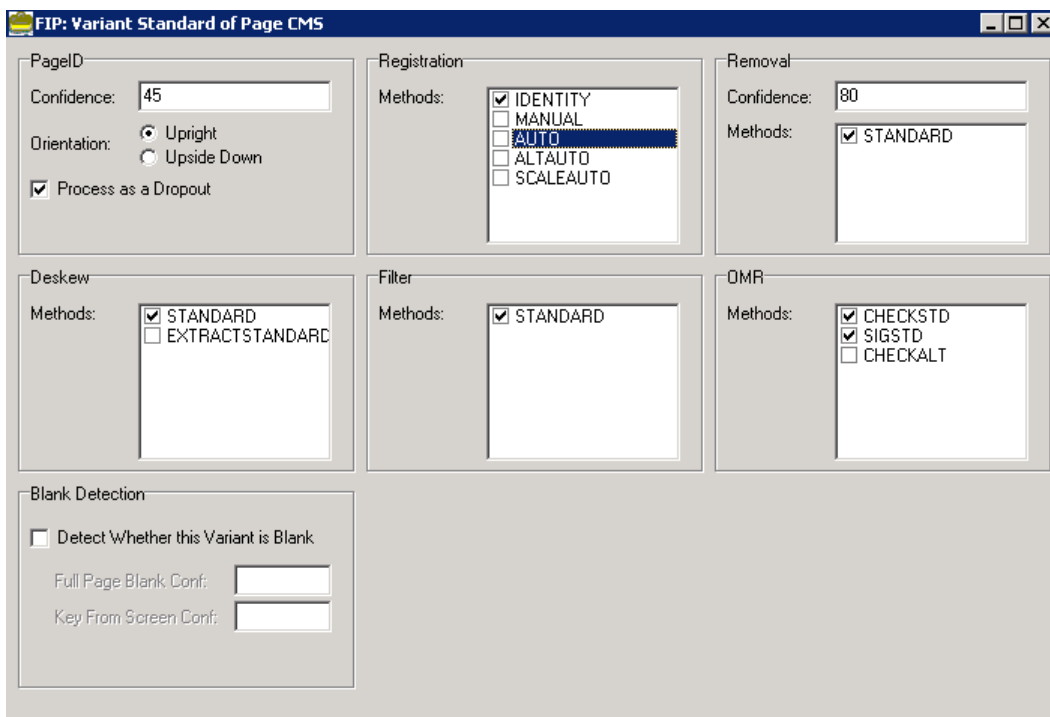
FIP: Variant Page Processing

FIP manipulates images before they are sent to the OCR Worker, creates the OCR Files, and performs OMR recognition.

To open the FIP Variant Page Processing window, **right-click** the **Variant** (Page) name, select **Processing > FIP > Edit Variant Processing**.



Listed below are some common components of the FIP Variant window and the various options available. The options that appear depend on which DLL(s) are specified in the [FIP Process](#).



PageID

Option	Description
Confidence	Threshold for positively identifying the form. If FIP's confidence level is below this number, the page is rejected.
Orientation	Orientation of scanned image (Upright or Upside down).
Process as Dropout	This option MUST be selected to correctly process dropout forms. Then define dropout regions or registration lines (see Dropout Regions , and Other Form Definition Options) for details.

Registration

Option	Description
AltAuto	Similar to Auto, but enhanced in functionality.
ScaleAuto	Similar to AltAuto, but can scale claims up to 13%
Manual	Select Manual when using registration lines to define a form. See Registration Lines , for more information.
Auto	Select Auto when using dropout regions or tags and lines to define the form. See Defining Non-dropout Forms , and Dropout Regions , for more information.

Removal

Option	Description
Confidence	Threshold for removing tags and lines from a non-dropout form.
Methods	Currently the only method is Standard.

Deskew

The option selected here is used by the FIPDeskew step in the batch processing script.

Option	Description
Methods:	.Two options available to the user. STANDARD EXTRACTSTANDARD

Filter

Option	Description
Standard	Standard filtering performs image filtering based on the Filter Regions defined on the variant layout (see Filter Regions).

OMR

Option	Description
CHECKALT	Enhanced version of CheckStd.
CHECKSTD	Reads checkbox fields from the image.
SIGSTD	Reads signature fields from the image.

Blank Detection

Blank detection options are not currently used.

Option	Description
Detect Whether This Variant is Blank	Checkbox
Full page Blank Cont.	<i>Note: Blank detection options are not currently used.</i>
Key From Screen Conf.	<i>Note: Blank detection options are not currently used.</i>

Advanced Options

Additional options are displayed when the user presses **CTRL + SHIFT** while double-clicking in any **empty gray area** of the FIP Variant Processing window.

Note: Advanced options are denoted by the Reserved box. When the FIP Variant window is closed, the Advanced Options automatically minimize and are not displayed the next time the FIP Variant window is opened.

The screenshot shows the 'FIP: Variant Standard' window. It contains several sections with checkboxes and 'Reserved' fields:

- PageID:** Confidence: 0, Orientation: Upright (selected), Upside Down, Process as a Dropout (-1), Reserved: 0.
- Registration:** Methods: MANUAL, AUTO, ALTAUTO, SCALEAUTO, Reserved: 0.
- Removal:** Confidence: 0, Methods: STANDARD, Reserved: 0.
- Desker:** Methods: STANDARD, EXTRACTSTANDARD, Reserved: 0.
- Filter:** Methods: STANDARD, Reserved: 0.
- OMR:** Methods: CHECKSTD, SIGSTD, CHECKALT, Reserved: 0.
- Blank Detection:** Detect Whether this Variant is Blank (checked), Full Page Blank Conf: 0, Key From Screen Conf: 0.
- Extract:** Methods: (empty), Reserved: 0.

Extract (Advanced Option)

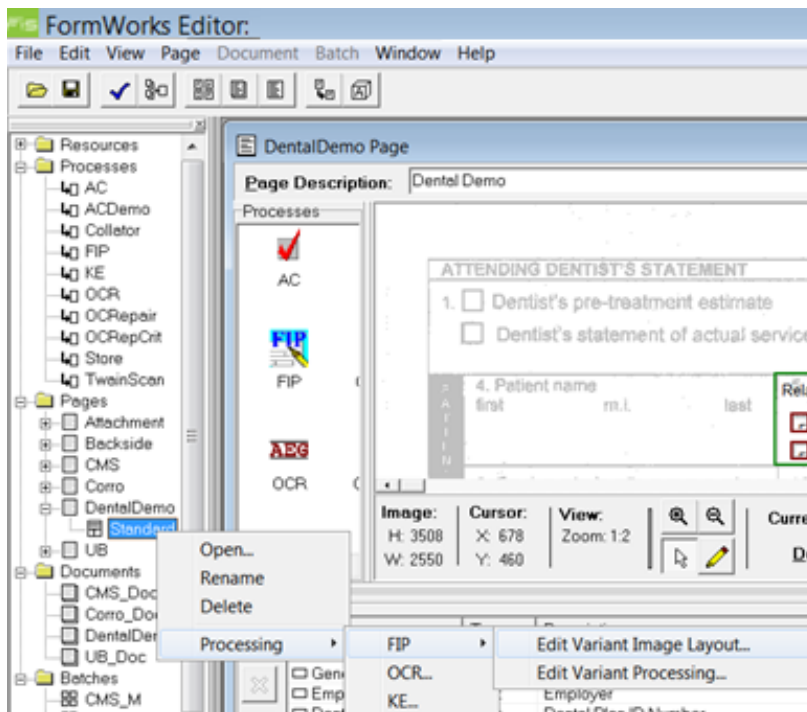
Option	Description
Reserved	

FIP: Variant Image Layout

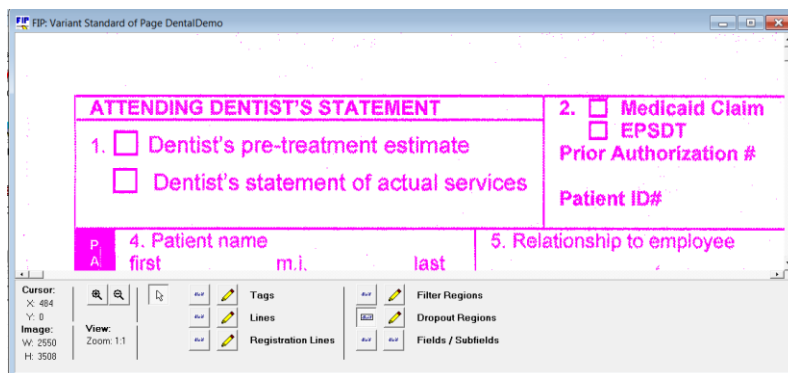
In addition to defining field mapping for imaged fields (see [Defining Bounding Rectangles](#), each Page Variant must be further defined for FIP processing. The lines and regions specified on the layout allow the FIP worker to identify pages and register the data by “fitting” a scanned image into the layout.

For example, users must specify which lines or areas on non-dropout forms need to be removed, and where data will and will not appear on a dropout form.

- Right-click the Variant Name, select Processing > FIP > Edit Variant Image Lay-out.



- The variant image window opens.



The image can become quite cluttered when viewing all of the defined areas on the form. Each option has an on/off toggle button so users may selectively view each defined item. (In the case of Fields/Subfields, either or both can be turned on or off.)

- For each option, select the drawing tool  and then drag or select to define the item.
- To switch back to the regular mouse pointer, select the arrow button .

Important: Beside each drawing tool is an On/Off button that when selected activates all Tags, Fields, Subfields, Lines and Regions currently within the form . To turn off this function the user must select the arrow tool then select the On/Off button .

Color Coding:

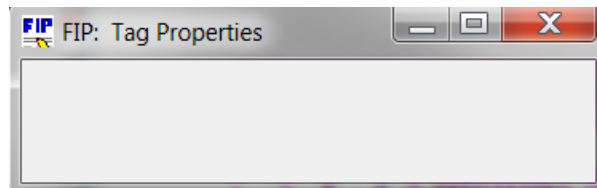
- Blue: Lines – (To be removed by FIP)
- Green – Registration Lines, Dropout Regions
- Red – Currently selection item and region
- Dark Blue – Tags (areas to be removed by FIP)
- Dark Purple – Filter Regions

How to Define Non-Dropout Forms

Tags

Tags are used to define areas of a non-dropout form that should be removed at FIP, such as a company logo, or other blocks of data.

- Select the **drawing tool** (pencil ) next to Tags and draw a rectangle around the area to be removed. Once the rectangle has been drawn, upon release of the mouse the FIP: Tag Properties displays.



Note: Close the Property box (there are currently no property settings for tags).

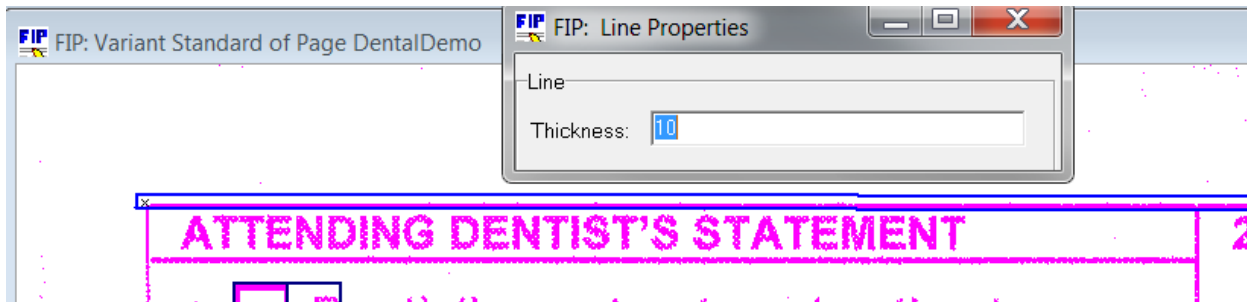
Lines

Lines are used to define the lines between fields to be removed from non-dropout forms.

- Select the **drawing tool** (pencil ) next to Lines.
- Draw the line around the area to be defined.

Note: In some cases, users may have to select and drag to define the line (especially for short line segments), but generally you can just select the line and the Editor will find the ends. To open the property box for any defined area, simply select in the area on the image layout.

Within the Properties box, FWEEditor automatically sets a line width (in pixels). As a best practice, it is recommended that the default setting be accepted. If after processing has begun, it is recognized that the tagged lines are not dropping out as expected, return to the image and modify the pixel value accordingly until the desired result is achieved.



- Close the Line **Property** box.

Deleting Tags And Lines

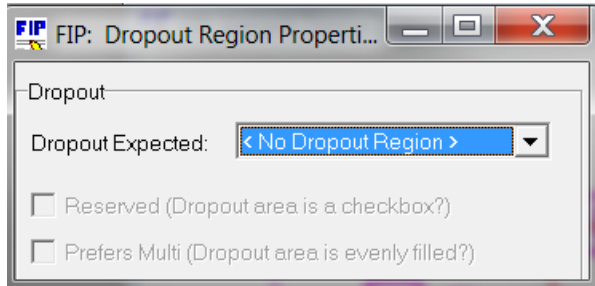
- Close any open **Property** boxes.
- Click the selection **tool button**,  (if necessary), to return to the regular mouse pointer.
- Select the **on/off**  **button**.
- Click the item to be removed. The item's boundary turns red.
- Press **Delete**.

Important: Upon selection, many fields will display the applicable Property box. To delete tags or lines simply press the Delete key to remove the desired item. Both the tagged item and the Property box will be removed upon the user pressing the Delete key.

Dropout Regions

For complex dropout forms, where registration lines are unreliable, use Dropout Regions to define where data will or will not be found on the form. The dropout region is usually defined as an area slightly smaller than the field box, tightening the region to include just the area in which data will appear (not field names or instructions).

- Select the **drawing tool** (pencil ) next to Dropout Regions.
- Draw the line around the area to be defined.
- The FIP: Dropout Region Property box window displays.



Populating the "Dropout Region Properties" field with the selectable values has the effect and action defined.

Dropout Expected

- **May Be Empty** – These areas are less precise, in terms of being used to register the form. These areas may be defined as a grouping of several fields; for example, a dropout region on a HCFA can include Insured Name, address, city, state, and zip in a single may be empty area.
- **Will Not Be Empty** – These are fields that are required on the form. FIP can reliably use these areas in matching up data on a scanned image with an area known to contain data in the layout.
- **Will Be Empty** – Parts of the form that should never contain data. Areas may be defined at the sides of the form as "Will be Empty" so that if FIP sees data in the areas defined.

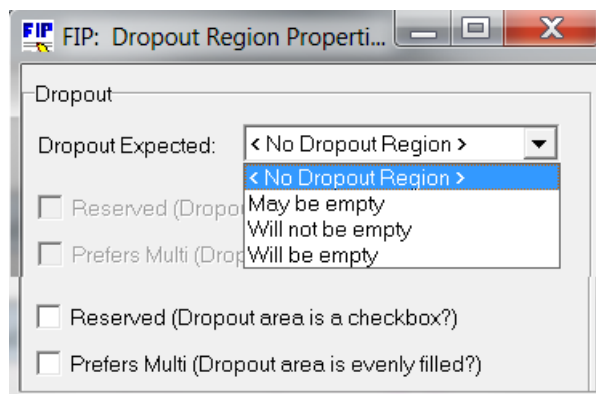
Note: The system will push the form left or right to fit into the field boxes.

If the provider completes the form with all the data printed slightly to the right, a "will be empty" area tells FIP that the area to the right of the field boxes should be empty, so it brings the data into alignment to fit in the defined field boxes.

- **Reserved (Dropout area is a checkbox?)** – If the dropout region defines a checkbox, select **Reserved** (Dropout area is a checkbox).
- **Prefers Multi (Dropout area is evenly filled?)** – If the dropout region defines an evenly filled area, select **Prefers Multi** (Dropout area is a checkbox).

To Define Dropout Regions

- Select the **pencil**  next to the desired Dropout Region type.
- Drag a box around the dropout area.
- The Dropout Properties box opens. Specify whether this region Will be Empty, Will Not be Empty, or May be Empty.



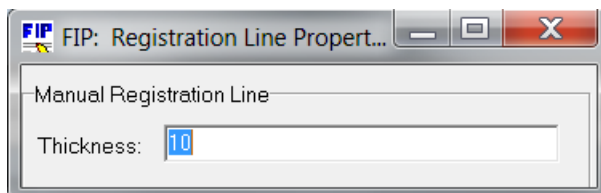
Other Form Definition Options

Registration Lines

Registration Lines define boundaries or landmarks on a form are used to assist FIP in identifying a form when processed. For example, registration lines at the corners of a form may be defined in order to identify the outer boundaries of the data.

Note: For complex forms, such as HCFA, UB-92, and most ADA Dental claim forms, use Dropout Regions to define the location of data on the form.

- Select the drawing tool  next to Registration Lines.
- Click on a line in the form, or drag to draw a line. A property box opens.



Within the Properties box, FWEditor automatically sets a line width (in pixels). As a best practice, it is recommended that you accept the default setting. If after processing has begun, it is recognized that the tagged lines are not dropping out as expected, return to the image and modify the pixel value accordingly until the desired result is achieved

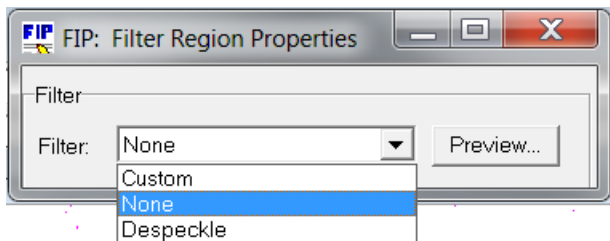
- When completed, close the property box to save changes .

Filter Regions

Use Filter Regions to specify small image cleanups where necessary. A variety of cleanup options are available to enhance image quality in the filter area.

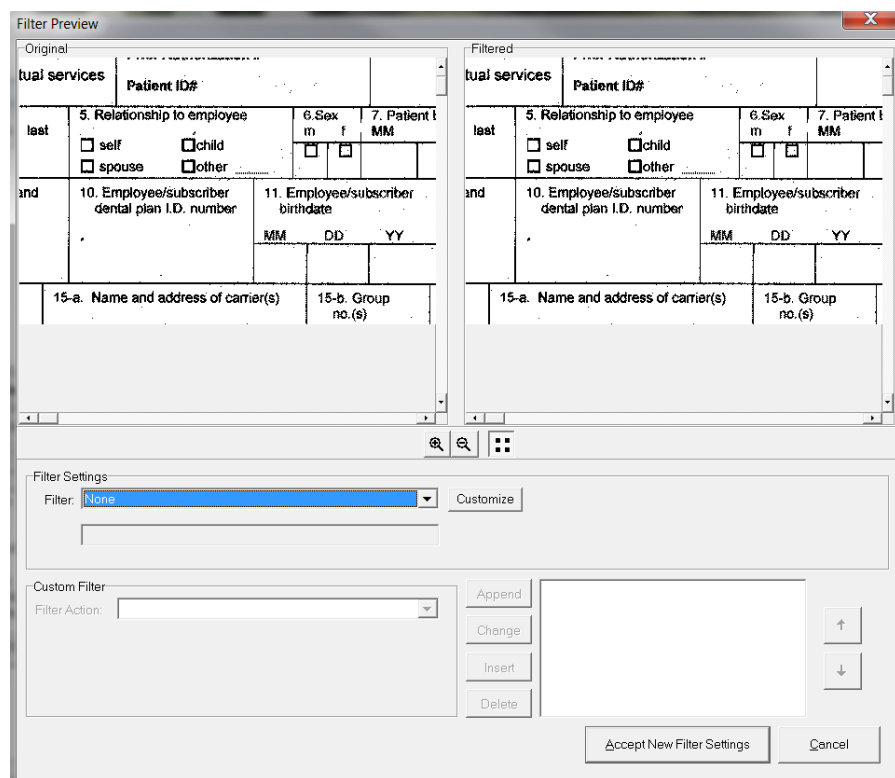
- Select the **drawing tool** (pencil) next to Filter Regions.
- Drag a box around the area to be filtered.

- In the property box, select **Despeckle**, **Custom** or **None**



- Select **Preview**
- The Filter Preview box is displayed. The left window (Original) displays the image as it currently is. The right window (Filtered) previews to the user the image with the Chosen Filter Settings applied.

Filter Preview



Note: The user may within the Filter Preview window change the Filter Settings mode. Dependent upon the selection, the Custom Filter displays the applicable Filters.

Selection of the Customize button will by default change the Filter Settings to Custom

Filter – None

No filter(s) applied.

The screenshot shows the 'Filter Settings' dialog box. The 'Filter' dropdown is set to 'None'. The 'Customize' button is visible. Below, the 'Custom Filter' section is empty, with 'Filter Action' set to 'None'. Buttons for 'Append', 'Change', 'Insert', and 'Delete' are present, along with 'Accept New Filter Settings' and 'Cancel' at the bottom.

Filter – Custom

Available for selection are a variety of enhancements for Filter Action and two selections for Filter Mode.

The screenshot shows the 'Filter Settings' dialog box with 'Custom' selected. The 'Filter' dropdown is set to 'Custom', and the text field below it contains 'sp,0.sp,1.sp,2.'. The 'Custom Filter' section shows 'Filter Action' set to 'Spot' and 'Filter Mode' set to '1 x 1'. The list of filter actions includes 'Spot (1 x 1)', 'Spot (2 x 2) (Weak)', and 'Spot (2 x 2) (Strong)'. Buttons for 'Append', 'Change', 'Insert', and 'Delete' are present, along with 'Accept New Filter Settings' and 'Cancel' at the bottom.

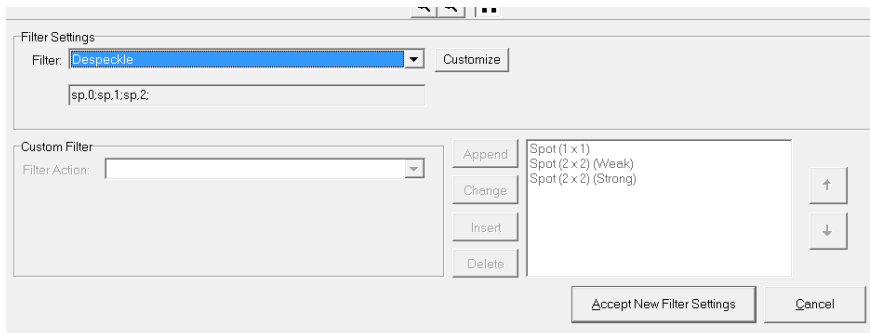
Filter Action	Filter Mode
- Spot - Spot - Dilation - Erosion - Majority - Majority Dilate - Majority Erode - Vertical Bump - Horizontal Bump	8 Neighbor 4 Neighbor 8 Neighbor

- Spot – Reduces noise (spots).

- **Dilation** – Generally to increase the area of the character(s). Used to eliminate noise in the window or to increase the thickness of the characters.
- **Erosion** – Generally, to reduce the area of the character(s). Used to eliminate noise in the window and/or to decrease the thickness of the characters.
- **Majority Dilate** – To increase the area (thickness) of the character(s) as a whole (Dilation x 2).
- **Majority Erode** – To decrease the area (thickness) of the characters as a whole (Erosion x 2).
- **Vertical Bump** – Useful in instances where when parts of a character come close to vertically touching each other, or occasionally touch, resulting in an unreadable character. The filter "bumps" the parts of the character vertically.
- **Horizontal Bump** – Useful in instances where when parts of a character come close to horizontally touching each other, or occasionally touch, resulting in an unreadable character. The filter "bumps" the parts of the character vertically.

Filter – Despeckle

Select **Despeckle** to remove speckles form images, thus reducing OCR error rates.



Despeckle Custom Filter Area

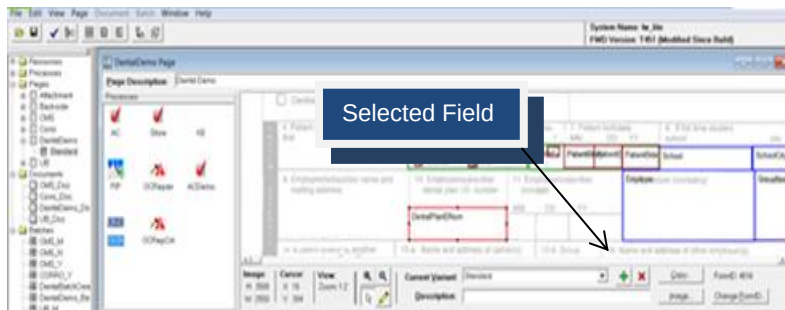
- Select a **Filter Action** and **Filter Mode** from the drop down lists.
- Click **Append** to add this filter to the list at the right side of the window. The preview area now shows the original image on the left, the filtered image on the right.
- Select and append any additional filter actions.
- Click **Accept New Filter Settings** to save changes.
- After the desired Filter Settings have been configured, select **Accept New Filter Settings**. The system returns the user to the FIP: Filter Region Properties window.
- Select **Close Window** to save changes .

Please refer to FIP documentation for more information about specific filters and their actions.

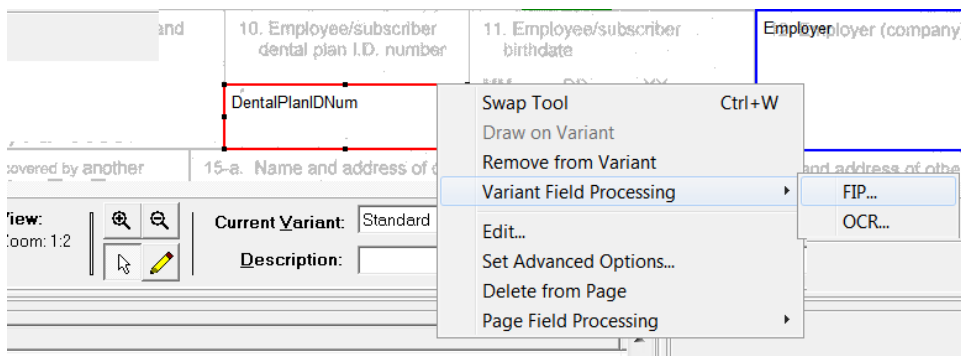
FIP: Variant Field Processing

There are two ways to open the Variant Field Processing window:

- The first way to open the Variant Field Processing window is to open the page to be worked, then select the field for which the Variant to be modified.

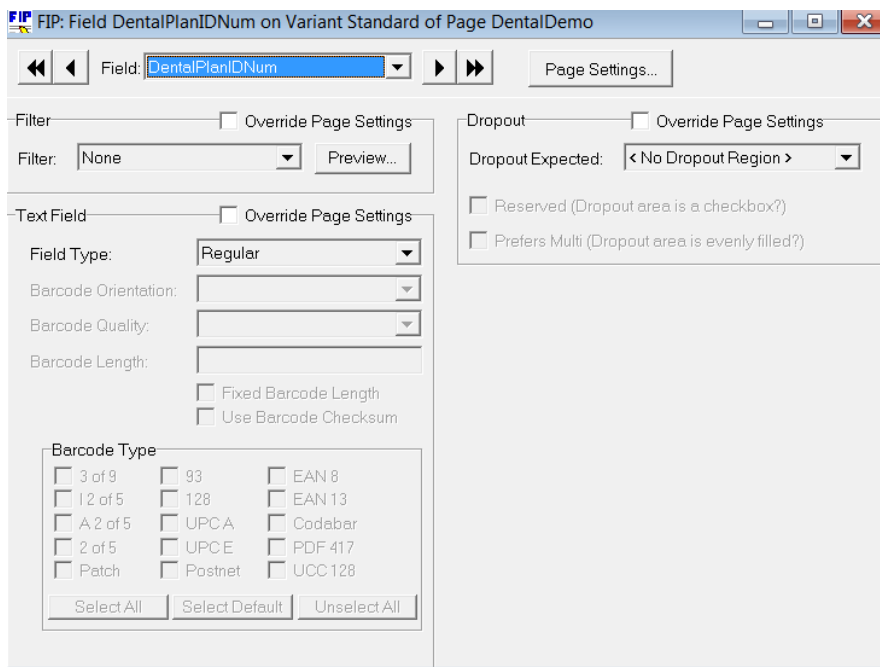


- Right-click on the field and select Variant Field Processing>FIP.

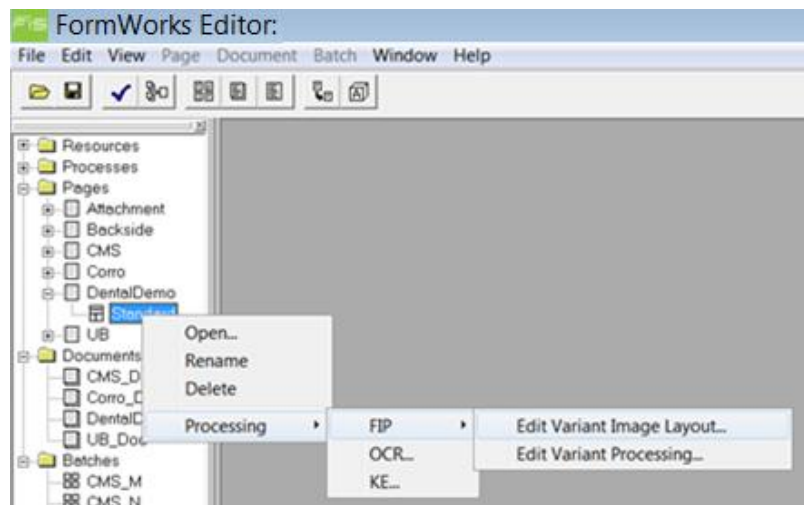


Note that the menu has separate options for both FIP and OCR. As a best practice, generally, options should be configured the page level.

- The FIP Variant dialog box is displayed and is for the variant listed in the selected field chosen in page window.

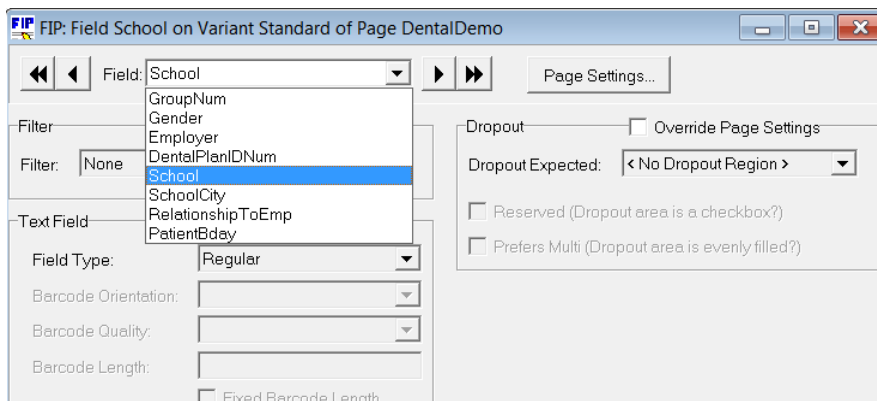


- The second method to access the Variant Field Processing window is to select the **Variant Page** from the Tree.
- Next, select **Processing>FIP>Edit Variant Image Layout**.



- The FIP Variant dialog box is displayed and is for the variant listed in the selected field chosen in page window.
- From the Variant Layout window, select the desired field to set FIP field processing options. If a tag, line, or dropout region set is needed, assign a property box for the item.

Note: A best practice to verify that the field display is on is to turn the display of the items off and on (effectively resetting the field display).



Warning: The configuration windows for FIP Variant Field Processing and FIP Page Field Processing are very easy to confuse. At the variant level, users may make changes only after selecting Override page settings. In the event options are set for a field at both the page and variant levels, the Editor uses the variant level setting.

As a best practice, users should generally make changes at the page level, so that all variants are affected (including variants that might be added at a later date).

With the exception of the Page Settings button, the options available in the FIP:Variant Field Processing dialog box (as applicable to Field(s)) are identical to those in the FIP:Page Field Processing (applies to Page) box.

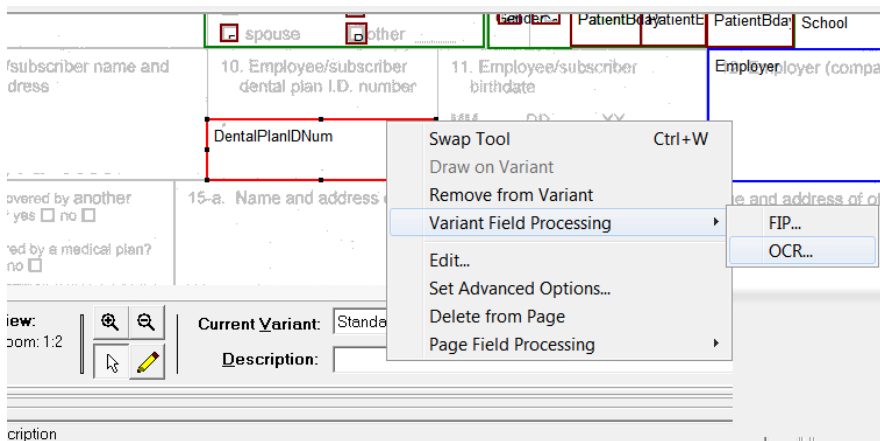
OCR: Variant Field Processing

There are two ways to open the OCR: Variant Field Processing dialog box:

The first way to open the Variant Field Processing window is to open the page to be worked, then select the field for which the Variant to be modified.



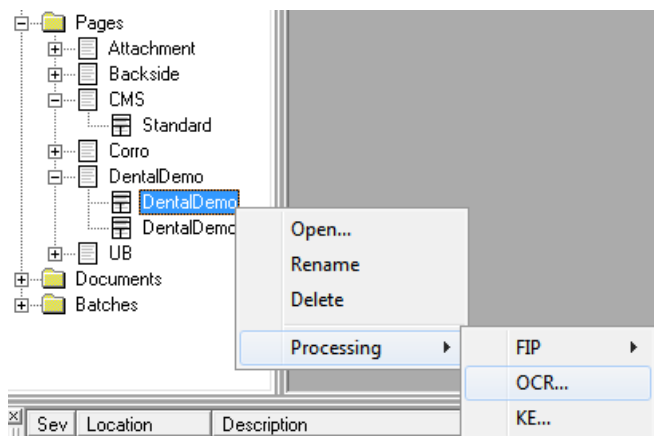
- Right-click on the field and select Variant Field Processing > OCR.



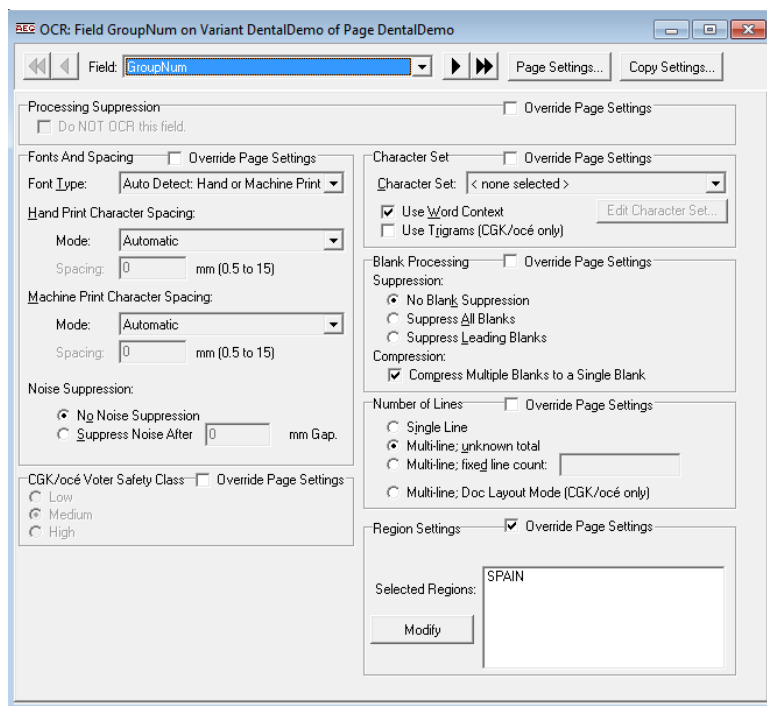
Notice that the menu has separate options for Variant Field Processing and Page Field Processing. As a best practice, options should generally be set at the page level.

In the Region Settings section, users can select Override Page Settings checkbox only if the system setting is overridden at the Page Field level.

The alternate method for access the OCR Field Variant screen is to right-click the **Field Variant**, select **Processing**, then select **OCR**.



The OCR Variant Page displays



Note: With the exception of the Override Page Settings button, the options available in the OCR:Field Variant Processing dialog box(applicable to the Variant) are identical to those in the OCR Page Field window.

To reduce duplication, the options replicated in both OCR Field Variant Page and OCR Field Page configuration windows are defined within the OCR Page Field Processing section of this document.

Configuration of Document and Batch Processing

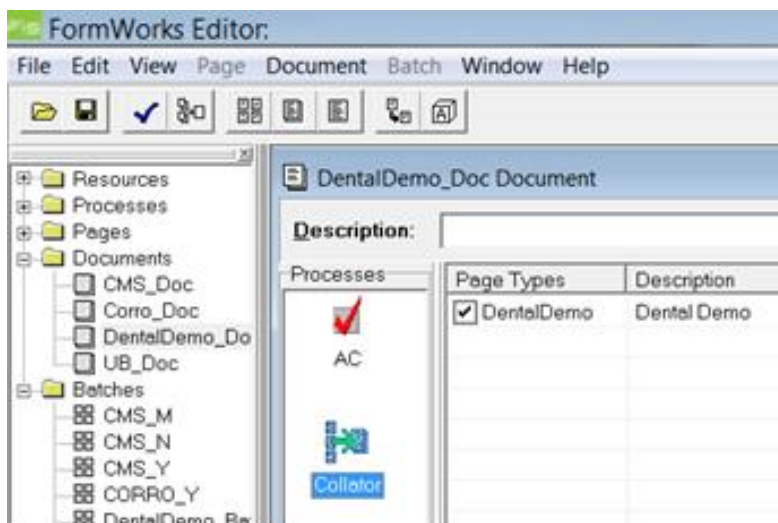
Important: PCMs are updated frequently. Every effort is made to keep this document current with the most recent versions of each PCM. See the PCMs listed in [Documented Components](#) before following any configuration instructions in this chapter.

Collator: Document Processing

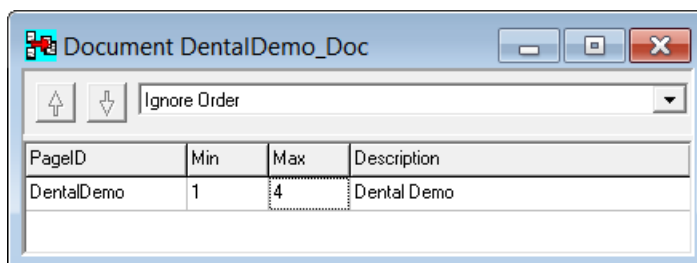
If the batch level Collator configuration for Document ID is set to Compare to Document Definitions for the batch type, then the batch must be configured at the document level.

To setup, configure the Collator with the minimum and maximum number of pages allowed for each page type in the document.

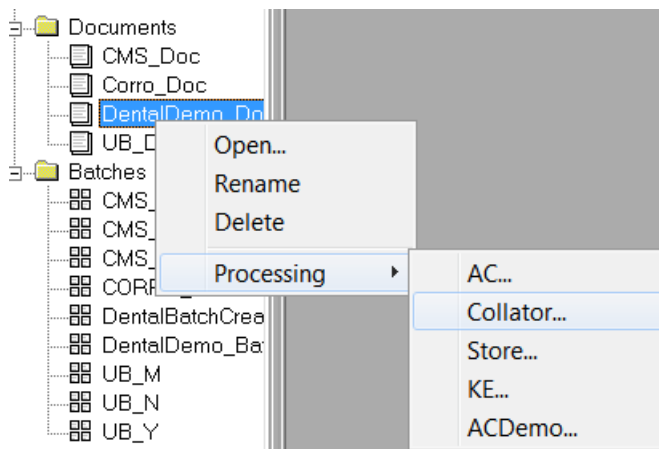
- Display the Document Collator window by selecting the **Document > Collator**.



- The Document Collator window is displayed.

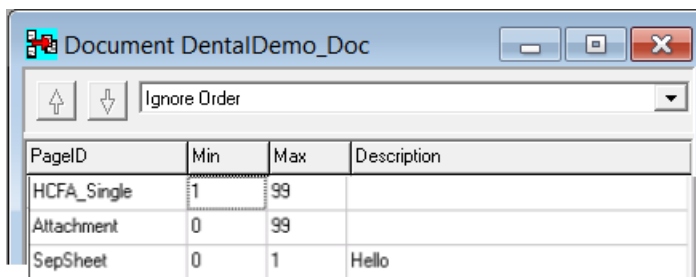


- Or, if the document is not open, right-click the **Document** name in the FWD tree, select **Processing**, then select **Collator**.

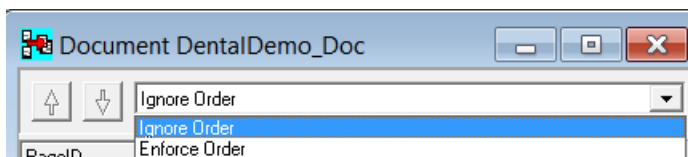


- In the Collator document configuration window, select the **cell** to be changed then enter a **new value** (Min/Max value).

Note: When defining the allowable numbers of pages, consider all possible combinations of pages. For example, the following example allows for HCFA single-page documents, multipage (up to 99 pages) documents, and single or multipage with or without attachments.



- When multiple fields are present, the user may select to either **Ignore Order** or **Enforce Order** of the fields populating the configuration window.

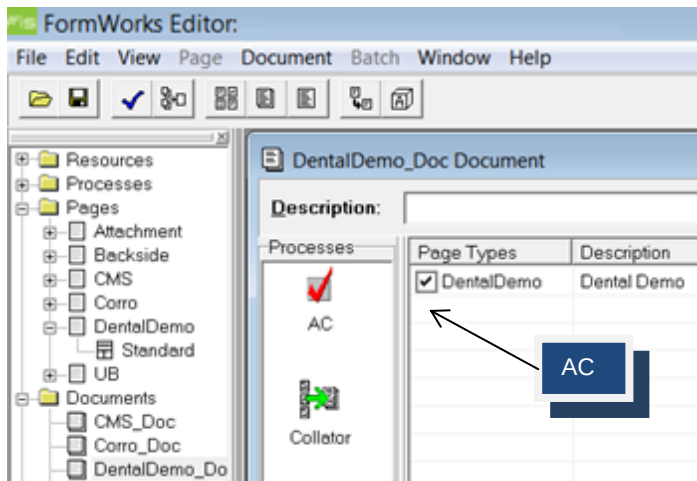


- Close  the window to save changes.

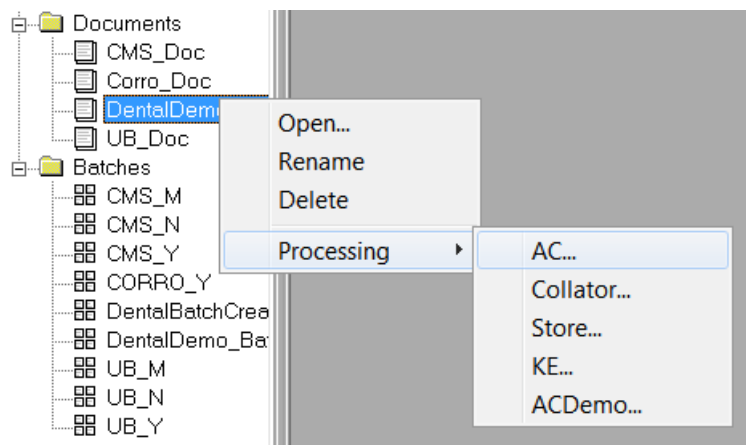
AC: Document Processing

To Open the AC Configuration Window

- If the document window is open, double-click AC in the processes panel of the document window.



- If the document is not open, right-click the document name in the FWD tree, select Processing, then select AC.



Using the instructions within the [AC: Page Processing](#), (if required), create any document level AC/DV/KE rules needed.

Note: Even if document rules are not present, as a best practice, the default document rule list should include the "_IEvalPages" rule.

The AC interface for document processing is identical to the one for setting page level rules illustrated in [AC: Document Processing](#).

[illegible]

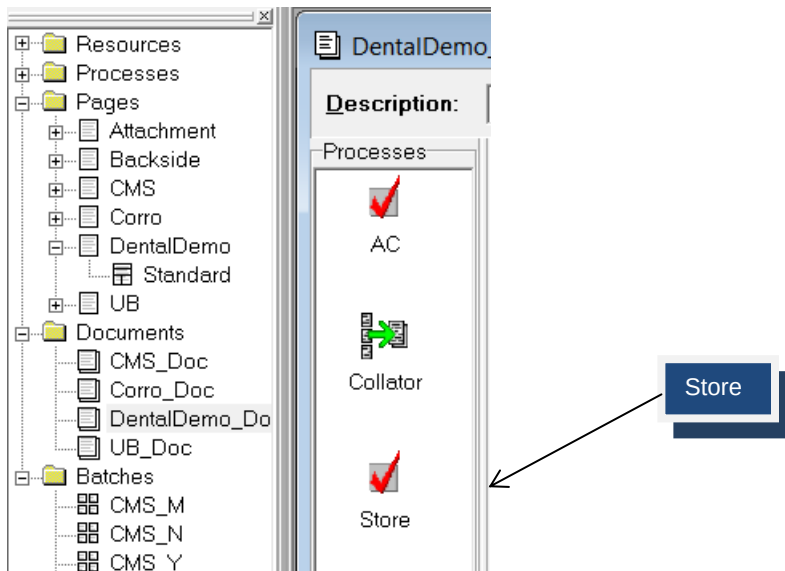
Important: Before creating or modifying any rules be familiar and experienced with the AC Rules Tester.

The AC Rules Tester enables users to test a rule or a user-defined function by specifying inputs to the rule, and executing the rule to view the results. This functionality is integrated with the FormWorks Editor. Contact your FIS representative for details.

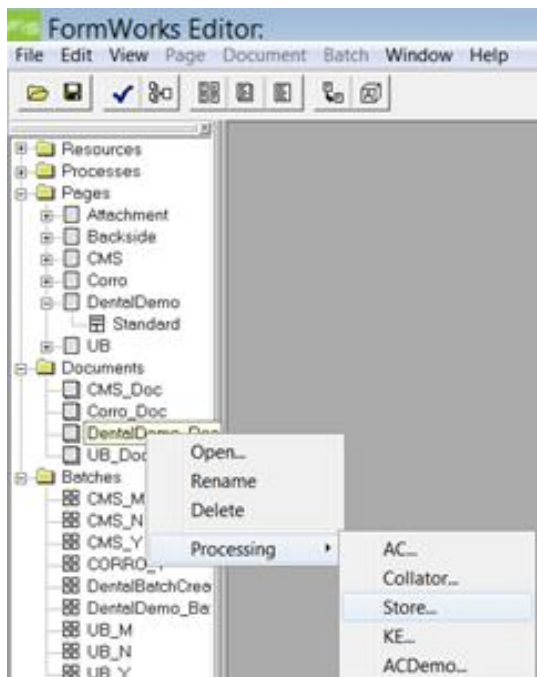
Store: Document Processing

To Open the Store Configuration Window

- From an open document window, double-click **Store** in the processes panel of the document window.



- If the document is not open, right-click the Document Name in the FWD tree, **select Processing**, then select **Store**.



Using the instructions at [AC: Page Processing](#), create and store rules for the document.

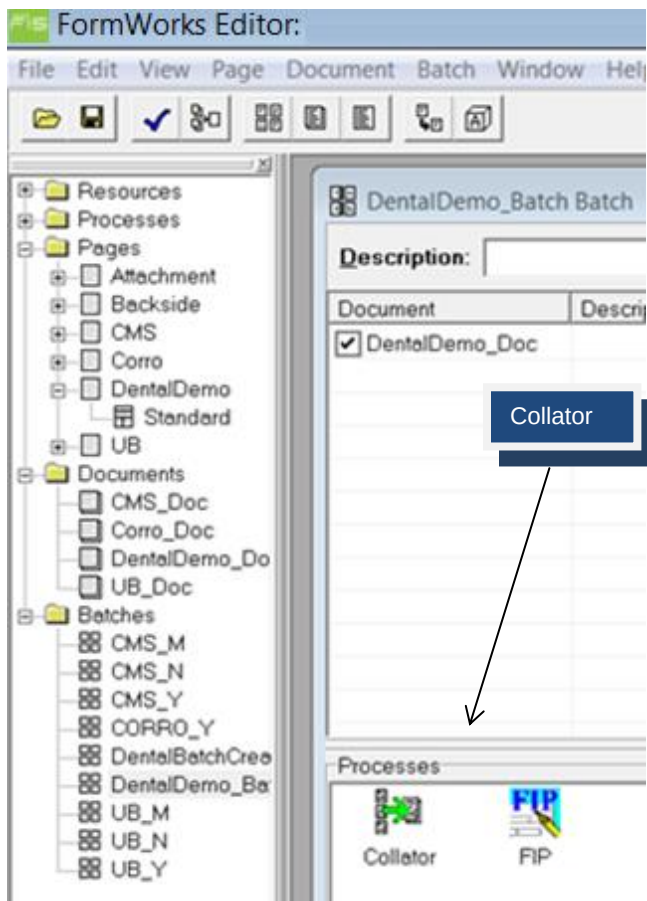
Note: The Store interface for document processing is identical to the AC rules interface.

Collator: Batch Processing

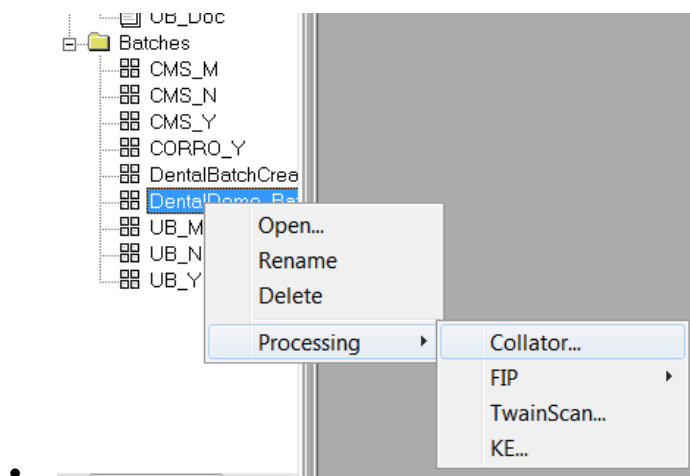
Most Collator options are set at the system level (see To [Configure the Collator Process](#)). However, on occasion, batch types may exist for which a different collation technique is preferred. For this scenario, the options set at the batch level override the system level settings for the selected batch.

To Open the Batch Collator Configuration Window

- If the batch window is open, double-click **Collator** in the processes panel.



- If the batch window is not open, right-click the **Batch Name** in the FWD tree, select **Processing**, then select **Collator**.

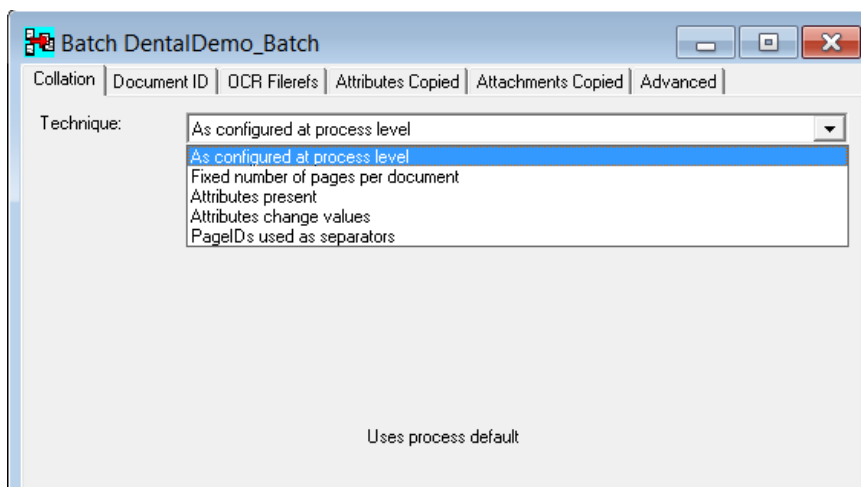


Batch Collator Tabs

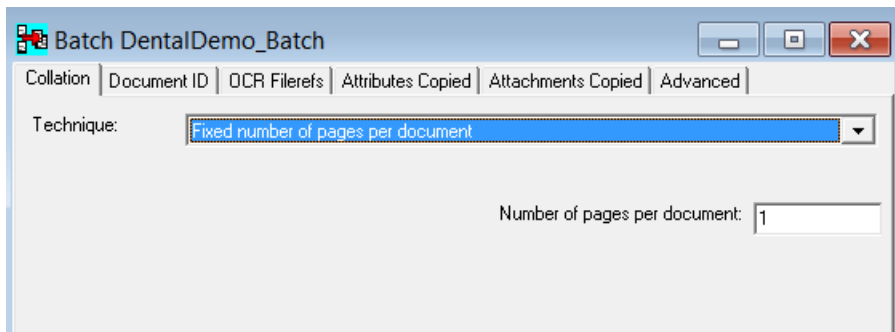
The Collator configuration window consists of six tabs.

Collation

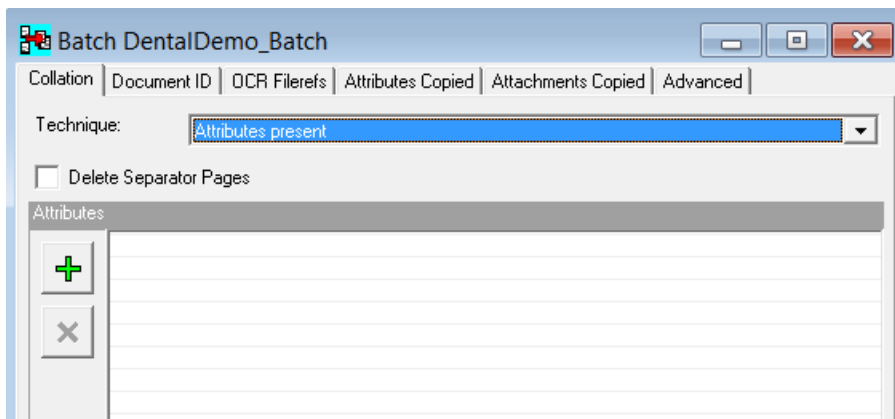
- As configured at process level – Uncollated pages as determined at the process level are moved into documents consisting of a fixed number of pages. If there are not enough pages to make an entire document, the pages available are placed into a document.



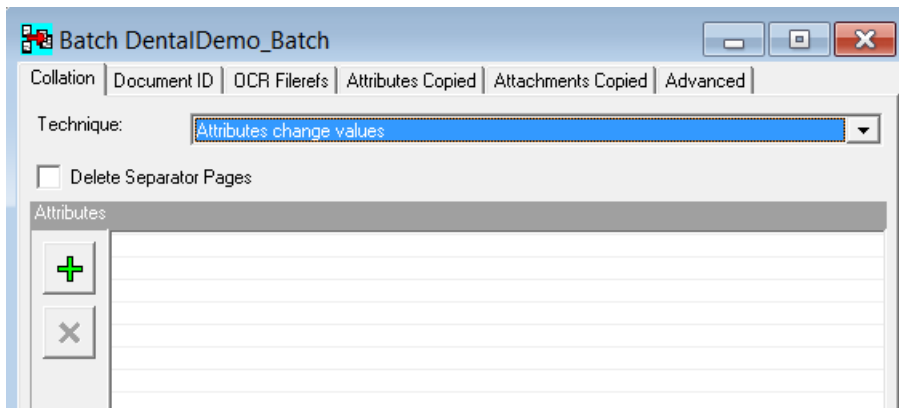
- Fixed number of pages per document – Uncollated pages are moved into documents consisting of a fixed number of pages. If there are not enough pages to make an entire document, the pages available are placed into a document, but no error is set. This technique should only be used when you know that all documents in the batch contain n pages, and are sorted correctly.



- **Attributes present** – Collator looks for the presence of one or more specified attributes to determine if a page is a separator page, and creates a new document each time a separator is found. Each successive page not having the specified attribute(s) is moved into the new document. The separator page can be moved into the new document or deleted based on the attribute found.

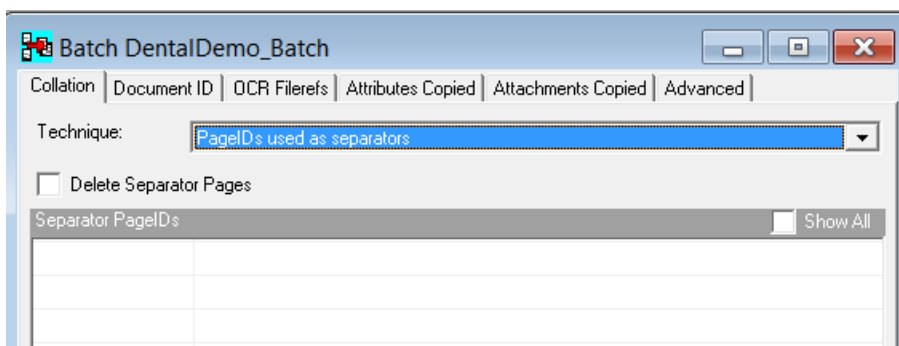


- **Attributes change values** – Collator compares the values of an attribute or set of attributes, usually DCN. The first uncollated page is assumed to be the first page of a document; a new document is created when the value of the specified attribute(s) changes. All pages with the same value for all specified attributes are moved into a document. The separator page (if there is one) can be deleted based on the attribute that changed.

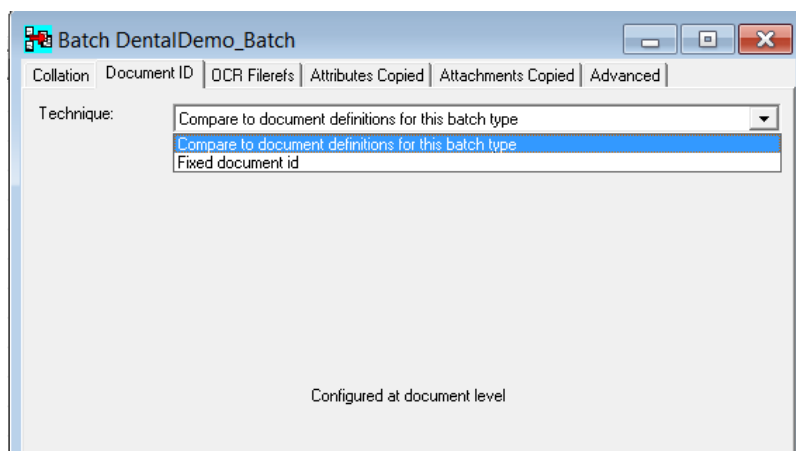


- PageIDs used as separators – The collator is configured with a set of PageIDs that are separator sheets. When it finds a page with a separator PageID, collator starts a new document. All successive pages that are not separators are moved into the document.

Note: Batch-level option only. Any Uncollated pages that occur before the first separator sheet in the work record will remain Uncollated.

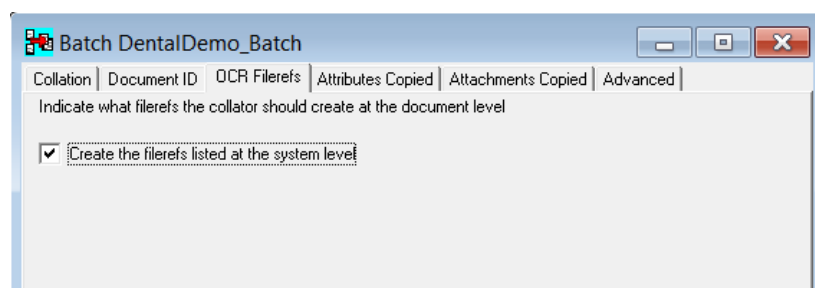


- DocumentID – After pages are assembled into documents, the Collator assigns a Document ID (DocID). One method of assigning the DocID is to compare each document to document templates. By comparing a document to document templates an algorithm is used to compare the pages (and quantities of those pages) in the template with the pages (and quantities) in a given document. These comparisons are based on the number of matching pages, extra pages, missing pages, and quantities of each page. Collator then assigns the DocID of the template to the most closely matching document in question.
 - Compare to document definitions for this batch type – As a best practice, **Compare to document definitions for this batch type** is normally selected. This selection uses the Collator's document level definitions of pages allowed in the document.
 - Fixed document id – All documents in the batch are of the same document type.



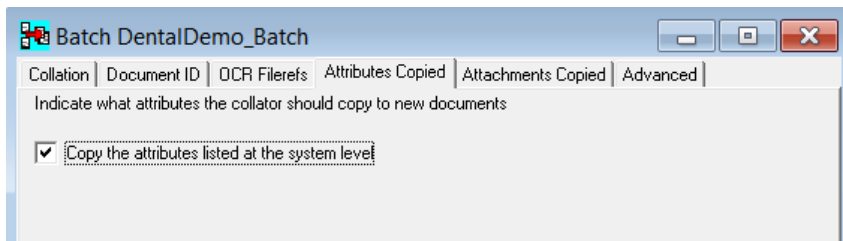
- OCR Filerefs – Indicate what filerefs the collator should create at the document level.
 - Create the filerefs listed at the system level – If all of the batch types in the system are to have the same OCR filerefs, it is a best practice to set the Create the filerefs listed at the system level option. This way, if changes are needed at a later date, the change will be required only once.

However, if one batch should utilize a different fileref, specify that fileref at the batch level on the OCR Filerefs tab (below).

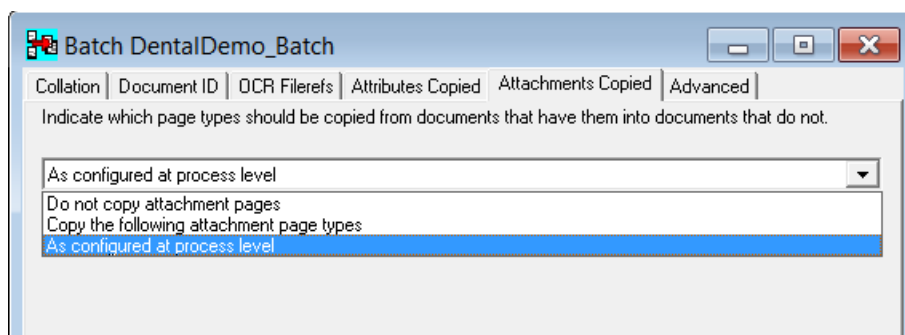


Note: Filerefs must first be defined as Resources to appear in any filerefs list. If the desired fileref does not appear when Show all is selected, the resource has not been defined.

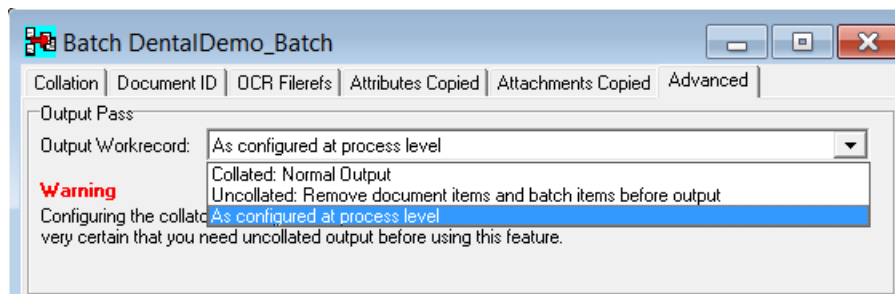
- Attributes Copied
 - Copy the attributes listed at the system level – Copies attributes defined at the system level.



- Attachments Copied
 - As configured at process level – Copies from the process level.
 - Copy the following attachment page types – Copies attachment and page types.



- Advanced
 - As configured at process level – Output Work record is configured at the process level.
 - Uncollated: Remove document items and batch items before output – For Uncollated items, removes items before output (documented and batch items).



Warning: Configuring the collator to output Uncollated work records may have unexpected side effects. Be very certain that uncollated output is required before using this feature. The FormWorks 5.0 systems currently cannot handle uncollated pages. Uncollated pages are ignored by the system. Users must select a collation technique that is reliable (identifies all pages as part of a document) for the system being configured.

A discussion of available [Collation Techniques](#) may be found earlier in this document within the [Configuring A Process](#) section. To use the system level default for the current batch, select **As configured at system level** from the selection list.

For healthcare systems processing HCFA or UB single-page documents in batches separate from multipage and attachment batches, and using Attributes Change Values collation, do not delete separator pages at the system level. Set this option at the batch level for the appropriate batch types only.

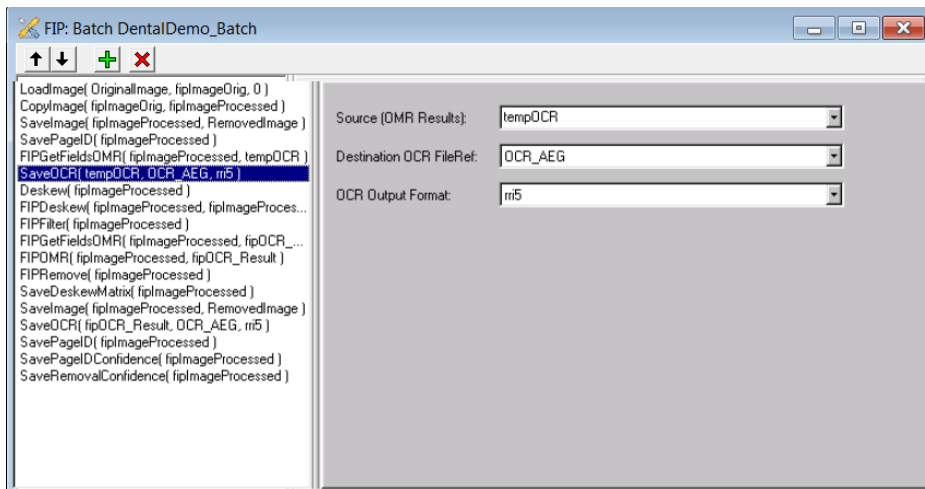
FIP: Batch Processing

Currently, there are two parts to FIP batch level configuration: a FIP Batch Script and [FIP Batch Processing \(PageID\)](#).

- Script – Is a list of steps FIP performs in processing a given batch.
- PageID – Is the method of page identification for the batch.

FIP Batch Script

Every batch must have a script to tell FIP how to process the batch. The majority of systems may utilize the same script (processing dropout forms) for all batches. The generic script most commonly used is shown below.



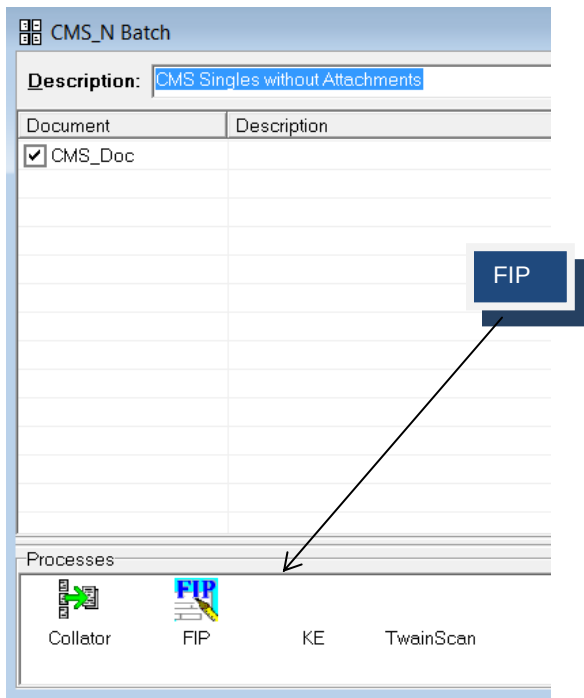
Note: PageID is not performed here; for this example the PageID is determined by the batch type (HCFA single, UB multi, and so on).

If a batch in the system may be comprised of different types (kinds) of forms (for example, both HCFA and UBs), or there are multipage forms where page 1 and page 2 are different forms that are not known to always be in the same order at scan time, one of the first steps in the script should be PageID. This PageID step uses the PageID method set in the FIP batch processing window ([FIP Batch Processing \(PageID\)](#)).

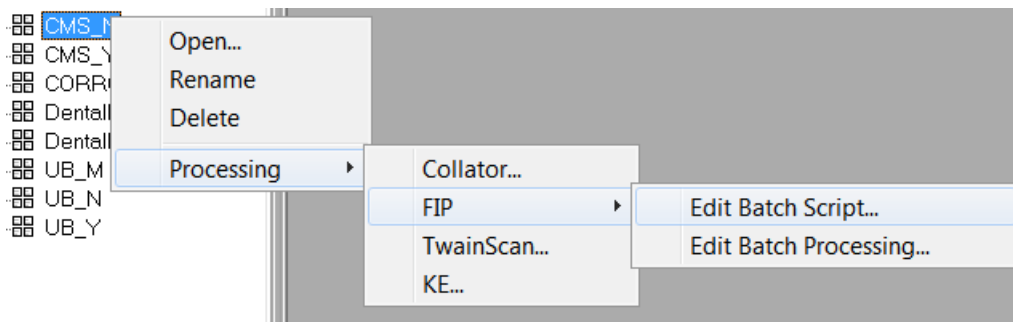
The actions that FIP performs depend on the options specified for FIP Variant Processing. For example, in the step “FIPDeskew” FIP deskews the image using the method previously specified. If a method has not been selected, this step in the script does not execute. See [FIP Variant Page Processing](#) for additional details.

To Open the Batch FIP Configuration Window (FIP Script)

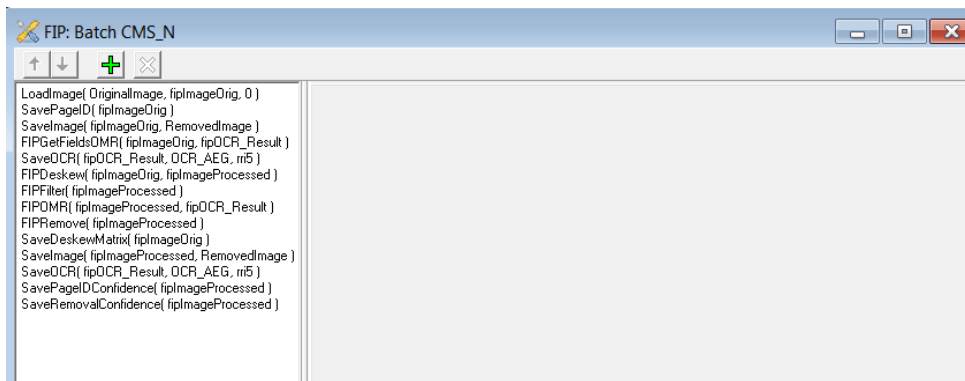
- If the batch window is open, double-click **FIP** in the processes panel



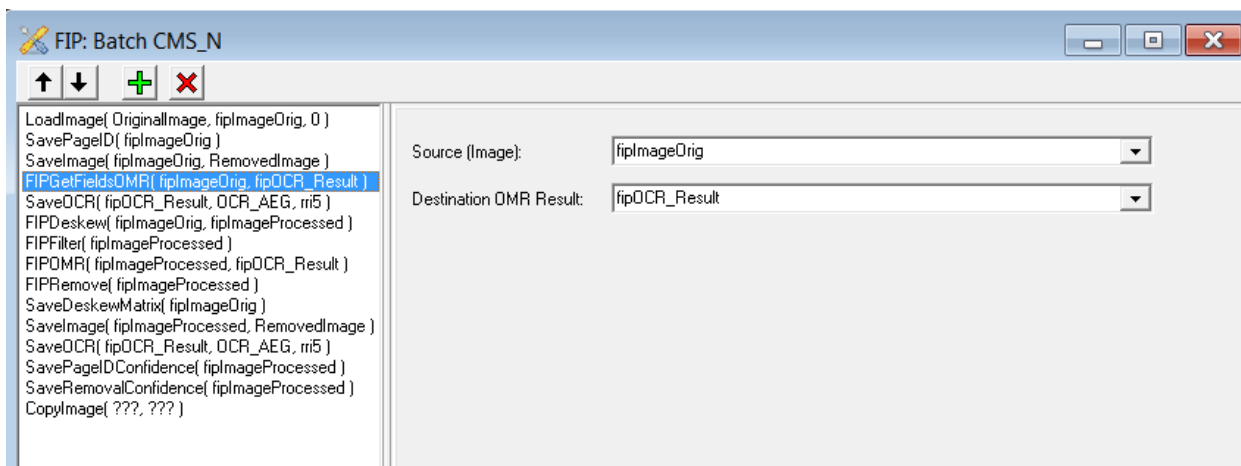
-
- If the batch window is not open, right-click the **Batch Name** in the FWD tree, select **Processing > FIP > Edit Batch Script**.



- Select Edit Batch Script, the FIP Batch window displays.



For each item in the script, parameters and attributes must be configured in the right side of the screen. For most items, at least the Source and Destination filerefs must be specified. The defined parameters appear in parentheses next to the steps. In the event parameters have not been defined, the parameter field displays question marks (see CopyImage in the graphic below).



Note: When a batch is copied from another FWD the batch script along with other processing specifications for the batch are also copied.

Description of Typical FIP Steps

- **LoadImage** – Load the scanned image from a source (usually OriginalImage) to memory (fipOriginal).
- **CopyImage** – Copy the image from memory to a temporary file (fipImageProcessed); this temporary file will be updated by subsequent steps.
- **SaveImage** – Saves the in-memory image to the specified fileref (usually RemovedImage).

Important: Be sure to include the “SaveImage” step to create a RemovedImage file. This file is required by other processing steps.

- SavePageID – Saves the PageID. Set by Scan in this example (or by a PageID step if the script has one) to the FormID attribute. Some workers, such as OCRRepair and KE, require a FormID.

Note that SavePageID does not actually perform PageID.

- FIPGetFieldsOMR – This step gets all fields that should be on the page, based on the PageID, and stores them in memory. T

Note: This step also initializes OMR (Optical Mark Recognition, or checkboxes) to an unread state.

- SaveOCR – Saves the fields obtained from FIPGetFieldsOMR to the OCR fileref (OCR_AEG).
-

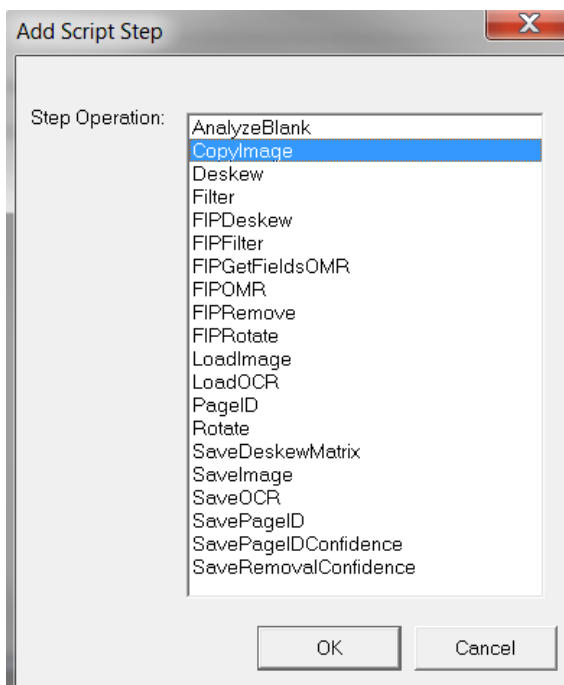
Important: As a best practice, make sure that script(s) include the steps “FIPGetFieldsOMR” (to get all the fields that should be on the page, based on the PageID), followed by “SaveOCR” (to save the fields to the designated OCR fileref).

- Deskew – Straightens the image as much as possible, regardless of form type.
- FIPDeskew – Deskews using the method specified for the variant in FIP Variant Processing. The source and destination are typically the same, writing the results to fiplImageProcessed.
- FIPFilter – Filters the image using the method and regions specified for the variant. The source and destination are typically the same, writing the results to fiplm-ageProcessed.
- FIPGetFieldsOMR – Same step performed prior to saving the OCR file; this step can be done twice for backup (if processing does not progress to this point an OCR file is still retained).
- FIPOMR – This step reads the OMR fields (checkboxes) from fiplImageProcessed and stores the results in memory (fiPCR_result).
- FIPRemove – Performs selective image feature removal based on the method and regions (tags, lines) specified for the variant.
- SaveDeskewMatrix – Stores a record of what actions were performed to straighten the image in FIPDeskew. For example, the number of pixels moved left or right, and the degree of rotation.
- SaveImage – Saves the processed image (fiplImageProcessed) to RemovedImage.
- SaveOCR – Saves OMR results from FIPOMR step to the OCR fileref (OCR_AEG).

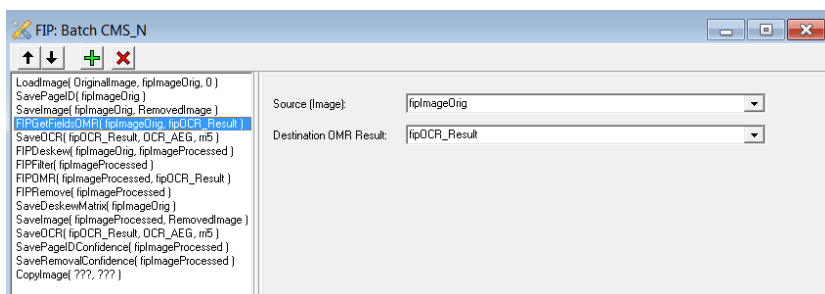
- **SavePageID** – Same as the previous step (SaveOCR); saves both the PageID and FormID attribute from fipIm-ageProcessed.
- **SavePageIDConfidence** – Records the confidence level at which the PageID was made; if the PageID was set at Scan, the confidence is normally 100%.
- **SaveRemovalConfidence** – Records the confidence level at which removal was performed.

To Add Steps to The Script

- To add a new step to the script, select the Add button .
- Select the new step from the Add Script Step window, and select OK.



- Specify the Source, Destination and in the right side of the window.



- Use the up and down arrow buttons  to place the steps in the correct order.

Note: New steps are automatically added to the bottom of the list.

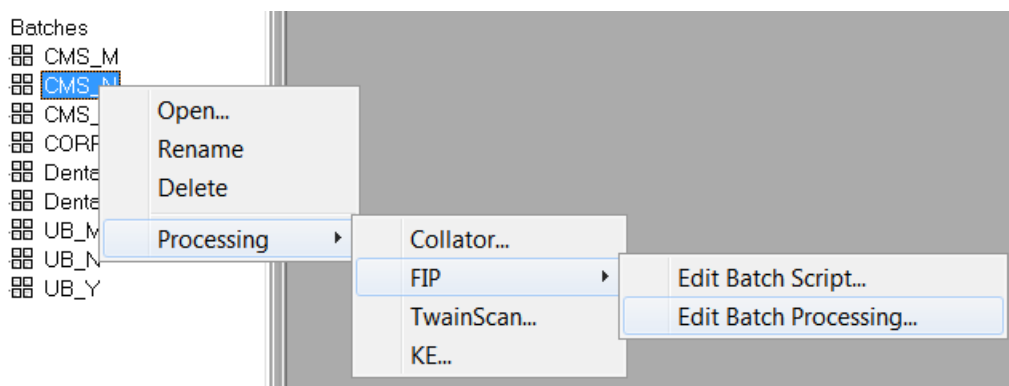
FIP Batch Processing (PageID)

If PageID is not determined at Scan, for example if the batch contains different kinds of forms, or documents with different page1 and page2 definitions, select the appropriate PageID method for the batch.

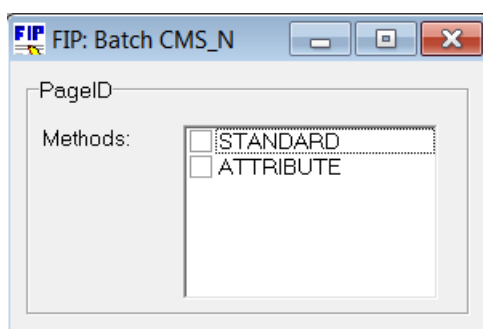
Note: For most healthcare systems using dropout forms, PageID is set prior to FIP processing. There are usually separate batch types for HCFA, UB, Dental, and so on. If FIP is not performing PageID, it is not necessary to choose a method here.

To Open the Batch FIP Processing Window (PageID)

- Right-click the Batch Name in the FWD tree, select Processing > FIP > Edit Batch Processing.



- The FIP Batch Processing window displays presenting with two options, Standard and Attribute.



- Click Close  to save and close the window.

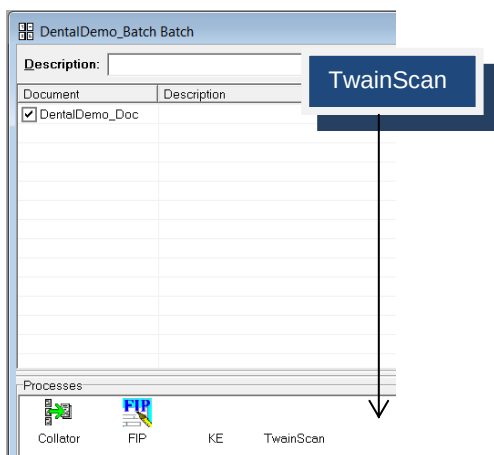
Scan: Batch Processing

A scan configuration must be set for each batch. If multiple batches have identical configurations, all scan configurations may be copied using the Copy Settings button on the General tab.

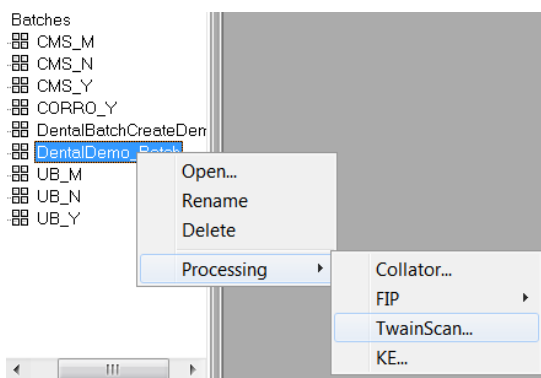
Scan worker maintains two work records while it is working on a batch: an internal work record that contains all information about a batch, including user attributes received from Inventory, and the output work record. A complete list of attributes in the internal work record can be found on RRI's internal website. The output work record is a subset of the internal work record, as configured on the Saved Attributes tab in Scan's configuration window.

To Open the TwainScan Batch Configuration Window:

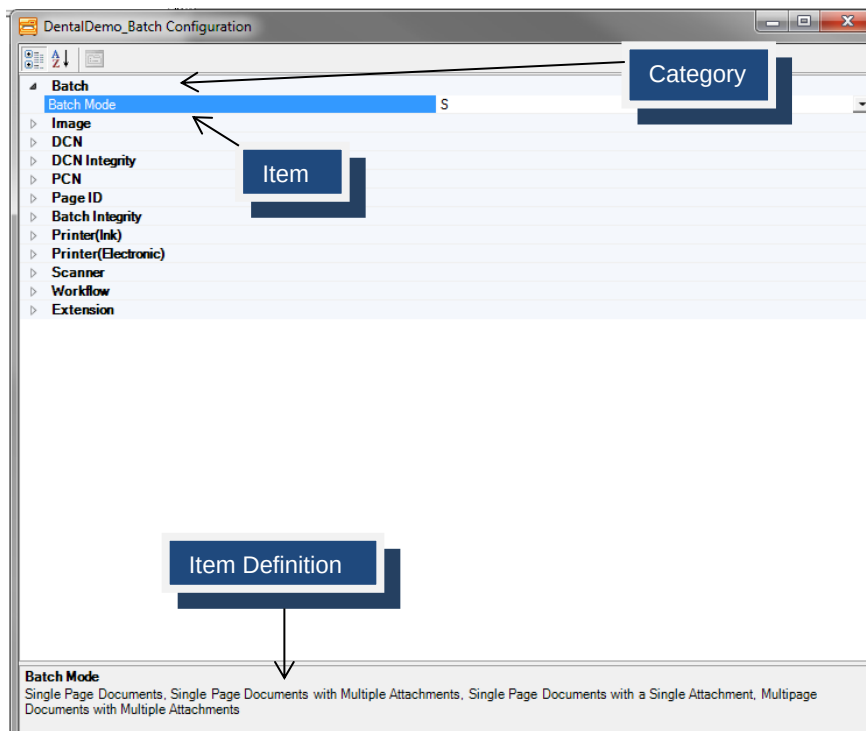
- If the batch is open, double-click **TwainScan** in the processes panel.



- If the batch is not open, right-click the batch in the **FWD tree**, select **Processing**, then select **TwainScan**.

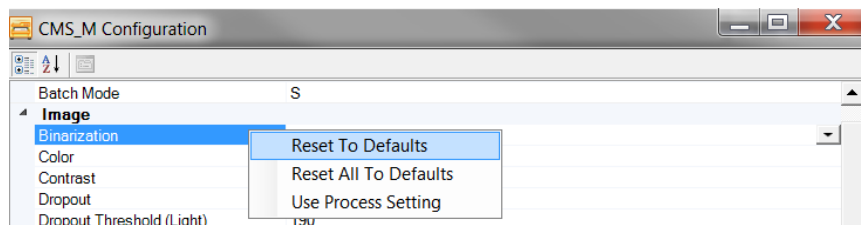


- Presented to the user is the Configuration window from which Items in the following categories may be configured.

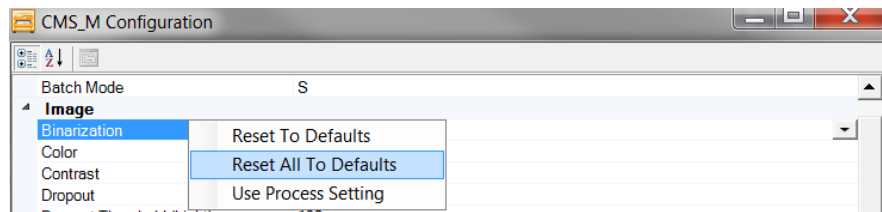


Note: Items within some categories are fixed and may not be edited.

- To edit a category within the Configuration window.
 - Select the Configuration category.
 - Right-click on an Item within the category.
 - Three options are presented to the user, Select one.
 - Reset To Defaults – Restores the individual item to its default.

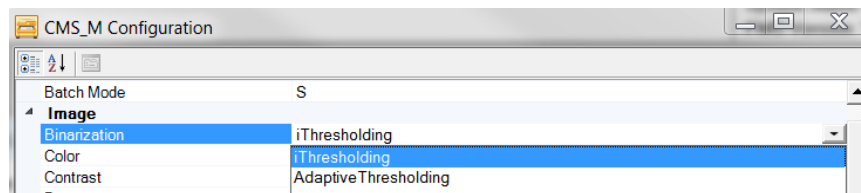


- Reset All To Defaults – Restores all items, all categories to their defaults.



Important: Restores all Categories to their default values. Once executed values may not be restored to their previous values.

- Use Custom Batch Setting or Use Process Setting – Where available, presents to the user a drop down box from which a selection may be made.



To use the Custom Batch Setting or Use Process Setting, select the Item and right-click the Item from the left column. The right column will present to the user a drop-down box from which to select or present a field for data entry.

Scan Batch Configuration Category/Item Definitions

Defined for each Item, (organized by Category) are the available Custom Batch Settings or Use Process Settings accompanied by a brief definition.

Important: TwainScan Process Configuration values may be modified from the Processes -> TwainScan -> TwainScan Process Configuration window.

Batch

- Batch Mode
 - S – Single Page Documents.
 - A – Single Page Documents with Multiple Attachments.

- A2 – Single Page Documents with a Single Attachment (used minimally).
- M – Multipage Documents with Multiple Attachments.

Image

- Binarization
 - Thresholding – Automatically determines the best threshold to separate text from a background.
 - Adaptive Thresholding – Allows manually setting the threshold.
- Color
 - True – Set to True to capture bitonal images and color. Color is generally bigger and slower
 - False – Set to False to only capture bitonal images.
- Dropout
 - Flag – The Dropout out respects the flag setting in the TWAINScan GUI for operator control.
 - True/False – Set to True to drop out (red) from image electronically or False to scan without dropping out. This overrides the TWAINScan GUI operator setting.

Note: Global GUI flag defaults may be changed in the config file.
- Dropout threshold (Light) – Custom configuration not available.
- Dropout Threshold (Regular) – Custom configuration not available.
- Page Sides
 - True/False – Set to True to capture the front and back of the paper.
- Paper Orientation –
 - Landscape – The fastest way to promote paper through the scanner. This option allows the paper to be scanned with its top oriented to the right when rotated to landscape. It is most compatible with scanner printers.
 - Landscape Top To Left – This is a legacy option less compatible with printers as printing must be done backwards when paper is pulled underneath.
 - Portrait – Portrait mode, vertical orientation vs. Landscape horizontal orientation
- Resolution (Bitonal) – Resolution of black and white images.
 - 200
 - 300
- Resolution (Color) – Resolution of color images.

- 100
- 150
- 200
- Binarization
 - iThresholding – Automatically determines the best threshold to separate text from a background.
 - Adaptive Thresholding – Allows manually setting the threshold.
- Threshold – If Adaptive Thresholding is selected, this determines the pixel gray color that differentiates text from background.
- Contrast – Custom configuration not available.

Note: In the event a value entered is not valid, the system will display a Property value is not valid message.

DCN

- Image Address (Fixed) Length
 - Values 1- 9, Length of the fixed part of the image address.
- Image Address (Fixed) Format
 - "<JDay,%s,YYJJ>" is the default value. Format string (ATTR_SPrintf) of the fixed part of the image address.

Note: ATTR_SPrintf fields may contain static text, or may contain a special string that indirectly gets the value from another source, such as the JDay attribute.
- Image Address (Fixed2) Length
 - Values 1- 9. Length of the Fixed2 part of the image address.
- Image Address (Fixed2) Format
 - Format string (ATTR_SPrintf) of the Fixed2 part of the image address.
- Image Address (DocIdx) Length
 - Values 1- 9. Length of the document index part of the image address.
- Image Address (DocIdx) Format
 - "<SequenceNum, %d,1>" is the default value. Format string (ATTR_SPrintf) of the document index part of the image address.
- Patch Scheme
 - 21 – Patch 2 starts a Doc and Patch 1 starts attachments (best practice).

- 32 – Patch 3 starts a Doc and Patch 2 starts attachments.
- Other Patches Used
 - Other patches that should be recognized (used for Super Batching).
 - In addition to the values specified in the Patch Scheme, valid values are 1, 2, 3, 4, 5, T.
- DCN Prefix
 - This may be used to specify an alphanumeric prefix to be prepended to the DCN (ATTR_SPrintf).
- DCN Suffix
 - Used to specify an alphanumeric suffix to be appended to the DCN (ATTR_SPrintf)

DCN Integrity

- First DCN Format
 - "<DCN_Prefix,%s,><JDay,%05s,YYJJJ><SequenceNum,%07d,0000001><DCN_Suffix,%s,> " is the default entry. The format string (ATTR_SPrintf) of the first DCN of the batch. This setting ensures that the image address has been set correctly in the scanner.

PCN

- PCN Prefix
 - Default value "<BatchNum,%08d,BBBBBBBB>PCN". This is a static prefix that is printed and added to the beginning of the Page Control Number.
- PCN Counter Width
 - Values 1- 9. This is the number of digits in the counter used in the Page Control Number.
- PCN Format
 - Default value "<PCN_Prefix><TWEI_TS_COUNTER,%0<CounterWidth>d,001>"

Page ID

- Page 1 Front ID – Page Variant ID of the main document page, commonly "HCFA05.Standard". "UB04.Standard", etc.
 - User may select from any of the available page variants currently configured for the system:
 - CMS Standard
 - UB Standard
 - Attachment Standard
 - Backside Standard

- Corro Standard
- DentalDemo Standard
- Page 1 Back ID – Page.Variant ID of the second document page backside if scanning both sides.
 - User may select from any of the available page variants currently configured for the system:
 - CMS Standard
 - UB Standard
 - Attachment Standard
 - Backside Standard
 - Corro Standard
 - DentalDemo Standard

Note: Page 2 Front through Page 5 Front and Page 2 Back through Page 5 back are additional Items available to the user for configuration. These values have been combined to reduce repetitiveness.

- Page 2 Front ID through Page 5 Front ID – The Page.Variant ID of X document page (X= page 2, 3, 4 or 5) backside if scanning both sides.
 - User may select from any of the available page variants currently configured for the system:
 - CMS Standard
 - UB Standard
 - Attachment Standard
 - Backside Standard
 - Corro Standard
 - DentalDemo Standard
- Page 2 Back ID through Page 5 Back ID – The Page.Variant ID of the X document page (X= page 2, 3, 4 or 5) backside if scanning both sides.

Batch Integrity

- Action When Too Many Docs
 - Error – When there are more documents scanned than indicated in Inventory; display an error (unoverridable) message (Best Practice).
 - Warn – When there are more documents scanned than indicated in Inventory; display an error (overridable) message.

- Action When Too Few Docs
 - Error – When there are more documents scanned than indicated in Inventory; display an error (unoverridable) message (Best Practice).
 - Warn – When there are more documents scanned than indicated in Inventory; display an error (overridable) message.
- Ultrasonic Double-Feed Detection
 - Disabled – Ultrasonic double feed detection is disabled.
 - Low – Low sensitivity, ultrasonic double-feed detection detects if two pieces of paper feed together.
 - Medium – Medium sensitivity, ultrasonic double-feed detection detects if two pieces of paper feed together.
 - High – High sensitivity, ultrasonic double-feed detection detects if two pieces of paper feed together.

Printer(Ink)

- Printer
 - Flag – Controls the physical printer to respect the GUI flag (Best Practice).
 - True – The physical printer is enabled.
 - False – The physical printer is disabled, the GUI flag is ignored,
- Note: Global GUI flag defaults may be changed in the config file.*
- Printer Offset
 - The default value is "4.5". Measured in fractional inches, this specifies far off the leading edge of the paper to start printing. Valid values ????
 - Image Address Prefix
 - The Image Address Prefix (enter value here) is a static prefix that is printed before (but not part of) the DCN.
 - Image Address Suffix
 - The Image Address Suffix (enter value here) is a static suffix that is printed after, (but not part of) the DCN.

Printer (Electronic)

- ePrinter
 - Flag (Best practice)
 - True
 - False

- ePrinter Format
 - Default value "<TWEI_TS_DCN>". Format string (ATTR_Sprintf) to be electronically imprinted on the image.
- ePrinter X-Offset
 - Default value "-1.0". The X offset (in inches) to start electronic imprinting, -1 centers the text on the image.
- ePrinter Y-Offset
 - Default value "0.3". The Y offset (in inches) to start electronic imprinting. Negative numbers indicate offset off the bottom of the image.
- ePrinter Font Size
 - Default value "36". The font size of imprinted text.
- ePrinter Font
 - Defaults to the system font if undefined, name of the font style for imprinted text.
- ePrinter Opacity
 - Transparent – The background opacity of the imprinted text. For Transparent, the text is XOR'd onto the image.
 - Opaque – The background opacity of the imprinted text. For Opaque, the area around the text is erased for legibility.

Scanner

- Paper Feeder Level
 - 100
 - 250
 - 500

The Paper Feeder Level controls how many pages to allow on the feeder at startup. This setting (100/250/500) affects how far down the feeder carriage is at startup.
- Transport Timeout (Seconds)
 - Default value "10". Time (in seconds) after which the transport will automatically timeout and return control back to the application.
- Flatbed Attachment
 - True – Allows (turns on) the control to allow an attached flatbed to be used. (i2xx)
 - False – Prohibits (turns off) the control not allowing an attached flatbed to be used, (i2xx)
- Scanner Mode

- Default value "4". User selectable from a value of "0" through "18". This specifies a pre-configured scanner mode when using a Kodak Digital Science or ImageLink scanner (i55xx, 75xx, 95xx).

Workflow

- Calculated work Record Attributes
 - Default value, "{DocCount", "<TWAINSCAN_DocCount>", "JDay", "<JDay,%s,00000>", "Production", "<Production,%s,0>", "RouteSelect", "<RouteSelect>"}". These are attributes that are calculated set into the record level of the work record.
- Calculated Item Attributes
 - Default value, "{DCN", "<TWEI_TS_DCN>", "Patch", "<TWEI_TS_PATCHCODE>", "PageSide", "<TWEI_HDR_PAGESIDE,%1d,0>", "PageSide_FileSize", "<TWEI_HDR_PAGESIDE,%1d,0>_<TWEI_TS_FILESIZE>", "BarCode", "<TWEI_TS_BARCODE>", "PCN", "<PCN_FormatStr,%s,>"}". Attributes that are calculated and set into each item of the work record.
- Page Deletion Regular Expressions
 - Default value, "{Patch", "[<DocSeparator_Patch,%s,3><IntraDocSeparator_Patch,%s,2>]", "PageID", "^Separator.Back\$|^Backside.Standard\$"}". Regular expressions that specify what pages to delete from the scanned stream.

Extension

- Extra Attribute Group Name
 - If specified and this group is present in the config, the group will be loaded. The purpose is to load extra attributes not available in the PCM. Valid values are attribute names that may be specified in the TWAINScan config file: a maximum of 31 alphanumeric characters long including underscores.
- Extra Capability Group Name
 - If specified and this group is present in the config, the group will be loaded. The purpose is to load extra attributes not available in the PCM. Valid values are attribute names that may be specified in the TWAINScan config file: a maximum of 31 alphanumeric characters long including underscores.

Appendix A: New System Checklist, Tips And Troubleshooting

New System Checklist

The list contained in this section is intended to be a general guideline, including the steps typically needed for a new system. As **each system is unique, some steps defined may not be required. If creating a new system by copying an existing system, many of these elements are already in place, but configurations and changes should be verified.**

Note: Not all steps and processes are defined within this checklist.

Preliminary System Setup

There are several preliminary tasks that may be done in preparation to receiving the business rules and other configuration information from the customer:

- Create Processing steps. (See [Creating a New Processing Step](#).)
 - Inventory
 - Scan
 - FIP
 - OCR
 - Collator
 - AC
 - OCRepair
 - Store
- Create Shared Resources.
 - Filerefs (See [Fileref Resource](#))
 - OCR_AEG (type OCR)
 - OriginalImage (type Image)
 - RemovedImage (type Image)
- AC Rule DLLs (See [Rule DLL Resource](#))
 - acsrules
 - acaeg
 - acstore
 - actcl
 - acengine
- Store output driver DLLs ([See Store DLL Resource](#))
 - store2out_tcl
 - store2out_odbc

- OCRRepair Colors
- OCRRepair Field Classes
- Inventory (See [Creating/Configuring an Inventory Resource](#))
- Configure Inventory Server host, port, type, etc.
- Return to Option Groups and enabling attributes in batches after have created the batches. Attributes and DCN groups will probably have to wait until you have detailed information from the customer.

Configure System-Level Settings for Processing Steps.

Scan (See [TWAINScan](#))

- Fileref Name (OriginalImage), device type (Scanner), Inventory Server (Inventory), inventory server login info

FIP (See [FIP Process](#))

- FIP Method DLLs; default is "f2stdplugin.dll"; probably good for most systems.
- FIP Worker DLL – This is the fipworker2_bws.dll; it is the source for the list of batch script steps.

OCR (See [OCR Process](#))

- OCR Output Format (rri5)
- OCR Output Fileref (OCR_AEG)
- Removed Image Fileref: (RemovedImage)
- Collator (See [Collator](#))
 - Collation Type: Attribute Change – use DCN attribute
 - OCR Filerefs: OCR_AEG
- AC (See [AC](#))
 - Rule DLLs – select the ones you use; DO NOT select acengine.dll here.
 - Default Sources – OCR_AEG
 - Do not select "Evaluate Pages of Rejected Documents" unless you know you need it.

Note that AC-I, AC-II, DV, KE, KFI all share this system-level configuration.

- Store (See [Store](#)).
 - Rule DLLs – select the ones you use; **DO NOT select acengine.dll** from here.

- Default Sources – OCR_AEG
- Store Configuration – The Store Configuration cannot be properly setup at this time, return to Store configuration after creating and defining the Store Template Sets (the output format).
- Inventory Server – Select the Inventory Server; fill in user info. Define Retire Attributes if necessary. Finish configuring Inventory after the batch types have been created.
- Retrieval – Fill in retrieval configuration.

System-Specific Configuration

Once detailed information from the customer has been provided, more specific system-level settings may be defined:

Complete Inventory configuration. (See [Creating/Configuring an Inventory Resource](#)).

- Configure [Inventory Attributes, DCN Groups](#).
- Configure Inventory Client (See also [Inventory \(Worker\)](#)).
- Create DateFormat Resources. (See [DateFormat Resource](#)).
 - The format of the dates you will need to convert from or to. Usually a MMDDYYYY and a YYYYMMDD, and sometimes a YYJJJ if needed to convert to or from a Julian date.
 - Create Table Resources. (See [Table Resources](#)).
 - Character Sets – As required.

Configure AEG with the proper information regarding what information to consider for a field (numbers and slashes and dashes for dates, numbers and decimal points for amounts, etc.).

Use in AC rules to test a field (make sure the name does not contain certain special characters, make sure the amount contains only numbers, etc.)

- Use in AC rules to filter (remove) characters (remove punctuation and spaces from account numbers, etc.).
- Tables (IndexedFile – fixed-width customer tables).

Use in AC Rules for various lookups.

- Template Sets (Output Format Definitions).
 - Usually a definition of an output record and its component slots for a flat file output.
 - For ODBC output, the name of the ODBC table, slots are the names of the columns.

- Selection List Resources
 - Used by AC at KE and DV for table lookups and plugging. These might be automatically created by the lookup rules in AC.
- Create Output Driver Resources. (See [Output Driver Resource](#))
 - Depending on the output format type, a FixedWidthTCL or ODBC output driver may be required. After configuring the template sets, configure the output driver. For fixed width flat file output, configure the width of each slot (column) and its padding and justification (left or right).
- Configure Store (Revisited). (See Store Process).
 - After creating output drivers and the template sets used by the store process (most commonly all) and the output drivers being used for the store step (again, most commonly all).
 - Reject Document Attributes: add “DCN”.
 - Reject Batch Attributes: add “BatchNum”.
 - As a best practice, leave “Evaluate Rejected Documents” unchecked, unless confirmation is received that the evaluation of rejected documents is required.
- Configure Retrieval settings, if using the FormWorks retrieval system.
 - Configure OCRepair processing steps. (See [OCRepair: Page Processing](#))
 - Select the fields to be processed at each OCRepair step
 - Generate a configuration file (system level).

Create Pages and Variants

For each page, complete the following steps:

- Create the page – When the page is created, the FormWorks Editor automatically assigns the default page variant name “Standard”. Choose the blank form (“overlay image” for dropouts) .TIFF to attach to the variant. (See [Creating Pages](#)).
- Create all the fields that might be used on the page – Start at the top left corner and proceed left to right and down the page. If a field on the form requires parsing into components, create the field components at this time.

Example: For the field “PatientName”, which will be parsed into “LastName”, “FirstName”, and “MiddleInitial”, create all four fields. Creating all four fields at this time saves from having to sort fields later on.

As a best practice, if fields on the form are numbered, create fields with this naming convention: “_2_PatientName”. (See [Adding Fields To A Page](#)).

Note: Be consistent in all naming conventions.

- Draw each field on the variant. (See [Defining Bounding Rectangles](#)).
- Create additional variants of the page, if any. (See [Adding Page Variants](#)).
- Configure processing steps for the page fields.
 - FIP – Only change configuration when necessary. Generally a change is required only for any bar-code fields or in some instances OMR fields. (See FIP: [Page Field Processing](#)).
 - OCR – Set the character set, number of lines, etc., for each field that is actually drawn on the image. Choose “Do NOT OCR this field” for any fields that are not to be captured. (See [OCR: Page Field Processing](#)).
- Configure the processing steps for each variant.
 - FIP – Image Processing: layout dropout regions for dropouts, registration lines for non-dropouts (See [FIP: Variant Image Layout, Defining Non-dropout Forms, and Dropout Regions](#)).
 - FIP – Variant Processing: choose “process as a dropout” if appropriate, choose methods for all steps that are appropriate.
 - Choose “STANDARD” or “AUTO” for everything if this is a form with data on it; leave items unchecked if it is an attachment that is not really processed.
 - Ignore the ‘Blank Detection’ setting. (See [FIP: Variant Page Processing](#)).
- Configure processing steps for the page itself.
 - AC – Configure the business rules. It may be useful to create some Function resources to re-use rules. (See [AC: Page Processing](#)).
 - Store – Configure the store rules; fill templates, queue templates. (See [Store: Page Processing](#)).
 - OCRepair – Select the fields that are to be processed at OCRepCrit (if using this step), then Numeric and other fields. (See [OCRepair: Page Processing](#)).

Create Documents

For each document, complete the following steps:

- Create the document. As a best practice, use the naming convention “XXX_Document” or “XXX_Doc”, where “XXX” is HCFA, UB92, Dental, etc.
- Utilizing this naming format, the document will not be easily confused with a page. When creating the document, select the page(s) that are a part of the document, see [Creating A New Document](#).
- Add any document fields to the document. Document fields are only needed if working multi-page claims, or if there may be a need for multi-page claims, see [Creating A New Document](#).
- Configure the processing steps for the document.
 - Collator – Set the number of expected pages for each page type. (See [Collator: Document Processing](#)).

- AC – If applicable, configure any AC/DV/KE rules.
 - The default document rule list should include an “_IEvalPages” rule; DO NOT MODIFY. (See [AC: Document Processing](#)).
- Store – Configure any Store rules for the document. (See [Store: Document Processing](#)).

Create Batches

For each batch, complete the following steps:

- Create the batch. When creating the batch, select the document(s) that should be in the batch (see [Creating a New Batch](#)).

Note: It is not uncommon for there to be only one possible document in a batch.

- Configure the processing steps for the batch.
 - Collator – When the batches are identical, set the batch to use the system-level collation settings. For batches that are non-identical, configure appropriately. (See [Collator: Batch Processing](#)).
 - Collation Technique – Select as configured at system level unless this batch is different than the system setting
 - Document ID – Compare to document definitions
 - Filerefs – Select create filerefs listed at the system level unless the filerefs for this batch are different than the system level
 - FIP Batch Script – Write the batch script. (See [FIP: Batch Processing](#)).
 - FIP Batch Processing – Choose a PageID method if this is a non-dropout batch. (See [FIP: Batch Processing](#)).
- Scan – Configure all the scan settings. (See [Scan: Batch Processing](#)).

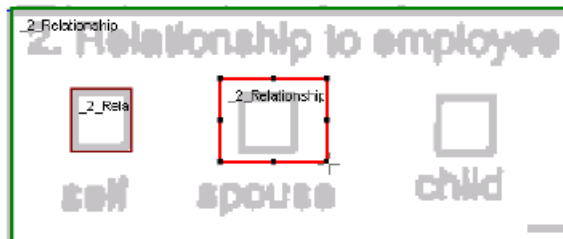
Finish configuring the Inventory Resource: Create a sequence of option groups and options to create each batch. If the batch selections do not fit into a decision tree format, create one option group called “BatchType”, then an option for each batch. Select the batch that corresponds to each option in the final group. After doing this, select on each option and press the "Configure" button next to the batch list. Set up each batch appropriately. (See [Create/Configure an Inventory Resource](#)).

Tips for Configuring a System

Defining Bounding Rectangles

- As a best practice, utilize the Aura option (OMR subfields to catch marks outside the checkbox). This option can be found in the FIP page level configuration window (see [FIP: Page Field Processing](#)).

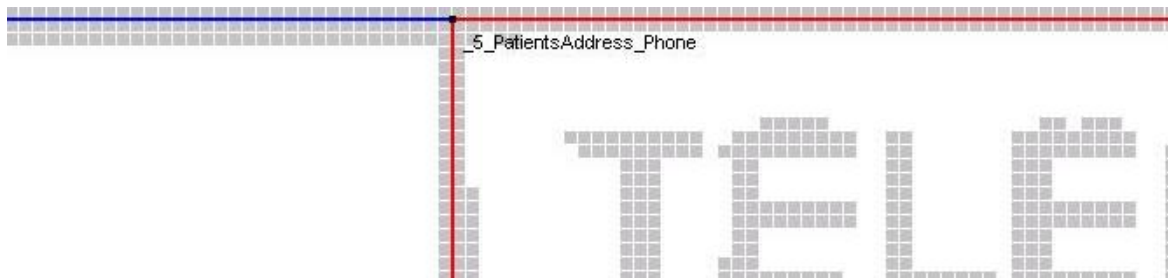
- Use the Snap feature (see Enabling/Disabling Snap Feature). When Snap is enabled, a box somewhat larger than the checkbox on the image may be drawn. See the Spouse subfield in the graphic below. After the mouse has been released, the rectangle “snaps” to the box on the image.



- Do not overlap fields – Overlapped bounding rectangles may result in the same character or characters being read in two different fields.

Don't leave gaps between adjacent fields. A character that falls exactly on the boundary between two fields will generally be picked up in one field or the other. If there is a gap, the character may not be read at all. Use the magnifier button to zoom in for a closer look.

[The](#) example below is a magnified view of properly drawn bounding fields (rectangles).



FIP

Important: Construction of the Batch script is imperative. If the Batch script is not correct, OCR files may be lost.

OCR

- **Lost Characters** – If characters are consistently being lost from the right side of a field, the noise suppression may have inadvertently be set. Right-click the page name (not the variant), Select Processing > OCR. In the OCR dialog box, select the field name, and check the noise suppression settings.
- **Misread Characters** – Select a character set that accurately reflects the expected contents of the field. For example, a field labeled Provider Number may actually (correctly) contain alphanumeric characters. If a numeric only character set is selected, OCR will read B as 8, D as 0, and so on. It is better to use an alphanumeric character set even if the character is read with a low confidence and must be keyed.

*Note: Verify that the page settings at the variant level have been overridden. Right click the **Variant Name**, select **Processing > OCR**. Remove any unnecessary overrides. The page level and variant level windows are very easy to confuse.*

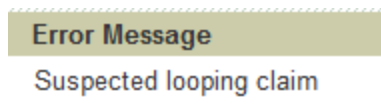
Troubleshooting

Looping

Looping, is in reference to when a set of instructions, rules or conditions contribute to items within the FormWorks system to continually repeat processing steps. Ideally a loop continues until a finite set of conditions are met, terminating the loop. However, if instructions, rules or conditions are not correctly defined, looping may continue unchecked.

Looping can cause the bridges to constantly process fields/item(s) while also attempting to service client requests. This scenario contributes to a decrease in the amount of keyable work available and a diminished work capacity. After an excessive number of work records are looped into the cycle, the effective capacity can be decreased to only a few work records, after which users will simply see the "Waiting for Work" message.

Looping is often initially identified with an error message displayed within Navigator (Alert Panel).



Within FWEEditor looping may principally be addressed within two primary areas.

1. Fields shown in OCRRepair that are not also shown in KE.
2. Batch looping between DV and KE.

As a best practice, contact your FIS technical representative before modifying or implementing changes to system configurations or functions.

Detection – Looping Fields In OCREpair Not Shown In KE

Most often, looping that is not reflected in KE may be detected by viewing the Alert Display Panel within Navigator, the error message Suspected looping claim is displayed.

Troubleshooting Looping Fields In OCREpair Not Shown In KE

The looping caused by fields identified in OCREpair that are not reflected in KE can generally be remedied by utilizing the function ClearCaptureError which removes capture errors from selected fields in order to prevent looping (see the AC Functions document, ClearCaptureError function for details).

Note: If the field in question is parsed into non-imaged fields, the OCREpair setting RejectTransferFields should be used to ensure the destination fields are populated at KE.

Batch Looping between DV And KE

Looping between a batch or batches may occur for multiple reasons.

- Tables on Back Ground Servers (BGServers) are not in sync with tables on the keying stations. An example of this scenario would be if DV did not find a lookup entry and put an error on the fields, then KE did find and remove the error, thus looping over and over.
- An error is on a field that is not defined in the EntryGrid, rendering the keyer unable to resolve the error.

Note: This scenario may often be caused by an OCREpair Operator rejecting a field with F7 line reject, or an AC Function, for example AlignInRows.

- A selection list lookup is looping. Occurring when the selection list inserts a value into a field, and that field changes value due to the application of a separate rule.